# Quantum Computing - Notes - v0.7.0

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# Preface

Every theory section in these notes has been taken from the sources:

- Course slides. [3]

About:

GitHub repository

These notes are an unofficial resource and shouldn't replace the course material or any other book on quantum computing. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

# Contents

# 1 Introduction

## 1.1 Complex Numbers

Let's not start from math. Let's start from **physics / computation intuition**. Classical bits are represented by two distinct states: 0 and 1. In quantum computing, we have **qubits**, which are essentially **vector of numbers**. But not just any numbers: a qubit is a vector of **complex numbers**. They are needed because quantum systems have **both magnitude and phase**, and complex numbers can represent both. Real numbers can only represent "*how much*" (magnitude), while complex numbers can represent "*how much*" and "*in which direction / phase*" (magnitude and phase).

### ❓ Why complex (and not just real) numbers?

1. In quantum computing, we don't store probabilities directly (e.g., 0.5 for a 50% chance). Instead, we store **probability amplitudes**, which are complex numbers. The probability of an outcome is the square of the magnitude of its amplitude:
$$P = |z|^2$$
2. **Interference**. Quantum states can **combine and cancel**. For example, if we have two states with amplitudes $z_1$ and $z_2$, the combined amplitude is:
$$z_{\text{combined}} = z_1 + z_2$$
If $z_1$ and $z_2$ have the same magnitude (*how much*) but opposite phases (*in which direction*), they can cancel each other out, leading to zero probability for that outcome. This is a key feature of quantum mechanics that allows for phenomena like quantum interference, which is essential for quantum algorithms.

In general, complex numbers are needed because **quantum states behave like waves**, not like classical values. Waves have both amplitude and phase, and complex numbers are the natural way to represent them mathematically. This is why quantum mechanics and quantum computing rely heavily on complex numbers.

### ❓ What do we actually need?

Not everything. Just a **minimal toolkit**. A complex number is:

$$z = x + iy$$

Where $x$ is the **real part** and $y$ is the **imaginary part**, and $i$ is the **imaginary unit** satisfying $i^2 = -1$. From a geometric interpretation, we can think of $z$ as a point in the 2D plane, where $x$ is the horizontal coordinate (real axis) and $y$ is the vertical coordinate (imaginary axis).

## Polar form

Instead of coordinates $(x, y)$, we can also represent a complex number in **polar form**:

$$z = re^{i\varphi}$$

Where:

- $r = |z| = \sqrt{x^2 + y^2}$ is the **modulus** (or *magnitude*) of the complex number, representing its distance from the origin in the complex plane.
- $\varphi = \arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$ is the **argument** (or *phase* angle), representing the angle between the positive real axis and the line connecting the origin to the point $z$ in the complex plane.
- The **exponential form** $e^{i\varphi}$ means that we can represent the complex number as a point on the unit circle in the complex plane, where $\varphi$ is the angle from the positive real axis. The modulus $r$ then scales this point to its actual distance from the origin.

  Using Euler's formula:

  $$e^{i\varphi} = \cos\varphi + i\sin\varphi$$

  We can rewrite $z$ as:

  $$z = r(\cos\varphi + i\sin\varphi)$$

  This form is particularly useful in quantum computing because it makes it easy to understand how complex numbers can represent both magnitude and phase, which are essential for describing quantum states and their evolution.

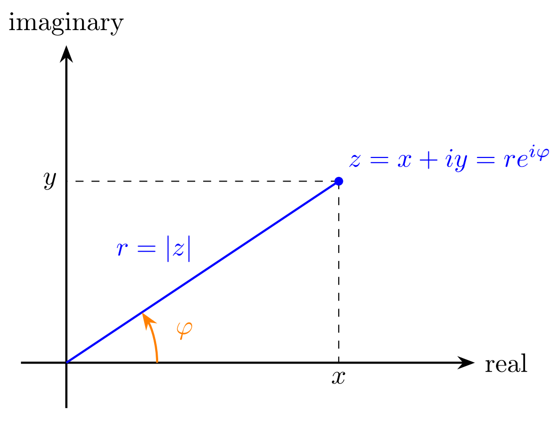


Figure 1: Geometric interpretation of a complex number in the complex plane. The arrow represents the complex number $z$ as a vector from the origin to the point $(x, y)$. The length of the vector is the modulus $r$, and the angle it makes with the positive real axis is the argument $\varphi$.

## Complex Conjugate

The **complex conjugate** of a complex number $z = x + iy$ is denoted as $\bar{z}$ and is defined as:

$$\bar{z} = x - iy$$

Geometrically, the complex conjugate $\bar{z}$ is the **reflection** of $z$ **across the real axis in the complex plane**. This means that if $z$ is represented by the point $(x, y)$, then $\bar{z}$ is represented by the point $(x, -y)$. The real part remains the same, while the imaginary part changes sign. See Figure 2 for a visual representation of this concept.

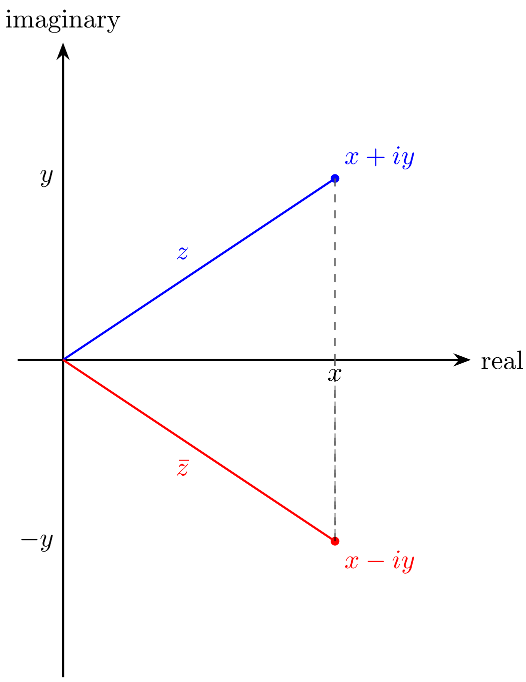


Figure 2: Geometric interpretation of the complex conjugate: reflection across the real axis.

## Magnitude

The **magnitude** (or **modulus**) of a complex number $z = x + iy$ is denoted as $|z|$ and is defined as:

$$|z| = \sqrt{x^2 + y^2}$$

And it can also be expressed using the complex conjugate:

$$|z|^2 = z \cdot \bar{z} = (x + iy)(x - iy) = x^2 + y^2$$

The magnitude represents the **distance** of the point $z$ from the origin in the complex plane. It is always a non-negative real number. The magnitude is important in quantum mechanics because the probability of an outcome is given by the square of the magnitude of the corresponding probability amplitude. For example, if a quantum state has an amplitude of $z$, the probability of measuring that state is $|z|^2$. This is why the magnitude of complex numbers is crucial in quantum computing, as it directly relates to the probabilities of different outcomes when measuring quantum states.

### Rotation in the complex plane

Multiplying a complex number by a complex exponential of the form $e^{i\theta}$ corresponds to a **rotation** in the complex plane:

$$z' = z \cdot e^{i\theta} = re^{i(\varphi+\theta)}$$

Geometrically, this operation:

- **Preserves the magnitude** $|z|$
- **Adds** $\theta$ **to the phase** $\varphi$

Therefore, multiplication by $e^{i\theta}$ rotates the complex number by an angle $\theta$ around the origin.

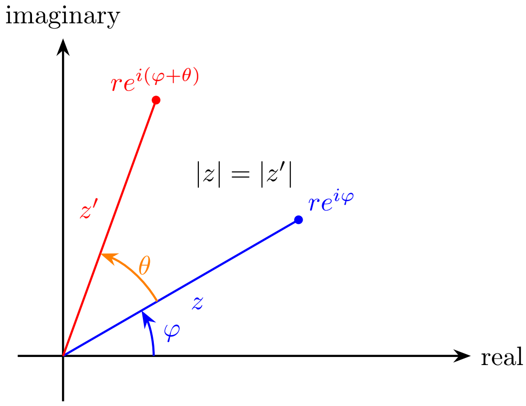


Figure 3: Rotation of a complex number: multiplying by $e^{i\theta}$ rotates the vector by angle $\theta$ while preserving its magnitude.

### Hermitian (Conjugate Transpose)

Given a complex vector:

$$\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$$

Its **Hermitian** (or **conjugate transpose**) is defined as:

$$\mathbf{v}^\dagger = \begin{bmatrix} \bar{a} & \bar{b} \end{bmatrix}$$

This operation consists of:

- Taking the **transpose** (column $\rightarrow$ row)
- Taking the **complex conjugate** of each element

## 1.2 Dirac's Notation

Dirac notation was introduced to make **linear algebra with complex vectors easier, cleaner, and more expressive**, especially for quantum systems.

Let's imagine our work only with standard linear algebra notation, where we have to write out all the vectors and matrices in their full form:

- Vectors:
$$\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$$
- Conjugate transpose:
$$\mathbf{v}^\dagger = \begin{bmatrix} \bar{a} & \bar{b} \end{bmatrix}$$
- Inner product:
$$\mathbf{v}^\dagger \mathbf{w} = \bar{a}c + \bar{b}d$$
- Matrix-vector multiplication:
$$M\mathbf{v} = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} m_{11}a + m_{12}b \\ m_{21}a + m_{22}b \end{bmatrix}$$

This quickly becomes **cumbersome and hard to read**, especially as we deal with more complex quantum systems. In quantum mechanics we constantly write expressions involving inner products, matrix operations, and state transformations.

**Dirac's idea.** Paul Dirac said "*let's give different roles **different symbols***". So he introduced:

- **Kets** $|v\rangle$ to represent **column vectors** (quantum states).
- **Bras** $\langle v|$ to represent row vectors (**conjugate transposes**).

Where the operators stay in the middle, and we can write everything in a much cleaner way:

- Vectors (**kets**):
$$|v\rangle = \begin{bmatrix} a \\ b \end{bmatrix} \tag{1}$$
Where the entries are complex numbers and usually we require $|v\rangle$ to have unit norm (normalized state).
- Conjugate transpose (**bra**):
$$\langle v| = \begin{bmatrix} \bar{a} & \bar{b} \end{bmatrix} \tag{2}$$
- Inner product:
$$\langle v|w\rangle = \bar{a}c + \bar{b}d \tag{3}$$
The result is a scalar (complex number) that tells us **how much the two states "overlap"**. If $\langle v|w\rangle = 1$ then $|v\rangle$ and $|w\rangle$ are **identical** (up to a global phase, they overlap completely). If $\langle v|w\rangle = 0$ then $|v\rangle$ and

$|w\rangle$ are **orthogonal** (completely different states, no overlap). Otherwise, if $|\langle v|w\rangle| < 1$ then $|v\rangle$ and $|w\rangle$ are partially **similar** (some overlap, but not identical). So the inner product is basically a notion of "distance" or "similarity" between two quantum states.

- Matrix-vector multiplication:

$$M|v\rangle = \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} m_{11}a + m_{12}b \\ m_{21}a + m_{22}b \end{bmatrix} \tag{4}$$

  An operator $M$ acting on a state $|v\rangle$ gives us a new state. The result is a new ket.

Even if we ignore physics for now, we can see that Dirac's notation is more expressive and easier to read than standard linear algebra notation.

### ❓ What is *Chaining* in Dirac notation?

**Chaining** means writing multiple operations together in a single expression without needing to write out intermediate steps. For example:

$$\langle x| M |y\rangle$$

It is just a **compact way to combine linear algebra operations** (matrix multiplication and inner products) in a single expression. It allows us to write complex quantum mechanical expressions in a much cleaner and more intuitive way.

Without Dirac notation, we would have to write:

$$\mathbf{x}^\dagger M \mathbf{y}$$

That is a **chain of operations**: take the conjugate transpose of $\mathbf{x}$, multiply it by $M$, and then multiply the result by $\mathbf{y}$. With Dirac notation, we can write this in a much more compact and readable way as $\langle x| M |y\rangle$.

To read $\langle x| M |y\rangle$, we can think of it as:

1. Start with the ket $|y\rangle$ (a quantum state).
2. Apply the operator $M$ to $|y\rangle$, which gives us a new state $M|y\rangle$.
3. Take the inner product of $\langle x|$ with the new state $M|y\rangle$, which gives us a scalar (complex number) that represents the probability amplitude of transitioning from state $|y\rangle$ to state $|x\rangle$ via the operator $M$. At first, this might seem abstract, but as we go through more examples and applications in quantum mechanics and quantum computing, it will become much clearer how powerful and intuitive Dirac's notation is for describing quantum systems.

# 2 Single Qubit States

Before introducing qubits, we briefly describe what we mean by a **quantum system**.

A **Quantum System** is a physical system (such as a photon, an electron, or an atom) whose behavior is described by the laws of quantum mechanics.

In this course, we will focus on the simplest case: systems that can produce **two possible outcomes** when measured. These are called **two-level systems**, and they form the basis of qubits.

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## 2.1 Building and Measuring Qubits (Intuition)

Before introducing the mathematical formalism of qubits, it is useful to develop an **intuitive understanding** of how a simple quantum system behaves when it is **prepared** and **measured**.

### A Simple Quantum Experiment

Consider a very simple experimental setup:

- A **source** emits a quantum particle (for example, a photon).
- The particle passes through a device.
- A **detector** measures the outcome.

Each time we perform the experiment, the detector returns one of two possible results:

$$0 \quad \text{or} \quad 1$$

This motivates the idea of a **two-level system**, which is the simplest type of quantum system.

### Key Observation

If we repeat the same experiment many times under identical conditions, we observe that:

- The outcome is **not deterministic**.
- Instead, it is **probabilistic**.

For example, we might obtain:

- Outcome 0 with probability $p$
- Outcome 1 with probability $1 - p$

This is fundamentally different from classical systems, where the outcome is determined in advance.

### What is a Measurement?

A **Measurement** is the process that:

- Extracts **classical information** from a quantum system;
- Produces a **definite outcome** (e.g., 0 or 1).

An important property of quantum measurement is that it **affects the system itself**. After a measurement is performed, the **system is no longer in its original configuration**.

### From Intuition to Formalization

This simple experiment suggests that:

- A quantum system can produce **two possible outcomes**
- The outcomes are governed by **probabilities**
- The system must therefore contain some form of **internal state** that determines these probabilities

In the following example, we will introduce a concrete physical example (photon polarization) that allows us to represent this internal state using **vectors with complex coefficients**.

**Example 1: Example of a Two-Level Quantum System**

To make the previous discussion more concrete, we introduce a simple physical system that behaves as a two-level quantum system: the **polarization of a photon**.

### What is a Photon?

A **Photon** is the elementary particle of light. It carries energy and behaves both as a particle and as a wave. In this course, we will use photons as a simple example of a **quantum system**, since their properties can be easily controlled and measured.

### What is Polarization?

**Polarization** describes the direction in which the electric field of a light wave oscillates. In simple terms, it can be thought of as the **orientation** of the light wave. For a photon, polarization can be visualized as a direction in space. In particular, we consider two basic polarization directions:

- **horizontal** ($\rightarrow$)
- **vertical** ($\uparrow$)

These two directions form a **two-level system**, since a measurement can distinguish between them (i.e., we can determine whether the photon is horizontally or vertically polarized).

### Superposition of Polarizations

A photon is not restricted to being only horizontal or vertical. In general, it can be in a **combination** of both states:

$$v = a \rightarrow + b \uparrow$$

Where:

- $a$ and $b$ are **complex coefficients**;
- They determine the behavior of the system when measured.

This is the first example of a **quantum state**: a system described by a linear combination of basic states.

### Vector Representation

We can represent these polarization states using vectors:

$$\rightarrow = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \uparrow = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

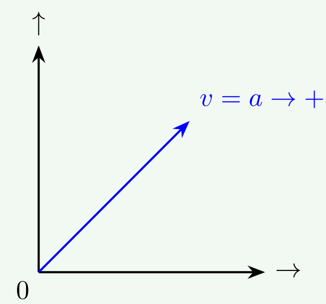
Therefore, a general polarization state becomes:

$$v = \begin{bmatrix} a \\ b \end{bmatrix}$$

This shows that the state of the system can be described as a **vector with complex components**, exactly as introduced in Dirac notation.

### Geometric Interpretation

The horizontal and vertical polarization states can be interpreted as two **orthogonal axes** in a two-dimensional space.




The point where the axes intersect represents the origin, and any polarization state can be represented as a **vector starting from the origin**.

Photon $\longrightarrow$ A ($\rightarrow$) $\longrightarrow$ B ($\uparrow$) $\longrightarrow$ Screen

In this representation:

- $\rightarrow$ and $\uparrow$ form a **basis**
- Any state is a **linear combination** of these basis states

### Measurement with Polarizers

A **Polarizer** is a device that **filters light based on its polarization**. A polarizer acts like a **filter** that allows only one polarization direction to pass and blocks the other. For example:

- A polarizer aligned with the horizontal direction ($\rightarrow$) will allow horizontally polarized light to pass and block vertically polarized light.
- A polarizer aligned with the vertical direction ($\uparrow$) will allow vertically polarized light to pass and block horizontally polarized light.

If the photon is a combination:

$$v = a \rightarrow + b \uparrow$$

And we use a:

- Horizontal polarizer ($\rightarrow$), the photon will pass with probability $|a|^2$ and be absorbed with probability $|b|^2$.
- Vertical polarizer ($\uparrow$), the photon will pass with probability $|b|^2$ and be absorbed with probability $|a|^2$.

This shows that:

- Probabilities are determined by the amplitudes (i.e., the coefficients $a$ and $b$).
- Measurement **modifies the state of the system** (e.g., if the photon passes through the horizontal polarizer, it will be in the state $\rightarrow$).

## 2.2 From Polarization to Qubits

The example of photon polarization (page 13) shows that a physical system can be described using two basic states and their linear combinations. We now **abstract** this idea and introduce a **general model that does not depend on a specific physical system**.

### From Physical States to Abstract States

In the polarization example, we used:

$$\rightarrow \quad \text{and} \quad \uparrow$$

As the two basic states of the system. We now replace them with an abstract notation:

$$\rightarrow \rightsquigarrow |0\rangle \qquad \uparrow \rightsquigarrow |1\rangle$$

These are called the **computational basis states**. They are just symbols (used in Dirac notation) that represent the two basic states of the system, without referring to any specific physical implementation.

### Definition of a Qubit

A **Qubit** is a **two-level quantum system** whose state can be written as:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

It is the **unit of quantum information**, analogous to a classical bit. However, unlike a classical bit that can only be in one of two states (0 or 1), a qubit can exist in **infinitely many possible states**. Indeed, any normalized vector of the form:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Represents a **valid state** of a qubit. Some **properties** of the qubit state are:

- The coefficients $a$ and $b$ are **complex numbers** (probability amplitudes). They determine the outcome of a measurement:
  - $|a|^2$ is the probability of measuring the state $|0\rangle$.
  - $|b|^2$ is the probability of measuring the state $|1\rangle$.

  Therefore, even though a qubit can be in infinitely many possible states, a measurement (in this basis) can only produce **two possible outcomes**: $|0\rangle$ or $|1\rangle$.

- The state is **normalized**:

  $$|a|^2 + |b|^2 = 1$$

  Because the probabilities must sum to 1 (i.e., we must get either $|0\rangle$ or $|1\rangle$ when we measure).

**Definition 1: Normalized Quantum State**

A quantum state $|\psi\rangle$ is said to be **normalized** if its norm is equal to one:

$$\||\psi\rangle\| = 1$$

Equivalently, using the inner product:

$$\langle\psi|\psi\rangle = 1$$

For a single-qubit state:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

This condition becomes:

$$|a|^2 + |b|^2 = 1 \tag{5}$$

This is required because $|a|^2$ and $|b|^2$ represent measurement probabilities, and the total probability of all possible outcomes must be equal to 1.

- The states $|0\rangle$ and $|1\rangle$ form a **basis** of the vector space describing the qubit.

**What is a *basis*?** A **Basis** is a set of vectors that allows us to **represent any vector in the space as a linear combination of them**. In quantum computing, $|0\rangle$ and $|1\rangle$ are the **basis vectors** and any qubit state is built from them.

In particular, the states $|0\rangle$ and $|1\rangle$ form an **Orthonormal Basis**, meaning that:

- They are **orthogonal**: $\langle 0|1\rangle = 0$.
  **Why is orthogonality important?** Orthogonality means that the two basis states are completely distinguishable by a measurement (not overlapping). If we measure a qubit in the state $|0\rangle$, we will never get $|1\rangle$, and vice versa.
- They are **normalized**: $\langle 0|0\rangle = \langle 1|1\rangle = 1$.
  **Why is normalization important?** Normalization ensures that the probabilities of measurement outcomes are well-defined and sum to 1. It also means that the basis states have unit length in the vector space, which is a fundamental property for quantum states.

### Why This Abstraction Matters

This abstraction is fundamental because it allows us to:

- Describe **any** two-level quantum system (not only photons);
- Use a unified mathematical framework based on vectors;
- Apply the tools of linear algebra to quantum systems.

From now on, we will work with qubits using Dirac notation, without referring to a specific physical implementation.

**Example 2: Different Choices of Basis**

The computational basis $\{|0\rangle, |1\rangle\}$ is not the only possible choice. A qubit can be described using **any orthonormal basis**.

For example, another important basis is the **Hadamard basis**, defined as:

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \qquad |-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

This shows that the same quantum state can be represented in different bases, depending on the context.

**Example 3: Orthogonality of the Hadamard Basis**

Show that the Hadamard basis states $|+\rangle$ and $|-\rangle$ are orthogonal.

*Proof.* Recall:

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \qquad |-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

Orthogonal means that their inner product is zero. So, we need to compute $\langle +|-\rangle$ and show that it equals zero:

$$\langle +|-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\end{bmatrix} \cdot \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix} = \frac{1}{2}(1-1) = 0$$

Therefore, $|+\rangle$ and $|-\rangle$ are orthogonal. QED

**Example 4: Change of Basis from Standard $\leftrightarrow$ Hadamard**

Given a qubit state in the standard basis:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

We want to express it in the Hadamard basis:

$$|\psi\rangle = x\,|+\rangle + y\,|-\rangle$$

*Proof.* Using the definitions of $|+\rangle$ and $|-\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} \quad |-\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

We can write $|\psi\rangle$ in terms of $|+\rangle$ and $|-\rangle$:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle = a\cdot\begin{bmatrix}1\\0\end{bmatrix} + b\cdot\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}a\\b\end{bmatrix}$$

Now, we express $|\psi\rangle$ as a linear combination of $|+\rangle$ and $|-\rangle$:

$$|\psi\rangle = x\,|+\rangle + y\,|-\rangle$$

Substituting the definitions of $|+\rangle$ and $|-\rangle$:

$$= x\cdot\frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} + y\cdot\frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\end{bmatrix}$$

Combining the terms:

$$= \frac{1}{\sqrt{2}}\begin{bmatrix}x+y\\x-y\end{bmatrix}$$

We want this to equal $\begin{bmatrix}a\\b\end{bmatrix}$, so we set:

$$\frac{1}{\sqrt{2}}\begin{bmatrix}x+y\\x-y\end{bmatrix} = \begin{bmatrix}a\\b\end{bmatrix}$$

This gives us the system of equations:

$$x+y = \sqrt{2}a, \qquad x-y = \sqrt{2}b$$

Adding the two equations:

$$2x = \sqrt{2}(a+b) \implies x = \frac{a+b}{\sqrt{2}}$$

Subtracting the second from the first:

$$2y = \sqrt{2}(a-b) \implies y = \frac{a-b}{\sqrt{2}}$$

It follows that:

$$|\psi\rangle = x\,|+\rangle + y\,|-\rangle = \frac{a+b}{\sqrt{2}}\,|+\rangle + \frac{a-b}{\sqrt{2}}\,|-\rangle$$

QED

**Example 5: Orthogonal State Construction**

Given:

$$|v\rangle = a\,|0\rangle + b\,|1\rangle$$

Define:

$$|w\rangle = \bar{b}\,|0\rangle - \bar{a}\,|1\rangle$$

Show that $|v\rangle$ and $|w\rangle$ are orthogonal.

*Proof.* To show that $|v\rangle$ and $|w\rangle$ are orthogonal, we need to compute their inner product $\langle w|v\rangle$ and show that it equals zero. First, we write the inner product converting $|w\rangle$ to its corresponding bra:

$$|w\rangle = \bar{b}\,|0\rangle - \bar{a}\,|1\rangle \implies \langle w| = b\,\langle 0| - a\,\langle 1|$$

Now we compute:

$$\langle w|v\rangle = (b\,\langle 0| - a\,\langle 1|) \cdot (a\,|0\rangle + b\,|1\rangle)$$

Expanding this expression:

$$= ba\,\langle 0|0\rangle + bb\,\langle 0|1\rangle - aa\,\langle 1|0\rangle - ab\,\langle 1|1\rangle$$

Using the orthonormality of the basis states:

$$\langle 0|0\rangle = \langle 1|1\rangle = 1, \quad \langle 0|1\rangle = \langle 1|0\rangle = 0$$

We get:

$$\langle w|v\rangle = ba\cancelto{1}{\langle 0|0\rangle} + \cancel{bb\,\langle 0|1\rangle} - \cancel{aa\,\langle 1|0\rangle} - ab\cancelto{1}{\langle 1|1\rangle} = ba - ab$$

Since $a$ and $b$ are complex numbers, multiplication is commutative, so $ba = ab$. Therefore:

$$\langle w|v\rangle = ba - ab = 0$$

Hence, $|v\rangle$ and $|w\rangle$ are orthogonal. QED

## 2.3 Single Qubit Measurement

After introducing the mathematical representation of a qubit, we now describe how information is extracted from it through **measurement**.

### ❓ What is *Single Qubit Measurement?*

We have a qubit:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Where $a$ and $b$ are complex numbers such that $|a|^2 + |b|^2 = 1$. Measurement answers the question: "*which classical value do we observe?*".

A measurement takes a **quantum state** as input and produces a **classical outcome** as output. In the case of a single qubit, the possible outcomes are 0 and 1.

### ⚠ Measurement rule (computational basis)

The probability of an outcome is the **squared magnitude of the coefficient of the basis vector**:

$$\mathrm{P}(0) = |a|^2, \quad \mathrm{P}(1) = |b|^2 \quad \text{(in the computational basis)}$$

*But which coefficients?* This is where the **basis** comes in. A state can be written in different ways depending on the basis. For example, take:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

- If we measure in basis $\{|0\rangle\,, |1\rangle\}$:
  - Outcome 0 with probability $|a|^2$
  - Outcome 1 with probability $|b|^2$
- If we measure in basis $\{|+\rangle\,, |-\rangle\}$:
  - Outcome $+$ with probability $|x|^2$
  - Outcome $-$ with probability $|y|^2$

The **coefficients depend on the chosen basis**. Let's analyze how a basis change affects the coefficients.

**❓ What is a basis *really?*** Any measurement device (e.g., a polarizer, page 13) has two preferred states (i.e., the states that it can distinguish). A basis is not just a mathematical construct; it corresponds to **what our measurements device can distinguish**. For example:

- Computational basis, the device distinguishes: *are you* $|0\rangle$ *or* $|1\rangle$*?*
- Hadamard basis, the device distinguishes: *are you* $|+\rangle$ *or* $|-\rangle$*?*

These are **different physical equations**.

**The math behind the measurement.** Measurement is projecting the state vector onto basis vectors. Forget qubits for a second and imagine a 2D real vector $\vec{v}$, and choose axes:

- **Standard axes**: we project onto $x$-axis and $y$-axis. We get components: $v_x$ and $v_y$.
- **Rotated axes**: we rotate axes by 45° and project onto new axes the same vector $\vec{v}$. We get new components: $v_+$ and $v_-$.

**Different numbers, same vector**. The components are **not intrinsic properties of the vector**, but depend on the chosen axes. Back to qubits, **measurement projects the vector** $|\psi\rangle$ **onto the measurement axes** (i.e., the basis vectors). So $\langle u|\psi\rangle$ is the projection of $|\psi\rangle$ onto $|u\rangle$, and $|\langle u|\psi\rangle|^2$ is the probability of observing outcome $u$.

**Example 6: Analogy - Measurement as projecting a vector onto different axes**

Consider a classical 2D vector pointing diagonally:

$$\vec{v} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

**Case 1: Standard axes (horizontal/vertical).** If we describe the vector using the standard basis:

$$\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\},$$

Then the vector has equal components along both axes:

$$\vec{v} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- Projection on horizontal axis: $\frac{1}{\sqrt{2}}$
- Projection on vertical axis: $\frac{1}{\sqrt{2}}$

**Case 2: Rotated axes (diagonal basis).** Now rotate the coordinate system by 45° and use the basis:

$$\left\{ \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$$

In this basis, the same vector becomes:

$$\vec{v} = 1 \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 0 \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

- Projection on first axis: 1
- Projection on second axis: 0

**Connection with qubits.** A qubit behaves exactly in the same way:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle = |+\rangle$$

- In basis $\{|0\rangle, |1\rangle\}$: equal projections $\Rightarrow$ probabilistic measurement.
- In basis $\{|+\rangle, |-\rangle\}$: full projection on $|+\rangle$ $\Rightarrow$ deterministic measurement.

The state does not change. What changes is the set of axes (basis) onto which the state is projected. Since measurement probabilities are given by these projections, the outcome depends on the chosen basis.

Given the state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

- If we measure in the **computational basis**, we get:
  - Outcome 0 with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$
  - Outcome 1 with probability $\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$

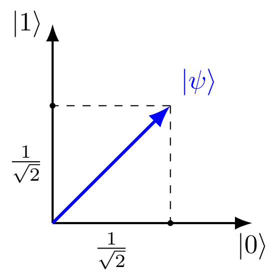



Figure 4: Measurement as projection onto basis vectors.

- If we measure in the **Hadamard basis**, we get:
  - Outcome + with probability $|\langle +|\psi\rangle|^2 = 1$
  - Outcome − with probability $|\langle -|\psi\rangle|^2 = 0$

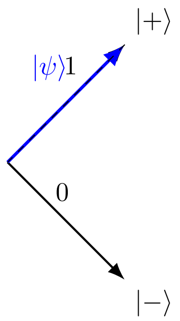



Figure 5: Measurement in the Hadamard basis. The state $|\psi\rangle$ is aligned with $|+\rangle$, so we get outcome + with probability 1. The projection onto $|-\rangle$ is zero, so we get outcome − with probability 0. The axes are rotated by 45° compared to the computational basis.

**What happens to the state after measurement?** The state **collapses** to the observed outcome. For example, if we measure $|\psi\rangle = a|0\rangle + b|1\rangle$ in the computational basis and observe 0, the state collapses to $|0\rangle$. If we observe 1, the state collapses to $|1\rangle$.

### Two possible situations

As we have seen in page 22, the same state can yield different measurement outcomes depending on the chosen basis. This leads to two important situations:

- **Probabilistic measurement**: If the **state is not aligned with the measurement basis**, we get a probabilistic outcome. For example:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

  Measured in the computational basis gives 0 or 1 with equal probability.

- **Deterministic measurement**: If the **state is aligned with one of the basis vectors**, we get a deterministic outcome. For example:

$$|\psi\rangle = |+\rangle$$

  Measured in the Hadamard basis gives + with probability 1 and − with probability 0.

**When should we prefer one situation to another?** It depends on the task. For example, if we want to encode a classical bit, we prefer deterministic measurement (e.g., $|0\rangle$ or $|1\rangle$). If we want to create superposition states for quantum algorithms, we might want probabilistic measurement (e.g., $|+\rangle$).

Now, if the same quantum state can lead to either deterministic or probabilistic measurement outcomes depending on the chosen basis, we can ask: "*how can be explained this phenomenon?*". It means that a qubit can simultaneously have non-zero components along multiple basis states. This is a fundamental property of quantum mechanics, and it is what allows quantum computers to perform certain computations more efficiently than classical computers. It is called **superposition**, and we will discuss it in the next section.

## 2.4 Superposition

We have already seen that:

1. **A qubit is represented as a vector**:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

   Where $a$ and $b$ are complex numbers, and $|0\rangle$ and $|1\rangle$ are the basis states of the qubit.

2. **Measurement probabilities depend on the chosen basis**:
   - In the computational basis $\{|0\rangle, |1\rangle\}$, the probabilities of measuring $|0\rangle$ and $|1\rangle$ are $|a|^2$ and $|b|^2$, respectively.
   - In another basis, such as the Hadamard basis $\{|+\rangle, |-\rangle\}$, the probabilities of measuring $|+\rangle$ and $|-\rangle$ depend on different coefficients, which can be calculated by expressing $|\psi\rangle$ in terms of the new basis states.

3. **The same state can lead to deterministic or probabilistic outcomes depending on the measurement basis**. For example, if $|\psi\rangle = |0\rangle$, measuring in the computational basis always yield $|0\rangle$ (deterministic):

$$|0\rangle = 1|0\rangle + 0|1\rangle = |0\rangle$$

   Whereas measuring in the Hadamard basis yields $|+\rangle$ or $|-\rangle$ with equal probability (probabilistic):

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

These observations raise an essential question: "*why can a single state exhibit such different behaviors under different measurements?*". The answer is **superposition**.

### ❓ What is Superposition?

**Superposition** is a fundamental principle of quantum mechanics that **applies to all quantum systems**, not just qubits.

**Definition 2: Superposition**

**Superposition** means that a **system can exist in several possible states at the same time until a measurement is made**. It is **basis-dependent**, meaning a state may be a superposition in one basis but not in another, for example:

$$|v\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Is a superposition with respect to the computational basis $\{|0\rangle, |1\rangle\}$, but it is **not** a superposition in the Hadamard basis $\{|+\rangle, |-\rangle\}$, where it can be written as:

$$|v\rangle = 1\,|+\rangle + 0\,|-\rangle = |+\rangle$$

Formally, let $\{|u\rangle, |u^{\perp}\rangle\}$ be an orthonormal basis of a single-qubit Hilbert space. A qubit is said to be in **superposition** with respect to this basis if it can be written as:

$$|\psi\rangle = a\,|u\rangle + b\,|u^{\perp}\rangle$$

Where both coefficients $a$ and $b$ are non-zero and satisfy $|a|^2 + |b|^2 = 1$ (normalization condition). If one of the coefficients is zero, the state coincides with one of the basis vectors and the measurements in that basis becomes deterministic, thus it is not a superposition.

More in general, a quantum state $|\psi\rangle$ in a system with multiple possible states can exist in a **linear combination** of these states:

$$|\psi\rangle = c_1\,|\psi_1\rangle + c_2\,|\psi_2\rangle + \cdots + c_n\,|\psi_n\rangle$$

Where:

- $|\psi_1\rangle, |\psi_2\rangle, \ldots, |\psi_n\rangle$ are **basis states** of the system.
- $c_1, c_2, \ldots, c_n$ are **complex probability amplitudes**.
- The system is in **all states at the same time**, with the probability of measuring each state given by $|c_i|^2$.
- The state is **normalized**, meaning:

$$|c_1|^2 + |c_2|^2 + \cdots + |c_n|^2 = 1$$

**Example 7: Single-Particle Systems (Quantum Mechanics)**

In general quantum mechanics, a particle can exist in **multiple positions simultaneously** as a wave function $\Psi(x)$:

$$|\Psi\rangle = \int \Psi(x)\;|x\rangle\;\mathrm{d}x$$

- The particle is in a **superposition of all possible positions** $|x\rangle$.
- Measurement collapses the wave function to a single position.

**How Can a Particle Exist in Multiple States at Once?** This question touches the heart of quantum mechanics, where our everyday intuition breaks down. The **key idea** is that a **quantum particle** is not just a tiny ball, it **is a wave function that spreads across multiple possibilities at once**.

1. **Quantum Particles Are Waves, Not Just Points**. In classical physics, we think of particles as tiny, solid objects, like a small ball that always has a precise position and velocity.

   In quantum mechanics, however, particles behave more like waves. These waves are described by a wave function $\Psi(x)$, which represents the **probability of finding the particle at different locations**.

   The key idea is:

   - The **wave function spreads across space**, meaning the **particle does not have a single location** before measurement.
   - Instead, it exists in a superposition of all possible locations.

2. **The Double-Slit Experiment: Proof That a Particle Can Be in Two Places at Once**. The double-split experiment was first performed by Thomas Young in 1801 to demonstrate the **wave nature of light**. Later, in the 20th century, it was repeated with **electrons**, **atoms**, and even **large molecules**, revealing that **matter itself exhibits wave-like behavior**. The classical expectation for the double-slit experiment is:

   - **Classical Particles** (e.g., tiny balls). Imagine firing small balls at a barrier with two slits:
     - Each ball passes through **either** slit 1 **or** slit 2.
     - On a screen behind the slits, we would observe **two bright bands**, corresponding to the two slits.

     The result is a simple pattern of two lines, no interference.
   - **Classical Waves** (e.g., Water or Light). If instead we send **waves** (like water waves or classical light):
     - Each wave passes through **both slits simultaneously**.
     - The waves emerging from the slits **interfere**: **constructive interference** creates bright bands, while **destructive interference** creates dark bands.

     The result is an **interference pattern** of alternating bright and dark bands.

   The quantum reality is even more surprising:

   - **Quantum Particles** (e.g., photons or electrons). When the experiment is performed with **single quantum particles** (photons or electrons), something remarkable happens:
     (a) **Particles are emitted one at a time**.
     (b) Each particle is detected as a **localized point** on the screen.
     (c) After many particles are detected, an **interference pattern gradually emerges**.

     This means that each particle behaves as if it passes through **both slits simultaneously**. The particle is described by a **wave function**, representing a **superposition of paths**:

     $$|\psi\rangle = \frac{1}{\sqrt{2}}(|\text{slit }1\rangle + |\text{slit }2\rangle)$$

The interference pattern arises from the **coherent combination** of these two possibilities.

Suppose we place detectors at the slits to determine **which slit** the particle passes through. The interference pattern **disappears** and the screen now shows **two bands**, just like classical particles. This is because measuring the particle's path **entangles** (i.e., correlates) it with the environment (the detector), leading to **decoherence** (loss of quantum coherence). The superposition:

$$\frac{1}{\sqrt{2}}\left(|\text{slit } 1\rangle + |\text{slit } 2\rangle\right)$$

Becomes effectively a **classical mixture**:

$$\frac{1}{2}\,|\text{slit } 1\rangle\,\langle\text{slit } 1| + \frac{1}{2}\,|\text{slit } 2\rangle\,\langle\text{slit } 2|$$

Without coherence, **interference cannot occur**.

In summary, the double-slit experiment demonstrates the principle of quantum superposition. When quantum particles such as photons or electrons are sent one at a time through two slits, they form an interference pattern on a detection screen. This indicates that each particle behaves as if it travels through both slits simultaneously, corresponding to the superposition:

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|\text{slit } 1\rangle + |\text{slit } 2\rangle\right)$$

If a measurement is performed to determine which slit the particle passes through, the interference pattern disappears due to decoherence, and the system behaves like a classical probabilistic mixture.

*Double Slit Experiment explained! by Jim Al-Khalili*

3. **Quantum Superposition: More Than Just Probability**. A common misconception is that a particle in superposition is just **an unknown state**, like a coin that is either heads or tails, but we just don't know which. This is wrong, because quantum superposition is much deeper.

   A quantum state is a **combination of all possibilities**. *Until measurement*, the system is in **all possible states at once**.

   Mathematically, for an electron in **two locations** $x_1$ and $x_2$:

   $$|\psi\rangle = a\,|x_1\rangle + b\,|x_2\rangle$$

   - The electron is **literally in both places simultaneously**.
   - The coefficients $a$ and $b$ are **complex numbers representing the probability amplitudes**.

- **Interference** between these amplitudes **creates quantum effects that cannot be explained by classical probability**.

4. **Superposition in Quantum Computing**. Quantum computing **directly uses** the fact that a particle can be in multiple states at once.

   - A **classical bit** can only be **0 or 1**.
   - A **qubit (quantum bit)** can be in a superposition:

   $$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

   This means a quantum computer can **perform many calculations simultaneously**. Therefore, **superposition allows quantum computers to process information exponentially faster than classical computers for certain tasks**.

5. **Why Don't We See Superposition in Everyday Life?** In our daily experience, objects are **not in multiple states at once** because of a process called **quantum decoherence**.

   Quantum superposition is **fragile**. When a quantum system interacts with the environment (air, light, etc.), the superposition **collapses into one definite state**. This is why large objects (like humans or cars) do not appear in multiple places at once.

   However, experiments (like the Double-Slit experiment) confirm that superposition is real at microscopic scales (electrons, photons, atoms, and even molecules).

The key **properties of Superposition** are:

1. **Linearity**: Any combination of valid quantum states is also a valid quantum state.

2. **Interference**: Quantum states in superposition can interfere, leading to constructive or destructive interference.

3. **Measurement Collapse**: When measured, the superposition collapses into a single outcome.

4. **Phase Information**: Unlike classical probabilities, quantum superpositions include complex phases that affect interference patterns.

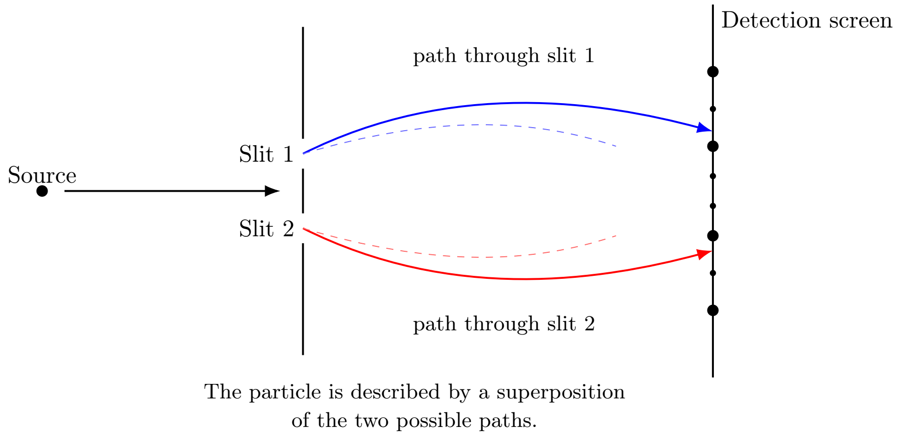



Figure 6: Double-slit experiment, superposition of paths. This figure illustrates the **quantum state of the particle before detection**. Instead of choosing a single path, the particle is described by a **superposition**:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|\text{slit } 1\rangle + |\text{slit } 2\rangle\right)$$

The particle is **not** traveling through only one slit, instead, it is described by a **coherent combination** of both possibilities. The diagram emphasizes the **quantum state** rather than the final inference pattern.

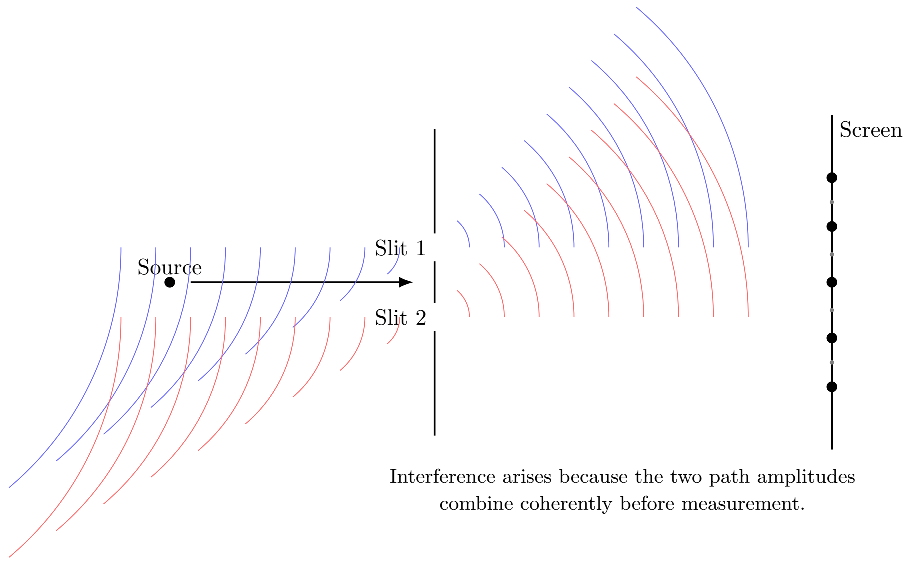



Figure 7: Wave-like representation of the double-slit experiment, interference of amplitudes. This figure focuses on the **wave nature** of the quantum particle. Each slit acts as a **coherent source**, producing wavefronts that overlap and interfere. If $A_1(x)$ and $A_2(x)$ are the probability amplitudes associated with the two slits, the probability of detecting the particle at position $x$ on the screen is:

$$P(x) = |A_1(x) + A_2(x)|^2 = |A_1(x)|^2 + |A_2(x)|^2 + 2\mathrm{Re}\left(A_1^*(x)A_2(x)\right)$$

The last term is the **inference term**, which can be positive (bright fringes, **constructive interference**) or negative (dark fringes, **destructive interference**). This illustrates how the superposition of paths leads to the characteristic interference pattern observed in the experiment.

## 2.5 Information Carried by a Single Qubit

This section addresses a fundamental question: ***if a qubit can exist in infinitely many states, how much information can it actually convey?*** A general single-qubit state is given by:

$$|\psi\rangle = a|0\rangle + b|1\rangle \qquad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1$$

Because $a$ and $b$ are complex numbers, a qubit can be prepared in **infinitely many distinct states**. This might suggest that a qubit can encode an infinite amount of information. However, this is **not** the case. When we perform a **measurement**, we obtain only a **single classical outcome** (either 0 or 1). Therefore, a single qubit can yield at most one classical bit of information when measured. This apparent paradox, **infinite possible states but limited measurable information**, is one of the most important conceptual aspects of quantum information theory.

### ❓ Why can't we determine the Amplitudes $a$ and $b$ of a single qubit state?

Suppose we are given a qubit in an unknown state $|\psi\rangle$. Can we determine $a$ and $b$ from a single measurement? The answer is **no**, because:

- ✗ **Measurement is probabilistic**. A single measurement provides only one outcome (0 or 1), which is insufficient to determine the amplitudes.
- ✗ **Measurement disturbs the state**. After measurement, the qubit collapses to the observed basis state, preventing further measurements on the original state.
- ✗ **State estimation requires many copies**. To estimate $a$ and $b$, we need **quantum state tomography**, which relies on performing measurements on **many identically prepared qubits**.

This **irreversible collapse** means that the original information encoded in the amplitudes is generally lost after measurement.

### ⚠ The No-Cloning Theorem

One might think of copying the qubit before measurement to extract more information. However, quantum mechanics forbids this through the **No-Cloning Theorem** (we will see more about this on page 173).

**Theorem 1 (No-Cloning).** *It is impossible to create an identical copy of an arbitrary unknown quantum state.*

*Proof.* Assume there exists a unitary operator $U$ that can clone any state $|\psi\rangle$:

$$U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \quad \forall |\psi\rangle$$

For two arbitrary states $|\psi\rangle$ and $|\phi\rangle$, linearity of quantum mechanics implies:

$$U(\alpha|\psi\rangle + \beta|\phi\rangle)|0\rangle = \alpha|\psi\rangle|\psi\rangle + \beta|\phi\rangle|\phi\rangle$$

Which is **not equal** to:

$$(\alpha |\psi\rangle + \beta |\phi\rangle)(\alpha |\psi\rangle + \beta |\phi\rangle) = \alpha^2 |\psi\rangle |\psi\rangle + \alpha\beta |\psi\rangle |\phi\rangle + \alpha\beta |\phi\rangle |\psi\rangle + \beta^2 |\phi\rangle |\phi\rangle$$

This contradiction proves that perfect cloning is impossible, since the two expressions cannot be equal for all $|\psi\rangle$ and $|\phi\rangle$. QED

Since we cannot clone a qubit, we cannot create multiple copies to determine its exact state from a single instance.

| Feature | Classical Bit | Qubit |
|---|---|---|
| Possible states | 0 or 1 | Qubit |
| Info from measurement | 1 bit | At most 1 bit |
| Measurement disturbance | No | Yes |
| Ability to copy | Yes | No (No-Cloning) |
| State determination | Single observation | Requires many identical copies |

Table 1: Comparison of classical bits and qubits in terms of information content and measurement properties.

## 2.6 State Space of a Single-Qubit System

The **State Space** of a physical system is the set of all possible states that the system can occupy. For a **single qubit**, the state space consists of all vectors of the form:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

- $a, b \in \mathbb{C}$ are **complex probability amplitudes**.
- $|0\rangle$ and $|1\rangle$ form an **orthonormal basis** (the computational basis).
- The **normalization condition** must hold: $|a|^2 + |b|^2 = 1$. This ensures that the total probability of all measurement outcomes is equal to 1.

### Degrees of Freedom of a Qubit

At first glance, the coefficients $a$ and $b$ appear to provide **4 real parameters**:

$$\begin{aligned} a = a_r + ia_i \quad &\Rightarrow \quad a_r, a_i \quad \text{(2 parameters)} \\ b = b_r + ib_i \quad &\Rightarrow \quad b_r, b_i \quad \text{(2 parameters)} \end{aligned}$$

So initially, it seems that a qubit has **four degrees of freedom**. However, not all of these correspond to physically distinguishable states.

- **Normalization constraint**. Quantum states must be normalized, which imposes the condition:

  $$|a|^2 + |b|^2 = a_r^2 + a_i^2 + b_r^2 + b_i^2 = 1$$

  Geometrically, it defines the surface of a **unit sphere in four dimension** (also known as a 3-sphere, $S^3$). This constraint effectively removes **one degree of freedom**, since the amplitudes must lie on the surface of a 3-sphere in 4-dimensional space.

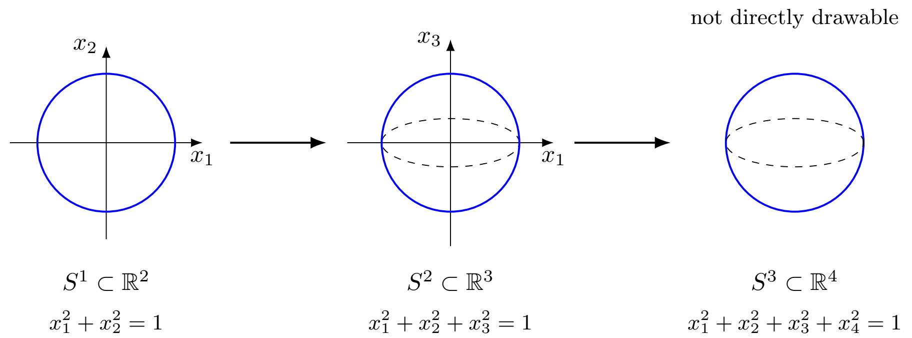



Figure 8: Analogy between the unit circle $S^1$, the ordinary sphere $S^2$, and the 3-sphere $S^3$. The normalized qubit state lives on $S^3$ because it is specified by four real coordinates constrained by one equation. In this figure, we can only visualize $S^1$ and $S^2$, while $S^3$ cannot be directly drawn in three-dimensional space, so we represent it with a dashed ellipse to indicate that it is a higher-dimensional object.

- **Global Phase Equivalence**. Two states that differ only by a **global phase** represent the **same physical qubit**:

$$|\psi'\rangle = e^{i\phi} |\psi\rangle$$

For example:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad e^{i\pi/3} |\psi\rangle = \frac{e^{i\pi/3}}{\sqrt{2}}(|0\rangle + |1\rangle)$$

❓ **Why is it physically irrelevant?** Measurement probabilities are given by:

$$P(i) = |\langle i|\psi\rangle|^2$$

If we multiply the state by a global phase:

$$\left|\langle i|e^{i\gamma}\psi\rangle\right|^2 = \left|e^{i\gamma}\langle i|\psi\rangle\right|^2 = |\langle i|\psi\rangle|^2$$

Because $\left|e^{i\gamma}\right| = 1$ for any real $\gamma$.

*Mathematical proof that* $\left|e^{i\gamma}\right| = 1$. We can express $e^{i\gamma}$ using Euler's formula:

$$e^{i\gamma} = \cos(\gamma) + i\sin(\gamma)$$

The magnitude of a complex number $z = x + iy$ is given by:

$$|z| = \sqrt{x^2 + y^2}$$

Applying this to $e^{i\gamma}$:

$$\begin{aligned}
\left|e^{i\gamma}\right| &= \sqrt{(\cos(\gamma))^2 + (\sin(\gamma))^2} \\
&\downarrow \text{using the trigonometric identity} \\
&= \sqrt{\cos^2\gamma + \sin^2\gamma} \\
&= \sqrt{1} \\
&= 1
\end{aligned}$$

QED

Therefore, **no experiment can distinguish between** $|\psi\rangle$ **and** $e^{i\phi}|\psi\rangle$, since **they yield the same measurement probabilities** (due to $\left|e^{i\phi}\right| = 1$).

❓ **How does this remove a degree of freedom?** Unlike normalization, the global phase does **not** impose a constrain equation. Instead, it introduces a **redundancy** in our description:

  - Many different vectors in $S^3$ correspond to the **same physical state**.
  - All vectors of the norm $e^{i\gamma}|\psi\rangle$ are considered **physically equivalent** to $|\psi\rangle$.

Mathematically, we are **identifying** all these vectors as a single point. This process is called forming a **quotient space**: the set of physically distinct pure qubit states is $S^3 \setminus S^1 \cong S^2$, which is a 2-dimensional surface (the Bloch sphere, see Figure 8, page 34).

After accounting for these constraints, we find that a single qubit state is fully described by **two real parameters** (two degrees of freedom). Therefore, any qubit state can be written in the form:

1. A normalized qubit can be written as:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle \qquad a, b \in \mathbb{C} \quad |a|^2 + |b|^2 = 1$$

2. Any complex number can be written using its **magnitude** and **phase**:

$$a = |a|\,e^{i\alpha} \qquad b = |b|\,e^{i\beta}$$

Where $|a|$ and $|b|$ are non-negative real numbers, and $\alpha$ and $\beta$ are real numbers representing the phases. Substituting these into the expression for $|\psi\rangle$ gives:

$$|\psi\rangle = \underbrace{|a|\,e^{i\alpha}}_{a}\,|0\rangle + \underbrace{|b|\,e^{i\beta}}_{b}\,|1\rangle$$

3. We can factor out the common global phase $e^{i\alpha}$:

$$|\psi\rangle = e^{i\alpha}\left(|a|\,|0\rangle + |b|\,e^{i(\beta-\alpha)}\,|1\rangle\right)$$

Since the global phase $e^{i\alpha}$ has **no physical effect**, we can omit it, leading to:

$$|\psi\rangle = |a|\,|0\rangle + |b|\,e^{i\phi}\,|1\rangle$$

Where we define also the **Relative Phase** $\phi = \beta - \alpha$. At this point, the state is characterized by two real magnitudes $|a|$ and $|b|$, and one relative phase $\phi$. Mathematically, if:

$$a = |a|\,e^{i\alpha} \qquad b = |b|\,e^{i\beta}$$

Then:

$$\text{Relative Phase} = \phi = \arg(b) - \arg(a) = \beta - \alpha \tag{6}$$

4. From normalization:

$$|a|^2 + |b|^2 = 1$$

We recognize a familiar trigonometric identity:

$$\cos^2(x) + \sin^2(x) = 1$$

Therefore, we can parametrize the magnitudes using a single angle $\theta$:

$$|a| = \cos\left(\frac{\theta}{2}\right) \qquad |b| = \sin\left(\frac{\theta}{2}\right)$$

With $0 \le \theta \le \pi$ to ensure that $|a|$ and $|b|$ are non-negative. The reason for using $\frac{\theta}{2}$ instead of $\theta$ will become clear when we discuss the Bloch sphere representation (Figure 9, page 38).

5. Substituting the parametrized magnitudes into the state:

$$|\psi\rangle = \underbrace{\cos\left(\frac{\theta}{2}\right)}_{|a|} |0\rangle + e^{i\phi} \underbrace{\sin\left(\frac{\theta}{2}\right)}_{|b|} |1\rangle \tag{7}$$

This is the **canonical parametrization** of a single qubit state.

Regardless of the specific values of $\theta$ and $\phi$ in Equation 7:

- $\theta$: controls the relative weights of $|0\rangle$ and $|1\rangle$ in the superposition, determining the probabilities of measuring each state.
- $\phi$: controls the relative phase between the $|0\rangle$ and $|1\rangle$ components. While it does not affect measurement probabilities, it is crucial for interference effects and the behavior of the qubit under quantum gates (which we will discuss later).

### Bloch Sphere Representation of a Qubit

After normalization and removal of the physically irrelevant global phase, a pure single-qubit state depends on only two real parameters. Therefore, it is natural to represent it geometrically as a point on a two-dimensional surface. This surface is the **Bloch sphere**.

Any pure qubit can be written in the form:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle ,$$

Where:

- $\theta \in [0, \pi]$ is the **polar angle**,
- $\phi \in [0, 2\pi)$ is the **Azimuthal Angle**, representing the **relative phase**.

This expression gives a one-to-one correspondence between pure qubit states and points on the surface of a unit sphere:

- $|0\rangle$ corresponds to the north pole, since $\theta = 0$:

$$|0\rangle = \cos\left(\frac{0}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{0}{2}\right) |1\rangle = 1 \cdot |0\rangle + 0 \cdot |1\rangle = |0\rangle$$

- $|1\rangle$ corresponds to the south pole, since $\theta = \pi$:

$$|1\rangle = \cos\left(\frac{\pi}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\pi}{2}\right) |1\rangle = 0 \cdot |0\rangle + e^{i\phi} \cdot 1 \cdot |1\rangle = e^{i\phi} |1\rangle \Rightarrow |1\rangle$$

**Attention!** $e^{i\phi} |1\rangle$ becomes $|1\rangle$ after **removing the global phase** (physically unobservable), so $|1\rangle$ is indeed at the south pole. In general, all states of the form $e^{i\phi} |1\rangle$ represent the **same physiical state** as $|1\rangle$. Hence, $|1\rangle$ corresponds to the **south pole** of the Bloch sphere.

- States on the equator correspond to equal-weight superpositions of $|0\rangle$ and $|1\rangle$.

In particular:

$$\text{Hadamard} \quad |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle),$$

$$\text{Imaginary} \quad |i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle), \quad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle),$$

All lie on the equator and differ only by their relative phase.

The Bloch sphere provides an intuitive visualization of a qubit: moving along the vertical direction changes the relative weights of $|0\rangle$ and $|1\rangle$, while rotating around the vertical axis changes the relative phase between them.

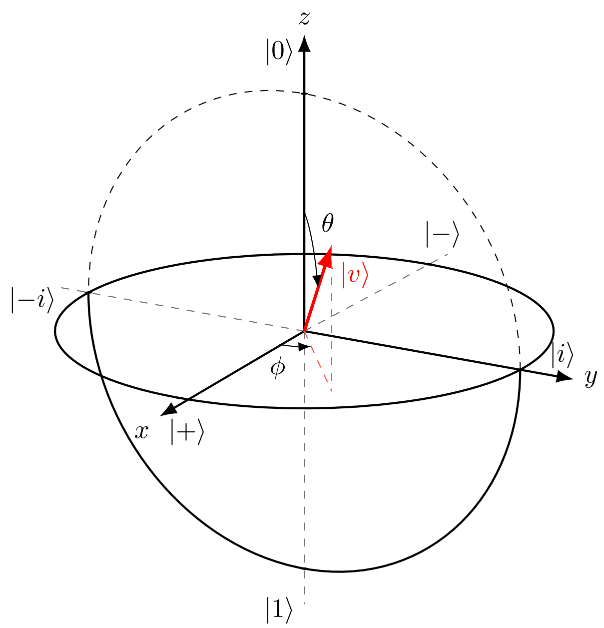



Figure 9: Bloch sphere representation of a pure qubit state. The red vector $|v\rangle$ represents a generic qubit state. Measuring $|v\rangle$ along the $z$-axis gives probabilities determined by $\theta$, while along $x$ or $y$-axes reveals information about the relative phase $\phi$.

On the Bloch sphere, the axes correspond to **three fundamental measurement bases** for a qubit. Each pair of opposite points represents the **eigenstates of one of the Pauli operators** (we will discuss the Pauli operators in more detail in 3.3.2.2, page 49):

- **$z$-axis**, **computational basis** $\{|0\rangle, |1\rangle\}$, eigenstates of $\sigma_z$. The states are:

  $$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \qquad \text{and} \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

  These states correspond to the classical bit values 0 and 1.

  - $|0\rangle$: North pole of the Bloch sphere $(0, 0, 1)$.
  - $|1\rangle$: South pole of the Bloch sphere $(0, 0, -1)$.

  Measuring along the $z$-axis answer the question: "*is the qubit* $|0\rangle$ *or* $|1\rangle$*?*".

- **$x$-axis**, **Hadamard basis** $\{|+\rangle, |-\rangle\}$, eigenstates of $\sigma_x$. The states are:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \qquad \text{and} \qquad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

  - $|+\rangle$: $+x$ direction on the equator $(1, 0, 0)$, with $\phi = 0$ (no relative phase).
  - $|-\rangle$: $-x$ direction on the equator $(-1, 0, 0)$, with $\phi = \pi$ (relative phase of $\pi$ between $|0\rangle$ and $|1\rangle$).

  These states represent **equal superpositions** of $|0\rangle$ and $|1\rangle$ with **relative phase** 0 and $\pi$.

- **$y$-axis**, **imaginary basis** $\{|i\rangle, |-i\rangle\}$, eigenstates of $\sigma_y$. The states are:

$$|i\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle) \qquad \text{and} \qquad |-i\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle)$$

  - $|i\rangle$: $+y$ direction on the equator $(0, 1, 0)$, with $\phi = \dfrac{\pi}{2}$ (relative phase of $\dfrac{\pi}{2}$ between $|0\rangle$ and $|1\rangle$).
  - $|-i\rangle$: $-y$ direction on the equator $(0, -1, 0)$, with $\phi = -\dfrac{\pi}{2}$ (relative phase of $-\dfrac{\pi}{2}$ between $|0\rangle$ and $|1\rangle$).

  These states also have equal probabilities of measuring $|0\rangle$ or $|1\rangle$ in the computational basis, but they differ by a **relative phase of** $\pm\frac{\pi}{2}$. This phase is crucial for **interference effects** in quantum algorithms.

# 3 Single Qubit Gates

## 3.1 Operations on Qubits

Quantum computing relies on three essential operations:

1. **Logic Gates**: transform quantum states.
2. **Measurements**: extract classical information from qubits.
3. **Initialization Procedures**: prepare qubits in known starting states.

### Logic Gates: Reversible Transformations

Quantum **Logic Gates** are the building blocks of quantum circuits. They **manipulate the state of qubits in a controlled and deterministic way**, analogous to classical logic gates (like AND or NOT), but with key quantum differences:

- **Operate on Single or Multiple Qubits**. They can affect one qubit (single-qubit gates) or multiple qubits simultaneously (multi-qubit gates):
  - **Single-Qubit Gates**: These gates, such as the Pauli-X, Y, Z, Hadamard (H), and Phase (S) gates, operate on a single qubit to change its state. We will explore these gates in detail in the next sections.
  - **Multi-Qubit Gates**: These gates, such as the CNOT (Controlled-NOT) and Toffoli gates, operate on multiple qubits and can create *entanglement* (we will discuss this in detail later) between them, which is a key resource in quantum computing.
- **Reversible Transformations**. Unlike many classical gates (e.g., AND), **all valid quantum gates are reversible**. Mathematically, this means they are represented by **unitary matrices** $U$, satisfying:

  $$U^\dagger U = UU^\dagger = I$$

  Where $U^\dagger$ is the conjugate transpose of $U$, $I$ is the identity matrix, and $U$ is a unitary matrix[1]. Because of this reversibility, quantum gates do not lose information and the original state can be recovered by applying $U^\dagger = U^{-1}$ to the output state. This is a fundamental difference from classical logic gates, which can be irreversible (e.g., AND gate loses information about the inputs).

  Finally, this property is related to fundamental principles such as the **no-cloning theorem** (page 32, page 173) and the **no-deleting theorem**

[1] A matrix $U$ is called **unitary** if its **conjugate transpose equals its inverse**:

$$U^\dagger U = UU^\dagger = I \tag{8}$$

We remind the reader that the definition of the inverse of a matrix is:

$$U^{-1}U = UU^{-1} = I$$

So for a unitary matrix, the conjugate transpose $U^\dagger$ plays the role of the inverse. This means that unitary transformations preserve vector norms, orthogonality, and therefore total probability in quantum mechanics.

(we cannot delete quantum information arbitrarily, since it is preserved under unitary transformations, page 182).

- **State Evolution**. If a qubit is in the state:

  $$|\psi\rangle = a|0\rangle + b|1\rangle$$

  Then after applying a gate $U$, the new state becomes:

  $$|\psi'\rangle = U|\psi\rangle = aU|0\rangle + bU|1\rangle$$

  This **transformation is deterministic** and **preserves the normalization of the state** (i.e., $|a|^2 + |b|^2 = 1$ remains true after applying $U$). The specific effect of the gate depends on its matrix representation, which we will explore in the next sections.

We can think of quantum logic gates as **rotations on the Bloch sphere**, smoothly transforming the qubit's state without destroying its quantum nature.

### Measurement: Irreversible Collapse of State

**Measurement** is the process of extracting information from a qubit. Unlike logic gates, measurement is **irreversible** and fundamentally probabilistic.

- **Irreversible Process**. Once a qubit is measured, its original quantum state is **lost**. This distinguishes measurement from unitary gate operations.

  **What is a unitary gate?** A **Unitary Gate** is a quantum gate represented by a **unitary matrix**. More precisely, a quantum gate $U$ is unitary if (see page 40):

  $$U^\dagger U = UU^\dagger = I$$

  Where $U^\dagger$ is the conjugate transpose of $U$, and $I$ is the identity matrix. Since a unitary matrix has an inverse equal to its conjugate transpose, then every quantum gate is **reversible**.

  **Important Property.** A fundamental property of **unitary gates** is that they **preserve similarity transformations**. This means that they preserve **geometric relationships** between quantum states, such as lengths (norms), angles, and inner products:

  $$\langle\psi|\phi\rangle = \langle U\psi|U\phi\rangle \tag{9}$$

  Where $|\psi\rangle$ and $|\phi\rangle$ are any two quantum states. For example, if two states are orthogonal (i.e., $\langle\psi|\phi\rangle = 0$), then after applying a unitary gate $U$, they remain orthogonal (i.e., $\langle U\psi|U\phi\rangle = 0$). In other words, if two states are "very close", they remain "very close" after applying a unitary transformation.

- **State Collapse**. Measuring a qubit in the computational basis $\{|0\rangle, |1\rangle\}$ causes the state:

  $$|\psi\rangle = a|0\rangle + b|1\rangle$$

  To **collapse** to:

  - $|0\rangle$ with probability $|a|^2$.
  - $|1\rangle$ with probability $|b|^2$.

- **Extraction of Classical Information**. The measurement outcome is a **classical bit** (0 or 1). After measurement, subsequent measurements in the same basis yield the same result with probability 1, since the state has collapsed to a definite state.

Measurement can be described using **projection operators**:

$$M_0 = |0\rangle\langle 0| \quad M_1 = |1\rangle\langle 1|$$

The probability of obtaining outcome $k$ (where $k \in \{0, 1\}$) when measuring $|\psi\rangle$ is given by:

$$\begin{aligned}
p_k &= \langle\psi| M_k |\psi\rangle \\
&\downarrow \text{substitute } M_k = |k\rangle\langle k| \\
&= \langle\psi| (|k\rangle\langle k|) |\psi\rangle \\
&\downarrow \text{using the associativity of inner products} \\
&= (\langle\psi|k\rangle)(\langle k|\psi\rangle) \\
&\downarrow \text{since } \langle\psi|k\rangle = (\langle k|\psi\rangle)^* \\
&\quad \text{where * denotes complex conjugation} \\
&= (\langle k|\psi\rangle)^* (\langle k|\psi\rangle) \\
&\downarrow \text{which is the definition of squared magnitude, page 8} \\
&= |\langle k|\psi\rangle|^2
\end{aligned}$$

In other words, the probability of measuring outcome $k$ is the **squared magnitude of the amplitude** of $|\psi\rangle$ along the direction $|k\rangle$. And the post-measurement state of the qubit, conditioned on obtaining outcome $k$, becomes:

$$|\psi_k\rangle = \frac{M_k |\psi\rangle}{\sqrt{p_k}}$$

Measurement is like **asking a question** to the qubit: "are you in state $|0\rangle$ or $|1\rangle$?". The answer is classical (i.e., 0 or 1), but the act of asking **changes the system**.

**Example 1: Calculating Measurement Probabilities**

Consider a qubit in the state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

With $|a|^2 + |b|^2 = 1$.

- Probability of measuring 0. Using the projector:

$$M_0 = |0\rangle\langle 0|$$

We compute:

$$\begin{aligned} p_0 &= \langle\psi| M_0 |\psi\rangle \\ &= \langle\psi|0\rangle \langle 0|\psi\rangle \\ &= (\langle\psi|0\rangle)(\langle 0|\psi\rangle) \\ &\downarrow \text{since } \ \langle\psi|0\rangle = (\langle 0|\psi\rangle)^* \\ &= |\langle 0|\psi\rangle|^2 \\ &= |a|^2 \end{aligned}$$

- Probability of measuring 1. Using the projector:

$$M_1 = |1\rangle \langle 1|$$

We compute:

$$\begin{aligned} p_1 &= \langle\psi| M_1 |\psi\rangle \\ &= \langle\psi|1\rangle \langle 1|\psi\rangle \\ &= (\langle\psi|1\rangle)(\langle 1|\psi\rangle) \\ &\downarrow \text{since } \ \langle\psi|1\rangle = (\langle 1|\psi\rangle)^* \\ &= |\langle 1|\psi\rangle|^2 \\ &= |b|^2 \end{aligned}$$

### Initialization: Preparing Known States

The **Initialization** is the process of preparing a qubit in a **known and controlled state**, typically $|0\rangle$ or $|1\rangle$. This step is essential because quantum algorithms require a well-defined starting point. It is implemented through various methods, including:

- **Physical Preparation**. Many quantum technologies naturally relax to the ground state $|0\rangle$. Additional gates (e.g., Pauli-X) can be applied to obtain $|1\rangle$ if needed.
- **Initialization via Measurement**. Measuring a qubit projects it into a basis state. If the outcome is not the desired one, a corrective gate can be applied. For example, if measurement yields $|1\rangle$ but $|0\rangle$ is needed, apply an X gate to flip it to $|0\rangle$.

Initialization is the **first step** in any quantum computation: (1) **prepare** qubits in known states, (2) **apply** them using quantum gates, and (3) **measure** to obtain classical results. It is analogous to **resetting a classical register** before starting a computation in classical computing.

## 3.2 Quantum Logic Gates Overview

In quantum computing, quantum logic gates are the primary tools used to **manipulate qubits**. These gates perform **unitary transformations** on the state of one or more qubits, meaning they **preserve the norm** of the quantum state and are **reversible**. The design of quantum algorithms and the execution of quantum circuits rely entirely on the sequential application of these gates.

### Types of Quantum Gates

**Quantum Gates** can be classified into:

- **Single-Qubit Gates**: These gates operate on **individual qubits**, modifying their state in isolation.
- **Multiple-Qubit Gates**: These gates act on **two or more qubits simultaneously**, allowing for entanglement and complex correlations between qubits.

In this section we will focus on single qubit gates.

### Mathematical Framework: Qubit as Vectors, Gates as Matrices

To describe quantum superposition mathematically, we require a framework that supports linear combinations of states, probabilities derived from inner products, complex amplitudes and phases, and linear state transformations. The mathematical structure satisfying all these requirements is a **complex Hilbert space**. Thus, a qubit is naturally represented as a **unit vector** in a two-dimensional complex Hilbert space $\mathbb{C}^2$.

Using the Dirac notation, the basis states $|0\rangle$ and $|1\rangle$ form an orthonormal basis for this space, satisfying $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$. Any qubit state can be expressed as a linear combination of these basis states:

$$|\psi\rangle = a|0\rangle + b|1\rangle \qquad \text{with } |a|^2 + |b|^2 = 1$$

Two state vectors that differ only by a global phase factor $e^{i\phi}$ represent the same physical qubit, since measurement probabilities remain unchanged (see page 35).

Choosing the computational basis $\{|0\rangle, |1\rangle\}$, we can represent these states as column vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Therefore, the qubit state can be written as:

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

This vector representation allows us to apply linear algebra techniques to analyze and manipulate qubits.

Once qubits are represented as vectors, transformations of these states must be **linear operators** acting on the vector space. In a chosen basis, linear operators are represented by **matrices**. To preserve normalization and probabilities, these matrices must be **unitary**:

$$U^\dagger U = UU^\dagger = I$$

Thus, quantum gates – the **building blocks of quantum circuits** – are represented as unitary $2 \times 2$ matrices that linearly transform the state vector:

$$|\psi'\rangle = U |\psi\rangle$$

**Why Matrices?** Because quantum mechanics is inherently linear, any physical evolution of a closed quantum system must be described by a **linear operator**. When expressed in a chosen basis (such as the computational basis $\{|0\rangle, |1\rangle\}$), this operator takes the form of a matrix.

### Unitary: The Fundamental Constraint

A matrix $U$ is **unitary** if it satisfies the condition:

$$U^\dagger U = UU^\dagger = I \tag{10}$$

Where:

- $U^\dagger$ is the **Hermitian transpose** (conjugate transpose) of $U$.
- $I$ is the **identity matrix**.

This property ensures that $U^{-1} = U^\dagger$, meaning every quantum operation is **reversible**, and the transformation preserves the **norm** of the state vector. So, quantum gate **must be unitary** to ensure that the evolution of quantum states is consistent with the principles of quantum mechanics (i.e., linearity, reversibility, and preservation of probabilities).

A general single-qubit unitary matrix can be expressed as:

$$U = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \quad \text{with} \quad \alpha, \beta, \gamma, \delta \in \mathbb{C} \quad \text{and} \quad U^\dagger U = I \tag{11}$$

### Why must gates be unitary?

1. **No-Cloning Theorem**: It is impossible to create an identical copy of an unknown qubit. Unitary transformations respect this by not duplicating information.
2. **No-Deleting Theorem**: We cannot delete quantum information arbitrarily. Since unitary gates are invertible, they do not erase information.
3. **Preservation of Probability**: The normalization condition $|a|^2 + |b|^2 = 1$ must hold after any operation, and only unitary matrices preserve this norm.

### Physical Intuition

In classical circuits, logic gates like AND, OR, NOT process bits deterministically. In quantum circuits:

- Gates **rotate** the quantum state on the **bloch sphere**.
- These rotations correspond to **unitary transformations**.
- The **physical implementation** of quantum gates may vary across hardware (e.g., superconducting qubits, trapped ions), but **mathematically**, the **same unitary matrix describes the gate**.

**Quantum gates do not correspond to physical gates** in the classical sense. They are **abstract operations** realized by **controlled interactions** in quantum systems.

## 3.3 Main Single-Qubit Gates

### 3.3.1 Identity Gate (I)

The **Identity Gate**, denoted $I$, is the most elementary gate in quantum computing. It performs **no change** to the state of a qubit. Its action on a qubit is trivial, yet it **serves important roles in quantum circuits**, particularly when managing multi-qubit systems or timing synchronization.

#### Matrix Representation

The Identity gate is represented by the $2 \times 2$ **identity matrix**:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \tag{12}$$

**Applying** the Identity gate to any qubit **leaves the state unchanged**. Formally, for a qubit in state $|\psi\rangle$:

$$I|\psi\rangle = |\psi\rangle$$

To better understand its effect, consider its action on the computational basis:

$$I|0\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$
$$I|1\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} = |1\rangle$$

Hence, the Identity gate **preserves the state** of the qubit, including any superposition or entangled state it may be in.

#### Physical Interpretation

Although it may appear useless at first glance, the Identity gate has **conceptual and practical utility**:

- It acts as a **placeholder** or **"*do nothing*" operation** when required by **circuit timing**.
- In **multi-qubit circuits**, it allows one **qubit to remain unchanged** while other qubits are operated on.
- In **matrix composition** of gates, it serves to **align dimensions** when performing **tensor products** (e.g., $I \otimes H$ applies Hadamard to the second qubit only).

#### Bloch Sphere Perspective

In the Bloch sphere (page 38), the **identity gate** represents a **rotation of zero degrees**, meaning the state vector remains exactly where it is. It is the neutral element of quantum transformations.

### 3.3.2 Pauli Gates

The **Pauli Gates** are a set of fundamental single-qubit quantum gates that play a central role in quantum computing. They are represented by the three **Pauli matrices** $X$, $Y$, and $Z$, and can be interpreted as the quantum analogues of basic operations on a qubit.

Each Pauli gate corresponds to a **rotation of $\pi$ radians** around one of the principal axes of the Bloch sphere:

- $X$: rotation around the $x$-axis (bit flip).
- $Z$: rotation around the $z$-axis (phase flip).
- $Y$: rotation around the $y$-axis (bit and phase flip).

These operators are **unitary** ($U^\dagger U = UU^\dagger = I$), **Hermitian** ($U = U^\dagger$), and **self-inverse** ($U^2 = I$), and their eigenstates define the three fundamental measurement bases of a qubit. Together, they form the building blocks for describing single-qubit transformations and quantum dynamics. Also, they have eigenvalues $\pm 1$ and satisfy the property:

$$X^2 = Y^2 = Z^2 = I$$

Meaning that applying the same Pauli gate twice returns the qubit to its original state.

---

#### 3.3.2.1 Rotations Generated by Pauli Operators

In quantum mechanics, rotations of a qubit are generated by the Pauli operators. A rotation around an axis $\hat{n}$ by an angle $\theta$ is given by:

$$R_{\hat{n}}(\theta) = e^{-i\frac{\theta}{2}(\hat{n}\cdot\vec{\sigma})} \tag{13}$$

Where $\vec{\sigma} = (X, Y, Z)$ are the Pauli matrices. We will see those matrices in more detail in the next sections, but for now we can just note that they are the generators of rotations around the $x$, $y$, and $z$ axes, respectively.

In particular:

$$R_x(\theta) = e^{-i\frac{\theta}{2}X} \tag{14}$$

$$R_y(\theta) = e^{-i\frac{\theta}{2}Y} \tag{15}$$

$$R_z(\theta) = e^{-i\frac{\theta}{2}Z} \tag{16}$$

Using the identity $\sigma^2 = I$, these can be expanded as:

$$R_\alpha(\theta) = \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right)\sigma_\alpha \tag{17}$$

#### 3.3.2.2 Pauli-X (NOT) Gate

The **Pauli-X Gate**, often called the **quantum NOT gate**, is one of the *fundamental* single-qubit quantum gates. It is denoted $X$ and **corresponds to the $\sigma_x$ Pauli matrix** in quantum mechanics. This gate **flips the state** of a qubit, **similar to how a classical NOT gate inverts a bit**.

**Remark: Pauli Matrix**

In quantum mechanics, the **Pauli matrices** are **a set of three** $2 \times 2$ **Hermitian and unitary matrices** that represent **spin operators** for spin-$\frac{1}{2}$ particles and are widely used in quantum computing to describe single-qubit gates. They are:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Key properties:

- Hermitian: $\sigma_i^\dagger = \sigma_i$ (observable-related)
- Unitary: $\sigma_i^\dagger \sigma_i = I$ (reversible transformations)
- Involution: $\sigma_i^2 = I$
- Non-commuting: $[\sigma_i, \sigma_j] = 2i\varepsilon_{ijk}\sigma_k$ (with $\varepsilon$ the Levi-Civita symbol)

**Matrix Representation**

$$X = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{18}$$

This **matrix swaps** the $|0\rangle$ and $|1\rangle$ **basis states**:

$$X|0\rangle = |1\rangle \qquad \text{and} \qquad X|1\rangle = |0\rangle \tag{19}$$

**Action on a General Qubit**

Let's consider a qubit in the general state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Applying the $X$ gate:

$$X|\psi\rangle = aX|0\rangle + bX|1\rangle = a|1\rangle + b|0\rangle = b|0\rangle + a|1\rangle$$

As a result, the **amplitudes** $a$ and $b$ are **swapped**.

### Key Properties of X Gate

The Pauli-X gate satisfies several important mathematical properties:

1. **Unitary**. The X gate is unitary, meaning it preserves the inner product and is reversible:
$$XX^\dagger = I$$
2. **Hermitian**. The X gate is also Hermitian, which means it is equal to its own conjugate transpose:
$$X = X^\dagger$$
3. **Self-Inverse**. The X gate is its own inverse, which means that **applying it twice returns the qubit to its original state**:
$$X^{-1} = X \quad \Rightarrow \quad X^2 = I$$

### Geometric Interpretation: Rotation on the Bloch Sphere

The Pauli-X gate corresponds to a **rotation around the $x$-axis by $\pi$ radians** (180°) on the Bloch sphere, denoted as $X = R_x(\pi)$.

**To understand** this, recall the **general qubit state in spherical coordinates**:
$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$
After a $\pi$ rotation around the $x$-axis:

- $\theta \to \pi - \theta$
- $\phi \to -\phi$

This **rotation mirrors the qubit across the $x$-axis**, flipping its position between $|0\rangle$ and $|1\rangle$, and adjusting the phase accordingly.

### Rotation Interpretation

From Equation 13 (page 48) and Equation 17 (page 48), a rotation around the $x$-axis by an angle $\theta$ is given by:

$$R_x(\theta) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)X$$

For $\theta = \pi$, we obtain:
$$R_x(\pi) = -iX$$
Therefore, the Pauli-X gate corresponds to a **rotation of $\pi$ around the $x$-axis, up to a global phase**:
$$X \equiv R_x(\pi)$$
Since global phases are physically irrelevant, $X$ and $R_x(\pi)$ represent the same physical transformation.

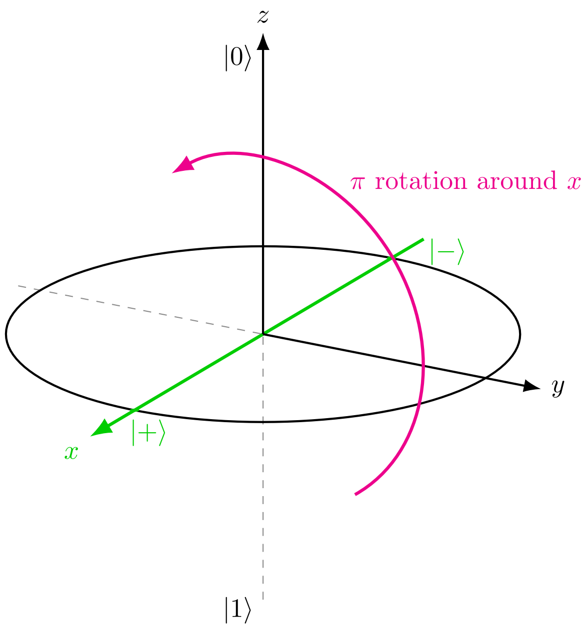



Figure 10: Eigenstates of the Pauli-$X$ gate. Since $|+\rangle$ and $|-\rangle$ lie on the $x$-axis, they are not moved by a rotation around that axis. The state $|-\rangle$ only acquires a global phase $-1$.

### Eigenstates

In linear algebra, an **eigenvector** of a matrix (or linear operator) $A$ is a non-zero vector $|v\rangle$ that, when the operator is applied, changes only by a **scalar factor** called the **eigenvalue** $\lambda$:

$$A\,|v\rangle = \lambda\,|v\rangle$$

In quantum mechanics, we use the term **Eigenstate** instead of eigenvector. So, an eigenstate of an operator $U$ satisfies:

$$U\,|v\rangle = \lambda\,|v\rangle$$

An **eigenstate** is a state that remains **unchanged in direction** after the operation is applied, except only its **magnitude** or **phase** may change by a factor of $\lambda$. For the Pauli-X gate, solving the eigenvalue equation:

$$X\,|v\rangle = \lambda\,|v\rangle$$

Yields two eigenstates:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad \text{with eigenvalue} \quad +1 \Rightarrow X\,|+\rangle = +1 \cdot |+\rangle = |+\rangle \tag{20}$$

Called the **Plus State**, and:

$$|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad \text{with eigenvalue} \quad -1 \Rightarrow X\,|-\rangle = -1 \cdot |-\rangle = -\,|-\rangle \tag{21}$$

Called the **Minus State**. These states are also known as **Hadamard basis states** (page 63), **X-basis states**, or **Eigenstates of the Pauli-X operator**.

**Example 2: Measurement in the X-basis**

For example, if a qubit is in the state $|x\rangle$ and we measure it in the X-basis, the outcome is 1 with probability 1.

*Proof.* We want to show that if the qubit is in the state:

$$|x\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

And we measure it in the X-basis $\{|+\rangle, |-\rangle\}$, then the outcome is $+1$ with probability 1.
Measuring in the X-basis means using the eigenstates of the Pauli-X operator:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These correspond to the new possible outcomes: $+1$ for $|+\rangle$ and $-1$ for $|-\rangle$.
If the system is in state $|\psi\rangle$, the probability of obtaining the outcome corresponding to basis state $|u\rangle$ is:

$$P(u) = |\langle u|\psi\rangle|^2$$

In our case, the system is in state $|\psi\rangle = |+\rangle$, so:

- Probability of outcome $+1$ (measuring $|+\rangle$):

$$P(+1) = |\langle +|+\rangle|^2$$

- Probability of outcome $-1$ (measuring $|-\rangle$):

$$P(-1) = |\langle -|+\rangle|^2$$

Since every normalized state has inner product 1 with itself, we have:

$$\langle +|+\rangle = 1$$

Therefore:

$$P(+) = |\langle +|+\rangle|^2 = |1|^2 = 1$$

So the probability of getting $+1$ is 1.
If we calculate the probability of getting $-1$, first write the bras and kets:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Changing the ket $|+\rangle$ to a bra:

$$\langle -| = \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)$$

Now we can compute the inner product:

$$
\begin{aligned}
\langle -|+\rangle &= \frac{1}{\sqrt{2}}\left(\langle 0| - \langle 1|\right)\frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \\
&\downarrow \text{ bring out the factors} \\
&= \frac{1}{2}\left(\langle 0| - \langle 1|\right)\left(|0\rangle + |1\rangle\right) \\
&\downarrow \text{ expand the product} \\
&= \frac{1}{2}\left(\langle 0|0\rangle + \langle 0|1\rangle - \langle 1|0\rangle - \langle 1|1\rangle\right) \\
&\downarrow \text{ use orthonormality } (\langle 0|0\rangle = \langle 1|1\rangle = 1,\ \langle 0|1\rangle = \langle 1|0\rangle = 0) \\
&= \frac{1}{2}\left(1 + 0 - 0 - 1\right) \\
&= 0
\end{aligned}
$$

Therefore:

$$P(-1) = |\langle -|+\rangle|^2 = |0|^2 = 0$$

So the probability of getting $-1$ is 0. QED

#### 3.3.2.3 Pauli-Z (Phase Flip) Gate

The **Pauli-Z Gate**, also called the **Phase Flip Gate**, is one of the core **Pauli operators** used in quantum computing. It acts **only on the phase** of the qubit, **leaving the $|0\rangle$ amplitude unchanged** but **flipping the sign of the $|1\rangle$ amplitude**.

**Matrix Representation**

$$Z = \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{22}$$

**Action on a General Qubit**

For a general state:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Applying $Z$:

$$Z\,|\psi\rangle = aZ\,|0\rangle + bZ\,|1\rangle = a\,|0\rangle - b\,|1\rangle$$

As a result, $Z$ **flips the sign of the $|1\rangle$ amplitude**, this is why it's called a **phase flip**. Finally, it's trivial to show that:

$$Z\,|0\rangle = |0\rangle \quad \text{(unchanged)}, \qquad Z\,|1\rangle = -\,|1\rangle \quad \text{(sign flip)}$$

So the **magnitudes** of the amplitudes remain unchanged and the **probabilities of measurement outcomes** in the standard basis are unaffected, but the **relative phase** between $|0\rangle$ and $|1\rangle$ is altered (since $b$ changes to $-b$).

**Example 3: Effect on Superposition States**

The Hadamard (or X-basis) states are:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

These two states differ **only by their relative phase**:

- $|+\rangle$: relative phase 0.
- $|-\rangle$: relative phase $\pi$.

Applying $Z$ on $|+\rangle$:

$$\begin{aligned} Z\,|+\rangle &= Z\left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right) \\ &= \frac{Z\,|0\rangle + Z\,|1\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ &= |-\rangle \end{aligned}$$

Applying $Z$ on $|-\rangle$:

$$\begin{aligned} Z|-\rangle =& Z\left(\frac{|0\rangle - |1\rangle}{\sqrt{2}}\right) \\ =& \frac{Z|0\rangle - Z|1\rangle}{\sqrt{2}} \\ =& \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ =& |+\rangle \end{aligned}$$

The Pauli-Z gate **does not change the probabilities** of measuring $|0\rangle$ or $|1\rangle$ in the computational basis. However, it **changes the relative phase**, transforming $|+\rangle$ into $|-\rangle$ and vice versa.

### Geometric Interpretation: Rotation around the $z$-axis

On the Bloch sphere, the Pauli-Z gate performs a $\pi$ **radians (180°) rotation around the $z$-axis**. In spherical coordinates:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

After Z rotation:

- The angle $\phi \to \phi + \pi$
- This phase shift **changes the sign of the complex amplitude for** $|1\rangle$

Since **measurement probabilities depend on** $|a|^2$ and $|b|^2$, which remain unchanged, the Z gate **does not affect measurement outcomes** in the standard basis. It only affects **interference and future gate interactions**.

### Rotation Interpretation

From Equation 13 (page 48) and Equation 17 (page 48), a rotation around the $z$-axis by an angle $\theta$ is given by:

$$R_z(\theta) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)Z$$

For $\theta = \pi$, we obtain:

$$R_z(\pi) = -iZ$$

Therefore, the Pauli-Z gate corresponds to a **rotation of $\pi$ around the $z$-axis, up to a global phase**:

$$Z \equiv R_z(\pi)$$

Since global phases are physically irrelevant, $Z$ and $R_z(\pi)$ represent the same physical transformation.

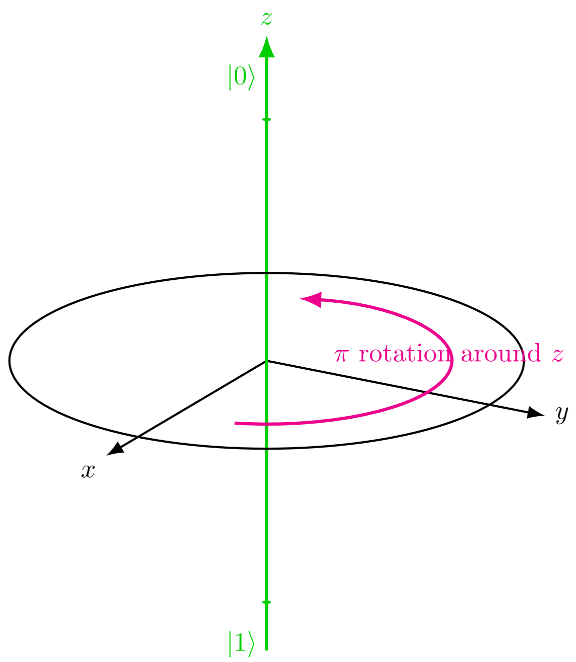



Figure 11: Eigenstates of the Pauli-$Z$ gate. Since $|0\rangle$ and $|1\rangle$ lie on the $z$-axis, they are not moved by a rotation around that axis. The state $|1\rangle$ only acquires a global phase $-1$.

### Key Properties

1. **Unitary**: $Z^\dagger = Z^{-1} = Z$. Ensuring normalization is preserved.
2. **Hermitian**: $Z^\dagger = Z$. Meaning it is equal to its own conjugate transpose.
3. **Self-Inverse**: $Z^{-1} = Z \Rightarrow Z^2 = I$. Applying the Pauli-Z gate twice returns the original state.
4. **Eigenvalues and Eigenstates**. The Pauli-Z operator has *eigenvalues* $+1$ and $-1$, with corresponding *eigenstates* $|0\rangle$ and $|1\rangle$:

$$Z|0\rangle = +1 \cdot |0\rangle, \qquad Z|1\rangle = -1 \cdot |1\rangle$$

   This means that $|0\rangle$ and $|1\rangle$ are **unchanged in direction** by the action of $Z$ (up to a possible phase factor). In particular:

   - $|0\rangle$ is completely unchanged.
   - $|1\rangle$ acquires only a phase factor $-1 = e^{i\pi}$.

   Therefore, these states are **invariant under the action of** $Z$ (up to global phase), and represent the **natural states of the operator**.

**Geometric Insight.** On the Bloch sphere, the Pauli-Z gate corresponds to a rotation around the $z$-axis. The eigenstates $|0\rangle$ and $|1\rangle$ lie exactly on this axis (north and south poles). As a consequence, they are **not affected by the rotation**, which explains why they are eigenstates of $Z$.

In contrast, any state that is not aligned with the $z$-axis (for example $|+\rangle$, Figure 9) is rotated by the action of $Z$. This highlights the difference between:

- **Eigenstates**: unchanged (up to phase)
- **Superposition states**: geometrically rotated

#### 3.3.2.4 Pauli-Y Gate

The **Pauli-Y Gate**, denoted $Y$, is one of the three fundamental **Pauli matrices** in quantum mechanics. It is less intuitive than Pauli-X and Z, but equally important, especially for its role in complex rotations and quantum interference.

**Matrix Representation**

$$Y = \sigma_Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

**Action on a General Qubit**

At first glance, this looks similar to the Pauli-X gate because it exchanges $|0\rangle$ and $|1\rangle$. But there is something extra: $X$ swaps the two basis states (i.e., $|0\rangle \leftrightarrow |1\rangle$), while $Y$ swaps them **and** multiplies them by a phases of $\pm i$. So Pauli-Y is not just a bit flip (like Pauli-X), it is a bit flip together with a phase modification.

Given a qubit in state:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Apply the Y gate:

$$Y|\psi\rangle = aY|0\rangle + bY|1\rangle = a(-i|1\rangle) + b(i|0\rangle) = ib|0\rangle - ia|1\rangle$$

So the **amplitudes are exchanged** (as with Pauli-X), but each one also acquires a **phase factor of $i$ or $-i$**.

**Geometric Interpretation: Rotation on the Bloch Sphere**

The Pauli-Y gate corresponds to a **rotation around the $y$-axis by $\pi$ radians** (180°). The effect on Bloch sphere coordinates:

- $\theta \to \pi - \theta$
- $\psi \to \pi - \psi$

This rotates the qubit across the $y$-axis, **altering both amplitudes and their phases**.

**Rotation Interpretation**

From Equation 13 (page 48) and Equation 17 (page 48), a rotation around the $y$-axis by an angle $\theta$ is given by:

$$R_y(\theta) = \cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) Y$$

For $\theta = \pi$, we obtain:

$$R_y(\pi) = -iY$$

Therefore, the Pauli-Y gate corresponds to a **rotation of $\pi$ around the $y$-axis, up to a global phase**:

$$Y \equiv R_y(\pi)$$

Since global phases are physically irrelevant, $Y$ and $R_y(\pi)$ represent the same physical transformation.

### Pauli-Y as a Composite Gate

A very useful identity is:

$$Y = iXZ \tag{23}$$

This means that Pauli-Y can be understood as the combination of:

1. A Pauli-Z gate (phase flip);
2. Followed by a Pauli-X gate (bit flip);
3. Up to a global phase factor of $i$.

$$XZ = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Now, multiply by $i$:

$$iXZ = i \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = Y$$

This identity confirms that **Pauli-Y can be viewed as a combination of Pauli-X and Pauli-Z with a global phase factor** $i$, which does not affect measurement outcomes.

### Eigenstates of Y Gate

To find the eigenstates, we solve:

$$Y |v\rangle = \lambda |v\rangle$$

Where $\lambda$ is the eigenvalue. The result is:

$$|i\rangle = \frac{1}{\sqrt{2}} (|0\rangle + i |1\rangle) \tag{24}$$

$$|-i\rangle = \frac{1}{\sqrt{2}} (|0\rangle - i |1\rangle) \tag{25}$$

With eigenvalues:

$$Y |i\rangle = +1 |i\rangle \qquad Y |-i\rangle = -1 |-i\rangle \tag{26}$$

### Key Properties of Y Gate

1. **Unitary**: $Y^\dagger Y = I$. So it is a valid quantum gate that preserves normalization.
2. **Hermitian**: $Y^\dagger = Y$. So it is also an observable, meaning it corresponds to a measurable quantity in quantum mechanics.
3. **Self-inverse**: $Y^2 = I$. Applying the Y gate twice returns the qubit to its original state.

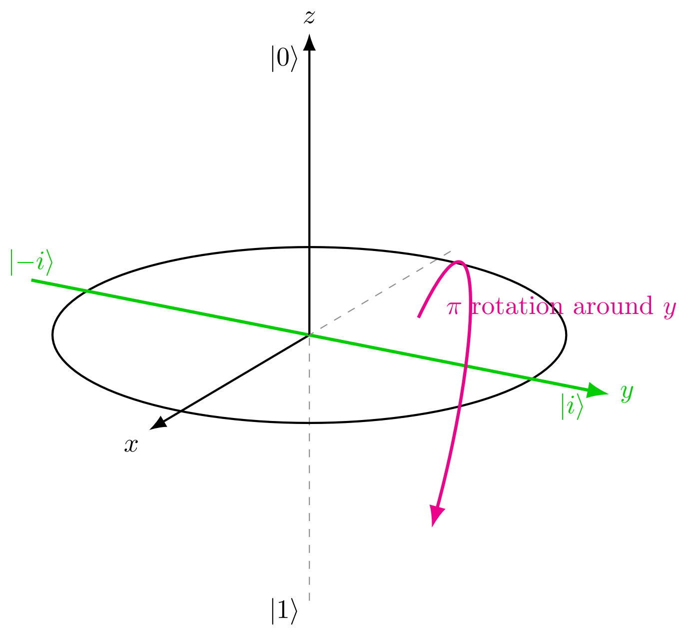



Figure 12: Eigenstates of the Pauli-$Y$ gate. Since $|i\rangle$ and $|-i\rangle$ lie on the $y$-axis, they are not moved by a rotation around that axis. The state $|-i\rangle$ only acquires a global phase $-1$.

### 3.3.3 Phase Gate (S)

The **Phase Gate (S)** is a single-qubit gate that **adds a phase shift of $\frac{\pi}{2}$ (90°) to the amplitude of the $|1\rangle$ state**. It **leaves $|0\rangle$ unchanged**, like the Pauli-Z gate, but instead of a sign flip ($\pi$ phase), it **introduces a complex phase factor** $i$.

#### Matrix Representation

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \tag{27}$$

#### Action on a General Qubit

For a state:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

Apply $S$:

$$S\,|\psi\rangle = a\,|0\rangle + ib\,|1\rangle$$

Only the $|1\rangle$ **amplitude gains a phase of** $i$, affecting interference but **not measurement probabilities** in the computational basis.

$$S\,|0\rangle = |0\rangle \quad \text{(unchanged)}, \qquad S\,|1\rangle = i\,|1\rangle \quad \left(\text{phase } +\frac{\pi}{2}\right)$$

#### Geometric Interpretation: Rotation around $z$-axis $\left(\frac{\pi}{2}\right)$

On the Bloch sphere, $S$ performs a rotation of $\frac{\pi}{2}$ radians around the $z$-axis. This rotation **shifts the phase angle** $\phi \to \phi + \frac{\pi}{2}$, altering **how the qubit interferes** in later operations, especially when **Hadamard gates** are involved. In Figure 13, we see how $S$ transforms the state $|+\rangle$ (on the $x$-axis) into $|i\rangle$ (on the $y$-axis) through a $\frac{\pi}{2}$ rotation around the $z$-axis, demonstrating the phase shift's effect on the qubit's state.

#### Relationship to Pauli-Z

$S$ is often called the **square root of Z** because applying it twice yields the Pauli-Z gate: $S = \sqrt{Z}$. Pauli-Z is a **full phase flip**:

$$e^{i\pi} = -1$$

The $S$ gate is a **partial phase shift**:

$$e^{i\frac{\pi}{2}} = i$$

So:

$$Z = e^{i\pi\cdot|1\rangle\langle 1|} \quad \text{and} \quad S = e^{i\frac{\pi}{2}\cdot|1\rangle\langle 1|}$$

Same transformation, but different phase angles. Let's verify $S^2 = Z$:

$$S \cdot S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & i^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z$$

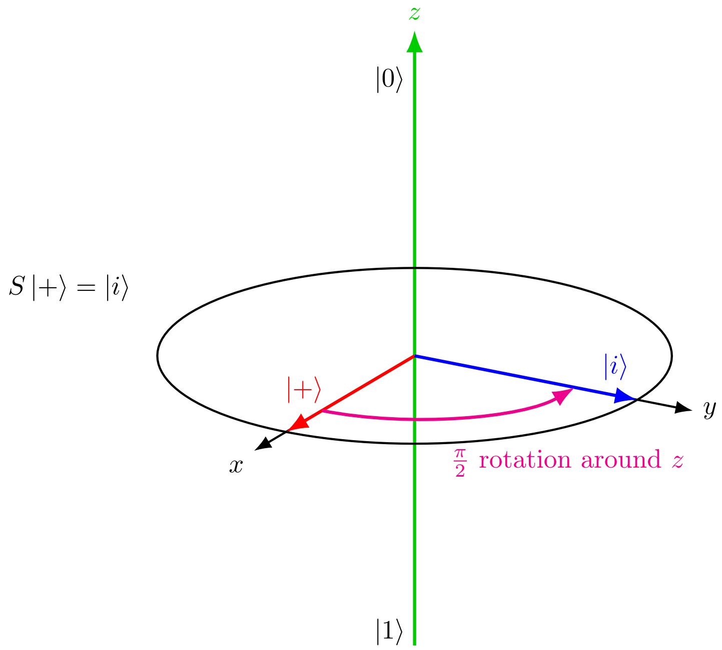



Figure 13: Geometric interpretation of the Phase gate $S$. The gate performs a rotation of $\frac{\pi}{2}$ around the $z$-axis, transforming $|+\rangle$ into $|i\rangle$.

### Eigenvalues and Eigenstates

The Phase gate $S$ is diagonal in the computational basis, so its eigenstates are $|0\rangle$ and $|1\rangle$:

$$S|0\rangle = 1 \cdot |0\rangle, \qquad S|1\rangle = i \cdot |1\rangle$$

Thus, the eigenvalues of $S$ are:

$$\lambda_0 = 1, \qquad \lambda_1 = i$$

This means that $|0\rangle$ remains unchanged, while $|1\rangle$ acquires a phase factor $i = e^{i\frac{\pi}{2}}$.

### Key Properties

1. **Unitary**: $S^\dagger = S^{-1}$, so normalization is preserved.
2. **Not Hermitian**: $S^\dagger \neq S$, therefore $S$ is not an observable, it is only a transformation.
3. **Inverse**:
$$S^{-1} = S^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & -i \end{pmatrix}$$
4. **Relation with Pauli-Z**: $S^2 = Z$, meaning that $S$ is the square root of the Pauli-Z gate.
5. **Periodicity**: $S^4 = I$, since $i^4 = 1$.
6. **Phase Shift**: $S$ applies a relative phase of $\frac{\pi}{2}$ to the $|1\rangle$ component.

### 3.3.4 Hadamard Gate (H)

The **Hadamard Gate (H) changes the basis representation of the qubit**. More concretely:

$$H : \{|0\rangle, |1\rangle\} \longleftrightarrow \{|+\rangle, |-\rangle\}$$

It maps computational basis states into equal superpositions and vice versa, allowing phase information to be transformed into measurable probabilities.

#### Matrix Representation

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{28}$$

#### Action on Basis States

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle \tag{29}$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle \tag{30}$$

The $|+\rangle$ and $|-\rangle$ **states** are known as the **Hadamard basis**, or **superposition states**, with **equal probability amplitudes** for $|0\rangle$ and $|1\rangle$.

#### Superposition and Measurement

After applying $H$ to $|0\rangle$, the resulting qubit is:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This state is a **superposition** of $|0\rangle$ and $|1\rangle$, meaning that the qubit does not have a definite classical value prior to measurement. Instead, it is described by **probability amplitudes** associated with each basis state.

Upon measurement in the computational basis, the state collapses probabilistically:

- **Probability of 0**:

$$\left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

- **Probability of 1**:

$$\frac{1}{2}$$

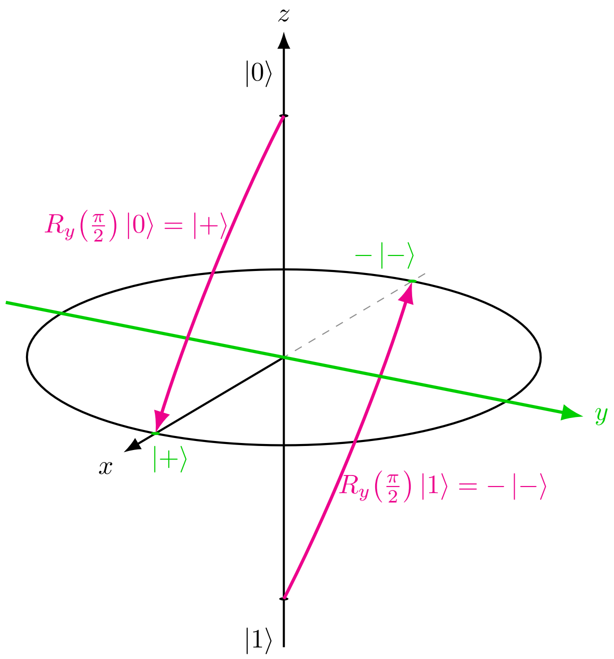
After the measurement:

- The qubit collapses to $|0\rangle$ with probability $\frac{1}{2}$, or to $|1\rangle$ with probability $\frac{1}{2}$.
- The original superposition is **destroyed**, and cannot be recovered.

### ❓ Bloch Sphere Interpretation

On the Bloch sphere, the **Hadamard gate performs two sequential rotations**:

1. Rotation around the $y$-axis by $\frac{\pi}{2}$ radians (90°): $R_y\left(\frac{\pi}{2}\right)$.




The $y$-axis stays fixed. So the states on the $y$-axis $|i\rangle$ and $|-i\rangle$ do not move. Instead, the $z$-axis rotated toward the $x$-axis: $|0\rangle \to |+\rangle$ and $|1\rangle \to -|-\rangle$. So the first step already moves the computational basis from the $z$-axis to the $x$ axis.

We can verify the matrix representation of $R_y\left(\frac{\pi}{2}\right)$ by using the exponential of the Pauli $Y$ matrix (see page 48):

$$\begin{aligned} R_y(\theta) &= e^{-i\frac{\theta}{2}Y} \\ &= \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right) Y \\ &= \cos\left(\frac{\theta}{2}\right)\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} - i\sin\left(\frac{\theta}{2}\right)\begin{pmatrix}0 & -i\\ i & 0\end{pmatrix} \\ &= \begin{pmatrix}\cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right)\\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right)\end{pmatrix} \end{aligned}$$

Now, substituting $\theta = \frac{\pi}{2}$ gives us the expected matrix for the first step of the Hadamard decomposition:

$$R_y\left(\frac{\pi}{2}\right) = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}$$

- Apply it to $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$:

$$\begin{aligned} R_y\left(\frac{\pi}{2}\right)|0\rangle &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}}\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right) \\ &= \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \\ &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ &= |+\rangle \end{aligned}$$

So $|0\rangle$, the north pole of the Bloch sphere, is rotated to the $+x$-axis, which is the $|+\rangle$ state.

- Apply it to $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$:

$$\begin{aligned} R_y\left(\frac{\pi}{2}\right)|1\rangle &= \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}}\left(\begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \\ &= \frac{1}{\sqrt{2}}\left(|1\rangle - |0\rangle\right) \\ &= -\frac{|0\rangle - |1\rangle}{\sqrt{2}} \\ &= -|-\rangle \end{aligned}$$

This means that $|1\rangle$, the south pole of the Bloch sphere, moves to the $-x$-axis, but with a global phase of $-1$. Since global phases do not affect measurement outcomes, we can ignore it and say that $|1\rangle$ is effectively rotated to the $-x$-axis, which is the $|-\rangle$ state. In simple terms: $-|-\rangle \equiv |-\rangle$ up to a global phase.

2. Rotation around the $x$-axis by $\pi$ radians (180°). This step is represented by the operator $R_x(\pi)$, which can be expressed as:

$$R_x(\pi) = e^{-i\frac{\pi}{2}X} = \cos\left(\frac{\pi}{2}\right)I - i\sin\left(\frac{\pi}{2}\right)X = -iX$$

Applying $R_x(\pi)$ to the states on the $x$-axis ($|+\rangle$ and $|-\rangle$) does not change their position on the Bloch sphere, but it does introduce a phase factor (which becomes a global phase only when applied to the entire state):

$$R_x(\pi)\,|+\rangle = -iX\,|+\rangle = -i\,|+\rangle$$
$$R_x(\pi)\,|-\rangle = -iX\,|-\rangle = -i(-\,|-\rangle) = i\,|-\rangle$$

Applying $R_x(\pi)$ to the superposition state $a\,|+\rangle - b\,|-\rangle$ (obtained from the first rotation) gives us:

$$\begin{aligned} & a\,|+\rangle - b\,|-\rangle \\ R_x(\pi)(a\,|+\rangle - b\,|-\rangle) &= a(-i\,|+\rangle) - b(i\,|-\rangle) \\ &= -ia\,|+\rangle - ib\,|-\rangle \\ &= -i(a\,|+\rangle + b\,|-\rangle) \end{aligned}$$

Since global phases $-i$ do not affect measurement outcomes, we can ignore them and say that $|+\rangle$ and $|-\rangle$ remain on the $x$-axis. The second rotation does not change the position of the states on the Bloch sphere, but it is necessary to ensure that the transformation matches the Hadamard gate at the operator level, preserving the correct relative phase for all superpositions.

**Why is the second rotation necessary?** We **cannot** simply change $-\,|-\rangle$ to $|-\rangle$ inside a superposition, because that would change only one component and therefore change the **relative phase**. The fix comes from applying $R_x(\pi)$. After the first rotation, we have:

$$a\,|+\rangle - b\,|-\rangle$$

Now, applying the Pauli-X operator using the rotation $R_x(\pi)$ (page 48):

$$R_x(\pi) = -iX = -i\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{with} \quad X\,|+\rangle = |+\rangle\,, \quad X\,|-\rangle = -\,|-\rangle$$

Applying $R_x(\pi)$ to the superposition gives us:

$$\begin{aligned} R_x(\pi)(a\,|+\rangle - b\,|-\rangle) &= -iX(a\,|+\rangle - b\,|-\rangle) \\ &= -i\,(aX\,|+\rangle - bX\,|-\rangle) \\ &= -i\,(a\,|+\rangle - b(-\,|-\rangle)) \\ &= -i\,(a\,|+\rangle + b\,|-\rangle) \end{aligned}$$

Now the factor $-i$ multiplies the **entire state**, so it is a global phase:

$$-i(a\,|+\rangle + b\,|-\rangle) \equiv a\,|+\rangle + b\,|-\rangle$$

So the important point is:

$$a\,|+\rangle - b\,|-\rangle \not\equiv a\,|+\rangle + b\,|-\rangle$$

But:

$$R_x(\pi)(a\,|+\rangle - b\,|-\rangle) = -i(a\,|+\rangle + b\,|-\rangle) \equiv a\,|+\rangle + b\,|-\rangle$$

**Insight.** The second rotation **does not affect the geometric position** of the states, but **corrects the relative phase** structure, ensuring that the transformation is a valid unitary operator acting consistently on the entire Hilbert space.

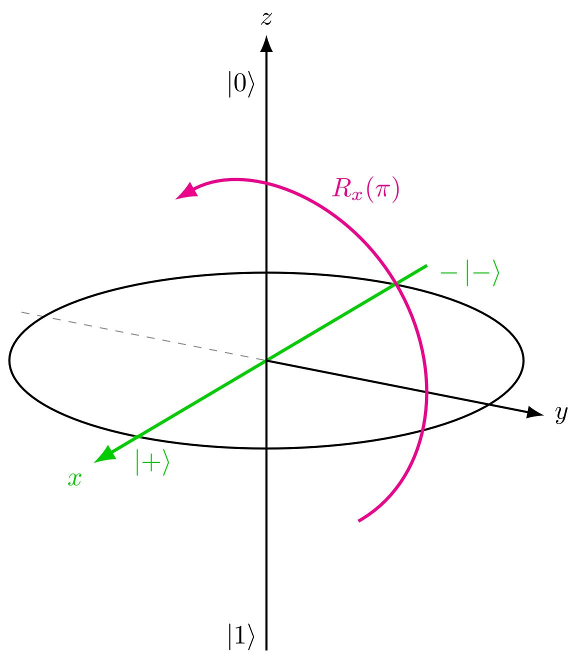



Figure 14: Second step of the Hadamard decomposition: rotation by $\pi$ around the $x$-axis. States on the $x$-axis are not moved on the Bloch sphere; only global phases are introduced.

This brings $|0\rangle$ to the $+x$-axis ($|+\rangle$) and $|1\rangle$ to the $-x$-axis ($|-\rangle$). As a result, **qubit moves from pole to equator**, entering **maximum superposition**.

### Eigenvalues and Eigenstates

The Hadamard operator has eigenvalues:

$$\lambda = +1, \qquad \lambda = -1$$

with corresponding eigenstates given by the solutions of:

$$H|v\rangle = \lambda |v\rangle$$

These eigenstates are superpositions of $|0\rangle$ and $|1\rangle$, and are aligned along an axis halfway between the $x$ and $z$ directions on the Bloch sphere. Precisely, the eigenvectors are:

$$|v_+\rangle = \cos\left(\frac{\pi}{8}\right)|0\rangle + \sin\left(\frac{\pi}{8}\right)|1\rangle$$
$$|v_-\rangle = \sin\left(\frac{\pi}{8}\right)|0\rangle - \cos\left(\frac{\pi}{8}\right)|1\rangle$$

Where $|v_+\rangle$ corresponds to the eigenvalue $+1$ and $|v_-\rangle$ corresponds to the eigenvalue $-1$; the cosine and sine factors ensure normalization of the eigenstates.

Unlike Pauli gates, the Hadamard eigenstates lie along a **diagonal axis between the computational basis** $z$ **and the Hadamard basis** $x$, reflecting its role as a **basis transformation** rather than a simple bit or phase flip.

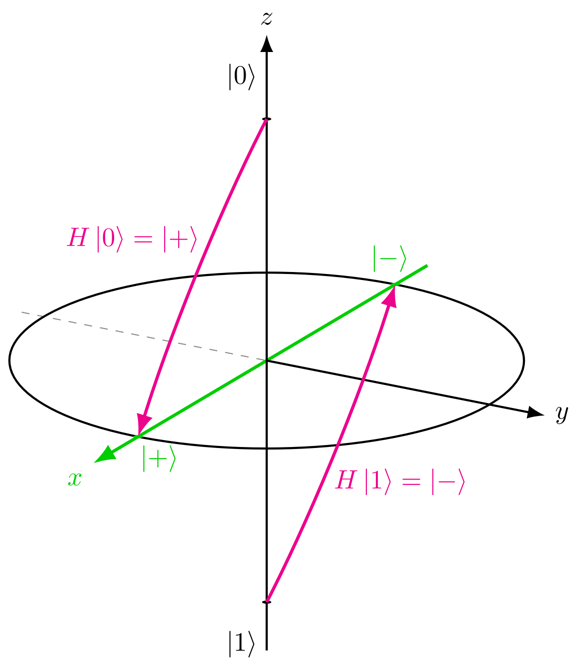



Figure 15: Bloch sphere interpretation of the Hadamard gate. The gate maps $|0\rangle$ to $|+\rangle$ and $|1\rangle$ to $|-\rangle$, moving computational basis states from the $z$-axis to the $x$-axis.

### Key Properties

1. **Unitary**: $H^\dagger H = I$, so normalization is preserved.
2. **Hermitian**: $H^\dagger = H$, therefore it corresponds to an observable.
3. **Self-inverse**: $H^2 = I$. Applying the Hadamard gate twice returns the original state.
4. **Basis transformation**: H maps the computational basis $\{|0\rangle, |1\rangle\}$ to the Hadamard basis $\{|+\rangle, |-\rangle\}$:
$$H|0\rangle = |+\rangle, \qquad H|1\rangle = |-\rangle$$
5. **Interference enabler**: $H$ converts phase differences into amplitude differences, making them observable through measurement.
6. **Axis mixing**: $H$ transforms Pauli operators as:
$$HZH = X, \qquad HXH = Z$$

## 3.4 Properties

Single-qubit gates exhibit a rich set of mathematical and geometric properties, all of which stem from their nature as unitary transformations on a two-level quantum system. Understanding these properties is key to **predicting gate behavior**, **designing quantum circuits**, and **analyzing quantum algorithms**.

### 1. All Quantum Gates Are Equivalent to Rotations

A profound insight in quantum computing is that **every single-qubit unitary gate corresponds to a rotation of the qubit's state vector on the Bloch sphere**.

- **Bloch Sphere** is a geometric representation of a qubit's state, where any point on the sphere represents a valid pure state.
- A **unitary operation** on qubit results in **rotating the vector** on this sphere **without changing its length** (preserving probability).

Some examples of rotations:

- `X Gate` → Rotation $\pi$ around x-axis (section 3.3.2.2, page 49)
- `Z Gate` → Rotation $\pi$ around z-axis (section 3.3.2.3, page 54)
- `Y Gate` → Rotation $\pi$ around y-axis (section 3.3.2.4, page 58)
- `S Gate` → Rotation $\frac{\pi}{2}$ around z-axis (section 3.3.3, page 61)
- `H Gate` → Rotation $\frac{\pi}{2}$ around y, then $\pi$ around x (section 3.3.4, page 63)

**Why is this important?** Quantum computation is fundamentally about **state rotations**, understanding gates as rotations helps visualize **interference**, **phase shifts**, and **entanglement creation**.

### 2. Reversibility: Applying a Gate Twice

Many single-qubit gates, especially the Pauli gates, exhibit the property of **involution**: applying the same gate twice **returns the qubit to its original state**. Mathematically it can be expressed as:

$$X^2 = Y^2 = Z^2 = I$$

This is due to the fact that **two** 180° **rotations around the same axis bring the vector back** to its original position on the Bloch sphere.

*Proof that X Gate has involution.*

$$X^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

These identities confirm that **Pauli gates are their own inverses**, and **all unitary gates are reversible**. QED

#### 3. Global Phases Do Not Matter

Quantum gates may **introduce a global phase** (a constant complex factor $e^{i\phi}$) to a state. Physically, **global phases are unobservable**, meaning they do **not affect measurement probabilities**. For example:

- States $|\psi\rangle$ and $e^{i\phi}|\psi\rangle$ are physically indistinguishable.
- The Y gate $= iXZ$, where $i$ is a global phase that can be ignored in practice.

#### 4. Composite Gates and Commutativity

- Gates can be **combined** by **matrix multiplication** (note: non commutative in general).
- **Order matters**: $AB \neq BA$ for most gates.
- Exception: **Gate Z and Gate S** are **commuted** because they are both **diagonal phase gates**.

| **Property** | Pauli Gates (X, Y, Z) | Phase Gate (S) | Hadamard (H) |
|---|---|---|---|
| **Rotation Axis** | x, y, z ($\pi$ radians) | z $\left(\frac{\pi}{2}\text{radians}\right)$ | y $\left(\frac{\pi}{2}\right)$ + x $(\pi)$ |
| **Involution** $G^2 = I$ | ✔ | ✘ ($S^2 = Z$) | ✔ |
| **Self-adjoint** $G = G^\dagger$ | ✔ | ✘ | ✔ |
| **Unitary** | ✔ | ✔ | ✔ |
| **Phase Introduction** | Z, S | $i$ for $\|1\rangle$ | ✔ (interference) |
| **Creates Superposition?** | ✘ (X, Y, Z alone) | ✘ | ✔ |

Table 2: Summary of Properties.

## 3.5 When Does a Gate Create Superposition?

In this section, we will answer the question: *when does a quantum gate create superposition?* More precisely, if we start from a **basis state** $|0\rangle$ or $|1\rangle$, when does a gate produce:

$$a\,|0\rangle + b\,|1\rangle \quad \text{with} \quad a, b \neq 0\,?$$

**What does "*create superposition*" really mean?** It means that the output must contain **both basis states**. So:

- $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is not a superposition, it's just $|0\rangle$.
- $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is not a superposition, it's just $|1\rangle$.
- $\begin{pmatrix} a \\ b \end{pmatrix}$ with $a, b \neq 0$ is a superposition, because it contains both $|0\rangle$ and $|1\rangle$ components. Indeed, we can write it as:

$$a\,|0\rangle + b\,|1\rangle = a\begin{pmatrix} 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \end{pmatrix}$$

**How does a matrix act on a basis state?**

The main idea is that each column of the matrix tells us where a basis vector goes. For example, for a matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

The first column $\begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix}$ tells us where $|0\rangle$ goes, and the second column $\begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$ tells us where $|1\rangle$ goes. In general, a gate creates superposition if one of its columns has **two non-zero entries**.

**Constraint: Unitarity**

Not every matrix can be a valid quantum gate. Quantum gates must be **unitary**, meaning:

$$A^\dagger A = I$$

This imposes strong constraints on the matrix entries. In particular, the columns must be:

- **Normalized**, total probability is 1: $\|\psi\|^2 = |a|^2 + |b|^2 = 1$
- **Orthogonal**, inner product is zero: $\langle v|w\rangle = 0$

As a consequence, some matrices that appear to create superposition are **not physically allowed**, because they are not unitary.

### 💡 Key Insight

Superposition arises from the **mixing of basis states**. Therefore:

- A gate **creates superposition** if at least one of its columns contains **two non-zero entries**.
- A gate **cannot create superposition** if its columns contain only one non-zero entry (as in diagonal or triangular matrices).

In particular:

- Gates like $H$ (Hadamard) create superposition because they mix $|0\rangle$ and $|1\rangle$.
- Gates like $X$, $Y$, $Z$ or $S$ do not create superposition because they only modify phases and do not mix basis states.

***Proof that $A$ cannot create superposition.*** Consider the gate:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix}$$

To create superposition from a basis state, at least one column of $A$ must contain two non-zero entries.

Applying $A$ to $|0\rangle$ gives:

$$A|0\rangle = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} a_{11} \cdot 1 + a_{12} \cdot 0 \\ 0 \cdot 1 + a_{22} \cdot 0 \end{pmatrix} = \begin{pmatrix} a_{11} \\ 0 \end{pmatrix} = a_{11}|0\rangle$$

Therefore, $|0\rangle$ is not mapped to a superposition because the second entry is zero.

Applying $A$ to $|1\rangle$ gives:

$$A|1\rangle = \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a_{11} \cdot 0 + a_{12} \cdot 1 \\ 0 \cdot 0 + a_{22} \cdot 1 \end{pmatrix} = \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix} = a_{12}|0\rangle + a_{22}|1\rangle$$

This would be a superposition only if both $a_{12} \neq 0$ and $a_{22} \neq 0$.

However, since $A$ is a quantum gate, it must be unitary (property of quantum gates):

$$A^\dagger A = I$$

Assuming $A$ is real, we have $A^\dagger = A^T$, so:

$$A^T = \begin{pmatrix} a_{11} & 0 \\ a_{12} & a_{22} \end{pmatrix}$$

Hence:

$$A^T A = \begin{pmatrix} a_{11} & 0 \\ a_{12} & a_{22} \end{pmatrix} \cdot \begin{pmatrix} a_{11} & a_{12} \\ 0 & a_{22} \end{pmatrix} = \begin{pmatrix} a_{11}^2 & a_{11}a_{12} \\ a_{11}a_{12} & a_{12}^2 + a_{22}^2 \end{pmatrix}$$

Since $A$ is unitary (i.e., $A^\dagger A = I$), this matrix must equal the identity:

$$\begin{pmatrix} a_{11}^2 & a_{11}a_{12} \\ a_{11}a_{12} & a_{12}^2 + a_{22}^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Therefore, to satisfy this equality, the following equations must hold:

$$\begin{cases} a_{11}^2 = 1 \\ a_{11}a_{12} = 0 \\ a_{12}^2 + a_{22}^2 = 1 \end{cases}$$

From $a_{11}^2 = 1$, we get $a_{11} = \pm 1$, so $a_{11} \neq 0$. Therefore, from $a_{11}a_{12} = 0$, it follows that:

$$a_{12} = 0$$

Substituting this into the last equation gives:

$$a_{22}^2 = 1 \quad \Rightarrow \quad a_{22} = \pm 1$$

Hence the matrix becomes diagonal:

$$A = \begin{pmatrix} \pm 1 & 0 \\ 0 & \pm 1 \end{pmatrix}$$

Since $A$ is diagonal, it only modifies the phases of $|0\rangle$ and $|1\rangle$, without mixing them. Thus, when we apply $A$ to $|0\rangle$ and $|1\rangle$, we get:

$$A\,|0\rangle = \pm\,|0\rangle\,, \qquad A\,|1\rangle = \pm\,|1\rangle$$

So neither $|0\rangle$ nor $|1\rangle$ is mapped to a superposition, because the output states are just $|0\rangle$ or $|1\rangle$ (up to a global phase). Thus, $A$ cannot create superposition (i.e., it cannot produce a state with both $|0\rangle$ and $|1\rangle$ components). QED

## 3.6 Single-Qubit Quantum Circuits

A **Quantum Circuit** is a **mathematical model** for quantum computation, **composed of sequential operations applied to qubits**. In other words, a quantum circuit is a **model of computation** that describes **how a quantum state evolves step by step**. The operations in a quantum circuit include initializations, unitary gates, and measurements. In this section, we focus on circuits involving a single qubit, which form the foundation for understanding multi-qubit systems.

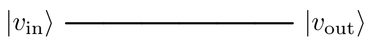
### Circuit Elements

A quantum circuit typically consists of three main elements:

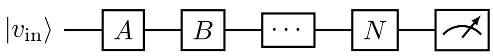
1. **Initialization** (initial state)
    - Qubit is prepared in a **known state**, typically $|0\rangle$ or $|1\rangle$.
    - Essential **starting point** for any computation.
2. **Gates**
    - Single-qubit **unitary operations** (e.g., X, Z, H, S).
    - Transform the qubit's state **reversibly**.
3. **Measurement**
    - **Irreversible process**.
    - Collapses the qubit to $|0\rangle$ or $|1\rangle$ with probabilities dictated by the quantum state's amplitudes.

### Quantum Circuit Diagram: Visual Language

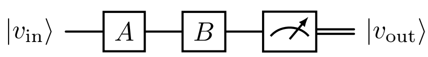
- **Horizontal lines** represent the **timeline of a single qubit**.




- **Boxes** on the line represent **gates** (letters) or **measurements** (tachometer icon).




- **Double lines** (if present) represent **classical information** (e.g., measurement result).




Left to right: The **circuit evolves over time from left (input) to right (output)**.

### Gate Sequence: Serial Gates

Consider two gates A and B applied in sequence:

$$|v_{\text{in}}\rangle \;—\boxed{A}—\boxed{B}—\; |v_{\text{out}}\rangle$$

1. First apply A, then B.
2. **Attention**: the **order of matrix multiplication** is **reversed** when we write the combined effect of the gates as a single matrix. The gate that is applied **last** (B) appears **first** in the product.

$$v_{\text{out}} = BAv_{\text{in}}$$

**Why?** Because first apply A to the input state $v_{\text{in}}$, then apply B to the result of $Av_{\text{in}}$.

3. The **combined effect** of the two gates is equivalent to a single gate C = BA:

$$|v_{\text{in}}\rangle \;—\boxed{BA}—\; |v_{\text{out}}\rangle \quad \equiv \quad |v_{\text{in}}\rangle \;—\boxed{A}—\boxed{B}—\; |v_{\text{out}}\rangle$$

$$v_{\text{out}} = Cv_{\text{in}} = BAv_{\text{in}}$$

Obviously, if we merge the gates into a single gate $C$, we must respect the order of operations, which is why $C = BA$ and not $AB$.

**Example 4: Hadamard followed by Hadamard**

Let's compute H followed by H to $|0\rangle$:

$$|0\rangle \;—\boxed{H}—\boxed{H}—\; |v_{\text{out}}\rangle$$

1. Apply H:

$$\begin{aligned} H\,|0\rangle &= \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \left( \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \\ &= \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle \end{aligned}$$

2. Applying H again, we **restore the original state due to the involution properties** ($H^2 = I$):

$$H\,(H\,|0\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = |0\rangle$$

## 3.7 Outer Product of Kets

In quantum mechanics, **Bra-Ket notation** (introduced by Dirac) is a concise and powerful tool for representing quantum states and operations. It uses **kets** $|\psi\rangle$ for **vectors** (states), and **bras** $\langle\psi|$ for their **Hermitian adjoints** (dual vectors).

Given two kets:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

The **Outer Product** $|a\rangle\langle b|$ is a **matrix**, formed by:

$$|a\rangle \langle b| = |a\rangle \otimes \langle b| = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \begin{pmatrix} \overline{b_1} & \overline{b_2} \end{pmatrix} = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix} \tag{31}$$

Where the symbol $\otimes$ is the **tensor product operator**. This operation **maps a vector to a matrix**.

**Why is the outer product useful?** Because **all quantum gates can be written using outer products of kets**. For example, the **Pauli-X gate** can be expressed as:

$$X = |0\rangle \langle 1| + |1\rangle \langle 0| = \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix}}_{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}} + \underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \end{pmatrix}}_{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

The **Inner Product** $\langle b|a\rangle$ is a scalar:

$$\langle a|b\rangle = \begin{pmatrix} \overline{a_1} & \overline{a_2} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \overline{a_1}b_1 + \overline{a_2}b_2 \tag{32}$$

So:

- **Inner product** $\rightarrow$ **projection**, result is number (scalar, see also page 10).
- **Outer product** $\rightarrow$ **matrix**, result is operator.

### Associativity between outer and inner products

A key property of the outer product is how it interacts with inner products. Specifically:

$$(|a\rangle \langle b|) |c\rangle = |a\rangle (\langle b|c\rangle) \tag{33}$$

This identity shows that the outer product acts as an operator: it first computes the inner product $\langle b|c\rangle$ (a scalar), and then scales the vector $|a\rangle$.

***Proof the Associativity.*** In this proof, we will demonstrate that applying the outer product $|a\rangle \langle b|$ to a ket $|c\rangle$ is equivalent to scaling $|a\rangle$ by the inner product $\langle b|c\rangle$.

Let's define:

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \qquad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \qquad |c\rangle = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

1. Compute **inner product** $\langle b|c\rangle$:

$$\langle b|c\rangle = \overline{b_1}c_1 + \overline{b_2}c_2 = \alpha$$

   Where $\alpha$ is a **scalar** value.

2. Multiply **scalar** $\alpha$ with $|a\rangle$ (vector):

$$|a\rangle\, \alpha = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \alpha = \begin{pmatrix} a_1 \cdot \alpha \\ a_2 \cdot \alpha \end{pmatrix} = \begin{pmatrix} a_1 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \\ a_2 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \end{pmatrix}$$

3. Compute outer product $|a\rangle\langle b|$ (matrix):

$$|a\rangle\, \langle b| = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \begin{pmatrix} \overline{b_1} & \overline{b_2} \end{pmatrix} = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix}$$

4. Multiply $|a\rangle\langle b|$ by $|c\rangle$:

$$(|a\rangle\, \langle b|)\, |c\rangle = \begin{pmatrix} a_1\overline{b_1} & a_1\overline{b_2} \\ a_2\overline{b_1} & a_2\overline{b_2} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} a_1 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \\ a_2 \cdot \left(\overline{b_1}c_1 + \overline{b_2}c_2\right) \end{pmatrix}$$

The second and fourth steps yield the same result. This proves that **outer product applied to a ket is equivalent to inner product $\langle b|c\rangle$ scaled by $|a\rangle$**. QED

## 3.8 Measurement

In quantum computing, **measurement** is a special type of operation that differs fundamentally from unitary gates. While gates perform reversible transformations, **measurement is irreversible**. It allows us to **gain information about the state of a qubit**, but at the price of **collapsing the quantum state**.

### Measurement Operator: Matrix Form

A **measurement operator** is a **non-unitary**, **non-invertible** matrix. For a measurement **along a specific direction**, represented by a ket $|k\rangle$, the associated **projector** is (outer product):

$$M_k = |k\rangle\langle k| = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} \begin{pmatrix} \overline{k_1} & \overline{k_2} \end{pmatrix} = \begin{pmatrix} k_1\overline{k_1} & k_1\overline{k_2} \\ k_2\overline{k_1} & k_2\overline{k_2} \end{pmatrix} \tag{34}$$

This is called a **Projection Operator**, it projects any vector **onto the direction of** $|k\rangle$.

**Definition 1: Projection Operator**

A **Projection Operator** (or simply **Projector**) is a linear operator that **extracts the component of a quantum state along a specific direction** (or subspace).

For a normalized state $|k\rangle$, the projector onto $|k\rangle$ is defined as:

$$P_k = |k\rangle \langle k| \tag{35}$$

When applied to an arbitrary state

$$|v\rangle = \sum_j c_j |j\rangle,$$

The projector returns only the component of $|v\rangle$ parallel to $|k\rangle$:

$$P_k |v\rangle = (|k\rangle\langle k|) |v\rangle = \langle k|v\rangle |k\rangle$$

For example, let's suppose the state:

$$|v\rangle = a|k\rangle + b|k_\perp\rangle$$

Where $|k\rangle$ is the direction we want to keep and $|k_\perp\rangle$ is the direction we want to discard that is orthogonal to $|k\rangle$ (i.e., $\langle k|k_\perp\rangle = 0$). The projector $P_k$ will keep the component $a|k\rangle$ and remove the component $b|k_\perp\rangle$:

$$\begin{aligned}
P_k |v\rangle &= |k\rangle\langle k| (a|k\rangle + b|k_\perp\rangle) \\
&= a|k\rangle \langle k|k\rangle + b|k\rangle \langle k|k_\perp\rangle \\
&\downarrow |k\rangle \text{ is normalized, then } \langle k|k\rangle = 1 \\
&\quad \text{and } |k_\perp\rangle \text{ is orthogonal to } |k\rangle \text{, then } \langle k|k_\perp\rangle = 0 \\
&= a|k\rangle \underbrace{\langle k|k\rangle}_{=1} + b|k\rangle \underbrace{\langle k|k_\perp\rangle}_{=0} \\
&= a|k\rangle + b|k\rangle \cdot 0 \\
&= a|k\rangle
\end{aligned}$$

Therefore, a **projector acts as a filter**: it keeps the part of the state aligned with $|k\rangle$ and removes all orthogonal components. For example, suppose:

$$|v\rangle = 0.8\,|0\rangle + 0.6\,|1\rangle$$

Project onto $|0\rangle$:

$$|0\rangle\langle 0|\,|v\rangle = |0\rangle\langle 0|\,(0.8\,|0\rangle + 0.6\,|1\rangle) = 0.8\,|0\rangle$$

The $|1\rangle$ part disappears because it is orthogonal to $|0\rangle$, while the $|0\rangle$ part is preserved (but scaled by the coefficient 0.8).

Projection operators satisfy the **idempotence property**:

$$P_k^2 = P_k$$

Meaning that applying the same projector twice has the same effect as applying it once.

### Effect of Measurement

Given a qubit $|\psi\rangle$, applying $M_k$ yields the **(unnormalized) projected vector**:

$$|\psi_k\rangle = M_k\,|\psi\rangle = |k\rangle\,\langle k|\psi\rangle = (|k\rangle\,\langle k|)\,|\psi\rangle \tag{36}$$

To obtain the **new normalized state**, we **divide by the square root of the probability**:

$$|\psi_k^{\text{norm}}\rangle = \frac{M_k \cdot |\psi\rangle}{\sqrt{\langle\psi\,|M_k|\,\psi\rangle}} = \frac{\langle k|\psi\rangle \cdot |k\rangle}{\sqrt{|\langle k|\psi\rangle|^2}} = |k\rangle \tag{37}$$

So after measurement, the qubit **collapses** to $|k\rangle$, the **eigenstate corresponding to the measurement outcome**.

### Why Normalization is Needed

After applying the measurement operator $M_k = |k\rangle\,\langle k|$ (Equation 34) to a state $|\psi\rangle$ (Equation 36), we obtain:

$$|\psi_k\rangle = M_k\,|\psi\rangle = |k\rangle\,\langle k|\psi\rangle$$

This vector is generally **not normalized**. Its norm is:

$$\|\,|\psi_k\rangle\,\|^2 = \langle\psi|\,M_k\,|\psi\rangle = |\,\langle k|\psi\rangle\,|^2 = p_k$$

Where $p_k$ is the **probability of obtaining outcome** $k$.

**Deepening: Why $\|\,|\psi_k\rangle\,\|^2 = p_k$?**

We want to show:

$$\||\psi_k\rangle\|^2 = \langle\psi|\,M_k\,|\psi\rangle = |\langle k|\psi\rangle|^2 = p_k$$

1. **Definition of squared norm**. The squared norm of a ket is:

$$\||\psi_k\rangle\|^2 = \langle\psi_k|\psi_k\rangle$$

Since $|\psi_k\rangle = M_k |\psi\rangle$, we can write:

$$\||\psi_k\rangle\|^2 = \underbrace{(M_k |\psi\rangle)^\dagger}_{\langle\psi_k|} \underbrace{(M_k |\psi\rangle)}_{|\psi_k\rangle}$$

We have only expanded the definition of the squared norm and substituted $|\psi_k\rangle$.

2. **Take the dagger** †. The dagger of a product of operators is the product of the daggers in reverse order:

$$(AB)^\dagger = B^\dagger A^\dagger$$

Applying this to $(M_k |\psi\rangle)^\dagger$:

$$(M_k |\psi\rangle)^\dagger = \langle\psi| M_k^\dagger$$

And substituting back into the expression for the squared norm:

$$\||\psi_k\rangle\|^2 = \langle\psi| M_k^\dagger M_k |\psi\rangle$$

3. **Use the Hermitian property of** $M_k$ (explained later page 82). Since $M_k$ is Hermitian, we have $M_k^\dagger = M_k$. Therefore:

$$\||\psi_k\rangle\|^2 = \langle\psi| M_k M_k |\psi\rangle$$

4. **Use the idempotent property of** $M_k$ (explained later page 83). Since $M_k$ is idempotent, we have $M_k^2 = M_k$. Therefore:

$$\||\psi_k\rangle\|^2 = \langle\psi| M_k |\psi\rangle$$

5. **Substitute the definition of** $M_k$. Since $M_k = |k\rangle\langle k|$, we can write:

$$\||\psi_k\rangle\|^2 = \langle\psi| (|k\rangle\langle k|) |\psi\rangle = \langle\psi|k\rangle\langle k|\psi\rangle$$

6. **Use the property of inner products**. The inner product $\langle\psi|k\rangle$ is the complex conjugate of $\langle k|\psi\rangle$:

$$\langle\psi|k\rangle = \overline{\langle k|\psi\rangle}$$

Therefore (page 8):

$$\||\psi_k\rangle\|^2 = \overline{\langle k|\psi\rangle}\langle k|\psi\rangle = |\langle k|\psi\rangle|^2 = p_k$$

This is exactly the probability $p_k$ of obtaining outcome $k$ when measuring the state $|\psi\rangle$ with the measurement operator $M_k$.

**Important.** The result of a measurement must be a valid quantum state, and therefore it must be **normalized** (i.e., have unit norm).

To restore normalization, we divide the projected vector by its norm:

$$|\psi_k^{\text{norm}}\rangle = \frac{|\psi_k\rangle}{\| \, |\psi_k\rangle \, \|} = \frac{|k\rangle \langle k|\psi\rangle}{\sqrt{|\, \langle k|\psi\rangle \,|^2}}$$

Observe that:

$$\frac{\langle k|\psi\rangle}{|\, \langle k|\psi\rangle \,|}$$

Is a complex number of magnitude 1, i.e., a **global phase**. Since global phases do not affect physical states, we conclude:

$$|\psi_k^{\text{norm}}\rangle = |k\rangle$$

**Interpretation** Measurement can be understood as a two-step process:

1. **Projection**: extract the component of $|\psi\rangle$ along $|k\rangle$ (this determines the probability $p_k$).
2. **Renormalization**: rescale the result to obtain a valid quantum state.

### Probability of Outcome

The **probability of measuring the qubit in the state** $|k\rangle$ is:

$$p_k = \langle \psi \,| M_k |\, \psi \rangle = |\langle k|\psi\rangle|^2 = \underbrace{\left|\overline{k_1} \cdot \psi_1 + \overline{k_2} \cdot \psi_2\right|^2}_{|\langle k|\psi\rangle|^2 \text{ expanded}} \tag{38}$$

This is the **Born rule**, a fundamental postulate of quantum mechanics. It states that the probability of obtaining a specific measurement outcome is given by the **squared magnitude of the inner product between the state being measured and the eigenstate corresponding to that outcome**. In other words, it quantifies how much the state $|\psi\rangle$ (*the state being measured*) overlaps with the state $|k\rangle$ (*the eigenstate corresponding to the outcome*). The more $|\psi\rangle$ is aligned with $|k\rangle$, the higher the probability of measuring $k$. Conversely, if $|\psi\rangle$ is orthogonal to $|k\rangle$, the probability of measuring $k$ is zero. For example:

$$p_0 = \langle \psi \,| M_0 |\, \psi \rangle = |\langle 0|\psi\rangle|^2 = |1 \cdot \psi_1 + \cancel{0 \cdot \psi_2}| = |\psi_1|$$
$$p_1 = \langle \psi \,| M_1 |\, \psi \rangle = |\langle 1|\psi\rangle|^2 = |\cancel{0 \cdot \psi_1} + 1 \cdot \psi_2| = |\psi_2|$$

These probabilities measure the likelihood of obtaining the outcomes 0 and 1 when measuring the qubit $|\psi\rangle$ in the computational basis.

**Remark:** $\langle \psi | M_k | \psi \rangle$

This is called a quadratic form in linear algebra, and in quantum mechanics it represents the **probability of the measurement outcome** $k$ when **measuring the qubit** $|\psi\rangle$ with **measurement operator** $M_k$. Suppose a qubit $\psi$:

$$|\psi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \qquad \text{with } |a|^2 + |b|^2 = 1 \text{ (normalization condition)}$$

If we apply the measurement operator $M_0$:

$$M_0 |\psi\rangle = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

And we finally calculate the bra $\langle \psi |$:

$$\langle \psi | \begin{pmatrix} a \\ 0 \end{pmatrix} = \begin{pmatrix} \overline{a} & \overline{b} \end{pmatrix} \begin{pmatrix} a \\ 0 \end{pmatrix} = \overline{a} \cdot a = |a|^2$$

In other words, this is the probability of measuring 0: $p_0$.

### Standard Basis Measurement

In quantum computing, we usually measure in the **computational basis** $\{|0\rangle, |1\rangle\}$. The **projectors** are:

$$M_0 = |0\rangle \langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \qquad M_1 = |1\rangle \langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \tag{39}$$

For a **general qubit** $|\psi\rangle = a|0\rangle + b|1\rangle$, the **probabilities** are:

$$p_0 = |a|^2 \qquad p_1 = |b|^2 \tag{40}$$

After measurement, the qubit **collapses to** $|0\rangle$ **with probability** $|a|^2$, or to $|1\rangle$ **with probability** $|b|^2$.

### Key Properties of Measurement Operator

- **Hermitian**: $M_k^\dagger = M_k$, which means that $M_k$ **is equal to its conjugate transpose**. This implies that $M_k$ has real eigenvalues and orthogonal eigenvectors.

$$\begin{aligned} M_k^\dagger &= (|k\rangle \langle k|)^\dagger \\ &\downarrow \text{ use the rule } (AB)^\dagger = B^\dagger A^\dagger \\ &= (\underbrace{|k\rangle}_{A} \underbrace{\langle k|}_{B})^\dagger \\ &= (\langle k|)^\dagger (|k\rangle)^\dagger \\ &= |k\rangle \langle k| \\ &= M_k \end{aligned}$$

- **Idempotent**: once applied, **applying it again does nothing**:

  $$M_k^2 = M_k$$

  A simple proof:

  $$M_k^2 = |k\rangle\,|k\rangle\langle k|\,\langle k| = |k\rangle\,\langle k| = M_k$$

  This reflects that **after projection, the qubit is already in the state** $|k\rangle$.

- **Non-unitary** and **Non-reversible**: measurement **destroys superposition** and **cannot be undone**.

# 4 Multiple Qubit Gates

## 4.1 Multiple Qubit States

Now that we have a solid understanding of single qubits and their states, we can transition to discussing many qubits as one joint system, rather than just one vector with one qubit.

With **one qubit**, life is simple because the state of the system can be described by a single vector in a two-dimensional complex vector space:

$$|v\rangle = a|0\rangle + b|1\rangle$$

With **two qubits**, something radical happens. We can no longer describe the system by looking at each qubit separately. Instead, we must describe **one global state** that captures the behavior of both qubits together. For example:

$$|v\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

This is not just a simple combination of two single-qubit states.

**Exponential Explosion.** The first thing to notice is that the number of amplitudes needed to describe the state of a system grows exponentially with the number of qubits.

- For 1 qubit, we need 2 amplitudes: $a$ and $b$.
- For 2 qubits, we need 4 amplitudes: $c_0$, $c_1$, $c_2$, and $c_3$.
- For 3 qubits, we need 8 amplitudes: $d_0$, $d_1$, $d_2$, $d_3$, $d_4$, $d_5$, $d_6$, and $d_7$.
- For $n$ qubits, we need $2^n$ amplitudes to fully describe the state of the system.

This is why quantum computing is so powerful: it can **process many possibilities at the same time** and **manipulate them so that only the correct ones survive**. This will become clear when we introduce entanglement (page 93).

### The Key New Phenomenon: Entanglement

**Entanglement is the key new phenomenon when dealing with multiple qubits.** It refers to quantum states that cannot be expressed as a tensor product of individual qubit states. For example, the state:

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Is an entangled state. In such cases, the system must be described globally, and it is not possible to assign independent quantum states to each qubit.

If one qubit is measured, the outcome is random, but it determines the outcome of the other qubit due to the structure of the global state. This leads to strong correlations between measurements results. However, this does not imply any form of information transfer between the qubits.

Entanglement is a fundamental resource in quantum computing, enabling correlations that have no classical counterpart, especially when combined with interference effects in quantum algorithms.

- Entanglement does not mean that one qubit sends information to the other. The observed correlations arise from the collapse of a **global quantum state**, not from communication between subsystems.
- Before measurement, individual qubits in an entangled state do **not have a well-defined state**. Only the joint system is well-defined.
- Each qubit individually appears **random**, but measurement outcomes are **correlated**.
- Entanglement alone is not the source of quantum computational power; it becomes powerful when combined with **interference** and quantum operations.

### New tools we need

To handle this new world of multiple qubits, we need new mathematical tools. We will introduce the **tensor product**, which allows us to construct the joint state of multiple qubits; we will also introduce the **multi-qubit gates**, which are operations that can manipulate the joint state of multiple qubits simultaneously; and we will discuss how to **measure** multiple qubits and understand the resulting probabilities.

### 4.1.1 Tensor Product

Let's start with something simple. As in previous sections, we have a simple one-qubit state. Now, we want to **add a second qubit to the system** to create a multi-qubit state. For simplicity, we will use only two qubits. *How do we describe the **combined system**?* Simply keeping them separate is not enough because we would miss correlations between the two qubits. We need a way to combine the two qubits into a **single object**. The tool for doing so is the **tensor product**.

**Example 1: Dice Analogy**

Imagine we have two dice. Each die can be in one of six states (1, 2, 3, 4, 5, or 6). If we want to describe the combined state of both dice, we can't just list their individual states separately. Instead, we need to consider all possible combinations of their states. The tensor product allows us to create a new space that includes all these combinations, giving us a complete description of the system.

The **Tensor Product** is a **mathematical operation** that takes two vectors (or more generally, two mathematical objects):

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And produces a new vector that represents the combined state of the two systems:

$$\begin{aligned} |v_A\rangle \otimes |v_B\rangle &= (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle \end{aligned} \tag{41}$$

Here, the resulting state $|v_A\rangle \otimes |v_B\rangle$ is a **superposition of all possible combinations of the states of the two qubits**.

**❓ Why amplitudes multiply? For example, why is the amplitude of $|00\rangle$ equal to $a_0 b_0$?** A qubit is a vector:

$$|v_A\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \quad |v_B\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$$

The tensor product is **defined** as:

$$|v_A\rangle \otimes |v_B\rangle = \begin{pmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{pmatrix} \tag{42}$$

So amplitudes multiply because **that's how the tensor product is defined**. *But why is it defined like that?* Because we want a rule that satisfies two requirements:

1. **Linearity**. Quantum mechanics is linear:

$$(a_0 |0\rangle + a_1 |1\rangle) \otimes |v_B\rangle = a_0 (|0\rangle \otimes |v_B\rangle) + a_1 (|1\rangle \otimes |v_B\rangle)$$

The tensor product must be **bilinear**, i.e., linear in each argument, so that superpositions are preserved when combining quantum states.

2. **Basis consistency**:

$$|0\rangle \otimes |0\rangle = |00\rangle\,, \quad |0\rangle \otimes |1\rangle = |01\rangle\,, \quad |1\rangle \otimes |0\rangle = |10\rangle\,, \quad |1\rangle \otimes |1\rangle = |11\rangle$$

The tensor product must map basis states of individual systems to a consistent basis of the joint system (e.g., $|i\rangle \otimes |j\rangle = |ij\rangle$), preserving the structure of the state space.

This structure ensures that probabilities factorize correctly for independent systems.

**Remark: Inner Product of Tensor Products**

For composite quantum systems (i.e., systems composed of multiple qubits), the **inner product between tensor product states factorizes**:

$$(\langle a| \otimes \langle b|)\,(|c\rangle \otimes |d\rangle) = \langle a|c\rangle \cdot \langle b|d\rangle \tag{43}$$

This means that the "similarity" (overlap) between two composite states is the product of the similarities of their individual subsystems (page 10).

For example:

$$(\langle 0| \otimes \langle +|)\,(|0\rangle \otimes |-\rangle)$$

Has overlap:

$$\langle 0|0\rangle \langle +|-\rangle = 1 \cdot 0 = 0$$

So even though the first qubit is identical, the total states are orthogonal because the second qubits are orthogonal!

**Intuition.** A tensor product state combines independent subsystems. Therefore, the total overlap must take into account the overlap of *each* subsystem simultaneously. If one subsystem is perfectly distinguishable (overlap 0), then the entire composite states become perfectly distinguishable.

### Not all states can be factored

Given two qubits, their joint state lives in a 4-dimensional vector space. However, **not all states** in this 4-dimensional space can be written as a tensor product of two single-qubit states. Precisely, if a multi-qubit state can be written as a **tensor product**, then it is **separable**, and we can recover the individual qubit states. But if it is **not a tensor product**, then it is **entangled**, and we cannot assign well-defined quantum states to each qubit individually.

In general, exists a **necessary (and sufficient) condition** for being a tensor product for two qubits. Given a 2-qubit state:

$$|\psi\rangle = (c_0, c_1, c_2, c_3)$$

Where $c_0, c_1, c_2, c_3$ are the amplitudes of the basis states $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, respectively. The state $|\psi\rangle$ can be written as a tensor product of two single-qubit states if and only if:

$$c_0c_3 = c_1c_2 \tag{44}$$

Equivalently, if we reshape the coefficients into a matrix

$$C = \begin{pmatrix} c_0 & c_1 \\ c_2 & c_3 \end{pmatrix}$$

The state is separable if and only if $\det(C) = 0$. More generally, for $n$-qubit states, there is **no single simple scalar condition** that characterizes separability. A state is separable **if and only if it can be written as a tensor product of $n$ single-qubit states**, which is a more complex condition to verify. So, the 2-qubit case is simpler to analyze, but for larger systems, determining whether a state is separable or entangled can be more challenging.

Let's make this more concrete with two examples:

- ✔ Let's start from a 4D vector:

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

We can check if it is a tensor product by verifying the condition in Equation 44:

$$c_0c_3 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}, \quad c_1c_2 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Since $c_0c_3 = c_1c_2$, the state is a tensor product and thus separable.

Now, we can find the two single-qubit states whose tensor product gives us $|\psi\rangle$. We try:

$$|v_A\rangle = \begin{pmatrix} a_0 \\ a_1 \end{pmatrix}, \quad |v_B\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix}$$

We want:

$$(a_0b_0, a_0b_1, a_1b_0, a_1b_1) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right) \quad \Rightarrow \quad \begin{cases} a_0b_0 = \frac{1}{2} \\ a_0b_1 = \frac{1}{2} \\ a_1b_0 = \frac{1}{2} \\ a_1b_1 = \frac{1}{2} \end{cases}$$

Solving this system of equations, we find:

$$a_0 = b_0 = \frac{1}{\sqrt{2}}, \quad a_1 = b_1 = \frac{1}{\sqrt{2}}$$

And this works because:

$$\begin{cases} a_0b_0 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_0b_1 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_1b_0 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \\ a_1b_1 = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} \end{cases}$$

Therefore:

$$|\psi\rangle = \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right] \otimes \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right]$$

The global 2-qubit state can be written as **two independent qubit states**, confirming that it is separable.

$$|\psi\rangle = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \end{pmatrix} = \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right] \otimes \left[\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right]$$

✘ Let's start from a different 4D vector:

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

We check the condition for separability:

$$c_0 c_3 = \frac{1}{2}, \quad c_1 c_2 = 0$$

Since $c_0 c_3 \neq c_1 c_2$, the state is not a tensor product and thus is **not separable**. This state is an example of an **entangled state** (**Entanglement** phenomenon), where the two qubits are correlated in such a way that we cannot describe them independently. We will see more about entanglement in the next section, but for now, the key takeaway is that not all multi-qubit states can be factored into tensor products of single-qubit states, and this is a fundamental aspect of quantum mechanics that leads to phenomena like entanglement.

In summary, the tensor product generates only a subset of the full 4-dimensional space of two-qubit states. Separable states have a special multiplicative structure and lie on a lower-dimensional manifold within this space. Most states in the full space do not satisfy this structure and are therefore entangled.

### 4.1.2 Building a Two-Qubit System

In the previous sections, we have seen how to represent the state of a single qubit using a two-dimensional complex vector. Now, we want to extend this representation to a system of two qubits. To do this, we will use the concept of the tensor product seen in the previous section.

We start from two independent qubits:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad \text{and} \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And our **goal is to find a state that encodes all joint possibilities** of the two qubits:

- Both qubits are in state $|0\rangle$: $|00\rangle$
- The first qubit is in state $|0\rangle$ and the second in state $|1\rangle$: $|01\rangle$
- The first qubit is in state $|1\rangle$ and the second in state $|0\rangle$: $|10\rangle$
- Both qubits are in state $|1\rangle$: $|11\rangle$

To represent these joint possibilities, we will use the tensor product of the two qubits' states:

$$|v_A v_B\rangle = |v_A\rangle \otimes |v_B\rangle$$

Expanding this expression, we get:

$$\begin{aligned} |v_A v_B\rangle &= (a_0 |0\rangle + a_1 |1\rangle) \otimes (b_0 |0\rangle + b_1 |1\rangle) \\ &= a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle \end{aligned} \tag{45}$$

This state $|v_A v_B\rangle$ is a linear combination of the four possible joint states of the two qubits. Each term in the expansion corresponds to one of the joint possibilities, and the coefficients $a_0 b_0$, $a_0 b_1$, $a_1 b_0$, and $a_1 b_1$ represent the probability amplitudes of each joint state.

We are no longer describing two systems separately, we are describing **one system with 4 possible outcomes**.

#### General Two-Qubit State

In Equation 45 we built a two-qubit state starting from a **very specific structure**, where the coefficients of the state were products of the coefficients of the individual qubits (i.e., $a_i b_j$). However, we can also consider a more **general two-qubit state**, where the coefficients are not necessarily products of the individual qubits' coefficients:

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle \tag{46}$$

Where $c_0$, $c_1$, $c_2$, and $c_3$ are **arbitrary complex numbers** that represent the probability amplitudes of each joint state.

Obviously, this general two-qubit state **must satisfy the normalization condition**:

$$|c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2 = 1 \tag{47}$$

This means that the sum of the probabilities of all joint states must equal 1, ensuring that the state is properly normalized.

We can also generalize deeper to $n$ qubits, where the state of the **system is a linear combination** of $2^n$ **basis states**, and the **coefficients** are arbitrary **complex numbers** that satisfy the normalization condition. Mathematically, we can express this as:

$$|v\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle \tag{48}$$

Where $|k\rangle$ represents the basis states of the $n$-qubit system, and $c_k$ are the corresponding probability amplitudes. As we have already discussed at page 84, this general form illustrates the exponential growth of the state space as the number of qubits increases. However, we can write our one- and two-qubit states as special cases from this general form:

- For $n = 1$ qubit, we have $2^1 = 2$ basis states: $|0\rangle$ and $|1\rangle$, and the state can be written as:
$$|v\rangle = c_0 |0\rangle + c_1 |1\rangle$$
- For $n = 2$ qubits, we have $2^2 = 4$ basis states: $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, and the state can be written as:
$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

In summary, a general two-qubit state is a normalized superposition (i.e., the *coefficients satisfy the normalization condition*) of the four basis states (i.e., $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$). Unlike tensor product states, its coefficients are arbitrary and *may not* factorize, leading to entanglement and non-classical correlations between the qubits (like the state in Example at page 89).

### 4.1.3 Separable vs Entangled States

Let the general state of a two-qubit system be (as defined in Equation 46, page 90):

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

The question is: ***can we interpret this as two independent qubits?*** The answer is **not always**. As we saw on page 86, the answer depends on whether the states are separable or entangled.

#### Separable States

A state is **separable if it can be written as a tensor product of two single-qubit states**. In other words, there exist single-qubit states $|v_A\rangle$ and $|v_B\rangle$ such that:

$$|v\rangle = |v_A\rangle \otimes |v_B\rangle$$

So the system can be split into one state for qubit $A$ and one state for qubit $B$, and the global state is just the combination of these two states. As consequence, the coefficients factorize as:

$$c_0 = a_0 b_0, \quad c_1 = a_0 b_1, \quad c_2 = a_1 b_0, \quad c_3 = a_1 b_1$$

#### Entangled States

A state is **entangled if it cannot be written as a tensor product of two single-qubit states**. In other words, there do not exist single-qubit states $|v_A\rangle$ and $|v_B\rangle$ such that:

$$|v\rangle \neq |v_A\rangle \otimes |v_B\rangle$$

In this case, the state cannot be separated into two independent qubits, and the coefficients do not factorize as above. The system is simply one single entity. For example, the Bell state (we will see in future sections, page 116) is an entangled state:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}} \left(|00\rangle + |11\rangle\right)$$

And the demonstration was written on page 89. In other words, we can only say that the system is in state $|\Phi^+\rangle$, but we cannot say that qubit $A$ is in some state and qubit $B$ is in some state, because the state of the system does not factorize into two independent states.

At this stage, entanglement is introduced as a structural property of quantum states. Its physical meaning and implications become clear only when studying measurement and quantum circuits in later sections.

**Difference.** With **separable** states, the information is stored in the individual qubits. In contrast, with **entangled** states, the information is stored in the correlations between the qubits.

### 4.1.4 Why Entanglement Matters

Let's answer to the question "*why entanglement matters*" step by step. First of all, the **complexity of quantum states grows exponentially** with the number of qubits, as we have introduced at page 84. As consequence, to describe the system, we need $2^n$ complex numbers, where $n$ is the number of qubits. This means that for a system of 50 qubits, we would need $2^{50}$ complex numbers to describe its state, which is an astronomically large number (approximately $1.13 \times 10^{15}$). It could be a **problem for classical computers**, but quantum computers can handle this complexity naturally.

The reason why quantum computers can handle this complexity naturally is not that they explicitly store all these amplitudes as a classical computer would. Rather, the quantum system itself physically evolves in this exponentially large state space.

Entanglement matters because **most quantum states cannot be decomposed into independent single-qubit states**. In fact, the set of separable states, i.e., states that can be written as a tensor product of individual qubit states, is **very small compared to the set of all possible states**. For two qubits, separable states must satisfy:

$$c_0 c_3 = c_1 c_2$$

Which is a strong structural constraint. Therefore, most two-qubit states do not satisfy this condition and are entangled.

**❓ But why does this matter?** Because **entanglement prevents the system from being compressed into simpler independent components**. The state must be described globally. This global structure is difficult to simulate classically, but quantum mechanics allows us to manipulate it directly through quantum operations. This is one of the reasons why quantum computers can solve certain problems more efficiently than classical computers.

In summary, entanglement is a fundamental feature of quantum mechanics that prevents a multi-qubit system from being decomposed into independent subsystems. As the number of qubits increases, the state space grows exponentially, and entanglement forces a global description of the system. This global structure can be exploited by quantum algorithms to achieve computational advantages over classical approaches.

### 4.1.5 Exercise: Normalization of Tensor Product States

Given two qubits in the states $|v_A\rangle$ and $|v_B\rangle$ respectively:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle, \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

And the combined state of the two qubits is given by the tensor product:

$$|v_A v_B\rangle = |v_A\rangle \otimes |v_B\rangle = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

Show that:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2 = 1$$

*Proof.* To show that the sum of the squares of the coefficients equals 1, we start expanding the expression:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2$$

Using the property:

$$|xy|^2 = |x|^2 |y|^2$$

Applying this property to each term, we get:

$$\begin{aligned} &= |a_0|^2 |b_0|^2 + |a_0|^2 |b_1|^2 + |a_1|^2 |b_0|^2 + |a_1|^2 |b_1|^2 \\ &= |a_0|^2 \left(|b_0|^2 + |b_1|^2\right) + |a_1|^2 \left(|b_0|^2 + |b_1|^2\right) \end{aligned}$$

Since $|v_B\rangle$ is a valid quantum state and must be normalized, we have:

$$|b_0|^2 + |b_1|^2 = 1$$

Substituting this back into our expression, we get:

$$\begin{aligned} &= |a_0|^2 \cdot 1 + |a_1|^2 \cdot 1 \\ &= |a_0|^2 + |a_1|^2 \end{aligned}$$

Since $|v_A\rangle$ is also a valid quantum state and must be normalized, we have:

$$|a_0|^2 + |a_1|^2 = 1$$

Therefore, we conclude that:

$$|a_0 b_0|^2 + |a_0 b_1|^2 + |a_1 b_0|^2 + |a_1 b_1|^2 = 1$$

The combined state $|v_A v_B\rangle$ is also a valid quantum state and must be normalized, which is consistent with our result. QED

This exercise demonstrates that the **tensor product of two normalized quantum states is also normalized**, because the squared amplitudes factorize and the normalization conditions of the individual states ensure that the total probability remains 1.

## 4.2 Introduction to Multiple Qubit Gates

Until now, we learned that a quantum state is a vector in a complex vector space, and for 2 qubits, the state is a vector in $\mathbb{C}^4$:

$$|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$$

Now the question is: ***how do we change this state?***

**The key idea.** A **quantum gate** is just a rule that transforms one state into another. For 1 qubit we already know (page 40):

$$|v'\rangle = U |v\rangle$$

Where $U$ is a $2\times2$ unitary matrix. For example, the Hadamard gate $H$ (page 63) transforms the state $|0\rangle$ into $|+\rangle$:

$$H |0\rangle = |+\rangle$$

**What about 2 qubits?** For 2 qubits, we need a $4 \times 4$ unitary matrix to transform the state:

$$|v'\rangle = U |v\rangle \quad \text{with} \quad |v\rangle = \begin{pmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{pmatrix}, \quad U = \begin{pmatrix} u_{00} & u_{01} & u_{02} & u_{03} \\ u_{10} & u_{11} & u_{12} & u_{13} \\ u_{20} & u_{21} & u_{22} & u_{23} \\ u_{30} & u_{31} & u_{32} & u_{33} \end{pmatrix}$$

This means that the gate can change **all amplitudes at once**, not just one at a time as in the single-qubit case. This is the important difference between single and multi-qubit gates: with 1 qubit, we modify only 2 amplitudes, but with 2 qubits, we can modify all 4 amplitudes **simultaneously**. And now, the transformation can mix amplitudes, create correlations, and even generate entanglement between the qubits.

In simple terms, a multi-qubit gate is just a transformation of a larger state vector. Nothing more, nothing less. However, this simple idea has important implications: it allows us to manipulate the global state of a quantum system, including correlations between qubits, which can lead to computational advantages over classical approaches.

## 4.3 Parallel Gates

Now that we have a way to represent the state of multiple qubits, we can also represent the operations (gates) that act on them. In particular, if we have two qubits $|v_A\rangle$ and $|v_B\rangle$, and two gates $A$ and $B$, what happens when we want to apply $A$ to $|v_A\rangle$ and $B$ to $|v_B\rangle$ simultaneously?

The answer is given by the **tensor product of quantum gates**:

$$(A \otimes B)\left(|v_A\rangle \otimes |v_B\rangle\right) = (A\,|v_A\rangle) \otimes (B\,|v_B\rangle) \tag{49}$$

This means that the combined operation of applying $A$ to $|v_A\rangle$ and $B$ to $|v_B\rangle$ is represented by the tensor product of the two gates, $A \otimes B$. So, **applying gates in parallel** corresponds to the tensor product of the individual gates.

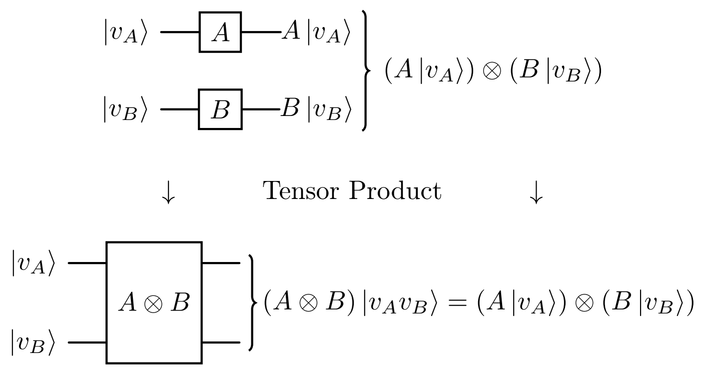



Figure 16: Parallel gates as a tensor product operation.

In simple terms, instead of applying $A$ and $B$ separately, we build one big matrix $A \otimes B$ that acts on the **whole system**. Note that a state is represented as a column vector (e.g., size 4 for two qubits), and a gate is represented as a square matrix (e.g., size $4 \times 4$ for two qubits). So *parallel gates* are simply one big transformation of the global state of the system.

### ❓ What if we want to apply a gate to only one qubit?

Suppose we want to **apply a gate** $A$ to the **first qubit** $|v_A\rangle$ while **leaving the second qubit** $|v_B\rangle$ **unchanged**. We can represent this as:

$$(A \otimes I)\left(|v_A\rangle \otimes |v_B\rangle\right) = (A\,|v_A\rangle) \otimes |v_B\rangle \tag{50}$$

Here, $I$ is the **identity gate** (page 47) that acts on the second qubit, meaning it **leaves it unchanged**. So, to apply a gate to just one qubit in a multi-qubit system, we use the tensor product of that gate with the identity operator on the other qubits.

**Example 2:** $H \otimes X$

Let's consider the gates $H$ (Hadamard, page 63):

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

And $X$ (Pauli-X, page 49):

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

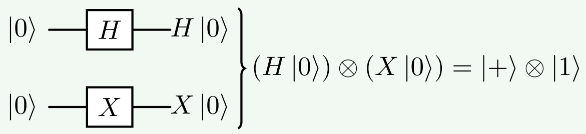
The tensor product $H \otimes X$ is calculated as follows:

$$\begin{aligned}
H \otimes X =& \frac{1}{\sqrt{2}} \left( \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) \\
=& \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \cdot X & 1 \cdot X \\ 1 \cdot X & -1 \cdot X \end{pmatrix} \\
=& \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & -1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix} \\
=& \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix}
\end{aligned}$$

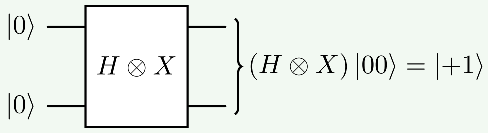
Now, if we apply $H \otimes X$ to a simple two-qubit state, say $|00\rangle$:

$$\begin{aligned}
(H \otimes X) |00\rangle =& \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\
=& \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 0 + -1 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 + -1 \cdot 0 + 0 \cdot 0 \end{pmatrix} \\
=& \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \\
=& \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \\
=& \frac{1}{\sqrt{2}} |01\rangle + \frac{1}{\sqrt{2}} |11\rangle
\end{aligned}$$

This can be visually represented as parallel gates acting on the two qubits:




Or as a single global gate:




Where $|{+}1\rangle$ is $|+\rangle \otimes |1\rangle$. Note that applying the Hadamard gate to the qubit $|0\rangle$ creates the superposition state $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, while applying the $X$ gate to $|0\rangle$ flips it to $|1\rangle$.

**Example 3: $I \otimes X$**

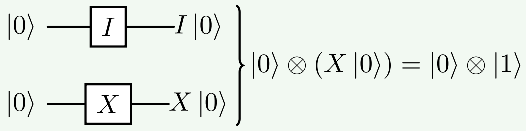
Now, let's consider applying the identity gate $I$ to the first qubit and the $X$ gate to the second qubit. The tensor product is:

$$
\begin{aligned}
I \otimes X &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 0 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ 0 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} & 1 \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \end{pmatrix} \\
&= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}
\end{aligned}
$$

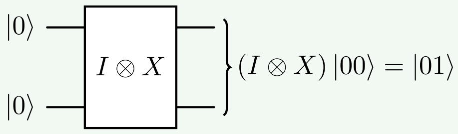
Applying this to the state $|00\rangle$ gives:

$$
\begin{aligned}
(I \otimes X)|00\rangle &= \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 + 0 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 + 1 \cdot 0 + 0 \cdot 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \\
&= |01\rangle
\end{aligned}
$$

This means that applying $I$ to the first qubit leaves it unchanged as $|0\rangle$, while applying $X$ to the second qubit flips it from $|0\rangle$ to $|1\rangle$, resulting in the combined state $|01\rangle$. Visually, this can be represented as:




Or as a single global gate:



## 4.4 Hadamard Transform on Multiple Qubits

Applying the Hadamard gate to every qubit is a useful tool in quantum computing, as it creates a superposition of all possible states. Now, let's see how this is done.

We already know that applying the Hadamard gate to a single qubit in the state $|0\rangle$ gives us:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This means that $H$ creates a **balanced superposition** of the two basis states $|0\rangle$ and $|1\rangle$.

Now, if we apply this idea to two qubits, we can apply the Hadamard gate to each qubit individually. This is done using the tensor product of the Hadamard gates, for example:

$$(H \otimes H)|00\rangle = H|0\rangle \otimes H|0\rangle$$

So it is equivalent to applying the Hadamard gate to each qubit separately. Expanding this, we get:

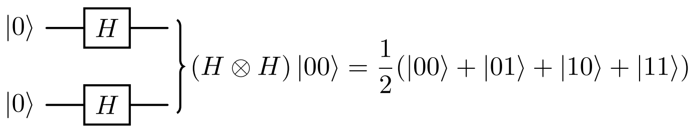
$$\begin{aligned}
(H \otimes H)|00\rangle &= H|0\rangle \otimes H|0\rangle \\
&= \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \otimes \left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) \\
&= \frac{1}{2}(|0\rangle + |1\rangle) \otimes (|0\rangle + |1\rangle) \\
&= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)
\end{aligned}$$

This means that applying Hadamard gate in parallel **generates all possible combinations of the basis states** for the two qubits, creating a superposition of all four possible states.

In general, for $n$ qubits, applying the Hadamard gate to each qubit will create a superposition of all $2^n$ possible states. The resulting state can be expressed as:

$$\left(H^{\otimes n}\right)|0\rangle^{\otimes n} = \frac{1}{2^{\frac{n}{2}}}\sum_{k=0}^{2^n-1}|k\rangle = \frac{1}{\sqrt{2^n}}\sum_{k=0}^{2^n-1}|k\rangle \tag{51}$$

Where $|k\rangle$ represents the basis states of the $n$-qubit system, and the **sum runs over all possible combinations of the basis states**.




In summary, **Hadamard on all qubits spreads the state over the entire space of possible states**, creating a uniform superposition that is fundamental in many quantum algorithms.

## 4.5 Main Multi-Qubit Gates

### 4.5.1 Controlled NOT (CNOT) Gate

The **Controlled NOT (CNOT) Gate** is a **two-qubit gate** composed of:

- A **control qubit** that determines whether the NOT operation is applied or not.
- A **target qubit** that is flipped (NOT operation) if the control qubit is in the state $|1\rangle$, and remains unchanged if the control qubit is in the state $|0\rangle$.

In simple terms, it is a gate that performs a NOT operation on the *target* qubit only when the *control* qubit is in the state $|1\rangle$.

#### Action on Basis States

The CNOT gate acts on the computational basis states as follows:

| **Input** | **Output** |
|---|---|
| $\|00\rangle$ | $\|00\rangle$ |
| $\|01\rangle$ | $\|01\rangle$ |
| $\|10\rangle$ | $\|11\rangle$ |
| $\|11\rangle$ | $\|10\rangle$ |

If the first qubit (control) is $|0\rangle$, the second qubit (target) remains unchanged. If the first qubit is $|1\rangle$, the second qubit is flipped: $|0\rangle \leftrightarrow |1\rangle$.

#### Matrix Representation

The CNOT gate can be represented as a $4 \times 4$ unitary matrix in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{52}$$

This matrix indicates that the first two basis states $|00\rangle$ and $|01\rangle$ are unchanged, while $|10\rangle$ and $|11\rangle$ are swapped, reflecting the action of the CNOT gate on the target qubit based on the state of the control qubit.

#### Action on General States

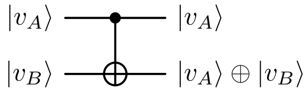
For a general two-qubit state $|v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$, the CNOT gate transforms it as follows:

$$\text{CNOT} |v\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |11\rangle + c_3 |10\rangle \tag{53}$$

The amplitudes $c_0$ and $c_1$ remain unchanged, while $c_2$ and $c_3$ are **swapped**, demonstrating the conditional nature of the CNOT gate based on the state of the control qubit.

### Circuit Representation

In quantum circuit diagrams, the CNOT gate is typically represented with a solid dot on the control qubit line and a plus sign $\oplus$ (indicating the NOT operation) on the target qubit line, connected by a vertical line.




Here, $|v_A\rangle$ is the control qubit and $|v_B\rangle$ is the target qubit. The output state of the target qubit is the result of the XOR operation between the control and target qubits, reflecting the conditional flipping of the target qubit based on the state of the control qubit.

In summary, if the control qubit is $|1\rangle$, apply NOT (Pauli-X) to the target qubit; otherwise do nothing.

### Properties

- **Unitary**

$$\text{CNOT}^\dagger \cdot \text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}^\dagger \cdot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = I$$

  CNOT gate is unitary, so it is a valid quantum gate that preserves normalization of quantum states, and the total probability of all outcomes remains 1.

- **Reversible**

$$\text{CNOT}^{-1} = \text{CNOT}^\dagger = \text{CNOT}$$

  The CNOT gate is reversible, meaning that applying the CNOT gate twice will return the system to its original state:

$$\text{CNOT}\,(\text{CNOT}\,(|\psi\rangle)) = |\psi\rangle$$

  In other words, we can **always undo a CNOT operation without loss of information**.

- **Conditional Operation**

$$\text{CNOT}\,(|v_A\rangle \otimes |v_B\rangle) \neq \text{CNOT}\,|v_A\rangle \otimes \text{CNOT}\,|v_B\rangle$$

  The CNOT gate is a conditional operation, meaning that the **action on the target qubit depends on the state of the control qubit**. This is different from parallel gates, where each qubit is acted upon independently:

$$(H \otimes X)\,|00\rangle = (H\,|0\rangle) \otimes (X\,|0\rangle)$$

  CNOT **breaks parallelism** because it introduces interaction: the evolution of one qubit depends on another, so the **system can no longer be described as independent parts**.

- **Entangling**. CNOT creates *entanglement* **if and only if** the input state has superposition on the control qubit (in the computational basis).

  For example:

  $$(H \otimes I)|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

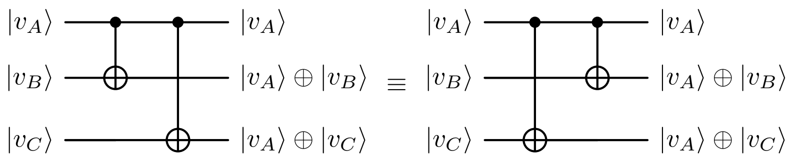
  Applying CNOT to this state:

  $$\text{CNOT}\left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

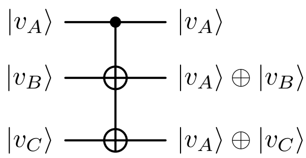
  The resulting state is an entangled state (cannot be factored into a tensor product, page 87), demonstrating that CNOT can create entanglement from a superposition state of the control qubit.

## Multiple CNOT Gates

Multi-CNOT notation means applying several CNOT gates that share the same control qubit, and these can be represented equivalently as sequential gates or as a single control connected to multiple targets. Visually, this can be represented as:




But it can also be represented as:




The order of the CNOT gates does not matter because they **commute when they share the same control qubit**, meaning that the final state of the system will be the same regardless of the order in which the CNOT gates are applied.

### 4.5.2 Generic Controlled Gate (CU)

The **Generic Controlled Gate (CU)** is a **two-qubit gate** composed of:

- A **control qubit** that determines whether the operation is applied or not (first qubit).
- A **target qubit** on which the operation is applied if the control qubit is in the state $|1\rangle$ (second qubit).
- A **unitary operation** $U$ that is applied to the target qubit when the control qubit is in the state $|1\rangle$.

It **generalizes the CNOT gate** (page 101, $U = X$) by allowing **any unitary operation** $U$ to be applied to the target qubit, rather than just the NOT operation.

#### Action on Basis States

The action of the CU gate on the computational basis states can be described as follows:

| Input | Output |
|---|---|
| $\|00\rangle$ | $\|00\rangle$ |
| $\|01\rangle$ | $\|01\rangle$ |
| $\|10\rangle$ | $\|1\rangle \otimes U\|0\rangle$ |
| $\|11\rangle$ | $\|1\rangle \otimes U\|1\rangle$ |

If the control qubit is $|0\rangle$, the target qubit remains unchanged. If the control qubit is $|1\rangle$, the unitary operation $U$ is applied to the target qubit, transforming $|0\rangle$ to $U|0\rangle$ and $|1\rangle$ to $U|1\rangle$.

#### Matrix Representation

The CU gate can be represented as a $4 \times 4$ unitary matrix in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$:

$$\mathrm{CU} = C_U = \begin{bmatrix} I & 0 \\ 0 & U \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix} \tag{54}$$

Where $U_{00}, U_{01}, U_{10}, U_{11}$ are the elements of the $2 \times 2$ unitary matrix $U$ that acts on the target qubit when the control qubit is $|1\rangle$. For the first two basis states $|00\rangle$ and $|01\rangle$, the identity operation $I$ is applied, leaving them unchanged. For the last two basis states $|10\rangle$ and $|11\rangle$, the unitary operation $U$ is applied to the target qubit, resulting in the transformation defined by $U$.

## Action on General States

Let's consider two general single-qubit states:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

Combining these states into a two-qubit state, we have:

$$|v\rangle = |v_A\rangle \otimes |v_B\rangle = a_0 |0\rangle |v_B\rangle + a_1 |1\rangle |v_B\rangle$$

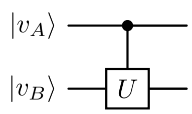
Applying the CU gate to this state, we get:

$$\begin{aligned} \mathrm{CU}\,|v\rangle &= \mathrm{CU}(a_0 |0\rangle |v_B\rangle + a_1 |1\rangle |v_B\rangle) \\ &= a_0 \mathrm{CU}(|0\rangle |v_B\rangle) + a_1 \mathrm{CU}(|1\rangle |v_B\rangle) \\ &= \underbrace{a_0 |0\rangle |v_B\rangle}_{\text{unchanged}} + \underbrace{a_1 |1\rangle U |v_B\rangle}_{\text{transformed}} \end{aligned}$$

This shows that the CU gate applies the unitary operation $U$ to the target qubit only when the control qubit is in the state $|1\rangle$, while leaving the target qubit unchanged when the control qubit is in the state $|0\rangle$. Each component of the superposition evolves independently according to the value of the control qubit, resulting in a new superposition where different branches have undergone different transformations.

## Circuit Representation

In quantum circuit diagrams, the CU gate is typically represented with a **control dot on the control qubit** and a **box labeled with the unitary operation $U$ on the target qubit**. The control dot indicates that the operation is conditional on the state of the control qubit. The circuit representation of the CU gate is as follows:




In summary, if the control qubit is $|1\rangle$, we apply $U$ to the target qubit; otherwise do nothing.

## Properties

The CU gate has the same properties as the CNOT gate because it is a generalization of the CNOT gate. Specifically:

- **Unitary**

$$\mathrm{CU}^\dagger \mathrm{CU} = I$$

- **Reversible**

$$\mathrm{CU}^{-1} = \mathrm{CU}^\dagger$$

- **Conditional Operation**

$$\mathrm{CU}\left(|v_A\rangle \otimes |v_B\rangle\right) \neq \mathrm{CU}\,|v_A\rangle \otimes \mathrm{CU}\,|v_B\rangle$$

It does not act independently on each qubit, but rather applies a transformation to the target qubit based on the state of the control qubit.

- **Entangling**. The CU gate can create entanglement between the control and target qubits if the unitary operation $U$ and the input state are chosen appropriately. For example, when $U = X$ (i.e., the CNOT gate), we have:

$$\mathrm{CNOT}(|+\rangle\,|0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

### 4.5.3 SWAP gate

The **SWAP gate** is a **two-qubit gate** that exchanges the states of two qubits and has no control/target distinction, it is **symmetric**. It simply swaps:

$$|v_A\rangle \otimes |v_B\rangle \xrightarrow{SWAP} |v_B\rangle \otimes |v_A\rangle \tag{55}$$

#### Action on Basis States

The SWAP gate acts on the basis states as follows:

| **Input** | **Output** |
|---|---|
| $\|00\rangle$ | $\|00\rangle$ |
| $\|01\rangle$ | $\|10\rangle$ |
| $\|10\rangle$ | $\|01\rangle$ |
| $\|11\rangle$ | $\|11\rangle$ |

It essentially **swaps the second and third basis states**, while leaving the first and fourth unchanged.

#### Matrix Representation

The matrix representation of the SWAP gate in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ is:

$$\text{SWAP} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{56}$$

This matrix clearly shows the **swapping of the second and third basis states**, while the first and fourth remain unchanged.

#### Action on General States

Let's consider two arbitrary single-qubit states:

$$|v_A\rangle = a_0 |0\rangle + a_1 |1\rangle, \quad |v_B\rangle = b_0 |0\rangle + b_1 |1\rangle$$

The tensor product of these states is:

$$|v_A\rangle \otimes |v_B\rangle = a_0 b_0 |00\rangle + a_0 b_1 |01\rangle + a_1 b_0 |10\rangle + a_1 b_1 |11\rangle$$

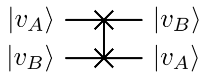
Applying the SWAP gate to this state gives:

$$\begin{aligned} \text{SWAP}(|v_A\rangle \otimes |v_B\rangle) &= a_0 b_0 |00\rangle + a_1 b_0 |01\rangle + a_0 b_1 |10\rangle + a_1 b_1 |11\rangle \\ &= (b_0 |0\rangle + b_1 |1\rangle) \otimes (a_0 |0\rangle + a_1 |1\rangle) \\ &= |v_B\rangle \otimes |v_A\rangle \end{aligned}$$

This confirms that the SWAP gate indeed **exchanges the states of the two qubits**, as expected.

### Circuit Representation

In quantum circuit diagrams, the SWAP gate is typically represented by two crossed lines with $\times$ symbols at the crossing point:




This notation visually emphasizes the **swapping of the two qubits' states**.

### Properties

- **Unitary**
$$\mathrm{SWAP}^\dagger \mathrm{SWAP} = I$$
This means that the SWAP gate preserves the inner product and is reversible.

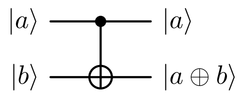
- **Reversible (self-inverse)**
$$\mathrm{SWAP}^2 = I$$
This means that applying the SWAP gate twice will return the system to its original state.

- **Symmetric (no control/target)**. The SWAP gate does not distinguish between the two qubits; it treats them symmetrically. This is in contrast to gates like CNOT, which have a clear control and target qubit.

- **Does NOT create entanglement**. The SWAP gate simply exchanges the states of two qubits and does **not introduce any new correlations** between them. If the input state is separable (not entangled), the output state will also be separable (and vice versa).

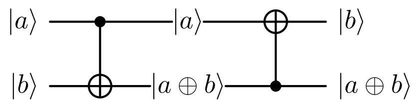
- **Can be decomposed into 3 CNOT gates**. The **SWAP gate can be implemented using three CNOT gates**, which is important for practical implementations on quantum hardware that may not have a native SWAP gate.

  *Proof.* We want to transform:
$$|a,\, b\rangle \to |b,\, a\rangle$$
  But CNOT only does:
$$|a,\, b\rangle \to |a,\, a \oplus b\rangle$$
  So we need to **combine multiple CNOTs** to achieve a SWAP. Let's track what happens to a generic input:
$$|a,\, b\rangle \quad \text{with } a, b \in \{0, 1\}$$
  - **First** CNOT (control: first qubit, target: second qubit):
$$|a,\, b\rangle \xrightarrow{\text{CNOT}} |a,\, a \oplus b\rangle$$



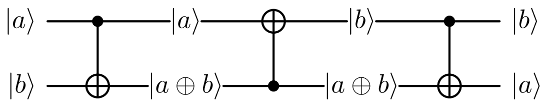

– **Second** CNOT (control: second qubit, target: first qubit):

$$|a,\, a \oplus b\rangle \xrightarrow{\text{CNOT}} |a \oplus (a \oplus b),\, a \oplus b\rangle = |b,\, a \oplus b\rangle$$

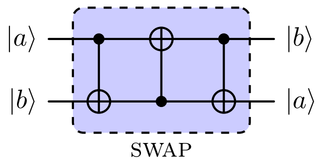



– **Third** CNOT (control: first qubit, target: second qubit):

$$|b,\, a \oplus b\rangle \xrightarrow{\text{CNOT}} |b,\, b \oplus (a \oplus b)\rangle = |b, a\rangle$$




In summary, the sequence of operations is:




QED

In general, the SWAP gate **does not create new correlations**; it only reorderes the information between qubits. In other words, it only changes where the information is stored.

### 4.5.4 Controlled-Controlled-NOT (CCNOT) Gate (Toffoli)

The **Controlled-Controlled-NOT (CCNOT) Gate**, also called the **Toffoli Gate**, is a **three-qubit gate** composed of:

- Two **control qubits** that determine whether the gate will be applied.
- One **target qubit** that is flipped (NOT operation) if both control qubits are in the state $|1\rangle$.

It generalizes the CNOT gate by adding an additional control qubit, making it a crucial component in quantum computing for implementing more complex operations and algorithms.

#### Action on Basis States

The CCNOT gate acts on the basis states as follows:

| Input | Output |
|---|---|
| $\|000\rangle$ | $\|000\rangle$ |
| $\|001\rangle$ | $\|001\rangle$ |
| $\|010\rangle$ | $\|010\rangle$ |
| $\|011\rangle$ | $\|011\rangle$ |
| $\|100\rangle$ | $\|100\rangle$ |
| $\|101\rangle$ | $\|101\rangle$ |
| $\|110\rangle$ | $\|111\rangle$ |
| $\|111\rangle$ | $\|110\rangle$ |

The target flips **only if both control qubits are** $|1\rangle$, demonstrating the conditional nature of the gate (last two rows of the table). For all other input combinations, the output remains unchanged.

#### Matrix Representation

The matrix representation of the CCNOT gate in the computational basis is an $8 \times 8$ (since it operates on three qubits) unitary matrix given by:

$$\text{CCNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \tag{57}$$

This matrix shows that the first six basis states are unchanged (diagonal entries of 1), while the **last two basis states are swapped** (off-diagonal entries of 1 in the last two rows and columns).

### Action on General States

For a general three-qubit state:

$$|\psi\rangle = \sum_{i,j,k \in \{0,1\}} \alpha_{ijk} |ijk\rangle$$

The CCNOT gate acts linearly on each component of the superposition and flips the target qubit (the third qubit) if and only if both control qubits are equal to $|1\rangle$. This can be expressed as:

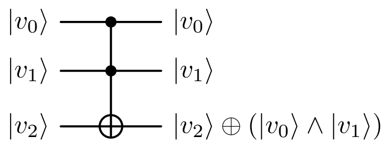
$$\text{CCNOT} |\psi\rangle = \sum_{i,j,k \in \{0,1\}} \alpha_{ijk} |ij\,(k \oplus (i \cdot j))\rangle \tag{58}$$

Where $\oplus$ denotes the XOR operation, and $i \cdot j = 1$ if and only if $i = j = 1$, and 0 otherwise.

In simple terms, applying CCNOT affects only the $c_6$ and $c_7$ coefficients (corresponding to $|110\rangle$ and $|111\rangle$) by swapping them, while all other coefficients remain unchanged.

### Circuit Representation

In quantum circuit diagrams, the CCNOT gate is typically represented by a symbol with two control dots connected to a NOT gate (X gate) on the target qubit. The control dots indicate which qubits are the controls, and the NOT gate indicates the operation performed on the target qubit when both controls are active. It is similar to the CNOT gate but with an additional control dot, making it visually distinct and easy to identify in quantum circuits.




In summary, if both control qubits are $|1\rangle$, apply NOT to the target; otherwise do nothing.

### Properties

- **Unitary**

$$\text{CCNOT}^\dagger \text{CCNOT} = I$$

- **Reversible**

$$\text{CCNOT}^{-1} = \text{CCNOT}^\dagger$$

- **Classical universal gate (can simulate AND)**. CCNOT gate is a **universal gate** for classical computation, meaning that **any classical**

**logic gate can be constructed using CCNOT gates and single-qubit NOT gates**. In particular, it can compute the AND operation in a reversible way:

$$|a,\, b,\, 0\rangle \xrightarrow{\text{CCNOT}} |a,\, b,\, 0 \oplus (a \cdot b)\rangle = |a,\, b,\, a \cdot b\rangle$$

Where $a$ and $b$ are the control qubits, and the target qubit is initialized to $|0\rangle$. After applying the CCNOT gate, the target qubit will be in the state $|1\rangle$ if and only if both $a$ and $b$ are $|1\rangle$, effectively computing the AND of the two control qubits.

Since AND and NOT are sufficient to express any Boolean function, the Toffoli gate can reproduce any classical computation.

- **Introduce multi-qubit interaction**. Multi-qubit interaction refers to operations in which the evolution of one qubit depends on the state of one or more other qubits. Unlike parallel gates, which act independently on each qubit, **interacting gates introduce conditional behavior and correlations between qubits**.

  Typical examples include controlled gates such as CNOT and CCNOT, where the transformation applied to a target qubit depends on the value of one or more control qubits. These interactions are essential for creating entanglement and for implementing complex quantum algorithms.

- **Can create entanglement**. Similar to the CNOT gate, also the CCNOT gate can create entanglement between the control and target qubits. For example, if the control qubits are in a superposition state, applying the CCNOT gate can result in an entangled state between the control and target qubits.

- **Not decomposable into a single tensor product**. An interacting multi-qubit gate cannot be expressed as a tensor product of single-qubit operators. That is, in general:

  $$U \neq U_A \otimes U_B$$

  This means that the gate introduces correlations between qubits and cannot be reduced to independent operations.

In summary, the Toffoli gate performs a conditional operation based on two qubits, enabling multi-qubit logic and more complex correlations.

## 4.6 Quantum Circuit Representation

In the previous sections, we have been using visual representations of quantum gates and their effects on qubits. However, a quantum circuit can be expressed in a more formal way using a **single unitary matrix** that represents the entire operation of the circuit. This matrix acts on the global state of the system and is obtained by composing the individual gates according to the structure of the circuit. This is known as the **quantum circuit representation**.

**Definition 1: Quantum Circuit Representation**

A **quantum circuit** can be **represented as a unitary matrix** $U$ that acts on the global state vector of the qubits. The overall operation of the circuit is obtained by composing the unitary matrices corresponding to each layer of gates in the circuit.

Mathematically, if a circuit consists of layers $G_1, G_2, \ldots, G_n$ with corresponding unitary matrices $U_1, U_2, \ldots, U_n$, then the overall unitary matrix $U$ representing the circuit is given by:

$$U = U_n U_{n-1} \cdots U_2 U_1$$

Where the rightmost operator is applied first.

Each $U_i$ is constructed by combining gates acting in parallel using tensor products, and by extending gates with identity operators on qubits where no operation is applied.

In simple terms, a **quantum circuit is just a structured way to build a single unitary matrix acting on the whole system**. So:

$$|\psi_{\text{out}}\rangle = U\,|\psi_{\text{in}}\rangle$$

Where $|\psi_{\text{in}}\rangle$ is the initial state of the qubits, $|\psi_{\text{out}}\rangle$ is the final state after applying the circuit, and $U$ is the **product of all the gates in the circuit**.

**⚠ The 3 Fundamental Rules of Quantum Circuit Representation**

1. **Parallel composition**. Gates acting on different qubits are combined using the tensor product:
$$U = A \otimes B$$
2. **Sequential composition**. Gates applied in sequence are combined using matrix multiplication, evaluated from right to left:
$$U = U_n \cdots U_2 U_1$$
So the first gate applied is the rightmost one in the product.
3. **Identity extension**. When a gate does not act on a qubit, the identity operator must be used to extend it:
$$U = I \otimes A$$

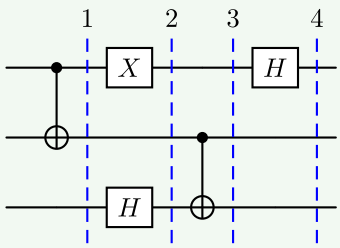
This keeps dimensions consistent when combining gates that act on different subsets of qubits.

**Example 4: Quantum Circuit Representation**

Let's consider the following quantum circuit representation:

$$\underbrace{(H \otimes I \otimes I)}_{4} \cdot \underbrace{(I \otimes \text{CNOT})}_{3} \cdot \underbrace{(X \otimes I \otimes H)}_{2} \cdot \underbrace{(\text{CNOT} \otimes I)}_{1}$$

First of all, the order of the gates is from right to left, so we start with the rightmost gate.



## 4.7 Entanglement

As discussed on page 93, we have already seen why entanglement exists and matters. In short, describing an $n$-qubit quantum state requires an exponential number of parameters, meaning that most multi-qubit states are entangled.

If qubits were independent (separable), we would only need $2n$ real numbers, since each qubit would require only 2 real parameters:

$$|v\rangle = a_0 |0\rangle + a_1 |1\rangle$$

Each qubit needs two real parameters, so for $n$ qubits, we would need $2n$ real numbers. However, a general quantum state of $n$ qubits is given by (see page 91):

$$|\psi\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle$$

This representation requires $2^n$ complex coefficients (subject to normalization and global phase constraints), corresponding to an exponential number of real degrees of freedom.

In conclusion, most multi-qubit states cannot be described as independent qubits. This phenomenon is called **entanglement** and is a fundamental feature of quantum mechanics with no classical analog.

**Definition 2: Entanglement**

A multi-qubit state is called **Entangled** if it **cannot** be written as a tensor product of individual qubit states (i.e., it is not separable). Otherwise, it is called **separable** or **unentangled**. Entanglement can be viewed as a “failure of factorization”, where the state of the whole system cannot be factored into states of its parts.

An important consequence of entanglement is that we cannot assign a separate quantum state to each qubit independently.

### Analysis of a Real Example: Bell States

To better understand the concept of entanglement, it is useful to analyze a concrete example. In particular, we consider a class of two-qubit states known as **Bell states**, which provide the simplest and most important examples of entangled states. These states cannot be written as a tensor product of individual qubit states, and therefore explicitly demonstrate the impossibility of decomposing certain multi-qubit states into independent subsystems.

**Definition 3: Bell States**

The **Bell States** are a set of 4 two-qubit states that form an orthonormal basis of maximally entangled states. They are defined as:

$$\begin{aligned} \left|\Phi^{+}\right\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) & \left|\Phi^{-}\right\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \\ \left|\Psi^{+}\right\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) & \left|\Psi^{-}\right\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \end{aligned} \tag{59}$$

These states are called **maximally entangled** because each qubit, when considered individually, is in a completely mixed (maximally uncertain) state. In other words, Bell states are characterized by the fact that each individual qubit is in a maximally mixed (*completely random*) state, while the joint state exhibits perfect correlations between the qubits.

This means that when we measure a single qubit of a Bell state, the outcome is completely random. In other words, there is a 50% chance of getting a 0 and a 50% chance of getting a 1. Therefore, we cannot say that it is mostly a 0 or a 1. However, once we measure a single qubit, a paradox occurs: even though each qubit alone is random, together they are perfectly correlated. In fact, if we measure the first qubit and get $|0\rangle$, we know with certainty that the second qubit is also $|0\rangle$ (in the case of $|\Phi^+\rangle$ and $|\Phi^-\rangle$), or $|1\rangle$ (in the case of $|\Psi^+\rangle$ and $|\Psi^-\rangle$). Thus, we have **maximum uncertainty individually but maximum correlation jointly**.

**Example 5: Proof of Entanglement in Bell States**

After introducing the Bell states, we would like to proof that decomposing them into independent qubits is impossible. In other words, we want to show that:

$$\left|\Phi^{+}\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Can be written as:

$$\left|\Phi^{+}\right\rangle = |v_A\rangle \otimes |v_B\rangle$$

*Proof.* Let's assume, for the sake of argument, that $|\Phi^+\rangle$ can be written as a tensor product of two single-qubit states:

$$\left|\Phi^{+}\right\rangle = |v_A\rangle \otimes |v_B\rangle$$

Where:

- $|v_A\rangle = a_0|0\rangle + a_1|1\rangle$ is the state of the first qubit.
- $|v_B\rangle = b_0|0\rangle + b_1|1\rangle$ is the state of the second qubit.

The tensor product of these two states is:

$$|v_A\rangle \otimes |v_B\rangle = a_0b_0|00\rangle + a_0b_1|01\rangle + a_1b_0|10\rangle + a_1b_1|11\rangle$$

For this to equal $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, we must have:

$$a_0b_0 = \frac{1}{\sqrt{2}}, \quad a_1b_1 = \frac{1}{\sqrt{2}}, \quad a_0b_1 = 0, \quad a_1b_0 = 0$$

But here's the problem, if $a_0b_1 = 0$, then either $a_0 = 0$ or $b_1 = 0$:

- ✗ Case $a_0 = 0$, then $a_0b_0 = 0$, which **contradicts** $a_0b_0 = \frac{1}{\sqrt{2}}$ for $|00\rangle$.
- ✗ Case $b_1 = 0$, then $a_1b_1 = 0$, which **contradicts** $a_1b_1 = \frac{1}{\sqrt{2}}$ for $|11\rangle$.

Therefore, there is no way to choose $a_0, a_1, b_0, b_1$ such that the above equations hold simultaneously. This means that $|\Phi^+\rangle$ cannot be written as a tensor product of two single-qubit states, and thus it is an entangled state. All we can say is that these qubits are correlated and cannot be described independently. QED

### Entanglement: Intrinsic or Basis-Dependent?

Having established that Bell states are entangled by showing that they cannot be written as a tensor product of individual qubit states, a natural question arises: is this property intrinsic to the state, or does it depend on the basis in which the state is expressed? In other words, could a change of basis make an entangled state appear separable? The answer is no. **Entanglement** is not a property of the representation, but of the **structure of the state itself**.

In contrast to entanglement, **superposition depends on the basis we choose**, for example:

$$|0\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$$

This shows that the state $|0\rangle$ is a **superposition** of $|+\rangle$ and $|-\rangle$ in the **Hadamard basis**, but it is **not a superposition in the computational basis**, indeed:

$$|0\rangle = 1 \cdot |0\rangle + 0 \cdot |1\rangle$$

It is not a superposition because only one basis vector is present.[2]

Differently, **entanglement** does **not** depend on basis, because it depends on **factorization**, not representation. So we can change basis, rewrite the state, but we **cannot make an entangled state separable**. This prevents a common misunderstanding: one might wonder whether a suitable change of basis could remove entanglement. This is not the case. Entanglement is preserved under local changes of basis, that is, transformations of the form $U_A \otimes U_B$ acting independently on each qubit. Therefore, entanglement is an intrinsic property of the quantum state and does not depend on the particular representation of each subsystem.

[2]A state is a superposition in a given basis if it is written as a combination of multiple basis vectors with non-zero coefficients.

### From States to Operations

So far, we have analyzed entanglement as a property of quantum states, showing that certain states cannot be decomposed into independent subsystems. A natural question now arises: *how can such entangled states be generated in practice?* In other words, **which quantum operations are capable of transforming separable states into entangled ones?**

The answer lies in a special class of quantum gates, known as **entangling gates**, which introduce interactions between qubits and allow the creation of correlations that cannot be reduced to independent components.

**Definition 4: Entangling Gates**

A multi-qubit gate is called **entangling** (**Entangling Gates**) if it cannot be written as a tensor product of single-qubit gates. Equivalently, a gate is entangling if there exists at least one separable input state that it transforms into an entangled state.

Put simply, an **entangling gate is a gate that makes qubits interact and can create entanglement**.

**Attention:** Not every input becomes entangled. There exists at least one input that becomes entangled, but not all inputs necessarily do.

In simple terms, an entangling gate is one that can take two or more qubits that are initially uncorrelated (separable) and produce a state where the qubits become correlated in such a way that they cannot be described independently (entangled). For example, the CNOT gate is an entangling gate because it can create Bell states from separable inputs.

**Example 6: CNOT as an Entangling Gate**

Consider the separable state:

$$|+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$$

This is a separable state, since it is written as a tensor product of two single-qubit states.

Now apply a CNOT gate (with the first qubit as control and the second as target, page 101):

$$\text{CNOT}\left(\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)\right) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

The resulting state is the Bell state $|\Phi^+\rangle$, which is entangled. Therefore, the CNOT gate is an **entangling gate**.

### Example 7: CNOT Does Not Always Create Entanglement

Not all inputs to an entangling gate produce entangled states. Consider the separable state:

$$|0\rangle \otimes |0\rangle = |00\rangle$$

Applying the CNOT gate:

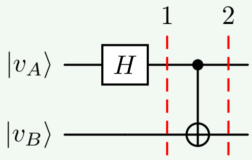
$$\text{CNOT}(|00\rangle) = |00\rangle$$

The state remains separable.

This shows that an entangling gate does not necessarily produce entanglement for every input. However, it is called **entangling** because there exists at least one separable input state that it transforms into an entangled state.

### Example 8: Bell-state generator

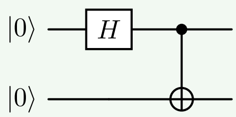
Exists a simple quantum circuit that produces **different Bell states depending on the input**. So, for the four input states $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$, the circuit creates the four Bell states $|\Phi^+\rangle$, $|\Phi^-\rangle$, $|\Psi^+\rangle$, and $|\Psi^-\rangle$ respectively. This circuit consists of a Hadamard gate followed by a CNOT gate:




$$(\underbrace{\text{CNOT}}_{2} \cdot \underbrace{(H \otimes I)}_{1})\,|(|v_A\rangle \otimes |v_B\rangle)\rangle$$

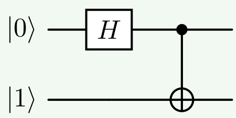
This circuit is a **Bell-state generator** because it can produce any of the four Bell states from the appropriate input.

- Input $|00\rangle$ produces $|\Phi^+\rangle$:



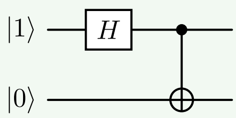

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- Input $|01\rangle$ produces $|\Phi^-\rangle$:



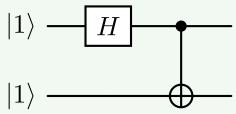

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$

- Input $|10\rangle$ produces $|\Psi^+\rangle$:




$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

- Input $|11\rangle$ produces $|\Psi^-\rangle$:




$$|\Psi^-\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right)$$

**Example 9: Bell States in the Hadamard Basis**

In this exercise, we express the Bell state $|\Phi^+\rangle$ in the Hadamard basis instead of the computational basis.

*Bell state $|\Phi^+\rangle$ in the Hadamard basis.* Given:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

We aim to re-express this state using the Hadamard basis $\{|+\rangle, |-\rangle\}$, where:

$$|+\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) \qquad |-\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right)$$

1. **Invert the definitions**. To convert from the computational basis to the Hadamard basis, we invert the above equations to express $|0\rangle$ and $|1\rangle$ in terms of $|+\rangle$ and $|-\rangle$:

$$|0\rangle = \frac{1}{\sqrt{2}}\left(|+\rangle + |-\rangle\right) \qquad |1\rangle = \frac{1}{\sqrt{2}}\left(|+\rangle - |-\rangle\right)$$

2. **Substitute into $|00\rangle$ and $|11\rangle$**. We compute the tensor products using the expression above.

   - Compute $|00\rangle$:

$$\begin{aligned}
|00\rangle &= |0\rangle \otimes |0\rangle \\
&= \left(\frac{1}{\sqrt{2}}\left(|+\rangle + |-\rangle\right)\right) \otimes \left(\frac{1}{\sqrt{2}}\left(|+\rangle + |-\rangle\right)\right) \\
&= \frac{1}{2}\left(|{+}{+}\rangle + |{+}{-}\rangle + |{-}{+}\rangle + |{-}{-}\rangle\right)
\end{aligned}$$

   - Compute $|11\rangle$:

$$\begin{aligned}
|11\rangle &= |1\rangle \otimes |1\rangle \\
&= \left(\frac{1}{\sqrt{2}}\left(|+\rangle - |-\rangle\right)\right) \otimes \left(\frac{1}{\sqrt{2}}\left(|+\rangle - |-\rangle\right)\right) \\
&= \frac{1}{2}\left(|{+}{+}\rangle - |{+}{-}\rangle - |{-}{+}\rangle + |{-}{-}\rangle\right)
\end{aligned}$$

3. **Add the two to get $|\Phi^+\rangle$.** Now add the two results:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right)$$

Substituting:

$$\begin{aligned}\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\Bigg[&\frac{1}{2}\left(|{+}{+}\rangle + |{+}{-}\rangle + |{-}{+}\rangle + |{-}{-}\rangle\right)\\ &+ \frac{1}{2}\left(|{+}{+}\rangle - |{+}{-}\rangle - |{-}{+}\rangle + |{-}{-}\rangle\right)\Bigg]\\ = \frac{1}{\sqrt{2}} \cdot \frac{1}{2}&\left(2\,|{+}{+}\rangle + 0\,|{+}{-}\rangle + 0\,|{-}{+}\rangle + 2\,|{-}{-}\rangle\right)\\ = \frac{1}{\sqrt{2}} \cdot \frac{1}{2}&\left(2\,|{+}{+}\rangle + 2\,|{-}{-}\rangle\right)\\ = \frac{1}{\sqrt{2}}&\left(|{+}{+}\rangle + |{-}{-}\rangle\right)\end{aligned}$$

Thus, in the Hadamard basis, we have:

$$\left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}\left(|{+}{+}\rangle + |{-}{-}\rangle\right)$$

Which is a Bell-like state expressed in the Hadamard basis instead of the computational basis. QED

### ❓ Why single-qubit gates cannot create entanglement?

Entanglement means that two qubits become so strongly correlated that they can no longer be described independently. To create this kind of correlation, the qubits must **interact**. Since single-qubit gates act on each qubit **independently**, they cannot introduce any interaction between qubits.

Suppose we have two qubits $|\psi_A\rangle$ and $|\psi_B\rangle$, and the combined state is:

$$|\psi\rangle = |\psi_A\rangle \otimes |\psi_B\rangle$$

The $|\psi\rangle$ is a **separable** state because it can be written as a tensor product of two single-qubit states (see above). In contrast, suppose we apply a gate $U_A$ to the first qubit, and a gate $U_B$ to the second qubit. The overall operation is:

$$U_A \otimes U_B$$

Applying this to the state $|\psi\rangle$ gives:

$$(U_A \otimes U_B)(|\psi_A\rangle \otimes |\psi_B\rangle) = (U_A\,|\psi_A\rangle) \otimes (U_B\,|\psi_B\rangle)$$

The result is **still a tensor product of two separate states**. That means the final state is still separable. Therefore, single-qubit gates preserve separability, and cannot create entanglement.

In general, to create entanglement, qubits must interact. We need a gate that acts on both qubits together, such as the CNOT gate. The CNOT gate is the most common example of an entangling gate because under the hood is very simple, but it can create complex correlations between qubits, including the Bell states. CNOT works because it makes the second qubit (the target) depend on the first qubit (the control), thus creating correlations that cannot be reduced to independent states. Instead, single-qubit gates never do this, because each qubit evolves independently.

**Deepening: Physical Meaning of Entanglement**

We have introduced entanglement as a mathematical property of quantum states, but it also has profound physical implications. We said that *a state is entangled when it cannot be factorized into tensor products of individual subsystem states.* But physics asks "***what does this impossibility mean physically?***". And the answer is entanglement means that the subsystems no longer possess independent physical states. That is the key idea.

**Before Entanglement: Independent Systems.** Suppose we have two independent qubits:

$$|\psi_A\rangle = a|0\rangle + b|1\rangle \qquad |\psi_B\rangle = c|0\rangle + d|1\rangle$$

The joint system is:

$$|\psi_{AB}\rangle = |\psi_A\rangle \otimes |\psi_B\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

In this case, we can assign a separate state to each qubit, and they evolve independently. The state of the whole system is just the combination of the states of the parts. Physically this means that qubit $A$ has its own state, and qubit $B$ has its own state, and they do not affect each other.

**Entanglement: The Global State Exists, Local Ones Do Not.** Now consider the Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

We already know mathematically that this state cannot be written as a tensor product of two single-qubit states (page 116):

$$|\Phi^+\rangle \neq |v_A\rangle \otimes |v_B\rangle$$

But physically this means something deeper:

- There is a state for the **pair**;
- There is NOT a complete independent state for each particle separately.

The **system only has a meaningful description as a whole**. This is the **physical meaning**.

For example, with the Bell state $|\Phi^+\rangle$, the possible outcomes are $|00\rangle$ and $|11\rangle$. So if Alice measures her qubit and gets $|0\rangle$, Bob must get $|0\rangle$ as well. If Alice gets $|1\rangle$, Bob must get $|1\rangle$. The outcomes are perfectly correlated, but neither qubit has a definite state on its own. The state of the whole system is well-defined, but the state of each part is not. This is the essence of entanglement: the parts do not have independent states; only the whole does. Entanglement creates strong correlations and the particles behave as parts of a single object.

In summary, entanglement means that multiple quantum particles no longer possess independent physical states; **only the global system has a complete physical description**. Physically this means that the particles are not separate "*objects with separate properties*" anymore, but rather the information is stored in the relationship between them and the measurements become strongly correlated.

## 4.8 Observables and Measurements

### 4.8.1 Observables: What Are We Measuring?

Let's start from the basics: ***what is an observable?*** In quantum mechanics, an **Observable** is a mathematical object representing a measurable physical quantity.

- For example, in classical physics, observables are physical quantities that we can measure, such as position, momentum, energy, etc. (e.g., the position of a particle, the energy of a system).
- In quantum mechanics, the state is no longer a point, but rather a vector (wavefunction) in a complex vector space (Hilbert space). Because of this, observables can no longer be ordinary functions of the state, but instead must be represented as **operators** that act on the state vector.

**Definition 5: Observable**

In quantum mechanics, an **Observable** is a **Hermitian operator** $O$ associated with a measurable quantity of a quantum system:

$$O = O^\dagger$$

Where $O^\dagger$ denotes the Hermitian adjoint (conjugate transpose) of $O$.

With *conjugate transpose*, we mean that if $O$ is represented as a matrix, then $O^\dagger$ is obtained by taking the **transpose** of the matrix and then taking the **complex conjugate of each element**. The condition $O = O^\dagger$ ensures that the eigenvalues of $O$ are real, which is necessary for them to correspond to measurable quantities in quantum mechanics.

#### ❓ Why does quantum mechanics specifically choose Hermitian operators for observables?

The answer is simple: because measurements must produce physically meaningful outcomes. Specifically, Hermitian operators guarantee: real eigenvalues and orthonormal eigenvectors.

1. **Real eigenvalues**. The eigenvalues of a Hermitian operator are always real numbers, which is **essential because measurement outcomes must be real values**. We cannot measure $3 + 2i$ as a physical quantity, but we can measure 3. Mathematically, if $O$ is a Hermitian operator, then:

   $$O |k\rangle = \lambda_k |k\rangle$$

   Where $\lambda_k$ is the possible measurement outcome (eigenvalue) corresponding to the eigenvector $|k\rangle$, and $|k\rangle$ is the state of the system after the measurement.

**Remark**

When we write:

$$O|k\rangle = \lambda_k |k\rangle$$

We are choosing $|k\rangle$ to be one of the eigenvectors of $O$, and $\lambda_k$ is its corresponding eigenvalue. It is the definition of the eigenvector/eigenvalue relationship.

Suppose $O$ is an operator, so a matrix. An eigenvector $|k\rangle$ of $O$ is a special vector such that, when $O$ acts on it, the result is the **same vector direction**, only multiplied by a number. That number is called the eigenvalue $\lambda_k$. So:

$$O|k\rangle = \lambda_k |k\rangle$$

Means applying the operator $O$ to $|k\rangle$ does not change its direction; it only scales it by $\lambda_k$.

**Example 10: Eigenvectors and Eigenvalues of the Pauli-Z Operator**

For example, let's take the Pauli-Z operator:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

And the state $|0\rangle$:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Applying $Z$ to $|0\rangle$:

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1 \cdot |0\rangle = \lambda_0 |0\rangle$$

Therefore, $|0\rangle$ is an eigenvector of $Z$ with eigenvalue $\lambda_0 = 1$. Similarly, applying $Z$ to $|1\rangle$:

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -1 \cdot |1\rangle = \lambda_1 |1\rangle$$

Therefore, $|1\rangle$ is an eigenvector of $Z$ with eigenvalue $\lambda_1 = -1$.

Beyond the mathematical definition, $|0\rangle$ is a special state of the observable $Z$, measuring $Z$ when the system is in $|0\rangle$ always gives the outcome $+1$. So, when we say that $|0\rangle$ and $|1\rangle$ are eigenvectors of $Z$, we are saying that these are the states in which the measurement of $Z$ is **perfectly predictable**, yielding the eigenvalues $+1$ and $-1$, respectively.

This is a fundamental aspect, because:

- If the **system is already in an eigenvector**: there is no uncertainty, and the measurement outcome is known with certainty.

- In contrast, if the **system is not in an eigenvector**: it is a combination of eigenvectors, and measurement becomes probabilistic. For example, suppose the qubit is in:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

This is **not** an eigenvector of $Z$, because applying $Z$:

$$\begin{aligned} Z|+\rangle =& \frac{1}{\sqrt{2}}\left(\begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}1 & 0\\ 0 & -1\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix}\right)\\ =& \frac{1}{\sqrt{2}}\left(\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\-1\end{bmatrix}\right)\\ =& \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\\ =& |-\rangle \end{aligned}$$

**Why is $|+\rangle$ not an eigenvector of $Z$?** If $|+\rangle$ were an eigenvector, we would have by definition:

$$Z|+\rangle = \lambda|+\rangle$$

For some eigenvalue $\lambda$. However, as we calculated, applying $Z$ to $|+\rangle$ gives us $|-\rangle$, which is a different state. Therefore, there is no scalar $\lambda$ such that $Z|+\rangle = \lambda|+\rangle$, confirming that $|+\rangle$ is not an eigenvector of $Z$. Physically, this means that if the system is in state $|+\rangle$, measuring $Z$ does not yield a definite outcome; instead, it yields $+1$ with probability 0.5 and $-1$ with probability 0.5, reflecting the superposition of eigenstates.

In summary, if applying an observable $O$ to a state $|\psi\rangle$ produces the same state multiplied by a scalar:

$$O|\psi\rangle = \lambda|\psi\rangle$$

Then $|\psi\rangle$ is an eigenvector of $O$, and $\lambda$ is its eigenvalue. In this case, the state already has a definite value for that observable, so measuring $O$ will return $\lambda$ with certainty.

**Physical consequence.** If the system is in an eigenvector: the measurement is deterministic, the result is known in advance, the outcome is the corresponding eigenvalue.

***Why Hermitian operators have real eigenvalues.*** Let $A$ be a Hermitian operator:

$$A = A^\dagger$$

Assume that $|x\rangle$ is an eigenvector of $A$ with eigenvalue $\lambda$:

$$A|x\rangle = \lambda|x\rangle$$

Consider the quantity:

$$\langle x| A |x\rangle$$

Using the eigenvalue equation (i.e., $A|x\rangle = \lambda|x\rangle$) on the right:

$$\langle x| \underbrace{A|x\rangle}_{\lambda|x\rangle} = \langle x| (\lambda|x\rangle) = \lambda \langle x|x\rangle$$

This is possible because $\lambda$ is just a scalar, so it can be moved outside bra-ket notation.[3] Now take the Hermitian conjugate of the eigenvalue equation:

$$A|x\rangle = \lambda|x\rangle \implies \langle x| A^\dagger = \overline{\lambda} \langle x|$$

Since $A$ is Hermitian:

$$A^\dagger = A$$

Therefore:

$$\langle x| A = \overline{\lambda} \langle x|$$

Applying this on the left of $\langle x| A |x\rangle$:

$$\underbrace{\langle x| A}_{\overline{\lambda}\langle x|} |x\rangle = \left(\overline{\lambda} \langle x|\right) |x\rangle = \overline{\lambda} \langle x|x\rangle$$

Combining the two expressions:

$$\lambda \langle x|x\rangle = \overline{\lambda} \langle x|x\rangle$$

Since:

$$\langle x|x\rangle > 0$$

Because the inner product of a vector with itself is always positive (unless the vector is the zero vector, which cannot be an eigenvector), we can divide both sides by $\langle x|x\rangle$:

$$\lambda = \overline{\lambda}$$

Therefore, $\lambda$ is real. QED

[3] Remember that bras and kets can be represented using vectors and matrices, so the usual rules of linear algebra apply. In particular, since $\lambda$ is a scalar, it can be factored out of an inner product:

$$\langle x| (\lambda|x\rangle) = \lambda \langle x|x\rangle$$

More generally, scalar multiplication commutes with matrix products. Therefore, for matrices $A$ and $B$:

$$\lambda AB = A\lambda B = AB\lambda$$

Provided that the order of the matrices themselves is preserved.

2. **Orthonormal eigenvectors**. A vector is:

   - **Orthogonal** if its inner product with another vector is zero (i.e., they are perpendicular in the vector space). For example, in a 2D space, the vectors $|0\rangle$ and $|1\rangle$ are orthogonal because:

   $$\langle 0|1\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

   - **Normalized** if its inner product with itself is equal to one (i.e., it has unit length). For example:

   $$\langle 0|0\rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 1$$

   This property is crucial because quantum measurement requires a basis whose states are both perfectly distinguishable (orthogonal) and probabilistically consistent (normalized).

   ✗ **If eigenvectors were not orthogonal**, different measurement states could partially overlap:

   $$\langle v_1|v_2\rangle \neq 0$$

   In this case, the measurement process would not be able to distinguish the states perfectly, leading to ambiguity in the interpretation of measurement outcomes.

   **Example 11: Why Orthogonality Matters**

   Consider the two states:

   $$|v_1\rangle = |0\rangle \qquad |v_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

   Their inner product is:

   $$\langle v_1|v_2\rangle = \langle 0| \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}}$$

   Since:

   $$\langle v_1|v_2\rangle \neq 0$$

   the two states are not orthogonal. This means that the **states partially overlap**, and therefore a **measurement cannot distinguish them perfectly**. For example, if the system is in state $|v_2\rangle$, there is still a non-zero probability of confusing it with $|v_1\rangle$ during a measurement. Therefore, the measurement outcomes would be ambiguous, and we would not be able to assign a clear physical meaning to the measurement results.

   ✗ **If eigenvectors were not normalized**:

   $$\langle v|v\rangle \neq 1$$

Then the Born rule (page 81) would not produce properly normalized probabilities, making the probabilistic interpretation of quantum mechanics inconsistent.

**Example 12: Why Normalization Matters**

Consider the state:

$$|v\rangle = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

Its norm is:

$$\langle v|v\rangle = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 4$$

Therefore, the state is not normalized.

If we tried to interpret the amplitudes probabilistically using the Born rule, we would obtain:

$$p_0 = |2|^2 = 4$$

Which is **impossible**, since **probabilities must lie between 0 and 1**, and their total sum must equal 1.

To obtain a valid quantum state, we must normalize the vector:

$$|v_{\text{norm}}\rangle = \frac{1}{2}\begin{pmatrix} 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Therefore, without normalization, we would end up with an invalid quantum state that cannot be interpreted probabilistically, and the measurement outcomes would not correspond to any physical reality.

***Why eigenvectors of Hermitian operators are orthogonal.*** Let $A$ be Hermitian and let:

$$A\,|x\rangle = \lambda\,|x\rangle\,, \qquad A\,|y\rangle = \mu\,|y\rangle$$

Where $\lambda \neq \mu$ are the eigenvalues corresponding to the eigenvectors $|x\rangle$ and $|y\rangle$, respectively. Consider the inner product:

$$\langle y|\,A\,|x\rangle$$

Because we want to show that $|x\rangle$ and $|y\rangle$ are orthogonal, we need to show that $\langle y|x\rangle = 0$. Using the eigenvalue equation on the right:

$$\langle y|\,A\,|x\rangle = \langle y|\,(\lambda\,|x\rangle) = \lambda\,\langle y|x\rangle$$

We moved $\lambda$ outside the inner product because it is a scalar (see the footnote in the previous proof for more details on this step). Now use the Hermitian property on the left:

$$\underbrace{\langle y|\,A}_{\langle y|A^\dagger} = \mu\,\langle y|$$

Therefore:

$$\langle y| A |x\rangle = \mu \langle y|x\rangle$$

Combining the two expressions:

$$\lambda \langle y|x\rangle = \mu \langle y|x\rangle$$

Thus:

$$(\lambda - \mu) \langle y|x\rangle = 0$$

Since $\lambda \neq \mu$ by assumption, we must have:

$$\langle y|x\rangle = 0$$

Therefore, the eigenvectors are orthogonal. QED

We can think of a Hermitian operator as a sort of "*machine*" that decomposes the quantum state into measurable components (eigenvectors) and assigns a real value (eigenvalue) to each component, which corresponds to the possible outcomes of a measurement. The orthonormality of the eigenvectors ensures that these outcomes are distinct and can be interpreted probabilistically in a consistent way. In summary, a Hermitian observable tells us:

- *What outcomes are possible when we measure a quantum system.*
- *With respect to which basis we are measuring the system.*
- *Into which states the system will collapse after the measurement.*

### Spectral Theorem

So far, we know that an observable $O$ is a Hermitian operator, it has eigenvectors $|k\rangle$, and each eigenvector has an eigenvalue $\lambda_k$. So if the system is in $|k\rangle$, measuring $O$ returns $\lambda_k$ with certainty. Now, a natural question arises: **what happens if the system is in an arbitrary state $|k\rangle$, not necessarily one of the eigenvectors?** To answer this question, the Spectral Theorem comes into play by showing that every observable can be written as a sum of its eigenvectors and eigenvalues.

Suppose the observable has a set of eigenvectors $|k\rangle$ with corresponding eigenvalues $\lambda_k$. Since the eigenvectors form an orthonormal basis, any quantum state $|v\rangle$ can be expressed as a linear combination of these eigenvectors (page 91):

$$|v\rangle = \sum_k c_k |k\rangle$$

Where $c_k$ are complex coefficients. The **Spectral Theorem** states that the observable $O$ can be expressed as:

$$O = \sum_k \lambda_k |k\rangle\langle k| \tag{60}$$

Where each term $\lambda_k |k\rangle\langle k|$ contains two pieces of information:

- **Projector** $|k\rangle\langle k|$ (page 78) onto the eigenvector $|k\rangle$, which extracts the component of the state along $|k\rangle$ during measurement. It is defined as:

$$|k\rangle\langle k| = |k\rangle \langle k|$$

- **Eigenvalue** $\lambda_k$, which assigns the measurement outcome associated with that component.

**Why the Spectral Theorem is important?** Suppose someone gives us an observable $O$ without telling us its eigenvectors or eigenvalues:

$$O = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Without the Spectral Theorem, the physical meaning of the observable would not be immediately apparent. However, by applying the Spectral Theorem, we can decompose $O$ into its eigenvectors and eigenvalues, allowing us to understand the measurement process and predict the outcomes based on the state of the system. For example, we can find the eigenvalues and eigenvectors of $O$:

- To find the eigenvalues, we solve the characteristic equation:

$$\begin{aligned} \det(O - \lambda I) &= 0 \\ \det\left(\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix}\right) &= 0 \\ (2-\lambda)(2-\lambda) - 1 \cdot 1 &= 0 \\ 4 - 4\lambda + \lambda^2 - 1 &= 0 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ (\lambda - 3)(\lambda - 1) &= 0 \end{aligned}$$

  Thus, the eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = 1$.

- Eigenvectors can be found by solving $(O - \lambda I)|k\rangle = 0$ for each eigenvalue:

  - For $\lambda_1 = 3$:

$$\begin{aligned} (O - 3I)|v_1\rangle &= 0 \\ \left(\begin{bmatrix} 2-3 & 1 \\ 1 & 2-3 \end{bmatrix}\right)|v_1\rangle &= 0 \\ \left(\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\right)|v_1\rangle &= 0 \end{aligned}$$

    Now to find the eigenvector, we need to solve the system of equations:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

    That is a homogeneous system of linear equations:

$$\begin{cases} -x + y = 0 \\ x - y = 0 \end{cases} = \begin{cases} y = x \\ x = y \end{cases}$$

So any vector of the form $(x, x)$ is an eigenvector. We choose the simplest one $(1, 1)$, and then we normalize it:

$$\left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Therefore, the normalized eigenvector is:

$$|v_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

– For $\lambda_2 = 1$:

$$(O - 1I)\,|v_2\rangle = 0$$
$$\left( \begin{bmatrix} 2-1 & 1 \\ 1 & 2-1 \end{bmatrix} \right) |v_2\rangle = 0$$
$$\left( \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \right) |v_2\rangle = 0$$

We need to solve the system of equations:

$$\begin{cases} x + y = 0 \\ x + y = 0 \end{cases} \Rightarrow x + y = 0 \Rightarrow y = -x \xrightarrow{\text{substitute from } \begin{bmatrix} x \\ y \end{bmatrix}} \begin{bmatrix} x \\ -x \end{bmatrix}$$

So any vector of the form $(x, -x)$ is an eigenvector. We choose the simplest one $(1, -1)$, and then we normalize it:

$$\left\| \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

Therefore, the normalized eigenvector is:

$$|v_2\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

Then we can express $O$ in terms of its spectral decomposition:

$$O = 3\,|v_1\rangle\langle v_1| + 1\,|v_2\rangle\langle v_2|$$

From this decomposition, we can immediately read off the possible outcomes from the eigenvalues (3 and 1), the measurement basis from the eigenvectors ($|v_1\rangle$ and $|v_2\rangle$), and the measurement operators from the projectors ($|v_1\rangle\langle v_1|$ and $|v_2\rangle\langle v_2|$). It also allows us to calculate the probabilities of obtaining each outcome when measuring $O$ on an arbitrary state $|\psi\rangle$ using the Born rule:

$$p_1 = |\,\langle v_1|\psi\rangle\,|^2 \qquad p_2 = |\,\langle v_2|\psi\rangle\,|^2$$

Where $p_1$ and $p_2$ are the probabilities of obtaining the outcomes 3 and 1, respectively, when measuring the observable $O$ on a state $|\psi\rangle$. And finally the expectation value of $O$ in the state $|\psi\rangle$ can be calculated as:

$$\langle\psi|\,O\,|\psi\rangle = \sum_k \lambda_k p_k$$

In this example, this becomes:

$$\langle\psi| O |\psi\rangle = \sum_{k=1}^{2} \lambda_k p_k = 3p_1 + 1p_2$$

In summary, the Spectral Theorem transforms an abstract Hermitian matrix into a complete description of a quantum measurement. It reveals:

- The **possible measurement outcomes** (eigenvalues),
- The **corresponding post-measurement states** (eigenvectors),
- The **projectors used to compute probabilities**,
- And the **expectation value** as a weighted average of the outcomes.

### 4.8.2 Measurement Operators and Projectors

In this section, we will explain **how measurement is actually performed mathematically**.

#### Measurement Operators

After introducing the Spectral Theorem, we know that any observable can be written as:

$$O = \sum_k \lambda_k |k\rangle\langle k|$$

Where:

- $\lambda_k$ are the possible measurement outcomes (eigenvalues);
- $|k\rangle$ are the corresponding eigenvectors;
- $|k\rangle\langle k|$ are projection operators onto the eigenvectors.

At this point, a natural question arises: what role do these projectors play in the measurement process?

The answer is that each projector:

$$M_k = |k\rangle\langle k|$$

Is called a **Measurement Operator** (page 78). It represents the mathematical operation associated with the measurement outcome $\lambda_k$.

**Definition 6: Measurement Operator**

Given an observable (page 124):

$$O = \sum_k \lambda_k |k\rangle\langle k|$$

The **Measurement Operator** associated with the outcome $\lambda_k$ is:

$$M_k = |k\rangle\langle k| = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} \begin{pmatrix} \overline{k_1} & \overline{k_2} \end{pmatrix} = \begin{pmatrix} k_1\overline{k_1} & k_1\overline{k_2} \\ k_2\overline{k_1} & k_2\overline{k_2} \end{pmatrix}$$

In other words, each possible measurement outcome has an associated projector that:

1. **Extracts the component** of the quantum state along the eigenvector $|k\rangle$;
2. **Determines the probability** of obtaining the corresponding eigenvalue $\lambda_k$;
3. **Produces the post-measurement** state after collapse.

For this reason, a measurement operator can be thought of as a **detector** tuned to a specific eigenstate. If the detector corresponding to $|k\rangle$ "clicks", then:

- The measurement outcome is $\lambda_k$;
- The system collapses to the state $|k\rangle$.

For example, suppose an observable has two eigenstates $|0\rangle$ and $|1\rangle$. Then there are two measurement operators:

$$M_0 = |0\rangle\langle 0|\,, \qquad M_1 = |1\rangle\langle 1|$$

We can imagine them as two detectors:

- Detector $M_0$ looks for the component of the state along $|0\rangle$.
- Detector $M_1$ looks for the component of the state along $|1\rangle$.

Suppose the state is:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

This means that part of the state points along $|0\rangle$, and part of the state points along $|1\rangle$. The detector $M_0 = |0\rangle\langle 0|$ asks "*how much of the state is aligned with* $|0\rangle$?" and extracts $\alpha\,|0\rangle$. The detector $M_1 = |1\rangle\langle 1|$ asks "*how much of the state is aligned with* $|1\rangle$?" and extracts $\beta\,|1\rangle$.

**Example 13: Example: Measurement Operators for Pauli-Z**

The Pauli-Z observable can be written as:

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

Therefore, the associated measurement operators are:

$$M_0 = |0\rangle\langle 0|\,, \qquad M_1 = |1\rangle\langle 1|$$

These operators correspond to the two possible outcomes:

- $+1$, associated with the eigenstate $|0\rangle$;
- $-1$, associated with the eigenstate $|1\rangle$.

Suppose the system is in the state:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

Applying the measurement operator $M_0$:

$$M_0\,|\psi\rangle = |0\rangle\,\langle 0|\,(\alpha\,|0\rangle + \beta\,|1\rangle) = \alpha\,|0\rangle$$

This means that the operator $M_0$ extracts only the component of the state aligned with $|0\rangle$ and removes the orthogonal component $|1\rangle$.

The probability of obtaining the corresponding outcome is:

$$p_0 = \langle\psi|\,M_0\,|\psi\rangle = |\alpha|^2$$

Similarly:

$$p_1 = \langle\psi|\,M_1\,|\psi\rangle = |\beta|^2$$

Therefore, **measurement operators** provide the complete mathematical description of the measurement process: they isolate the relevant component of the quantum state, determine the probability of each outcome, and specify the state after the measurement.

### Completeness of the Basis

In the previous example, we defined two measurement operators $M_0$ and $M_1$ corresponding to the two eigenstates of the Pauli-Z observable. But what if we had an observable with more eigenstates? Or what if we had a different observable with a completely different set of eigenvectors? Therefore, how do we know that the measurement operators we defined account for all possible outcomes of the measurement?

The answer is that the eigenvectors of a Hermitian observable form a **complete orthonormal basis**. This means that any quantum state can be expressed as a linear combination of these eigenvectors:

$$|\psi\rangle = \sum_k c_k |k\rangle$$

Mathematically, this property is expressed by the **Completeness Relation**:

$$\sum_k |k\rangle\langle k| = I$$

Where $I$ is the identity operator (page 47).

**Definition 7: Completeness Relation**

If $\{|k\rangle\}$ is a complete orthonormal basis, then:

$$\sum_k |k\rangle\langle k| = I \tag{61}$$

Since the measurement operators are defined as:

$$M_k = |k\rangle\langle k|$$

It follows that:

$$\sum_k M_k = I$$

This means that the set of measurement operators is complete and accounts for all possible outcomes of the measurement.

**Deepening: When a set of vectors $\{|k\rangle\}$ is a complete orthonormal basis?**

A set of vectors $\{|k\rangle\}$ is called a **Complete Orthonormal Basis** when it satisfies **three conditions**:

1. **Normalized**, each vector has length 1: $\langle k|k\rangle = 1$.
2. **Orthogonal**, different vectors are perpendicular: $\langle k|j\rangle = 0$ for $k \neq j$.
3. **Complete**, the vectors span the entire Hilbert space, meaning that every quantum state can be written as a linear combination of them. That is, for any state $|\psi\rangle$, there exist coefficients $c_k$ such that:
$$|\psi\rangle = \sum_k c_k |k\rangle$$

Compactly, the *normalized* and *orthogonal* conditions are written as:

$$\langle k|j\rangle = \delta_{kj} \tag{62}$$

Where $\delta_{kj}$ is the **Kronecker delta**, which is 1 if $k = j$ and 0 otherwise:

$$\delta_{kj} = \begin{cases} 1 & k = j \\ 0 & k \neq j \end{cases} \tag{63}$$

And the *completeness* condition is expressed by the completeness relation:
$$\sum_k |k\rangle\langle k| = I$$

**Example 14: Single-Qubit Computational Basis**

For a single qubit, the computational basis consists of the vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

These vectors are:

- Normalized: $\langle 0|0\rangle = 1$ and $\langle 1|1\rangle = 1$.
- Orthogonal: $\langle 0|1\rangle = 0$.
- Complete: any single-qubit state can be expressed as a linear combination of $|0\rangle$ and $|1\rangle$.
$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

This Equation 61 states that the **set of measurement operators is complete**: together, they describe every possible outcome of the measurement.

**Why is the completeness relation important?** To understand why this relation is important, let us apply both sides to an arbitrary state:

$$|\psi\rangle = \sum_k c_k |k\rangle$$

Where $c_k$ are the coefficients of $|\psi\rangle$ in the basis $\{|k\rangle\}$.

Now take the inner product with $\langle k|$ on both sides:

$$\langle k|\psi\rangle = \langle k| \left( \sum_j c_j |j\rangle \right)$$

Using linearity of the inner product:

$$\langle k|\psi\rangle = \sum_j c_j \langle k|j\rangle$$

Since the basis is orthonormal, $\langle k|j\rangle = \delta_{kj}$, so only the term where $j = k$ survives (Equation 62, page 137):

$$\langle k|j\rangle = \delta_{kj} = \begin{cases} 1 & k = j \\ 0 & k \neq j \end{cases}$$

Thanks to the Kronecker delta, all terms in the sum vanish except the one where $j = k$:

$$\langle k|\psi\rangle = c_k \langle k|k\rangle = c_k \cdot 1 = c_k$$

In other words, the bra $\langle k|$ acts as a coefficient extractor: it selects exactly the amount of $|\psi\rangle$ aligned with the basis vector $|k\rangle$.

Finally, apply the sum of all projectors to $|\psi\rangle$:

$$\left( \sum_k |k\rangle\langle k| \right) |\psi\rangle = \sum_k |k\rangle \langle k|\psi\rangle = \sum_k |k\rangle c_k = \sum_k c_k |k\rangle = |\psi\rangle$$

This shows that the **sum of all projectors reproduces the original state exactly**. In other words, each projector $|k\rangle\langle k|$ extracts one component of the state, and when all these components are added together, the original state is reconstructed without losing any information. Therefore:

$$\sum_k |k\rangle\langle k| = I$$

**Example 15: Computational Basis**

For a single qubit, the computational basis is:

$$|0\rangle, \qquad |1\rangle$$

The corresponding projectors are:

$$|0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Their sum is:

$$|0\rangle\langle 0| + |1\rangle\langle 1| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

The completeness relation also guarantees that the probabilities of all possible outcomes sum to one. If:

$$p_k = \langle\psi| M_k |\psi\rangle$$

Then:

$$\begin{aligned}
\sum_k p_k &= \sum_k \langle\psi| M_k |\psi\rangle \\
&= \langle\psi| \left(\sum_k M_k\right) |\psi\rangle \\
&= \langle\psi| I |\psi\rangle \\
&= \langle\psi|\psi\rangle \\
&= 1
\end{aligned}$$

Therefore, the **completeness relation** ensures that:

- **All** possible **measurement outcomes** are included;
- **No probability is lost**;
- The **total probability** is always **equal to one**.

### 4.8.3 Expectation Value

In the previous sections, we have seen how to compute the probability of obtaining a certain measurement outcome when measuring an observable on a quantum state. However, in many cases, we are interested in the average value of an observable over many measurements, rather than the probability of specific outcomes. This average value is known as the **expectation value** of the observable.

**Definition 8: Expectation Value**

Let $O$ be an observable represented by a Hermitian operator, and let $|\psi\rangle$ be a quantum state. The quantity:

$$\langle O \rangle = \langle \psi | O | \psi \rangle = \sum_k \lambda_k p_k \tag{64}$$

Is called the **Expectation Value** of the observable $O$ in the state $|\psi\rangle$.

**Physical Meaning.** The expectation value $\langle O \rangle$ can be interpreted as the average result obtained by measuring the observable $O$ many times on identical copies of the state $|\psi\rangle$.

1. Prepare many identical copies of the state $|\psi\rangle$.
2. Measure the observable $O$ on each copy.
3. Record the outcomes.
4. Compute the arithmetic mean of the recorded outcomes.

As the number of experiments increases, the average converges to the expectation value $\langle \psi | O | \psi \rangle$.

**Important Clarification**

The expectation value is **not necessarily** one of the possible measurement outcomes. It is the **average (more precisely, the probability-weighted average)** of many outcomes, and can take values that are not eigenvalues of the observable. For example, if possible outcomes are $+1$ with probability 50% and $-1$ with probability 50%, then the expectation value is:

$$(+1) \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$$

Although the actual measurement never returns 0, the expectation value is 0 because it represents the average of many measurements. This means that half of the measurements return $+1$ and the other half return $-1$, resulting in an average of 0.

The expectation value tells us the "center of mass" of the measurement distribution. It summarizes the statistical behavior of the measurement in a single number.

**Why are we interested in the average value of an observable rather than in the probability of specific outcomes?** Because in many real situations we do not **care about** the result of a single measurement, but about the **overall behavior of the system**. Also, it proves a stable, experimentally meaningful quantity that summarizes the statistical behavior of an observable over many measurements, even when individual outcomes are random.

### How can we compute the expectation value?

To compute the expectation value $\langle O \rangle$ of an observable $O$ in a state $|\psi\rangle$, we start from the Spectral Theorem (page 130):

$$O = \sum_k \lambda_k |k\rangle\langle k|$$

Substitute this into the expectation value formula (page 140):

$$\begin{aligned}\langle\psi| O |\psi\rangle &= \langle\psi| \left( \sum_k \lambda_k |k\rangle\langle k| \right) |\psi\rangle \\ &\downarrow \text{ use linearity of the inner product} \\ &= \sum_k \lambda_k \langle\psi|k\rangle \langle k|\psi\rangle\end{aligned}$$

Using the property of inner products (page 8):

$$\langle\psi|k\rangle = \overline{\langle k|\psi\rangle} \quad \Longrightarrow \quad \overline{\langle k|\psi\rangle} \langle k|\psi\rangle = |\langle k|\psi\rangle|^2$$

Here, the Born rule (page 81) states that $|\langle k|\psi\rangle|^2$ is the probability of measuring the outcome $\lambda_k$ when measuring $O$ on the state $|\psi\rangle$. Therefore, we can rewrite the expectation value as:

$$p_k = |\langle k|\psi\rangle|^2 \quad \Longrightarrow \quad \langle\psi| O |\psi\rangle = \sum_k \lambda_k p_k \tag{65}$$

This formula shows that the expectation value is the **weighted average** of the possible outcomes $\lambda_k$, where each outcome is weighted by its probability $p_k$. In other words, the expectation value is the sum of all possible outcomes multiplied by their respective probabilities, which is consistent with our interpretation of it as an average over many measurements.

### 4.8.4 Measuring $|+\rangle$ with the Pauli-Z Observable

After introducing observables, projectors, probabilities, and expectation values, it is useful to analyze a concrete example that combines all these concepts.

Consider the quantum state:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

And the Pauli-Z observable:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Calculate the possible measurement outcomes, their probabilities, the post-measurement states, and the expectation value of $Z$ in the state $|+\rangle$.

From the spectral decomposition (page 130) of $Z$, we know that:

- The **eigenvalues** of $Z$ are $+1$ and $-1$:

$$\begin{aligned} \det(Z - \lambda I) &= 0 \\ \det\left(\begin{bmatrix} 1-\lambda & 0 \\ 0 & -1-\lambda \end{bmatrix}\right) &= 0 \\ (1-\lambda)(-1-\lambda) &= 0 \\ \lambda &= \pm 1 \end{aligned}$$

- The **first eigenvector** $|v_1\rangle$ is $|0\rangle$:

$$\begin{aligned} (Z - 1I)|v_1\rangle &= 0 \\ \begin{bmatrix} 1-1 & 0 \\ 0 & -1-1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\ \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} &= 0 \end{aligned}$$

$$\begin{cases} 0 \cdot x + 0 \cdot y = 0 \\ 0 \cdot x - 2 \cdot y = 0 \end{cases} \implies y = 0$$

So any vector of the form $\begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{y=0} \begin{bmatrix} x \\ 0 \end{bmatrix}$ is an eigenvector. We choose the simplest one $(1, 0)$, and then we normalize it:

$$\left\| \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\| = \sqrt{1^2 + 0^2} = 1$$

Therefore, the normalized eigenvector is:

$$|v_1\rangle = 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

- Similarly, the **second eigenvector** $|v_2\rangle$ is $|1\rangle$:

$$
\begin{aligned}
(Z + 1I)\,|v_2\rangle &= 0 \\
\begin{bmatrix} 1+1 & 0 \\ 0 & -1+1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\
\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= 0 \\
\begin{cases} 2 \cdot x + 0 \cdot y = 0 \\ 0 \cdot x + 0 \cdot y = 0 \end{cases} & \quad \Longrightarrow \quad x = 0
\end{aligned}
$$

So any vector of the form $\begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{x=0} \begin{bmatrix} 0 \\ y \end{bmatrix}$ is an eigenvector. We choose the simplest one $(0, 1)$, and then we normalize it:

$$\left\| \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\| = \sqrt{0^2 + 1^2} = 1$$

Therefore, the normalized eigenvector is:

$$|v_2\rangle = 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

Therefore:

- The possible measurement outcomes are $+1$ and $-1$;
- The corresponding eigenstates are $|0\rangle$ and $|1\rangle$;
- The measurement operators are:

$$M_0 = |0\rangle\langle 0|\,, \qquad M_1 = |1\rangle\langle 1|$$

### Measurement Probabilities

The state $|+\rangle$ is an equal superposition of $|0\rangle$ and $|1\rangle$:

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

Therefore, the amplitudes are:

$$\alpha = \frac{1}{\sqrt{2}}, \qquad \beta = \frac{1}{\sqrt{2}}$$

Using the Born rule:

$$p_0 = |\alpha|^2 = \frac{1}{2}, \qquad p_1 = |\beta|^2 = \frac{1}{2}$$

Thus:

- There is a 50% probability of obtaining the outcome $+1$;
- There is a 50% probability of obtaining the outcome $-1$.

### Post-Measurement States

If the measurement outcome is:

- $+1$, the state collapses to $|0\rangle$;
- $-1$, the state collapses to $|1\rangle$.

### Expectation Value

The expectation value of $Z$ in the state $|+\rangle$ is:

$$\langle +| Z |+\rangle = (+1) \cdot \frac{1}{2} + (-1) \cdot \frac{1}{2} = 0$$

### Remark

The expectation value is 0, but this does **not** mean that the measurement can return the value 0. The only possible outcomes are:

$$+1 \quad \text{and} \quad -1$$

The value 0 is simply the probability-weighted average of these outcomes over many repeated measurements.

### What if the state to be measured is already an eigenstate of the observable?

Let's assume that now we want to calculate the measurement outcomes, probabilities, post-measurement states, and expectation value when the input state is $|0\rangle$ or $|1\rangle$ instead of $|+\rangle$. How we have seen, $|0\rangle$ and $|1\rangle$ are eigenvectors of $Z$ with eigenvalues $+1$ and $-1$, respectively. This means that they are already measurement states of the observable $Z$. Let's analyze what happens in these cases.

- **Measuring** $|0\rangle$: the measurement probabilities of $|0\rangle$ are:

  $$|0\rangle = 1 \cdot |0\rangle + 0 \cdot |1\rangle$$

  Therefore:

  $$p_0 = |\alpha|^2 = 1, \qquad p_1 = |\beta|^2 = 0$$

  Using observable $Z$, there is a 100% probability of obtaining the outcome $+1$ and a 0% probability of obtaining the outcome $-1$. Moreover, the (post-)measurement outcome is always $+1$, and the post-measurement state remains $|0\rangle$ because we have 100% probability of collapsing to $|0\rangle$ and 0% probability of collapsing to $|1\rangle$. Finally, the expectation value is:

  $$\langle 0| Z |0\rangle = (+1) \cdot 1 + (-1) \cdot 0 = +1$$

  And indeed, the expectation value confirms that the measurement outcome is always $+1$.

- **Measuring** $|1\rangle$: the measurement probabilities of $|1\rangle$ are:

  $$|1\rangle = 0 \cdot |0\rangle + 1 \cdot |1\rangle$$

  Therefore:

  $$p_0 = |\alpha|^2 = 0, \qquad p_1 = |\beta|^2 = 1$$

  Using observable $Z$, there is a 0% probability of obtaining the outcome $+1$ and a 100% probability of obtaining the outcome $-1$. Moreover, the (post-)measurement outcome is always $-1$, and the post-measurement state remains $|1\rangle$ because we have 0% probability of collapsing to $|0\rangle$ and 100% probability of collapsing to $|1\rangle$. Finally, the expectation value is:

  $$\langle 1| Z |1\rangle = (+1) \cdot 0 + (-1) \cdot 1 = -1$$

  And indeed, the expectation value confirms that the measurement outcome is always $-1$.

In summary, when a quantum state is already an eigenvector of the observable being measured:

- The **measurement** outcome is **deterministic**;
- The **probability** of the corresponding eigenvalue is **1**;
- The **state does not change** after the measurement;
- The **expectation value** is exactly the associated **eigenvalue**.

In contrast, when the state is a superposition of eigenvectors (as in the case of $|+\rangle$), the measurement outcome is probabilistic and the expectation value becomes a weighted average of the possible outcomes.

### 4.8.5 Recovering Probabilities from Expectation Values

In the previous sections, we have seen that the expectation value of an observable is the weighted average of its possible measurement outcomes. But, is possible to go in the opposite direction? That is, if we know the expectation value, can we determine the probabilities of the individual outcomes? Let's explore how to got **Probabilities from the Expectation Value**.

Let's consider the Pauli-Z observable as an example.

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

With (proof at page 142):

- Eigenvalue $+1$ associated with the state $|0\rangle$;
- Eigenvalue $-1$ associated with the state $|1\rangle$.

Let:

- $p_0$ be the probability of measuring the state $|0\rangle$;
- $p_1$ be the probability of measuring the state $|1\rangle$.

Since Pauli-Z is a valid observable, the probabilities must satisfy:

$$p_0 + p_1 = 1$$

Therefore, the expectation value of $Z$ is:

$$\langle Z \rangle = (+1)p_0 + (-1)p_1 = p_0 - p_1$$

Where $+1$ and $-1$ are the eigenvalues of $Z$ corresponding to the states $|0\rangle$ and $|1\rangle$, respectively. Instead, $p_0$ and $p_1$ are the probabilities of obtaining those states upon measurement.

Now, we are interested in the opposite direction: given the expectation value $m = \langle Z \rangle$ (we change the notation just to be general), can we determine $p_0$ and $p_1$? The answer is yes, and we can derive the formulas for $p_0$ and $p_1$ in terms of $m$ (or $\langle Z \rangle$). We have two relationships:

$$\begin{aligned} p_0 + p_1 &= 1 \\ p_0 - p_1 &= m \end{aligned}$$

We can solve this system of equations to express $p_0$ and $p_1$ in terms of $m$, and thus recover the probabilities from the expectation value. Let's do that step by step.

Let's try to solve this system of equations to express $p_0$ and $p_1$ in terms of $m$:

$$\begin{cases} p_0 + p_1 = 1 \\ p_0 - p_1 = m \end{cases} = \begin{cases} p_0 = 1 - p_1 \\ p_1 = -m + p_0 \end{cases}$$
$$= \begin{cases} p_0 = 1 - p_1 \\ p_1 = -m + (1 - p_1) \end{cases}$$
$$= \begin{cases} p_0 = 1 - p_1 \\ 2p_1 = -m + 1 \end{cases}$$
$$= \begin{cases} p_0 = 1 - p_1 \\ p_1 = \frac{1-m}{2} \end{cases}$$
$$= \begin{cases} p_0 = 1 - \frac{1-m}{2} \\ p_1 = \frac{1-m}{2} \end{cases}$$
$$= \begin{cases} p_0 = \frac{2-(1-m)}{2} \\ p_1 = \frac{1-m}{2} \end{cases}$$
$$= \begin{cases} p_0 = \frac{1+m}{2} \\ p_1 = \frac{1-m}{2} \end{cases}$$

So, we have:

$$p_0 = \frac{1+m}{2} \quad \text{and} \quad p_1 = \frac{1-m}{2} \tag{66}$$

**Why is this important?** This result shows that the expectation value of the Pauli-Z observable contains enough information to reconstruct the complete probability distribution of the measurement outcomes. In other words, by measuring only the average value $m = \langle Z \rangle$, we can determine:

- The probability $p_0$ of obtaining the state $|0\rangle$;
- The probability $p_1$ of obtaining the state $|1\rangle$.

### Not always true in general

For a general observable with more than two outcomes, one expectation value is usually **not enough**. In general, to determine $n$ unknown probabilities, we need $n$ independent equations. Normalization provides one equation, so we typically need additional information such as: more expectation values, higher moments (e.g., $\langle O^2 \rangle$), or other constraints to fully determine the probability distribution.

However, if an observable has only two distinct outcomes $a$ and $b$, then the same idea works. If:

- Outcome $a$ occurs with probability $p$,
- Outcome $b$ occurs with probability $1 - p$,

Then the expectation value is:

$$\langle O \rangle = a \cdot p + b \cdot (1-p) \xRightarrow{\text{solve for } p} p = \frac{\langle O \rangle - b}{a - b} \tag{67}$$

It finds **the probability of the first outcome** $a$, and the **probability of the second outcome** $b$ can be found by normalization:

$$1 - p = 1 - \frac{\langle O \rangle - b}{a - b} = \frac{a - \langle O \rangle}{a - b} \tag{68}$$

For example, for the Pauli-Z observable, we have $a = +1$ and $b = -1$, so we can recover the probabilities as:

$$p_0 = P(+1) = \frac{\langle Z \rangle - (-1)}{(+1) - (-1)} = \frac{\langle Z \rangle + 1}{2}$$

$$p_1 = P(-1) = \frac{(+1) - \langle Z \rangle}{(+1) - (-1)} = \frac{1 - \langle Z \rangle}{2}$$

## 4.9 Heisenberg Representation of Quantum Circuits

Up to now, we have described quantum computation using the **Schrödinger representation**. In this approach, the quantum state evolves over time, while the observables remain fixed.

However, there is an alternative but completely equivalent description called the **Heisenberg representation**. In this approach, the quantum state remains fixed, while the observables evolve instead.

Both descriptions produce exactly the same physical predictions and measurement outcomes. They are just different mathematical viewpoints of the same underlying physics.

### Schrödinger Representation

In the **Schrödinger Representation**, a quantum gate $U$ acts directly on the state:

$$|\psi_1\rangle = U\,|\psi_0\rangle \tag{69}$$

Where:

- $|\psi_0\rangle$ is the initial state;
- $U$ is the unitary operator representing the quantum circuit;
- $|\psi_1\rangle$ is the final state.

The observable does not change. For example, if we measure in the computational basis, we use the Pauli-Z observable $Z$, which **remains the same before and after applying the circuit**. The expectation value after applying the circuit is:

$$\langle\psi_1|\,Z\,|\psi_1\rangle$$

In this case:

- The **state changes**;
- The **observable stays fixed**.

### Heisenberg Representation

In the **Heisenberg Representation**, the state remains unchanged $|\psi_0\rangle$. Instead, the **observable is transformed** according to:

$$Z \;\longrightarrow\; U^\dagger Z U \tag{70}$$

Where:

- $U^\dagger$ is the conjugate transpose of the gate $U$;
- $Z$ is the original observable;
- $U$ is the gate itself.

The expectation value is then computed as:

$$\langle\psi_0| U^\dagger Z U |\psi_0\rangle$$

In this case:

- The **state stays fixed**;
- The **observable changes**.

### Why are the two representations equivalent?

Starting from the Schrödinger representation:

$$|\psi_1\rangle = U |\psi_0\rangle$$

The corresponding bra is:

$$\langle\psi_1| = \langle\psi_0| U^\dagger$$

Substituting into the expectation value:

$$\begin{aligned}\langle\psi_1| Z |\psi_1\rangle &= \left(\langle\psi_0| U^\dagger\right) Z \left(U |\psi_0\rangle\right) \\ &= \langle\psi_0| U^\dagger Z U |\psi_0\rangle\end{aligned}$$

Therefore (and this is the key point), the expectation value computed in the Schrödinger representation is exactly equal to the expectation value computed in the Heisenberg representation (see Equation 70):

$$\langle\psi_1| Z |\psi_1\rangle = \langle\psi_0| U^\dagger Z U |\psi_0\rangle \tag{71}$$

This proves that both representations yield exactly the same measurement result.

### Physical Interpretation

The Schrödinger and Heisenberg representations are two different mathematical viewpoints of the same physical process.

In the **Schrödinger** representation:

- We imagine the quantum state moving through the circuit;
- Gates transform the state;
- Measurements are performed using fixed observables.

In the **Heisenberg** representation:

- We keep the quantum state fixed;
- Gates transform the observables;
- Measurements are performed using transformed observables.

### Why is the Heisenberg representation useful?

The Heisenberg representation is useful because it lets us answer a very important question: ***what observable is this quantum circuit actually measuring?***

For example, in the Schrödinger representation, suppose we have an initial state $|\psi\rangle$, a quantum circuit $H$ (the Hadamard gate, page 63), and we measure in the computational basis using the observable $Z$ (page 54). *But what does this circuit actually measure?*

To find out, we can switch to the Heisenberg representation and compute the transformed observable:

$$H^\dagger Z H$$

This will give us the new observable that corresponds to the measurement performed by the circuit $H$. In this case, we find that:

$$H^\dagger Z H = \frac{1}{2}\left(\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}\begin{bmatrix}1 & 1\\1 & -1\end{bmatrix}\right) = \frac{1}{2}\begin{bmatrix}0 & 2\\2 & 0\end{bmatrix} = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix} = X$$

This means that the circuit $H$ actually measures the observable $X$ (the Pauli-X gate, page 49), not $Z$. In other words, applying the Hadamard gate before measuring in the computational basis effectively changes our measurement from $Z$ to $X$.

So, without the Heisenberg representation, we only know the sequence of gates. With the Heisenberg representation, we immediately understand: "*this circuit is actually measuring $X$!*" This is the real physical meaning of the circuit, and it is crucial for understanding how quantum algorithms work.

**Example 16: Analogy with Arithmetic Expressions**

Suppose someone gives us:

$$(3 + 5) \times 2 - 16$$

We could compute it step by step. Or we could simplify the whole expression to: 0. The Heisenberg representation does something similar:

- Instead of following every state transformation,
- It simplifies the entire circuit into an equivalent measurement.

Suppose a circuit contains many gates. In the Schrödinger picture, we must track the state after each gate. In the Heisenberg picture, we ask only: *what observable is measured at the end?* Sometimes the answer is a single operator such as $X$ or $Z$. Sometimes it is a more complicated operator that involves multiple qubits. But in either case, we get a clear understanding of what the circuit does without having to track the state at every step.

## 4.10 Clifford Gates

### 4.10.1 Definition

In the previous section, we introduced the Heisenberg representation of quantum circuits. There, we saw that applying a Hadamard gate $H$ and then measuring an observable $Z$ is equivalent to first transforming the observable $Z$ into $X$ and then measuring $X$:

$$Z \xrightarrow{H} X$$

This naturally leads to an important question: ***what kinds of gates transform simple observables into other simple observables?*** A particularly important class of gates with this property is known as the **Clifford gates**.

**Definition 9: Clifford Gate**

A unitary gate $U$ is called a **Clifford Gate** if, for every Pauli operator $P$ (i.e., $P \in \{I, X, Y, Z\}$), the transformed operator

$$U^\dagger P U = P' \tag{72}$$

Is still a Pauli operator (possibly with an overall sign $\pm 1$).

In other words, Clifford gates map Pauli operators to other Pauli operators under conjugation.

For a single qubit, the Pauli operators are:

$$I^4, \qquad X, \qquad Y, \qquad Z.$$

Therefore, a single-qubit Clifford gate must transform each of these operators into another operator from the same set (up to sign).

For example:

$$Z \longrightarrow X$$

Is allowed, because $X$ is still a Pauli operator. However:

$$Z \longrightarrow \frac{X+Y}{\sqrt{2}}$$

Is not allowed, because the **result is not a single Pauli operator**.

#### Multi-Qubit Extension

For $n$ qubits, Pauli operators are formed by taking tensor products of single-qubit Pauli matrices. For two qubits, examples of Pauli operators include:

$$X \otimes I, \qquad I \otimes Z, \qquad Y \otimes X, \qquad Z \otimes Z$$

[4] Technically, Wolfgang Pauli only defined $X$, $Y$, and $Z$ as the Pauli operators because they are the non-trivial ones. However, we include the identity $I$ in the set of Pauli operators for completeness.

For three qubits, examples include:

$$X \otimes I \otimes Z, \qquad Y \otimes Y \otimes X$$

In general, an $n$-qubit Pauli operator has the form:

$$P = P_1 \otimes P_2 \otimes \cdots \otimes P_n$$

Where each $P_i$ is one of the single-qubit Pauli operators $\{I, X, Y, Z\}$.

**Example 17: Hadamard Gate is a Clifford Gate**

The Hadamard gate $H$ is:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Because $H$ is both Hermitian and unitary, we have $H^\dagger = H$ and $H^{-1} = H$. Therefore, we can compute the conjugation of the Pauli operators by $H$:

$$HPH = P'$$

In practice, the Hadamard gate transforms the Pauli matrices as follows:

$$\begin{aligned} HXH &= \frac{1}{2} \left( \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left( \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left( \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} \right) \\ &= Z \end{aligned} \tag{73}$$

$$\begin{aligned} HZH &= \frac{1}{2} \left( \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left( \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left( \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix} \right) \\ &= X \end{aligned} \tag{74}$$

$$\begin{aligned} HYH &= \frac{1}{2} \left( \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left( \begin{bmatrix} i & -i \\ -i & -i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \right) \\ &= \frac{1}{2} \left( \begin{bmatrix} 0 & 2i \\ -2i & 0 \end{bmatrix} \right) \\ &= -Y \end{aligned} \tag{75}$$

Each result is still a Pauli operator (up to a sign), confirming that the Hadamard gate is indeed a Clifford gate.

**Example 18: CNOT Gate is a Clifford Gate**

The CNOT gate (page 101) is:

$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

As with the Hadamard gate, we can verify that the CNOT gate is a Clifford gate by checking how it transforms the two-qubit Pauli operators. To do this, we compute the tensor products of the single-qubit Pauli operators with the identity (to create two-qubit Pauli operators) and then conjugate them by the CNOT gate (since the CNOT gate is hermitian, we have $\text{CNOT}^\dagger = \text{CNOT}$):

$$\begin{aligned}
X \otimes I &= \begin{bmatrix} 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \\
\text{CNOT}(X \otimes I)\text{CNOT} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \\
&= X \otimes X
\end{aligned} \tag{76}$$

For the $Z$ operator, the CNOT gate does not change the first qubit, so we have $Z \otimes I$:

$$\begin{aligned}
Z \otimes I &= \begin{bmatrix} 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ 0 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & -1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
\text{CNOT}(Z \otimes I)\text{CNOT} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \\
&= Z \otimes I
\end{aligned} \tag{77}$$

Therefore, the CNOT gate transforms $X \otimes I$ into $X \otimes X$ and leaves $Z \otimes I$ unchanged. By performing similar calculations for all other two-qubit Pauli operators, we can confirm that the CNOT gate maps every two-qubit Pauli operator to another two-qubit Pauli operator (up to a sign):

- $\text{CNOT}\,(I \otimes X)\,\text{CNOT} = I \otimes X$
- $\text{CNOT}\,(I \otimes Z)\,\text{CNOT} = Z \otimes Z$

These results confirm that the CNOT gate is indeed a Clifford gate.

### 4.10.2 Non-Clifford Gates

While the Hadamard and CNOT gates are examples of Clifford gates, there are many other gates that do not have this property. The most important example of a non-Clifford gate is the $T$ gate.

The **T Gate** (also called the $\pi/8$ gate) is a **single-qubit gate that applies a specific phase shift to the state** $|1\rangle$, while leaving $|0\rangle$ **unchanged**. Its matrix representation is:

$$T = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{bmatrix} \tag{78}$$

This means:

$$T|0\rangle = |0\rangle \qquad T|1\rangle = e^{i\pi/4}|1\rangle \tag{79}$$

On the Bloch sphere, the $T$ gate corresponds to a rotation around the $Z$-axis by an angle of $\pi/4$ (or 45 degrees). In terms of the Pauli operators, the $T$ gate transforms them as follows:

$$\begin{aligned} TXT^\dagger &= \frac{1}{\sqrt{2}}(X+Y) \\ TYT^\dagger &= \frac{1}{\sqrt{2}}(Y-X) \\ TZT^\dagger &= Z \end{aligned} \tag{80}$$

Since the $T$ gate transforms the $X$ and $Y$ operators into linear combinations of $X$ and $Y$ (which are not single Pauli operators), the $T$ gate is **not a Clifford gate.**

**Deepening: The $T$ Gate is the Square Root of the $S$ Gate**

The $T$ gate is the square root of the Phase gate $S$ (which is a Clifford gate) because applying $T$ twice gives exactly $S$.

If we apply the $T$ gate twice, we have:

$$T^2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{bmatrix}$$

Where:

$$e^{i\pi/4} \cdot e^{i\pi/4} = \exp\left(\frac{i\pi}{4} + \frac{i\pi}{4}\right) = \exp\left(\frac{2i\pi}{4}\right) = \exp\left(\frac{i\pi}{2}\right) = e^{i\pi/2}$$

Now using Euler's formula, we can express $e^{i\pi/2}$ as:

$$\begin{aligned} e^{i\theta} &= \cos(\theta) + i\sin(\theta) \\ e^{i\pi/2} &= \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right) \\ &= 0 + i \cdot 1 \\ &= i \end{aligned}$$

Therefore, we have:

$$T^2 = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = S$$

Finally, we can also express $T$ as the square root of $S$:

$$T = \sqrt{S} = \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{i} \end{bmatrix}$$

In summary, applying the $T$ gate twice gives us the $S$ gate, confirming that $T$ is indeed the square root of $S$.

### ❓ Why are Non-Clifford Gates important?

The real question is: **why can some quantum circuits be simulated efficiently on a classical computer, while others cannot?** To answer this, we must compare two ways of describing a quantum state.

- The **full state vector representation**, where an $n$-qubit quantum state has the form (page 91):

$$|\psi\rangle = \sum_{k=0}^{2^n-1} c_k |k\rangle$$

  This requires storing $2^n$ complex amplitudes (e.g., 1 qubit requires 2 amplitudes, 2 qubits require 4 amplitudes, etc.), which **grows exponentially** with the number of qubits.

- The **stabilizer representation**. Certain special states can be described much more compactly. Instead of storing all amplitudes, we store a small set of Pauli operators (stabilizers) that uniquely characterize the state. Only $n$ independent stabilizers are needed for $n$ qubits. For example, 10 qubits can be described by just 10 stabilizers, which is a huge reduction compared to the $2^{10} = 1024$ amplitudes needed in the full state vector representation. This representation **grows only linearly** with the number of qubits, making it much more efficient for certain states.

Clifford gates use the stabilizer representation, because transform Pauli operators into other Pauli operators. Therefore, we start with $n$ stabilizers ($n$ Pauli operators) that describe the initial state, and after applying Clifford gates, we still have $n$ stabilizers that describe the new state. This allows to leave the state description compact and efficient.

**❓ What happens at a Non-Clifford Gate?** Consider the $T$ gate. It transforms, for example, the $X$ operator into a linear combination of $X$ and $Y$:

$$TXT^\dagger = \frac{1}{\sqrt{2}}(X + Y)$$

The result is not a single Pauli operator, but a sum of two Pauli operators. After another Non-Clifford transformation, each term may split again. So **repeated Non-Clifford gates can cause the number of terms to double repeatedly, leading to exponential growth**.

**Computational Consequence.** Classical simulation cost depends on how much information must be stored.

- Clifford circuits keep the description small.
- Non-Clifford circuits cause the description to grow rapidly.

Therefore, if a circuit contains only Clifford gates, a classical computer can efficiently simulate it. In contrast, if the circuit includes Non-Clifford gates such as $T$, simulation generally becomes exponentially more difficult.

**So, why are Non-Clifford gates important?** If Clifford circuits are classically simulable, then they cannot provide full quantum computational power. To achieve universal quantum computation, we need gates that escape the stabilizer framework. The $T$ gate does exactly this. By adding the $T$ gate (or any other suitable non-Clifford gate) to the Clifford gate set, we obtain a universal set of quantum gates. This allows us to implement arbitrary quantum algorithms, including computations believed to be infeasible for classical computers.

### 4.10.3 Stabilizer States

As anticipated in the previous section, a **Stabilizer State** is a **quantum state that can be described** not by listing all of its amplitudes, but by **specifying which Pauli operators leave it unchanged**. This is one of the most elegant ideas in quantum computing and is the foundation of the stabilizer formalism used in quantum error correction and efficient simulation.

**Definition 10: Stabilizer State**

An $n$-qubit state $|\psi\rangle$ is a stabilizer state if there exists a Pauli operator $P$ (or, more generally, a set of commuting Pauli operators) such that:

$$P|\psi\rangle = |\psi\rangle \tag{81}$$

This means:

- Applying $P$ does not change the state $|\psi\rangle$.
- The state $|\psi\rangle$ is an eigenvector of $P$.
- The corresponding eigenvalue is $+1$.

The operator $P$ is called a **Stabilizer** because it "*stabilizes*" the state $|\psi\rangle$.

**Meaning of "*stabilize*".** Suppose a transformation leaves an object unchanged. Some examples:

- Rotating a perfect sphere leaves it unchanged;
- Reflecting a symmetric figure may leave it unchanged;
- Applying certain Pauli operators may leave a quantum state unchanged.

In quantum mechanics, if the relation $P|\psi\rangle = |\psi\rangle$ holds, then $P$ is a symmetry of the state $|\psi\rangle$, and we say that $P$ stabilizes $|\psi\rangle$. This means that **the state $|\psi\rangle$ is invariant under the transformation represented by $P$.**

**Example 19: $|0\rangle$ and $|1\rangle$**

The Pauli-Z gate is:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Applying $Z$ to $|0\rangle$:

$$Z|0\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

Therefore, $|0\rangle$ is a stabilizer state and the Pauli-Z operator is one of its stabilizers.

By contrast, applying $Z$ to $|1\rangle$:

$$Z|1\rangle = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

Shows that $|1\rangle$ is not stabilized by $Z$ (since the eigenvalue is $-1$), and thus $|1\rangle$ is not a stabilizer state with respect to $Z$.

**Example 20: $|+\rangle$**

The state $|+\rangle$ is defined as:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Applying the Pauli-X operator to $|+\rangle$:

$$X|+\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = |+\rangle$$

Therefore, $|+\rangle$ is a stabilizer state and the Pauli-X operator is one of its stabilizers.

**Example 21: Bell State**

Consider the Bell state (page 116):

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Applying the operator $X \otimes X$ to $|\Phi^+\rangle$:

$$\begin{aligned}(X \otimes X)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}(X|0\rangle \otimes X|0\rangle + X|1\rangle \otimes X|1\rangle) \\ &= \frac{1}{\sqrt{2}}(|11\rangle + |00\rangle) = |\Phi^+\rangle\end{aligned}$$

And also applying $Z \otimes Z$:

$$\begin{aligned}(Z \otimes Z)|\Phi^+\rangle &= \frac{1}{\sqrt{2}}(Z|0\rangle \otimes Z|0\rangle + Z|1\rangle \otimes Z|1\rangle) \\ &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle\end{aligned}$$

Therefore, $|\Phi^+\rangle$ is a stabilizer state and both $X \otimes X$ and $Z \otimes Z$ are stabilizers of $|\Phi^+\rangle$.

### Multiple Stabilizers

The Equation 81 defines **a stabilizer state with respect to a single Pauli operator** $P$. However, many quantum states are stabilized by multiple Pauli operators. In fact, for an $n$-qubit stabilizer state, we typically use $n$ independent commuting stabilizers:

$$P_1, P_2, \ldots, P_n$$

Each of these operators satisfies:

$$P_i |\psi\rangle = |\psi\rangle \quad \text{for } i = 1, 2, \ldots, n \tag{82}$$

The **set of all such stabilizers forms a group** under multiplication, known as the **Stabilizer Group**.

### Why Stabilizers are powerful

A general $n$-qubit state requires $2^n$ complex amplitudes to be fully described. However, a stabilizer state can be completely characterized by its $n$ independent stabilizers, which are much fewer in number. Furthermore, each stabilizer is describable with $O(n)$ symbols (since it is a tensor product of $n$ Pauli operators), leading to an efficient representation of the state. Therefore, the description size grows only polynomially (i.e., $O(n)$) rather than exponentially.

For example, for 100 qubits, a full state-vector simulation requires $2^{100}$ amplitudes; in contrast a stabilizer formalism requires roughly 100 stabilizers. This is why large Clifford circuits can be simulated efficiently on classical computers.

In summary, instead of describing "*what are all amplitudes of the state?*", we describe "*which symmetries does the state satisfy?*". This is often dramatically more efficient.

## 4.11 Universal set of quantum gates

After studying Clifford and non-Clifford gates, the next question is: *what is the minimum set of gates needed to build* ***any quantum computation****?* This leads to the concept of a **universal set of quantum gates**, which is a collection of gates that can be combined to approximate any unitary operation on a quantum system to arbitrary precision.

**Definition 11: Universal Set of Quantum Gates**

A **set of quantum gates** is called universal if any unitary transformation can be approximated to arbitrary accuracy using only gates from that set.

In other words, a set of gates is universal if using only gates from that set, we can approximate any unitary transformation to arbitrary precision.

$$\text{Universal gate set} \implies \forall U,\ \exists \text{ a circuit approximating } U$$

### Why "Approximate" and not "Exactly"?

In quantum computing, every quantum gate is represented by a **unitary matrix**. For example the Hadamard gate, the $T$ gate, and the CNOT gate. However, the set of all unitary matrices is **continuous** (i.e., there are infinitely many unitary transformations). For example, a single-qubit rotation may involve any real angle $R_z(\theta)$ with infinitely many possible values of $\theta$:

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

Here, the angle $\theta$ can be 0.1, 0.12345, $\pi/7$ or any other real number. There are infinitely many real numbers, so there are infinitely many possible rotations and therefore infinitely many possible unitary matrices.

**Why not build hardware for every matrix?** Building a quantum gate for every possible unitary transformation is **impossible**. Imagine trying to crate one physical instruction for every possible angle:

- One gate for 0.1 radians
- One gate for 0.12345 radians
- One gate for $\pi/7$ radians
- One gate for 0.00001 radians

And so on forever. This is clearly impractical.

**The solution: a Finite Gate Set.** Instead of trying to build a gate for every possible unitary, we choose a **small set of standard gates**, for example $\{H, \text{CNOT}, T\}$. These are the only **primitive gates the machine needs to implement directly**. All other operations are constructed by combining them.

### Which gates are in the universal set?

To keep things simple, we could choose Clifford gates, such as $H$ and CNOT, as our set of gates. Although these gates are powerful, they can still be efficiently simulated on a classical computer (page 157). Therefore, Clifford gates alone are not sufficient for universal quantum computation. For this reason, in general, a **universal set requires at least one non-Clifford gate**, and the most common choice is the $T$ gate because of its simplicity and its relationship to the $S$ gate (page 156). A widely used universal set of quantum gates is:

$$\{H, \text{CNOT}, T\} \tag{83}$$

### Why these three gates are enough?

- **Hadamard** $H$ transforms basis states into superpositions. For example, the $|0\rangle$ state becomes the plus state (page 51):

$$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

- **CNOT** creates entanglement between qubits. For example, applying the CNOT to the state $|+0\rangle$ creates the Bell state $|\Phi^+\rangle$:

$$\text{CNOT}(|+0\rangle) = \left|\Phi^+\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- $T$ **gate** adds a 45° phase and breaks the Clifford-only restriction, allowing us to access a much larger set of unitary transformations. Because $T$ is a non-Clifford gate, it enables us to perform operations that cannot be efficiently simulated classically.

Hadamard changes basis and creates superposition, CNOT creates entanglement between qubits, and $T$ provides the non-Clifford component required for universality. Together, **they can approximate any unitary operation**.

Other universal gate sets exist, such as:

- Toffoli and Hadamard: $\{H, \text{Toffoli}\}$.
- Single-qubit rotations and CNOT: $\{R_x(\theta), R_y(\theta), R_z(\theta), \text{CNOT}\}$.

### How to Generate a Universal Set of Quantum Gates

A universal gate set is obtained by combining two ingredients:

1. A set of gates that can generate arbitrary single-qubit unitary matrices.
2. One entangling two-qubit gates, usually the CNOT gate (page 101), which can create entanglement between qubits.

If both conditions are satisfied, the gate set is universal. Let's see why this is the case:

1. **Choose Gates that Generate Arbitrary Single-Qubit Unitaries** (e.g., $H$ and $T$ gates). A single qubit can undergo any unitary transformation:
$$U \in U^2 = \left\{ U \in \mathbb{C}^{2\times 2} : U^\dagger U = I \right\}$$
To obtain universality, we need to approximate any such $2 \times 2$ unitary matrix. A common way is to use a Hadamard gate and a $T$ gate. The pair of gates $\{H, T\}$ can approximate any single-qubit unitary to arbitrary precision.

   ❓ **Why universality?** While mathematically there are infinitely many possible $2 \times 2$ unitary matrices, our hardware can implement only a **finite, discrete set** of gates exactly (similar to how a digital computer uses a finite instruction set). Therefore, the goal of a *universal* gate set is not to implement every unitary matrix exactly. Instead, for any target unitary matrix $U$ and for any desired accuracy $\varepsilon > 0$, there must exist a finite sequence of gates from the set whose overall operation $V$ satisfies:[5]
$$\|U - V\| < \varepsilon$$
In other words, we can make the approximation error as small as we want. This ability to approximate any unitary operation to arbitrary precision is what makes a gate set universal.

2. **Add an Entangling (two-qubit) Gate**. Single-qubit gates alone cannot create entanglement (page 121). To manipulate multiple qubits, we add an entangling gate such as CNOT. Once CNOT is available, qubits can interact and become entangled.

Combining these two ingredients, the resulting set is universal, meaning that any quantum computation can be approximated using just these gates.

[5] The matrix $U - V$ contains the element-by-element differences, but by itself it is still a matrix, not a single number. To obtain one number representing the size of the difference, we apply a **norm**. It plays exactly the same role as absolute value for numbers. In other words, it means that the implemented operation $V$ is within distance $\varepsilon$ of the desired operation $U$.

## 4.12 Fault-Tolerant Quantum Gates

So far, we have studied quantum computation in an idealized setting, where quantum gates are represented by perfect unitary matrices acting on perfectly isolated qubits. In practice, however, real quantum hardware is extremely fragile. Physical qubits are affected by noise, decoherence, imperfect control operations, and measurement errors, all of which can corrupt the quantum information stored in the system.

To perform reliable quantum computation, it is therefore necessary to protect quantum information against errors. This is the goal of **quantum error correction**, where a single logical qubit is encoded into many physical qubits in such a way that errors can be detected and corrected without destroying the quantum state.

However, protecting quantum information is not sufficient by itself. We must also be able to apply quantum gates directly on encoded logical qubits while preserving the fault-tolerant structure of the error-correcting code. This introduces the notion of **fault-tolerant quantum gates**, namely gates designed so that local physical errors do not spread uncontrollably through the computation.

In this context, Clifford gates play a particularly important role because many of them admit simple fault-tolerant implementations, often through **transversal operations**. Non-Clifford gates, on the other hand, are significantly more difficult to implement fault-tolerantly, even though they are essential for universal quantum computation. This creates one of the central challenges of modern quantum computing: the gates required for computational universality are also the most expensive and difficult to protect against errors.

In the following sections, we introduce the distinction between physical and logical qubits, the concept of transversal gates, and the reasons why non-Clifford gates such as the $T$ gate represent the main obstacle in fault-tolerant quantum computation.

### 4.12.1 Logical vs Physical Qubits

**Problem.** In an ideal mathematical model of quantum computation, qubits are assumed to evolve perfectly under unitary transformations. Real quantum hardware, however, is inherently **noisy**. Physical qubits interact with the surrounding environment and are affected by decoherence, thermal fluctuations, imperfect control pulses, and measurement errors. As a consequence, **quantum information** stored in a single physical qubit is **extremely fragile**.

**Solution.** To overcome this limitation, quantum computers do not directly perform large computations on individual physical qubits. Instead, they use **quantum error correction**, where the information of a single qubit is encoded across many physical qubits, forming a **logical qubit**.

- **Physical Qubit**. A **physical qubit** is the actual **hardware-level quantum system used to store information**. For example superconducting qubits, trapped ions, neutral atoms or photonic qubits[6]. Physical qubits are **directly manipulated by the hardware** and **implement the elementary quantum operations supported by the quantum processor**. However, they are noisy and have limited coherence times.

- **Logical Qubit**. A **logical qubit** is an **encoded qubit constructed from many physical qubits** in order to protect quantum information against errors.

  Instead of storing the quantum state in a single fragile qubit, the information is distributed redundantly across multiple physical qubits:

  $$\text{Many physical qubits} \xrightarrow{\text{Encoding}} \text{One logical qubit}$$

  This encoding allows the system to detect errors, correct certain errors and preserve the quantum state for a much longer time.

  **Why do we need logical qubits?** Without error correction, errors accumulate extremely rapidly during a quantum computation. Even a small error probability per gate becomes catastrophic when millions or billions of gates are executed.

  Logical qubits make large-scale quantum computation possible by continuously protecting the encoded information while the computation is running.

**The Main Challenge**

Encoding a logical qubit is only the first step. We must also be able to apply quantum gates, manipulate information, execute algorithms. Directly on logical qubits without destroying the error-correcting structure. This leads to the concept of **fault-tolerant quantum gates**, which are quantum gates specifically designed to prevent local physical errors from spreading uncontrollably through the encoded system.

[6] It's normal don't know the details of these different physical implementations at this stage. The key point is that they are all examples of physical qubits, which are the building blocks of quantum computers.

### 4.12.2 Transversal Gates

In fault-tolerant quantum computing, one of the main objectives is to apply logical quantum gates without allowing physical errors to spread uncontrollably through the encoded qubits. Let's suppose one logical qubit[7] is encoded into many physical qubits:

$$\text{Logical Qubit} \rightarrow q_1, q_2, \ldots, q_7$$

It means that a single logical qubit is represented by a collection of physical qubits. Now imagine, for example, that $q_3$ contains an error, and all the other qubits are fine. This situation is still manageable because we can detect and correct the error in $q_3$ without affecting the other qubits.

**The Dangerous Situation.** Now suppose we apply a complicated gate that strongly couples all qubits together. Then, if $q_3$ has an error, it can spread to all the other qubits, making the entire logical qubit corrupted and uncorrectable.

**The core goal of Fault Tolerance.** So the main objective becomes: if **one physical qubit fails, the failure should remain localized and not spread to other qubits**. This is where transversal gates come into play.

**Definition 12: Transversal Gates**

A **Transversal Gate** is a logical quantum **gate** implemented by **applying independent operations** to corresponding **physical qubits of an encoded logical qubit**.

Suppose a logical qubit is encoded into $n$ physical qubits. A transversal implementation applies the same unitary operation independently to each physical qubit:

$$U_L = U^{\otimes n}$$

Where:

- $U$ is the physical gate;
- $U_L$ is the resulting logical gate;
- $n$ is the number of physical qubits used in the encoding.

In simple terms, a transversal gate applies operations independently to each physical qubit. So no qubit interacts with another qubit inside the same block. Therefore if $q_3$ was wrong before, only $q_3$ will be affected by the gate, and **the error will *not* spread to the other qubits**.

[7] A logical qubit is an encoded qubit constructed from many physical qubits in order to protect quantum information against errors. Instead of storing the quantum state in a single fragile qubit, the information is distributed redundantly across multiple physical qubits.

### Error Containment and Fault Tolerance

The **most important property** of transversal gates is that they prevent error propagation inside the encoded block.

Suppose one physical qubit contains an error before the gate is applied. Since each qubit is manipulated independently, the error remains localized and does not spread to many other qubits.

Furthermore, transversal gates are **inherently fault-tolerant**. In general, a fault-tolerant operation must ensure that:

- A small physical error remains small;
- A single faulty qubit does not corrupt the entire logical qubit.

Transversal gates naturally satisfy this requirement because there are no interactions between qubits within the same code block during the gate operation. Therefore, even if one qubit experiences an error, it does not affect the others.

**Example 22: Logical Hadamard Gate**

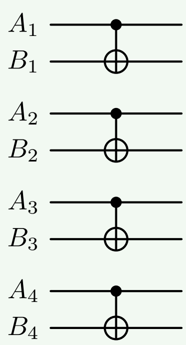
For many quantum error-correcting codes, a **logical Hadamard gate** can be implemented transversally by applying a Hadamard gate independently to every physical qubit:

$$H_L = H^{\otimes n} = \underbrace{H \otimes H \otimes \ldots \otimes H}_{n \text{ times}}$$

Thus, the logical operation is realized through many independent physical operations performed in parallel.

**Example 23: Logical CNOT Gate**

A logical CNOT gate between two encoded logical qubits is often implemented by applying **physical CNOT gates pairwise between corresponding qubits of the two blocks**:




Each pair of physical qubits interacts independently of the others.

### 4.12.3 Why Non-Clifford Gates Are Difficult

One of the central challenges of fault-tolerant quantum computing is that **most non-Clifford gates cannot be implemented transversally**. While many Clifford gates admit simple fault-tolerant implementations (we saw some examples, like the Hadamard gate and CNOT, in the last section on page page 168), non-Clifford gates generally require much more complex procedures.

#### Clifford Gates and Transversality

As discussed previously, transversal gates are highly desirable because they prevent physical errors from spreading across the encoded logical qubit (page 167).

For many quantum error-correcting codes, **Clifford gates** such as Hadamard gate, Phase gate $S$, and CNOT gate, can **often be implemented transversally**. This means that the logical operation can be realized by applying independent physical operations to the corresponding physical qubits of the encoded blocks.

#### The Problem with Non-Clifford Gates

However, **non-Clifford gates**, such as the $T$ gate, generally **do not admit transversal implementations** that preserve the structure of the error-correcting code. In fact, if we attempt to apply a non-Clifford gate transversally, two problematic situations may occur:

- ✗ The resulting operation does not correspond to the correct logical gate;
- ✗ Or the encoded structure of the code is broken, allowing errors to propagate uncontrollably through the logical qubit.

The difficulty of implementing non-Clifford gates in a fault-tolerant manner is not just a technological limitation of current quantum hardware; it is also a fundamental theoretical constraint on the structure of quantum error-correcting codes. The **Eastin-Knill theorem** formalizes this limitation by proving that no quantum error-correcting code (i.e., no encoding scheme) can realize a universal set of logical gates using only transversal operations.

Consequently, transversal gates are extremely useful for fault tolerance, but they are not sufficient to implement universal quantum computation. At least one non-Clifford gate must be implemented through more advanced fault-tolerant techniques.

**Deepening: Why non-Clifford gates do not admit transversal implementations?**

Briefly, because non-Clifford gates are “too powerful” to fit inside the simple independent structure of transversal operations. But let us build the intuition behind this statement more carefully.

**? What a Transversal Gate really does.** A transversal gate acts independently on each physical qubit: $U_L = U^{\otimes n}$, where $U$ is a single-qubit operation. Each physical qubit is treated separately, and this is **extremely restrictive** since it **cannot create** any **entanglement** between the physical qubits (since there are no multi-qubit interactions).

**? Why this restriction is good.** Because **independent operations prevent error spreading**. If one physical qubit is wrong, the gate acts independently, then only that qubit remains wrong. So transversal gates are naturally fault-tolerant.

**⚠ But there is a problem.** A universal quantum computer must be able to perform extremely rich transformations. In particular, it must be able to do arbitrary rotations on the Bloch sphere, complicated entangling dynamics, and non-stabilizer transformations. And transversal operations are simply too "*rigid*" to generate all of that while preserving the code structure.

**⚠ In fact, there is a no-go theorem.** The **Eastin-Knill theorem** [1] states that **no quantum error-correcting code can have a universal set of transversal gates**. This means that while we can have some transversal Clifford gates, we cannot have a transversal implementation of a non-Clifford gate like $T$ that would allow us to achieve universality. Therefore, we must look for more complex methods to implement non-Clifford gates in a fault-tolerant manner, such as magic state distillation or code switching, which are significantly more resource-intensive than transversal implementations.

In summary, transversal gates are simple, local and safe. Non-Clifford gates require transformations that are too complex to be realized by purely independent local operations while still preserving th error-correcting code. That is the real reason why non-Clifford gates generally cannot be implemented transversally.

### ✓ Solutions to the Non-Clifford Problem

To overcome the limitations imposed by the Eastin-Knill theorem, researchers have developed several techniques to implement non-Clifford gates in a fault-tolerant manner. Two standard approaches are: **magic state distillation** and **magic state injection**. The main idea is to avoid applying the non-Clifford gate directly to the encoded logical qubit. Instead, the quantum computer prepares a special ancillary quantum state, called a **magic state**, which can later be consumed to indirectly implement the desired logical gate.

However, **this course does not cover these techniques in detail**, as they are quite complex and require a deep understanding of quantum error correction and fault tolerance. The key takeaway is that while non-Clifford gates are essential for universal quantum computation, their implementation in a fault-tolerant manner is significantly more challenging than that of Clifford gates, and it requires sophisticated methods that go beyond simple transversal operations.

### 4.12.4 Physical vs Logical Gate Sets

In quantum computing, it is important to distinguish between:

- **Physical Gate sets**, which correspond to the elementary operations directly implemented by the hardware;
- **Logical Gate sets**, which are the fault-tolerant operations acting on encoded logical qubits (e.g., the Clifford gates).

Although the two descriptions correspond to the same underlying quantum computation, they differ significantly in terms of implementation complexity and fault-tolerant requirements.

#### Physical (Hardware-Level) Gate Sets

At the hardware level, quantum gates act directly on physical qubits. A typical universal physical gate set consists of:

- **Single-Qubit Gates**: These include rotations around the X, Y, and Z axes (e.g., $R_x(\theta)$, $R_y(\theta)$, $R_z(\theta)$).

  In general, single-qubit rotations are **easier to implement** than multi-qubit gates because they **act locally on only one physical qubit**. In particular, many hardware platforms can implement $R_z(\theta)$ very efficiently, sometimes even virtually through software phase updates rather than physical operations.

  At the physical level, also the $T$ gate is simply a rotation around the Z-axis by $\pi/4$, therefore it is not more difficult to implement than other single-qubit gates.

- **Two-Qubit Gates**: The most common two-qubit gate is the CNOT gate, which is essential for creating entanglement between qubits.

  Two-qubit gates such as CNOT are **significantly more difficult** because they require **interactions between qubits**. In general, the larger the number of qubits involved in a gate, the more complex and error-prone its physical implementation becomes.

These gates are **implemented through direct hardware control**, such as laser pulses in ion traps or microwave pulses in superconducting qubits.

#### Logical (Fault-Tolerant) Gate Sets

When quantum information is encoded into logical qubits using quantum error correction, the relevant gate set changes. The focus is no longer only on mathematical universality, but **also on fault-tolerant** implementability. A typical logical fault-tolerant gate set includes (page 162):

- **Clifford gates**:
$$\{H, S, \text{CNOT}\}$$
- **Non-Clifford gates** (e.g., the $T$ gate):
$$\{T\}$$

### ⚖ Key Distinction

The crucial observation is that the **complexity of a gate depends strongly on whether it acts on**: physical qubits or encoded logical qubits. For example:

- On physical qubits, $T = R_z(\pi/4)$ is a simple single-qubit rotation.
- On logical qubits, $T$ is a non-Clifford logical gate not usually transversally implementable, and therefore it requires magic state distillation and fault-tolerantly complex procedures to implement.

This distinction is one of the main practical challenges of large-scale quantum computing. At the mathematical level, the $T$ gate appears to be a simple phase rotation. However, once quantum information is encoded into logical qubits, implementing the same operation fault-tolerantly becomes extremely resource-intensive.

As a consequence, the cost of fault-tolerant quantum algorithms is often dominated by:

- The number of logical $T$ gates required;
- Magic state preparation;
- Error-correction overheads associated with non-Clifford operations.

In summary, the same gate may be easy at the hardware level, but extremely difficult at the logical fault-tolerant level.

# 5 Limits of Quantum Information

## 5.1 No-Cloning Principle

### 5.1.1 Why Cloning Matters

Briefly, the no-cloning principle states that it is **impossible to create a perfect copy of an arbitrary unknown quantum state**.

At first sight this may seem like a strange technical limitation, because in classical computing copying information is *trivial*: we can easily copy files, duplicate RAM, clone hard disks, send packets over the network, etc. However, in quantum computing, copying an arbitrary qubit is fundamentally impossible. This is **not a technological limitation**, instead it is a direct **consequence of the mathematical structure** of quantum mechanics.

In classical computing, a bit can be in one of two states: `0` or `1`. In quantum computing, a qubit instead can be in infinitely many states, because it can be in any superposition of `0` and `1`:

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle$$

With complex amplitudes $a$ and $b$ such that $|a|^2 + |b|^2 = 1$. So a qubit is not just "0" or "1", but it represents a point on the Bloch sphere, meaning there are infinitely many possible states. If arbitrary cloning were possible, we could take an unknown state $|\psi\rangle$ and create as many copies of it as we want. This would allow us to perform measurements on the copies to learn about the original state $|\psi\rangle$ without disturbing it, **violating the fundamental principles of quantum mechanics**.

**Relation with Unitary**

The no-cloning principle theorem is deeply connected to the fact that quantum evolution is **unitary**. Quantum gates are represented by unitary matrices:

$$U^\dagger U = UU^\dagger = I$$

Unitary operations preserve norms, probabilities and inner products between states. **Cloning would violate this preservation property**. Because, intuitively, if two partially similar states must remain partially similar after evolution (i.e., their inner product must be preserved), then cloning would instead make them "more distinguishable" (i.e., their inner product would change), which is a contradiction. We will see the formal proof of the no-cloning principle in the next section.

**Relation with No-Signaling**

The no-cloning principle is also related to the no-signaling principle, because **breaking the no-cloning principle would violate the no-signaling principle**.

Suppose Alice and Bob share an entangled Bell pair:

$$\left|\Phi^{+}\right\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Normally, Bob cannot determine whether Alice has measured her qubit or not, because his local measurements always appear random (50% probability of measuring `0` and 50% probability of measuring `1`). But *imagine* Bob could clone his qubit infinitely many times. Then, he could:

1. Create many identical copies of his qubit.
2. Perform measurements/statistics on all copies.
3. Infer whether the global quantum state collapsed (i.e., whether Alice measured her qubit). Using the principle of reductio ad absurdum, if Alice does not measure, then Bob will measure random results (0, 1, 0, 1, etc.). When Alice measures and collapses the state to $|0\rangle$, then Bob's qubit will instantly become $|0\rangle$, and therefore, all of his copies will be $|0\rangle$.

This would allow instantaneous communication between Alice and Bob, faster than the speed of light, which is forbidden by the no-signaling principle (violation of relativity, no information can travel faster than the speed of light). Therefore, the **impossibility of cloning helps preserve the no-signaling principle**.

### Relation with Quantum State Tomography

Another consequence of breaking the no-cloning principle would be that we would be able to store an infinite amount of classical information inside one qubit.

The idea is the following. Because a qubit can assume infinitely many states, one could theoretically encode a very long classical string into amplitudes (e.g., the binary expansion of $\pi$):

$$|\psi_{\text{message}}\rangle$$

But there is a problem: we **cannot directly read the amplitudes** from a single qubit, because measurement collapses the state. If we measure the qubit, we only get one classical outcome (e.g., `0` or `1`), and not the full amplitudes $a$ and $b$. So normally, the information is not accessible.

However, exists a technique called **quantum state tomography** that:

1. Prepares many identical copies[8] of the same state $|\psi_{\text{message}}\rangle$.
2. Measures them in different bases (e.g., X, Y, Z).

[8] Quantum State Tomography does not violate the no-cloning principle, because it does not create perfect copies of an unknown state. Instead, it prepares many identical copies of a **known** state $|\psi_{\text{message}}\rangle$, which is allowed by quantum mechanics. For example, if we have a quantum circuit that prepares the state $|\psi_{\text{message}}\rangle$, we can run that circuit many times to get many copies of $|\psi_{\text{message}}\rangle$, which is perfectly fine. The no-cloning principle only forbids creating perfect copies of an **unknown** state, not preparing many copies of a **known** state.

3. Reconstructs the amplitudes of the original state $|\psi_{\text{message}}\rangle$ statistically.

In simple terms, given many copies of the same unknown state, estimate what the state was.

Now, imagine there were **no no-cloning theorem**. Then we could do the following:

1. Take the single qubit $|\psi_{\text{message}}\rangle$ that encodes the message.
2. Clone it infinitely many times to get many copies of $|\psi_{\text{message}}\rangle$. This would be possible because we are assuming that the no-cloning principle does not hold.
3. Perform measurements/statistics on all copies to reconstruct the amplitudes of the original state $|\psi_{\text{message}}\rangle$, and therefore read the message encoded in the amplitudes.

So cloning would transform ONE unknown qubit into enough data for tomography, which would allow us to read the message encoded in the amplitudes. This would be a huge problem, because we could encode an infinite amount of classical information into one qubit and then read it using cloning and tomography.

**Example 1: Why Cloning Matters**

**✓ Normal world (no-cloning theorem exists).** Alice wants to send Bob one qubit:

$$|\psi_{\text{message}}\rangle = \sqrt{0.73148291}\,|0\rangle + \sqrt{0.26851709}\,|1\rangle$$

Suppose these decimals secretly encode a super secret message. Bob receives the qubit. Now Bob asks: "*what are the exact amplitudes of the state $|\psi_{message}\rangle$?*" Bob measures the qubit, but measurement only gives him one classical bit (0 or 1). For example, if Bob measures 0, then the state collapses to $|0\rangle$, and the original amplitudes are lost. So from ONE measurement, Bob cannot reconstruct 0.73148291 exactly. So the hidden information is inaccessible to Bob, and the message is safe.

**⚠ Hypothetical world (no-cloning theorem does not exist).** Now suppose cloning is allowed. Bob receives the qubit $|\psi_{\text{message}}\rangle$. Now Bob **can clone** the qubit, so he runs a mysterious cloning machine that creates many identical clones of $|\psi_{\text{message}}\rangle$:

$$|\psi\rangle \to |\psi\rangle\,|\psi\rangle\,|\psi\rangle\,|\psi\rangle\,|\psi\rangle \cdots$$

Now he has millions of copies of $|\psi_{\text{message}}\rangle$. Now Bob can decrypt the super secret message. How? Simple:

1. **Measure many copies in the computational basis**. Bob has millions of copies. Suppose he observes (after many measurements) that 731482 times he gets 0 and 268517 times he gets 1. So Bob can estimate the probabilities of measuring 0 and 1 as 0.731482 and 0.268517. Bob is already very close to the original amplitudes, but he can do better.

2. **Measure in other bases**. Bob is smart, and he knows that measuring in the computational basis only gives him the magnitudes of the amplitudes, but not their phases. So Bob also measures in the X and Y bases to get more information about the state. By combining all the measurement results, Bob can reconstruct the original state $|\psi_{\text{message}}\rangle$ with very high precision, and therefore read the super secret message encoded in the amplitudes.

**The Problem.** Alice effectively transmitted thousands, millions, theoretically infinite classical digits using only one qubit, that should not be physically possible (because in physics, the amount of extractable classical information from a physical system must be finite).

### Relation with Quantum Error Correction

In classical systems, redundancy is simple: we can copy bits to create redundancy for error correction. For example, we can encode a bit as three bits:

$$0 \to 000 \qquad 1 \to 111$$

Then, if one bit gets flipped, we can still recover the original bit by majority voting:

$$000 \to 000 \qquad 001 \to 000 \qquad 010 \to 000 \qquad 100 \to 000$$

Then majority voting allows us to correct single bit errors. However, in quantum computing we cannot simply duplicate qubits and we cannot create copies of arbitrary quantum states. So **quantum error correction cannot protect information by simply copying qubits**. Instead, it uses more complex techniques (e.g., entanglement, syndrome measurements, etc.) to protect quantum information without violating the no-cloning principle.

In summary, the no-cloning theorem is not an isolated curiosity. It is one of the foundational constraints that shape all of quantum information theory. It is connected to reversibility, unitarity, causality, measurement, entanglement, quantum cryptography, quantum error correction. Without the no-cloning theorem, the entire structure of quantum information would collapse.

### 5.1.2 Formal Statement

**Definition 1: No-Cloning Theorem**

The **No-Cloning Theorem** states that there **does not exist a universal quantum gate capable of perfectly copying an arbitrary unknown quantum state**.

Formally, suppose we have:

- A qubit in an unknown state: $|x\rangle$.
- And a second qubit initialized to: $|0\rangle$.

We would like a quantum gate $U$ such that:

$$U|x\rangle|0\rangle = |x\rangle|x\rangle \tag{84}$$

For every possible quantum state $|x\rangle$. The No-Cloning Theorem states that such a gate $U$ cannot exist.

*Proof.* We now prove that the cloning gate $U$:

$$U|x\rangle|0\rangle = |x\rangle|x\rangle$$

Cannot exist for an arbitrary quantum state $|x\rangle$.

1. **Assume that cloning is possible**. Suppose there exists a unitary gate $U$ such that:
$$U|x\rangle|0\rangle = |x\rangle|x\rangle$$
And also:
$$U|y\rangle|0\rangle = |y\rangle|y\rangle$$
So $U$ should clone both $|x\rangle$ and $|y\rangle$.

2. **Take the inner product before cloning**. Before cloning, the two input states are:
$$|x\rangle|0\rangle \quad \text{and} \quad |y\rangle|0\rangle$$
Their inner product is:
$$\langle x|y\rangle\langle 0|0\rangle$$
Since $|0\rangle$ is a normalized state, we have $\langle 0|0\rangle = 1$. Thus, the inner product before cloning is:
$$\langle x|y\rangle$$

3. **Take the inner product after cloning**. After cloning, the states become:
$$|x\rangle|x\rangle \quad \text{and} \quad |y\rangle|y\rangle$$
Their inner product is:
$$\langle x|y\rangle\langle x|y\rangle = (\langle x|y\rangle)^2$$

Now we use the tensor product property of inner products, which states that (page 87):

$$(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$$

Applying this to our inner product, we get:

$$(\langle x| \otimes \langle x|)(|y\rangle \otimes |y\rangle) = (\langle x|y\rangle) \otimes (\langle x|y\rangle)$$

Which simplifies to:

$$(\langle x|y\rangle)^2$$

4. **Use unitary**. Valid quantum gates must be unitary, which means they preserve inner products. Therefore, the inner product before and after applying $U$ must be the same:

$$\langle x|y\rangle = (\langle x|y\rangle)^2$$

This is only possible if $\langle x|y\rangle = 0$ or $\langle x|y\rangle = 1$. But for arbitrary states, this is not generally true. Thus, the assumption that cloning is possible leads to a contradiction.

Let's try to understand why this is a contradiction. Suppose two states are $|x\rangle$ and $|y\rangle$, and suppose $|\langle x|y\rangle|$ (i.e., the magnitude of the inner product) is 0.8. It means that the states are not identical, but they are still "quite similar". They are partially distinguishable, but not perfectly. Now, if cloning were possible, the inner product after cloning would be still 0.8. However, we found that the inner product after cloning would be $(0.8)^2 = 0.64$. This means that the states have become more distinguishable after cloning, which is a contradiction because unitary operations cannot change the inner product between states:

$$|\langle x|y\rangle| = 0.8 \xrightarrow{\text{clone}} |\langle x|y\rangle|^2 = 0.64 \xrightarrow{\text{clone}} |\langle x|y\rangle|^3 = 0.512\ldots$$

QED

**Deepening: Alternative Proof of No-Cloning Theorem**

We can also prove the no-cloning theorem using a different approach, by directly applying the unitarity of $U$ to the cloning operation. This alternative proof is often more concise and elegant.

*Proof.* Suppose, by contradiction, that there exists a cloning gate $U$ such that:

$$U|x\rangle|0\rangle = |x\rangle|x\rangle$$

For an arbitrary state $|x\rangle$.

Since $U$ is supposed to be universal, it must also clone another arbitrary state $|y\rangle$:

$$U|y\rangle|0\rangle = |y\rangle|y\rangle$$

At this point, we have two equations describing the action of $U$ on two different input states.

Now, the **only thing we know about unitary operators is that they preserve inner products** (page 41). So if we want to derive a contradiction from unitarity, we must compute an inner product. If the inner product before cloning is different from the inner product after cloning, then we have a contradiction, which means that $U$ cannot be unitary, and therefore cannot be a valid quantum gate.

Let's compute the inner product before and after cloning.

- **Before cloning**, the inner product between the two input states is:
$$(U|x\rangle|0\rangle)\cdot(U|y\rangle|0\rangle)$$
But the inner product between two states is written as (page 76):
$$\begin{aligned}(U|x\rangle|0\rangle)\cdot(U|y\rangle|0\rangle) &= (U|x\rangle|0\rangle)^\dagger(U|y\rangle|0\rangle)\\ &= \left(\langle 0|\langle x|U^\dagger\right)(U|y\rangle|0\rangle)\\ &\downarrow \text{Remove parentheses}\\ &= \langle 0|\langle x|U^\dagger U|y\rangle|0\rangle\\ &\downarrow \text{Use unitarity: } U^\dagger U = I\\ &= \langle 0|\langle x|y\rangle|0\rangle\end{aligned}$$

- **After cloning**, the inner product between the two output states is:
$$(|x\rangle|x\rangle)\cdot(|y\rangle|y\rangle)$$
Using the tensor product property of inner products:
$$\begin{aligned}(|x\rangle|x\rangle)\cdot(|y\rangle|y\rangle) &= (|x\rangle|x\rangle)^\dagger(|y\rangle|y\rangle)\\ &= (\langle x|\langle x|)(|y\rangle|y\rangle)\\ &= \langle x|y\rangle\langle x|y\rangle\\ &= (\langle x|y\rangle)^2\end{aligned}$$

Thus, **if cloning were possible**, we would have:
$$\langle 0|\langle x|y\rangle|0\rangle = (\langle x|y\rangle)^2$$

Using the tensor product property of inner products again:
$$\langle\psi|\langle\phi|\chi\rangle|\psi\rangle = \langle\phi|\chi\rangle\langle\psi|\psi\rangle$$

Remarkably, the inner product $\langle\phi|\chi\rangle$ is just a complex number (page 76), so we can pull it out of the inner product. So, we can simplify the left side:
$$\underbrace{\langle 0|\langle x|y\rangle|0\rangle}_{=\langle x|y\rangle\langle 0|0\rangle} = (\langle x|y\rangle)^2$$

Since $\langle 0|0\rangle = 1$, we have:

$$\langle x|y\rangle = (\langle x|y\rangle)^2 \quad \iff \quad \begin{cases} \langle x|y\rangle = 0 \\ \langle x|y\rangle = 1 \end{cases}$$

So unitarity forces the inner product to satisfy this equation. However, this equation is not true for all possible values of $\langle x|y\rangle$. It only holds if $\langle x|y\rangle = 0$ or $\langle x|y\rangle = 1$. But for arbitrary states, $\langle x|y\rangle$ can take any value between 0 and 1, so this is a contradiction. Therefore, a universal cloning gate $U$ cannot exist. QED

### Why CNOT Does NOT Clone Qubits

A common confusion is that the CNOT gate seems to copy the control qubit into the target qubit (page 101). This is true only when the control qubit is a computational basis state, i.e. $|0\rangle$ or $|1\rangle$. It is not true for a generic superposition.

Classically, CNOT is often written as:

$$\text{CNOT}\left(v_A, v_B\right) = \left(v_A, v_A \oplus v_B\right)$$

Where $\oplus$ denotes XOR. ⚠ So, if the second qubit is initialized to $|0\rangle$, one *may* be tempted to write:

$$\text{CNOT}\left(v_A, |0\rangle\right) = \left(v_A, v_A \oplus 0\right) = \left(v_A, v_A\right)$$

This is true because the XOR of any bit with 0 is the bit itself:

| $v_A$ | $\|0\rangle$ | $v_A \oplus 0$ |
|---|---|---|
| $\|0\rangle$ | $\|0\rangle$ | $\|0\rangle$ |
| $\|1\rangle$ | $\|0\rangle$ | $\|1\rangle$ |

For example, if we start with $v_A = |1\rangle$ and the second qubit is $|0\rangle$, applying CNOT gives:

$$\text{CNOT}\left(|1\rangle, |0\rangle\right) = \left(|1\rangle, |1\rangle \oplus |0\rangle\right) = \left(|1\rangle, |1\rangle\right)$$

This looks **like cloning**: the first qubit $v_A$ appears to be copied into the second qubit.

However, the mistake is that this notation is valid only for computational basis states $|0\rangle$ and $|1\rangle$, not for arbitrary superpositions. The informal notation:

$$\text{CNOT}\left(v_A, v_B\right) = \left(v_A, v_A \oplus v_B\right)$$

Can be used only when the inputs are basis states. For a generic superposition, we must use the full linear algebra representation of CNOT, which does not clone the state.

**Correct Quantum Computation.** Let the control qubit be a generic superposition:

$$v_A = a_0 |0\rangle + a_1 |1\rangle$$

And let the target qubit be initialized to $|0\rangle$. The joint input state is:

$$v_A |0\rangle = (a_0 |0\rangle + a_1 |1\rangle) |0\rangle$$

Using the tensor product:

$$v_A |0\rangle = a_0 |00\rangle + a_1 |10\rangle$$

Now apply CNOT linearly. CNOT leaves $|00\rangle$ unchanged and maps $|10\rangle$ to $|11\rangle$:

$$\text{CNOT} (a_0 |00\rangle + a_1 |10\rangle) = a_0 |00\rangle + a_1 |11\rangle$$

So the real output is:

$$a_0 |00\rangle + a_1 |11\rangle$$

If CNOT had truly cloned the qubit, the output should instead be:

$$v_A v_A = (a_0 |0\rangle + a_1 |1\rangle)(a_0 |0\rangle + a_1 |1\rangle)$$

Expanding:

$$v_A v_A = a_0^2 |00\rangle + a_0 a_1 |01\rangle + a_1 a_0 |10\rangle + a_1^2 |11\rangle$$

However, these two states are not the same:

$$a_0 |00\rangle + a_1 |11\rangle \neq a_0^2 |00\rangle + a_0 a_1 |01\rangle + a_1 a_0 |10\rangle + a_1^2 |11\rangle$$

Therefore, CNOT does not clone a superposition. More fundamentally, a true universal cloning operation cannot exist because it would violate the unitary structure of quantum mechanics. Indeed, in general:

$$a_0 |00\rangle + a_1 |11\rangle$$

Cannot be factorized as a tensor product of two independent single-qubit states. The second state $v_A v_A$ would be two independent qubits in the same state. This is the key distinction: CNOT can create correlation or entanglement, but it does not create an independent copy of an unknown qubit. In summary, the CNOT output is entangled, while the true cloned state would be two qubits in the same state.

In other words, CNOT preserves quantum coherence between basis components, while true cloning would require duplicating the amplitudes themselves, which is forbidden by unitary quantum evolution.

## 5.2 No-Deleting Principle

The **No-Deleting Principle** is the dual counterpart of the no-cloning theorem. While the no-cloning theorem states that it is impossible to perfectly copy an arbitrary unknown quantum state, the no-deleting principle states that it is also **impossible to perfectly delete quantum information through a unitary operation**.

In other words:

- Quantum information **cannot** be freely **copied**;
- Quantum information **cannot** be freely **destroyed**.

Like the no-cloning theorem, the no-deleting principle is deeply connected to the unitary structure of quantum mechanics.

**Definition 2: No-Deleting Principle**

Suppose we want a quantum gate $U$ that deletes a qubit state:

$$U\,|x\rangle = |0\rangle$$

For an arbitrary unknown state $|x\rangle$.

Equivalently, if we have two identical copies of a state, we may wish to delete one copy while leaving the other unchanged:

$$U\,|x\rangle\,|x\rangle = |x\rangle\,|0\rangle \tag{85}$$

The **No-Deleting Theorem** states that no universal unitary gate capable of performing this operation can exist for arbitrary unknown quantum states.

Intuitively, this means that **quantum information cannot** simply **disappear through reversible quantum evolution**.

As with the no-cloning theorem, we can prove the no-deleting principle by contradiction, using the fact that unitary operators preserve inner products (see page 178).

*Proof.* Suppose, by contradiction, that there exists a deleting gate $U$ such that:

$$U\,|x\rangle = |0\rangle$$

For an arbitrary quantum state $|x\rangle$.

Since $U$ is supposed to be universal, it must also delete another arbitrary state $|y\rangle$:

$$U\,|y\rangle = |0\rangle$$

At this point, we have two equations describing the action of $U$ on two different input states.

Now, the only thing we know about unitary operators is that they **preserve inner products**. Therefore, if $U$ is unitary, the inner product between the input states must be equal to the inner product between the corresponding output states.

- **Before deleting**, the inner product between the two input states is:

  $$(U|x\rangle) \cdot (U|y\rangle)$$

  Using the definition of inner product:

  $$\begin{aligned}(U|x\rangle) \cdot (U|y\rangle) &= (U|x\rangle)^\dagger (U|y\rangle) \\ &= \langle x| U^\dagger U |y\rangle \\ &\downarrow \text{since } U \text{ is unitary, } U^\dagger U = I \\ &= \langle x|y\rangle\end{aligned}$$

- **After deleting**, both output states become $|0\rangle$, so the inner product is:

  $$|0\rangle \cdot |0\rangle$$

  Again:

  $$\begin{aligned}|0\rangle \cdot |0\rangle &= |0\rangle^\dagger |0\rangle \\ &= \langle 0|0\rangle \\ &\downarrow \text{since } |0\rangle \text{ is a normalized state (page 17)} \\ &= 1\end{aligned}$$

Since unitary operators preserve inner products:

$$\langle x|y\rangle = 1$$

But this is **not true in general for arbitrary quantum states**. It is true only if:

$$|x\rangle = |y\rangle$$

Therefore, such a universal deleting gate $U$ cannot exist. QED

## 5.3 No-Signaling Principle

Quantum entanglement creates strong correlations between distant particles (i.e., the measurement of one particle instantaneously affects the state of the other, regardless of the distance between them, page 122). However, these correlations cannot be used to transmit information instantaneously.

This idea is known as the **no-signaling principle**: although measurements on **entangled systems are correlated**, **they cannot be exploited for faster-than-light communication**.

### ⚠ The Communication Problem

Suppose Alice and Bob are at opposite ends of the universe. They each possess one qubit of the Bell pair (page 116):

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The first qubit belongs to Alice, while the second belongs to Bob. More precisely, Alice owns the **first subsystem** of the bipartite state and Bob owns the **second subsystem**.

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B \qquad |11\rangle = |1\rangle_A \otimes |1\rangle_B$$

$$|\Phi^+\rangle = \frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}}$$

Because the two qubits are **entangled**, measuring one qubit instantaneously determines the state of the other. For example:

- If Alice measures $|0\rangle$, Bob's qubit immediately collapses to $|0\rangle$;
- If Alice measures $|1\rangle$, Bob's qubit immediately collapses to $|1\rangle$.

⚠ At first sight, this **seems to violate relativity**. One may wonder:

> "*Can Bob detect instantly whether Alice has already measured her qubit?*"

If the answer were yes, then Alice could communicate information to Bob instantaneously. For example, Alice could decide:

- If Alice measures her qubit $\Rightarrow$ transmit bit 1;
- If Alice does not measure her qubit $\Rightarrow$ transmit bit 0.

This **would allow faster-than-light communication**, contradicting special relativity.

The **no-signaling principle states that this is impossible**: although entanglement creates instantaneous correlations, Bob cannot locally determine whether Alice has performed a measurement or not.

**Definition 3: No-Signaling Principle**

The **No-Signaling Principle** states that quantum **entanglement cannot be used to transmit information instantaneously** between distant observers.

More precisely, local measurements performed on one subsystem of an entangled quantum system do not allow another observer to determine whether a measurement has already been performed on the distant subsystem.

Equivalently, although entangled measurements produce strong correlations, the local measurement statistics observed by one party are independent of the actions performed by the other party.

*Proof.* We want to understand whether Bob can locally detect whether Alice has already measured her qubit or not.

Alice and Bob share the Bell pair:

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}}$$

The question is whether Alice's measurement changes the probabilities observed by Bob. If Bob's probabilities changed, then **Alice could use her measurement choice to send information instantaneously**.

**Case 1: Alice does not measure.**
If Alice does not measure her qubit, the global state remains:

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

If Bob measures his qubit in the computational basis, he obtains:

$$P_B(0) = \frac{1}{2}, \qquad P_B(1) = \frac{1}{2}$$

So Bob sees a **completely random result**.

**Case 2: Alice measures.**
Now suppose Alice measures her qubit first in the computational basis. There are **two possible outcomes**:

- With probability $\frac{1}{2}$, Alice obtains $|0\rangle$, and the global state collapses to:

  $$|00\rangle$$

  In this case, Bob's qubit becomes $|0\rangle$.

- With probability $\frac{1}{2}$, Alice obtains $|1\rangle$, and the global state collapses to:

  $$|11\rangle$$

  In this case, Bob's qubit becomes $|1\rangle$.

Therefore, from Bob's point of view:

$$P_B(0) = \frac{1}{2}, \qquad P_B(1) = \frac{1}{2}$$

So **Bob again sees a completely random result**.

**Conclusion.**
In both cases:

$$P_B(0) = P_B(1) = \frac{1}{2}$$

Therefore, Bob's local measurement statistics are the same whether Alice has measured or not. This means Bob cannot infer Alice's action by measuring only his own qubit. Thus, no usable information is transmitted instantaneously, and the no-signaling principle is preserved. QED

❓ **What are we trying to prove?** We start with the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}}$$

Alice has the first qubit, while Bob has the second qubit. Now suppose Alice is billions of light-years away from Bob. The concern is that if Alice measures her qubit, the joint entangled state changes, and Bob's qubit can subsequently be described by a definite state correlated with Alice's outcome. Can Bob detect this change? If yes, then Alice could use her measurement choice to send information to Bob faster than light, for example:

- If Alice measures her qubit $\Rightarrow$ transmit bit 1;
- If Alice does not measure her qubit $\Rightarrow$ transmit bit 0.

This would violate the relativity principle.

❓ **What is the relativity principle?** It is not a topic of this course, but it is fundamental to understand the motivation behind the no-signaling principle. The **Einstein's Relativity Principle** core idea is: no information, influence, or signal can travel faster than the speed of light. This is not merely a technological limitation, but a fundamental law of spacetime.

Now, related to quantum mechanics, why is this principle relevant? Imagine Alice is on Earth and Bob is on Mars. Suppose Alice could send a message instantly. Then there would exist observes for whom: Bob receives the message before Alice sends it, which creates a paradox. In some reference frames, the **effect would occur before the cause**, but this contradicts our understanding of causality. Therefore, Einstein looked at this and thought[9]: "*wait, did*

[9] We will discuss the EPR paradox later, but it is interesting to note that Einstein historically disliked the idea that measuring one particle could apparently affect another distant particle instantaneously. He famously referred to this phenomenon as "spukhafte Fernwirkung" (spooky action at a distance), expressing his discomfort with the idea that quantum mechanics could allow for such non-local effects. As we will see later, Schrödinger coined the term "entanglement" to describe this phenomenon, and it has since become a fundamental aspect of quantum mechanics. Nevertheless, Einstein's concerns about entanglement's implications for locality and causality remain a topic of discussion and debate in the foundations of quantum mechanics.

*something travel from Alice to Bob faster than light?*". This is the motivation behind the no-signaling principle: yes, the correlations are real, but Alice cannot choose the outcome she obtains, so she cannot force Bob's qubit to collapse to a specific state. Therefore, Bob cannot detect whether Alice has measured her qubit or not, and no information is transmitted faster than light. In simple terms, **Entanglement creates correlations, but it does not allow for Communication**. Entanglement makes distant measurement outcomes correlated, but the no-signaling principle guarantees that these correlations cannot be controlled to transmit information.

## 5.4 EPR Paradox and Quantum Correlation

### 5.4.1 Definition of EPR Paradox

In 1935, Albert Einstein, Boris Podolsky, and Nathan Rosen published a paper entitled *Can Quantum-Mechanical Description of Physical Reality Be Considered Complete?*. [2] Their goal was not to disprove quantum mechanics, but to **show** that the **theory appeared to be incomplete**.

The source of their concern was a peculiar phenomenon predicted by quantum mechanics: two particles could become so strongly correlated that measuring one particle would instantaneously determine the state of the other, regardless of the distance separating them (explained in the proof of the no-signaling principle, page 184).

Einstein found this deeply troubling. According to special relativity, **no information can travel faster than the speed of light**. Yet entanglement seemed to imply an instantaneous connection between distant particles, a phenomenon Einstein famously called *"spooky action at a distance"* (*spukhafte Fernwirkung*).

The EPR paper sparked one of the most important debates in the history of physics. Shortly after its publication, Erwin Schrödinger analyzed the EPR argument and realized that it revealed what he considered the most distinctive feature of quantum mechanics. In a series of papers published the same year, he introduced the term *Verschränkung* ("entanglement"), which is now known as quantum entanglement (page 115).

Interestingly, Schrödinger largely shared Einstein's discomfort with the standard interpretation of quantum mechanics. However, while Einstein viewed the EPR argument as evidence that the theory was incomplete, **Schrödinger recognized entanglement as a fundamental property of quantum systems**.

The central question raised by the EPR paradox is therefore:

> *If measuring Alice's qubit immediately determines the state of Bob's qubit,* ***does quantum mechanics allow faster-than-light communication?***

As we have already seen through the no-signaling principle (page 184), the **answer is no**. Entanglement creates correlations between distant measurements, but these correlations cannot be used to transmit information.

Precisely, Einstein's concern was fundamentally:

> *If measuring Alice's qubit instantaneously determines the state of Bob's qubit, what is actually happening?*

In other words, **how can nature produce these instantaneous correlations?** Einstein was not primarily asking about the possibility of faster-than-light communication, but rather about the underlying mechanism that could explain the observed correlations without violating relativity. He suspected that Bob's particle already possessed some hidden information before the measurement, which would determine the outcome of Alice's measurement without any need for instantaneous influence.

### Modern Perception of the EPR Paradox

To understand the paradox in a modern quantum-computing setting, consider the Bell state:

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Suppose Alice and Bob each possess one qubit of this entangled pair and are separated by a very large distance.

If Alice measures her qubit in the computational basis and obtains:

$$|0\rangle$$

Then Bob's qubit immediately collapses to:

$$|0\rangle$$

Similarly, if Alice measures:

$$|1\rangle$$

Then Bob's qubit immediately collapses to:

$$|1\rangle$$

At first sight, this appears to imply an instantaneous influence between distant particles. How can Bob's qubit "know" which outcome Alice obtained if they may be separated by billions of light-years? This is the modern formulation of the EPR paradox.

**Definition 4: Einstein-Podolsky-Rosen (EPR) Paradox**

The **Einstein-Podolsky-Rosen (EPR) Paradox** is the **apparent contradiction** between quantum **entanglement** and the **classical principle of locality**.

More precisely, if two particles are prepared in an entangled state, a measurement performed on one particle appears to instantaneously determine the state of the other, regardless of the distance separating them.

This raises the question of whether quantum mechanics is a complete description of physical reality, or whether additional hidden information is required to explain these correlations without instantaneous influence.

### 5.4.2 Entanglement and Quantum Correlation

The EPR paradox highlights a fundamental feature of entangled systems: measurement outcomes may appear completely random to individual observers, yet exhibit strong correlations when compared afterward.

In simple terms, Alice and Bob each observe **random outcomes** when measuring their respective qubits. However, when they later compare their results, they discover that these outcomes are **correlated in a way that appears difficult to reconcile with classical intuition**.

**Alice and Bob obtain correlated results. Why is that surprising?** At first glance, it might seem that correlated results are not surprising at all. After all, classical systems can also exhibit correlations.

**Classical example.** Suppose two coins are prepared in such a way that they always show the same face. If one observer sees Heads, they immediately know that the other coin must also be Heads. Likewise, if one observer sees Tails, the other must see Tails. This is an example of a **classical correlation**.

In classical physics, such correlations are explained by assuming that the system possessed predetermined properties before any observation was made. In our example, the two coins were already both Heads or both Tails before anyone looked at them.

**Einstein's intuition.** Einstein believed that the correlations observed in entangled particles might be explained in exactly the same way. According to this view, the particles could carry some form of **hidden information that predetermines the outcome of future measurements**. If such hidden information existed, the apparent mystery of entanglement would disappear: measuring one particle would simply reveal information that was already present from the beginning.

Quantum entanglement exhibits a superficially similar phenomenon. Individual measurement outcomes appear random, yet they become strongly correlated when compared afterward.

**The difference.** Unlike classical correlations, which are completely determined by pre-existing properties of the system, **quantum correlations depend on the measurement basis chosen by the observers**. This means that changing how Alice and Bob perform their measurements can change the observed correlations in a way that has no classical analogue.

Therefore, the existence of correlations is not itself surprising. The real challenge is determining whether quantum correlations can be explained using the same classical intuition that explains correlated coins, cards, or dice, or whether they represent an entirely new physical phenomenon.

This question leads directly to the distinction between **classical correlation** and **quantum correlation**.

### 5.4.3 Classical vs. Quantum Correlation

#### Classical Correlation

A **Classical Correlation** arises from **information** that **already exists before any measurement** is performed.

In a classical system, correlated outcomes can be explained by assuming that the **system possesses predetermined properties**. These properties are often called **Hidden Variables**, because they determine the outcome of future observations even if the observers do not know their values.

**Example 2: Classical Correlation**

For example, consider again two coins prepared so that they always show the same face. Before any observation takes place, the system is already in one of two possible configurations:

- Heads - Heads;
- Tails - Tails.

The measurement does not create the correlation. It merely reveals information that was already present.

Therefore, knowing the outcome observed by one observer immediately allows the other outcome to be inferred. No instantaneous influence is required, because the **correlation existed before the measurement**.

#### Quantum Correlation

A **Quantum Correlation** arises from **measurements performed on an entangled quantum system**.

As in the classical case, each observer sees random outcomes individually. However, the observed **correlations** possess a remarkable property: they **depend on the measurement basis chosen by the observers**.

**Example 3: Quantum Correlation**

For example, if Alice and Bob share the Bell state

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

And both measure in the same basis, their outcomes are perfectly correlated. However, if they choose different measurement bases, the observed correlations change.

### ⚖ The Main Difference

In a classical system, changing how an observer looks at the system does not alter the underlying correlation. The correlation is determined entirely by the hidden variables that existed **before the measurement**.

In contrast, quantum correlations depend on the measurements that are actually performed. The observed correlations cannot always be explained by assuming that the particles carried predetermined answers before the measurement.

This distinction is what makes quantum correlation fundamentally different from classical correlation and lies at the heart of the EPR debate.

### Bell Pair Example

The distinction between classical and quantum correlation becomes clearer when we analyze an entangled Bell pair. Consider the Bell state:

$$\left|\Phi^+\right\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

This state can also be expressed in the Hadamard basis as (see the full derivation in section 4.7):

$$\left|\Phi^+\right\rangle = \frac{|{+}{+}\rangle + |{-}{-}\rangle}{\sqrt{2}}$$

Suppose Alice and Bob each possess one qubit of this entangled pair:

$$\left|\Phi^+\right\rangle_{AB} = \frac{|00\rangle_{AB} + |11\rangle_{AB}}{\sqrt{2}} = \underbrace{\frac{|+\rangle_A \otimes |+\rangle_B + |-\rangle_A \otimes |-\rangle_B}{\sqrt{2}}}_{\frac{|0\rangle_A \otimes |0\rangle_B + |1\rangle_A \otimes |1\rangle_B}{\sqrt{2}}}$$

**Example 4: Same Basis Measurements**

Suppose Alice and Bob both measure in the computational basis $\{|0\rangle, |1\rangle\}$.

Alice measures first and obtains:

$$|0\rangle \qquad \text{or} \qquad |1\rangle$$

With equal probability:

$$P(0) = P(1) = \frac{1}{2}$$

If Alice obtains:

$$|0\rangle$$

The Bell state collapses to:

$$|00\rangle$$

Therefore Bob's qubit becomes:

$$|0\rangle$$

Similarly, if Alice obtains:

$$|1\rangle$$

The Bell state collapses to:

$$|11\rangle$$

And Bob's qubit becomes:

$$|1\rangle$$

Consequently, Bob always obtains the same outcome as Alice.

Although each observer sees **random outcomes individually**, they discover **perfect correlation when they later compare their measurement results**.

**Example 5: Different Basis Measurements**

Now suppose Alice measures in the Hadamard basis:

$$\{|+\rangle, |-\rangle\}$$

Where

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Bob, however, continues measuring in the computational basis.

Alice obtains either:

$$|+\rangle \qquad \text{or} \qquad |-\rangle$$

With probability:

$$\frac{1}{2}$$

If Alice measures $|+\rangle$, Bob's qubit collapses to:

$$|+\rangle$$

If Alice measures $|-\rangle$, Bob's qubit collapses to:

$$|-\rangle$$

However, Bob is measuring in the computational basis. For both $|+\rangle$ and $|-\rangle$:

$$P(0) = P(1) = \frac{1}{2}$$

Therefore Bob obtains:

$$0 \qquad \text{or} \qquad 1$$

With equal probability, regardless of Alice's outcome.

When Alice and Bob later compare their results, they find that the **perfect correlation** observed in the previous experiment has **disappeared**.

In a classical hidden-variable model, the correlation would be determined entirely by pre-existing information carried by the particles. In contrast, the Bell-pair example shows that the **observed quantum correlation depends on the measurement basis chosen by the observers** (same basis, perfect correlation; different basis, no correlation). This basis dependence is one of the fundamental signatures of quantum correlation.

### 5.4.4 Entanglement vs. Quantum Correlation

Throughout the discussion of the EPR paradox, we have repeatedly encountered two closely related concepts: **entanglement** and **quantum correlation**. Although they are strongly connected, they describe different aspects of a quantum system.

**Entanglement.** We have already defined **entanglement** (page 115) as a property of the quantum state itself. A system is entangled when its state cannot be written as a tensor product of the states of its individual subsystems.

For example, the Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Is entangled because it cannot be expressed as:

$$|\psi_A\rangle \otimes |\psi_B\rangle$$

The important point is that entanglement is an intrinsic property of the system. It exists independently of whether anybody performs a measurement (see the discussion on page 117).

**Quantum Correlation. Quantum Correlation** refers to the statistical relationship observed between measurement outcomes. Unlike entanglement, quantum correlation is **not a property of the state** alone. It **depends on how the system is measured**. For instance, Alice and Bob may share the same entangled Bell pair, yet the observed correlations change depending on the measurement bases they choose.

#### Entanglement vs. Quantum Correlation

Entanglement does not depend on the measurement basis:

- Entanglement is a property of the quantum system.
- It exists before any measurement is performed.

Quantum correlation, on the other hand, depends on the measurement basis:

- Quantum correlation is a property of the measurement outcomes.
- Different measurement bases may produce different observed correlations.

In summary:

- **Entanglement** is a property of the quantum state and does not depend on how the system is measured.
- **Quantum correlation** is a property of the measurement outcomes and depends on the measurement basis chosen by the observers.

# 6 Transpiling

## 6.1 Quantum Fidelity

### 6.1.1 Why Quantum Computers Are Noisy

In all the previous chapters, we implicitly assumed that quantum gates exactly as their mathematical definitions. For example:

- Hadamard gate:
$$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$
Behaves as a perfect beam splitter, splitting the input state into an equal superposition of $|0\rangle$ and $|1\rangle$.

- Pauli-X gate:
$$X|0\rangle = |1\rangle$$
Behaves as a perfect NOT gate, flipping the state of the qubit from $|0\rangle$ to $|1\rangle$.

- CNOT gate:
$$\begin{aligned} \text{CNOT}|00\rangle = |00\rangle, \quad \text{CNOT}|01\rangle = |01\rangle \\ \text{CNOT}|10\rangle = |11\rangle, \quad \text{CNOT}|11\rangle = |10\rangle \end{aligned}$$
Behaves as a perfect Controlled-NOT gate, flipping the target qubit if the control qubit is in the state $|1\rangle$ and leaving it unchanged if the control qubit is in the state $|0\rangle$.

**Mathematically** these transformations **are exact**. However, in a real quantum computer, gates are implemented by physical hardware:

- Microwave pulses (superconducting qubits);
- Laser pulses (trapped ions);
- Electromagnetic fields;
- Control electronics;

Because the **hardware is imperfect** (e.g., due to manufacturing defects, environmental noise, or limitations in control precision), the **resulting operation is only an approximation of the ideal gate**. As a consequence:

$$U_{\text{physical}} \neq U_{\text{ideal}} \quad \xrightarrow{\text{better}} \quad U_{\text{physical}} \approx U_{\text{ideal}}$$

This discrepancy leads to **errors** in the quantum computation, which can accumulate over multiple gates and degrade the performance of quantum algorithms.

### ⚠ The Two Main Sources of Noise

- **Gate Infidelity**. Gate infidelity means that the gate physically applied by the hardware is not exactly the gate we intended to execute.

  Suppose we want to apply a Hadamard gate:

  $$U_{\text{ideal}} = H$$

  Ideally, we would like the physical implementation to match the ideal gate:

  $$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

  However, due to **imperfections** like inaccurate control pulses, calibration errors, electronic noise, manufacturing imperfections, the hardware may actually apply:

  $$U_{\text{physical}} = H + \varepsilon$$

  Where $\varepsilon$ is a **small error**. The resulting state is therefore slightly different from the desired one. After thousands of gates, these small errors can **accumulate** and significantly affect the final result.

  **❓ Why Multi-Qubit Gates Are Worse?** Intuitively, multi-qubit gates (like CNOT) are **more complex to implement** than single-qubit gates (like Hadamard) because they require precise control over interactions between multiple qubits. For this reason, the **fidelity of multi-qubit gates is typically lower** than that of single-qubit gates:

  $$\text{Fidelity}_{\text{CNOT}} < \text{Fidelity}_{\text{Hadamard}}$$

  This is one reason why transpilers try to minimize the number of CNOTs and other multi-qubit operations. However, we will understand this in more detail in the next chapters.

- **Decoherence**. A qubit is never perfectly isolated from the surrounding environment. Even if we do not intentionally measure it, the environment continuously interacts with it: electromagnetic radiation, thermal fluctuations, vibrations, surrounding particles, etc. These **interactions gradually destroy the quantum properties of the qubit**. In summary, decoherence is the process by which a quantum system progressively loses its "*quantumness*" and starts behaving more like a classical object (e.g., a bit instead of a qubit).

  For example, imagine preparing a qubit in the state $|+\rangle$:

  $$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

  Initially the qubit exhibits quantum superposition, interference, and phase relationships. However, as time passes, interactions with the environment disturb the phase information. Eventually, the qubit behaves more like a classical probabilistic bit than a true quantum state. The information is not necessarily destroyed instantly, but the quantum coherence gradually leaks into the environment, making it harder to maintain the desired quantum state over time.

**Why is Decoherence a serious problem?** A quantum algorithm often requires many sequential operations:

$$U_n U_{n-1} \cdots U_2 U_1$$

Where each operation takes a certain amount of time to execute. If the algorithm runs for too long, decoherence may occur before the computation finishes. **When decoherence becomes dominant**, the superposition is lost, the interference is lost, the entanglement is lost, and the **quantum advantage disappears**.

### Hardware Limitations

Both gate infidelity and decoherence originate from the same fundamental fact: **quantum computers are physical devices**. Unlike an ideal mathematical model, qubits are not perfectly isolated, gates are not perfectly accurate, and measurements are not perfectly reliable. Therefore, every real quantum computer has a finite error rate, coherence time, and computational depth. These limitations ultimately restrict how deep a quantum circuit can be before errors become overwhelming.

In summary, quantum computers are noisy because the **gates are imperfectly implemented** and the **qubits decohere over time**.

### 6.1.2 Gate Infidelity

In the mathematical model of quantum computing, quantum gates are represented by ideal unitary matrices that transform quantum states exactly as predicted by theory.

For example, the Hadamard gate is defined as:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

And when applied to the $|0\rangle$ state, it always produces the $|+\rangle$ state:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$$

In theory, this transformation is **exact**.

Real quantum computers, however, do not manipulate abstract vectors and matrices. They manipulate physical systems through microwave pulses, laser pulses, electromagnetic fields, and control electronics. As a consequence, the operation implemented by the hardware is never perfectly identical to the ideal gate (as we have seen in the previous section).

#### ⚖ Ideal Gate vs Physical Gate

It is useful to distinguish between:

- **Ideal gate**: the mathematical operation specified by the algorithm;
- **Physical gate**: the operation actually implemented by the hardware.

Ideally we would like:

$$U_{\text{physical}} = U_{\text{ideal}}$$

But in practice we have:

$$U_{\text{physical}} \approx U_{\text{ideal}}$$

But there is always a small discrepancy due to imperfections in the hardware. This **difference between the ideal and physical implementation** is called **Gate Infidelity**.

### 6.1.3 Decoherence

In the previous section, we introduced decoherence as the gradual loss of the quantum properties of a qubit due to interactions with the surrounding environment (page 196). Now, the natural question is: ***how does decoherence actually modify a quantum state?*** To understand this, we need to introduce two types of errors, because decoherence can manifest itself in different ways.

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#### 6.1.3.1 Phase-Flip Errors

One common consequence of decoherence is **Dephasing**, also called **Loss of Phase**.

Consider the qubit:

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

The probabilities are:

$$P(0) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2} \qquad P(1) = \left|\frac{1}{\sqrt{2}}\right|^2 = \frac{1}{2}$$

The important observation is that a quantum state contains more information than probabilities. It also contains the **relative phase** between the amplitudes. For example, the states:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

Have exactly the same probabilities:

$$P(0) = P(1) = \frac{1}{2}$$

However, they **behave differently** in interference experiments because their **relative phase is different**.

**What does Dephasing do?** And, **why is it hard to notice?** During dephasing, the **probabilities remain unchanged**, but the **relative phase becomes randomized**. This means that:

- The **coherence** between the states **is lost**.

  **What is Coherence?** A quantum state is **coherent** if the **different components** of the superposition **maintain a fixed phase relationship**.

  For example:

  $$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \qquad |\psi\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

  Are coherent states because the relative phase between $|0\rangle$ and $|1\rangle$ is well-defined. Also, the state:

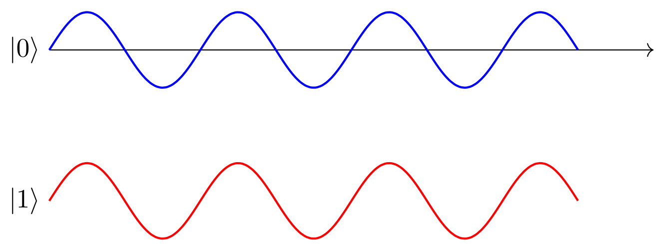
  $$|\psi\rangle = \sqrt{0.8}\,|0\rangle + e^{i\pi/3}\sqrt{0.2}\,|1\rangle$$

Is also coherent because the relative phase between $|0\rangle$ and $|1\rangle$ is well-defined. The **amplitudes don't matter** for coherence, the important thing is that the **phase relationship is known and stable**.

**Wave analogy for coherence.** The following figures should not be interpreted as the actual amplitudes of a qubit. They are only a visual analogy. The two waves represent the two components of a quantum superposition and illustrate whether their relative phase relationship remains stable over time.

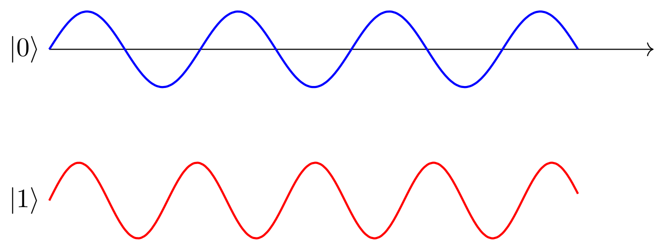
**Coherent state.** In a coherent quantum state, the relative phase between the **amplitudes remains well-defined and stable**. In the figure, the two waves remain perfectly synchronized: peaks align with peaks and valleys align with valleys. Although the amplitudes may evolve, their phase relationship is preserved. This stable phase relationship is what allows quantum interference to occur.

**Coherent**

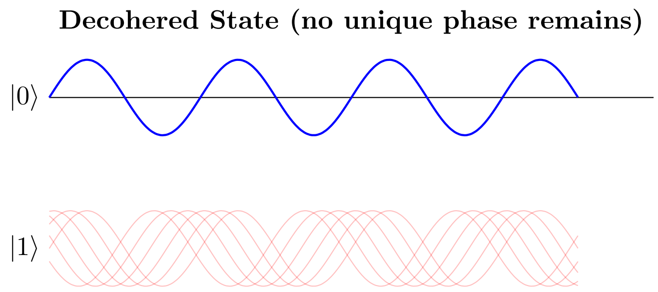



**Loss of Coherence.** When a qubit interacts with its environment, the relative phase between the amplitudes gradually becomes uncertain. In the figure, the **two waves lose their synchronization over time**. The peaks and valleys no longer align, representing the loss of a well-defined phase relationship. This **gradual loss of phase synchronization is called dephasing** and is one of the main mechanisms responsible for decoherence. Once coherence is lost, interference effects disappear and the quantum state behaves more like a classical probabilistic system.

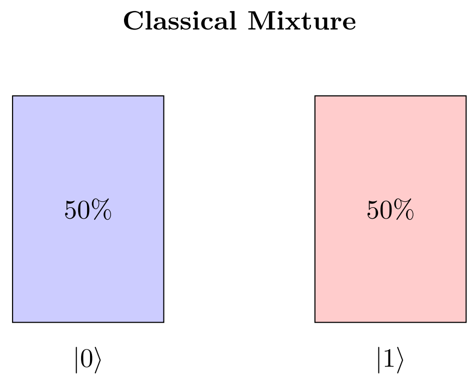
**Loss of Coherence (phase drifting)**




**Complete loss of coherence.** If the **dephasing process continues**, the **relative phase becomes completely randomized**. In the figure, the two waves are no longer synchronized at all, and there is no stable phase relationship between them. This represents a fully decohered state, where the phase information has been completely lost. The system can no longer exploit quantum interference and therefore behaves similarly to a classical probabilistic mixture.




**Classical Mixture.** After **complete decoherence**, **all phase information is lost**, and the **quantum state can be described as a classical mixture of states**. In this case, the system behaves as if it were in state $|0\rangle$ with probability 50% and in state $|1\rangle$ with probability 50%. The state no longer exhibits any quantum interference effects and can be treated as a classical probabilistic system. A great analogy is a coin that before the decoherence process was in a superposition of heads and tails, but after decoherence, it behaves like a classical coin that is either heads or tails with certain probabilities.




Notice that this is fundamentally different from the coherent state:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

Which also produces 50% probabilities when measured. The difference is that the **coherent state contains phase information and can exhibit interference**, whereas the classical mixture contains only probabilities.

In summary, a coherent quantum state and a classical probabilistic mixture **may produce the same measurement probabilities**, but only the coherent state contains phase information and can therefore exhibit interference effects.

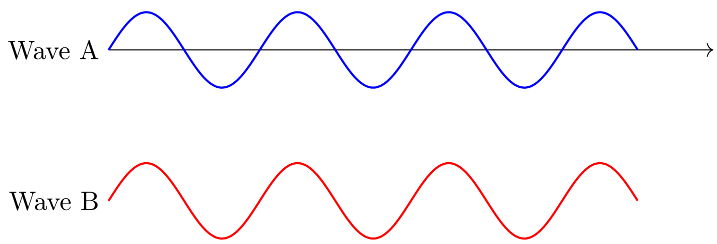
- Interference effects disappear.

  **What is Interference?** In the previous point, we mentioned that the loss of coherence leads to the disappearance of interference effects. *But what exactly are interference effects?* **Interference** is the **phenomenon by which quantum amplitudes combine**.

  Unlike classical probabilities, which simply add together, quantum amplitudes can be positive, negative or complex. Therefore, when **amplitudes are added** together, they can either **reinforce each other** (called **Constructive Interference**) or **cancel each other out** (called **Destructive Interference**).

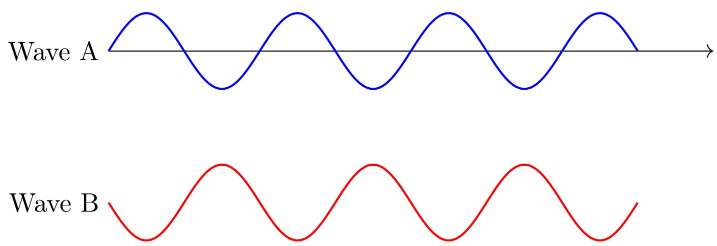
  An example of constructive interference is shown in the figure below. The two waves represent the two components of a quantum superposition. When they are in phase (peaks align with peaks and valleys align with valleys), their amplitudes reinforce each other, leading to a stronger overall amplitude, and therefore a higher probability of certain measurement outcomes (because probabilities are obtained by taking the square of the amplitude).

  **Constructive Interference**

  On the other hand, when the two waves are out of phase (peaks align with valleys), their amplitudes cancel each other out, leading to a weaker overall amplitude, and therefore a lower probability of certain measurement outcomes.

  **Destructive Interference**

  **Decoherence and interference.** Decoherence does not necessarily destroy the probabilities immediately. Instead, it destroys the **phase information**. Without phase information, interference effects disappear, and the quantum state behaves more like a classical probabilistic system.

- The state becomes more classical. As coherence is lost and interference effects disappear, the quantum state gradually behaves more like a classical probabilistic mixture. In the limit of complete decoherence, the state can be described as a classical mixture of states, where the system is in one state with a certain probability and in another state with another probability.

Dephasing is **invisible in the computational basis** because the probabilities of measuring $|0\rangle$ or $|1\rangle$ remain unchanged. However, the damage only becomes visible when the algorihtm relies on interference effects, which require coherence. This is one reason why decoherence is so dangerous. It can silently destroy the relative phases of a state without affecting the measurement probabilities in the computational basis, making it **difficult to detect until it's too late**.

**Which mathematical error changes the phase of a quantum state?** In other words, ***which mathematical error corresponds to dephasing?*** The answer is the Pauli-Z gate:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Which acts as follows:

$$Z|0\rangle = |0\rangle \qquad Z|1\rangle = -|1\rangle$$

Where the minus sign in front of $|1\rangle$ represents a phase shift of $\pi$ (or 180 degrees). It is more intuitive to see the effect of the Z gate on the Bloch sphere, page 56. However, with Pauli-Z, the probabilities of measuring $|0\rangle$ or $|1\rangle$ remain unchanged, but the relative phase between the two states is flipped. This is why it's called a **Phase-Flip Error**.

**Why is it dangerous?** Per se, the Pauli-$Z$ gate is not dangerous, and a **single application does not destroy coherence**. For example, applying a Pauli-$Z$ gate transforms $(|+\rangle)$ into $(|-\rangle)$, but the resulting state is still perfectly coherent.

However, in a real quantum computer the environment does not apply one neat Pauli-$Z$ gate. Instead, interactions with the environment introduce many random phase perturbations over time. These random phase changes can be modeled as phase-flip $(Z)$ errors. As they accumulate, the relative phase between the components of a superposition becomes increasingly uncertain, producing **dephasing**. The consequence is the loss of coherence and, therefore, the disappearance of the interference effects on which quantum algorithms rely.

**If the Pauli-Z gate is a valid quantum operation, why is it associated with errors?** Let's clarify this point with an analogy. For example, when we write $\text{NOT}(0) = 1$, there is nothing wrong with the NOT operation. The gate is doing exactly what it's supposed to do. However, suppose our program says to store the value 0 in a variable, but electrical noise flips the bit to 1. Now the bit has undergone a NOT operation, but this is not what we intended. The NOT operation is a valid operation, but in this context, it represents an error because it changes the value in an unintended way. Exactly the same thing happens in quantum computing.

Suppose our algorithm contains $Z\left|+\right\rangle = \left|-\right\rangle$. We intentionally changed the phase, and everything is correct. However, suppose our algorithm says "*do nothing*", but the environment interacts with the qubit and changes its phase. The effect on the state is $\left|+\right\rangle \rightarrow \left|-\right\rangle$, which is exactly what a $Z$ gate would have done. Now, we call this a **phase-flip error** because the phase changed even though we never requested that operation. The $Z$ gate is a valid quantum operation, but in this context, it represents an error because it changes the phase in an unintended way.

#### 6.1.3.2 Bit-Flip Errors

While dephasing affects the **relative phase** of a quantum state, amplitude dampig affects the **probabilities (populations)** of the basis states. In practice, **Amplitude Damping** occurs when the **qubit exchanges energy with the environment**. For example, initially the state:

$$|\psi\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

With probabilities:

$$P(0) = P(1) = \frac{1}{2}$$

After amplitude damping, the state might evolve into:

$$|\psi\rangle = \sqrt{0.8}\,|0\rangle + \sqrt{0.2}\,|1\rangle$$

With probabilities:

$$P(0) = 0.8 \qquad P(1) = 0.2$$

The population of $|1\rangle$ has decreased. This is amplitude damping. Specifically, this **form of amplitude damping** is called **Energy Relaxation** because the qubit loses energy to the environment. In this case, the state tends to relax towards the ground state $|0\rangle$.

**Why does Energy Relaxation occur?** Energy relaxation occurs because the qubit is not perfectly isolated from its surroundings. The qubit can interact with the environment, exchanging energy in the process. For example, if the qubit is implemented using a superconducting circuit, it can lose energy by emitting photons into the surrounding electromagnetic field. This process causes the excited state $|1\rangle$ to decay towards the ground state $|0\rangle$, leading to amplitude damping.

**Another type of amplitude damping: Thermal Excitation.** In some cases, the **qubit can also gain energy from the environment**, leading to **Thermal Excitation**. For example, if the qubit is in a warm environment (i.e., at a higher temperature), it can absorb thermal energy and transition from the ground state $|0\rangle$ to the excited state $|1\rangle$. This process also leads to amplitude damping because it changes the population of the basis states.

**Where does the Bit-Flip come from?** Suppose Energy Relaxation occurs:

$$|1\rangle \to |0\rangle$$

What happened is that the logical value changed from 1 to 0. Instead, suppose Thermal Excitation occurs:

$$|0\rangle \to |1\rangle$$

Now the logical value changed from 0 to 1. In both cases, the logical value of the qubit has flipped. This is exactly what the Pauli-$X$ gate does:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Which acts as follows:

$$X|0\rangle = |1\rangle \qquad X|1\rangle = |0\rangle$$

Therefore, we can model amplitude damping as a **Bit-Flip Error** because it changes the logical value of the qubit. Just like with phase-flip errors, a single application of the $X$ gate does not necessarily destroy the quantum state, but in a real quantum computer, interactions with the environment can introduce many random bit-flip errors over time, leading to significant amplitude damping and loss of quantum information.

⚠ **Important Caveat.** Amplitude damping is not literally a bit-flip; we use bit-flips as a convenient mathematical model. A Pauli-$X$ gate is a perfect bit-flip and treats the two states symmetrically $|0\rangle \iff |1\rangle$. Same behavior in both directions. Instead, the nature of amplitude damping is asymmetric. Usually, the qubit $|1\rangle$ tends to decay towards $|0\rangle$ (energy relaxation), and the reverse process (thermal excitation) is less common. Because $|1\rangle$ is typically the **higher-energy state**, while $|0\rangle$ is the **lower-energy state**, and the qubit naturally tends to lose energy to the environment rather than gain it. Therefore, while we can model amplitude damping as a bit-flip error for simplicity, it's important to remember that the underlying physical processes are more complex and not perfectly symmetric.

⚠ **Important Caveat.** Amplitude damping is not literally a bit-flip; rather, we use bit-flips as a convenient mathematical model. A Pauli-$X$ gate is a perfect bit-flip and treats the two states symmetrically:

$$|0\rangle \leftrightarrow |1\rangle$$

In other words, the behavior is identical in both directions. Amplitude damping, however, is generally asymmetric. Usually, the qubit $|1\rangle$ tends to decay towards $|0\rangle$ (energy relaxation), while the reverse process $|0\rangle \to |1\rangle$ (thermal excitation) is less likely. This happens because $|1\rangle$ is typically the **higher-energy state**, whereas $|0\rangle$ is the **lower-energy state**. Consequently, a qubit naturally tends to lose energy to the environment rather than gain it. Therefore, although amplitude damping is often modeled and corrected as an $X$-type (bit-flip) error, the underlying physical mechanism is more complex and not perfectly symmetric. In summary, a bit-flip describes *what happens* to the logical value of the qubit, while amplitude damping describes *why it happens* physically.

#### 6.1.3.3 Generic Errors

So far we have discussed two idealized errors:

- Phase-flip errors (Pauli-$Z$) that flip the phase of a qubit (page 200);
- Bit-flip errors (Pauli-$X$) that flip the state of a qubit (page 206).

However, real quantum hardware is messy (i.e., noisy) and a qubit may experience electromagnetic noise, thermal fluctuations, imperfect control pulses and interactions with the environment. Therefore, the resulting **disturbance** is generally **not** a perfect phase-flip or bit-flip error. Instead, the error can be some arbitrary transformation.

To describe this situation, we introduce an **Error Operator** $E$ that acts on the qubit exactly like a **quantum gate**:

$$|\psi\rangle \to E|\psi\rangle \tag{86}$$

But with the crucial difference that $E$ is **unwanted**. At first glance this *seems* disastrous because there are infinitely many possible error operators. Therefore, ***how could we possibly correct all of them?***

### Decomposition into Pauli Operators

The key insight is that **every** $2 \times 2$ complex matrix can be written as a linear combination of four special matrices:

$$I, \quad X, \quad Y, \quad Z$$

These matrices form a basis for the space of single-qubit operators (page 48). So **any error operator** $E$ **can be written as**:

$$E = \alpha_0 I + \alpha_1 X + \alpha_2 Y + \alpha_3 Z$$

Where the **coefficients** $\alpha_i$ with $i = 0, 1, 2, 3$ are complex numbers that **depend on the specific physical error**.

**Theorem 2** (Pauli Basis Decomposition)**.** *The set:*

$$\{I, X, Y, Z\}$$

*Forms a basis for the vector space of all $2 \times 2$ complex matrices. Therefore, any single-qubit operator $E$ can be uniquely written as:*

$$E = \alpha_0 I + \alpha_1 X + \alpha_2 Y + \alpha_3 Z \tag{87}$$

*Where $\alpha_0, \alpha_1, \alpha_2, \alpha_3 \in \mathbb{C}$ are complex coefficients that depend on the specific operator $E$.*

### Meaning of Each Term

Each term in the decomposition has a clear physical interpretation:

- **Identity term** ($I$): represents the case where no error occurs, and the qubit remains unchanged.
$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
- **Bit-flip term** ($X$): represents the probability amplitude of a bit-flip error, which flips the state of the qubit from $|0\rangle$ to $|1\rangle$ and vice versa (page 206).
$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
- **Phase-flip term** ($Z$): represents the probability amplitude of a phase-flip error, which flips the phase of the qubit from $|1\rangle$ to $-|1\rangle$ while leaving $|0\rangle$ unchanged (page 200).
$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$
- **Combined bit-and-phase-flip term** ($Y$): represents the probability amplitude of an error that combines both bit-flip and phase-flip effects, flipping the state of the qubit and changing its phase simultaneously, $Y = iXZ$.
$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

### Why Correcting $X$ and $Z$ Errors is Sufficient

Since $Y$ can be expressed as a combination of $X$ and $Z$ (i.e., $Y = iXZ$), if we can correct for $X$ and $Z$ errors, we can also correct for $Y$ errors. Therefore, **correcting for $X$ and $Z$ errors is sufficient to correct for any arbitrary error** $E$. This is a fundamental principle in quantum error correction, allowing us to design codes that can protect against a wide range of errors by focusing on correcting just these two types of errors. Depending on the source, the related theorem has many names. However, the important point is that if a quantum error-correcting code can correct all Pauli errors, then it can correct any arbitrary single-qubit error. We will discuss quantum error-correcting codes in future sections.

Theorem **3** (Discretization Pauli Error, Error Discretization, Pauli Error Expansion, Digitization of Quantum Errors)**.** *Let Equation 87:*

$$E = \alpha_0 I + \alpha_1 X + \alpha_2 Y + \alpha_3 Z$$

*Be an arbitrary error acting on a physical qubit. If a QECC (quantum error-correcting code) can correct each Pauli operator $I$, $X$, $Y$ and $Z$, individually, then it can correct the arbitrary error $E$ as well.*

### 6.1.4 Fidelity of a Quantum Gate

We have defined the types of errors that can occur in quantum gates. But ***how can we measure how well a quantum gate performs?*** The **fidelity** of a quantum gate is the answer to this question. It is a measure of how close the actual operation performed by the gate is to the ideal operation.

#### Motivation

In an ideal quantum computer, every gate would be implemented perfectly. If we want to apply a gate $U_I$ (where $I$ stands for "*ideal*"), then the **output should be exactly**:

$$y_I = U_I x \tag{88}$$

Where $x$ is the input qubit and the $y_I$ is the *ideal* output. Unfortunately, real hardware is noisy (as we have seen in the previous sections). The gate physically implemented on the processor is slightly different from the ideal one: $U_I \neq U_P$ (where $P$ stands for "*physical*"). Therefore, the **actual output** is:

$$y_P = U_P x \tag{89}$$

This output may differ from the ideal one due to imperfect control electronics, calibration errors, decoherence, and other sources of noise like environmental interactions. Given this situation, we ask ourselves: how close is $y_P$ to $y_I$? How closely does the physical output $y_P$ produced by the real gate $U_P$ resemble the ideal output $y_I$ we would get from a perfect gate $U_I$? We introduce the **fidelity** to answer this question.

**Definition 1: Fidelity**

The **Fidelity** of a quantum gate is a measure of **how closely the actual operation performed by the gate matches the ideal operation**. It is defined as:

$$F = \left|\langle y_I | y_P \rangle\right|^2 \tag{90}$$

Where $|y_I\rangle$ is the ideal output state and $|y_P\rangle$ is the actual output state produced by the physical gate. We use the inner product $\langle y_I | y_P \rangle$ to quantify the **overlap between the two states**, and the **square** of its magnitude gives us a **value between 0 and 1** that represents the fidelity. The fidelity ranges from 0 to 1, where:

- $F = 1$ means the **actual** output **perfectly matches the ideal** output (i.e., the gate is perfect).
- $F = 0$ means the **actual** output is **completely orthogonal to the ideal** output (i.e., the gate is completely wrong).
- $0 < F < 1$ indicates that the **actual** output is **partially close** to the ideal output, with higher values indicating better performance (i.e., the gate is somewhat good).

We can also write the fidelity more explicitly in terms of the gates. Using the Equation 88 and Equation 89:

$$y_I = U_I x \quad \text{and} \quad y_P = U_P x$$

Writing the conjugate transpose of $y_I$ (bra):

$$|y_I\rangle = U_I |x\rangle \quad \Rightarrow \quad \langle y_I| = \langle x| U_I^\dagger$$

Substituting this into the fidelity definition:

$$F = |\langle y_I|y_P\rangle|^2 = \left|\langle x| U_I^\dagger U_P |x\rangle\right|^2 \tag{91}$$

### Why isn't one fidelity value enough?

Suppose we build a terrible **physical gate**:

$$U_P = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

This gate does nothing to the input state. Now suppose the **ideal gate** should be:

$$U_I = X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Let's test the fidelity using only $|+\rangle$:

$$U_I |+\rangle = X |+\rangle = |+\rangle$$

Because $|+\rangle$ is an eigenstate of $X$. Meanwhile:

$$U_P |+\rangle = I |+\rangle = |+\rangle$$

Calculating the fidelity:

$$F = \left|\langle +| X^\dagger I |+\rangle\right|^2 = |\langle +| XI |+\rangle|^2 = |\langle +| X |+\rangle|^2 = |\langle +|+\rangle|^2 = 1$$

Perfect fidelity! But we know that $U_P$ is a terrible gate. In fact, if we test the fidelity with $|0\rangle$:

$$U_I |0\rangle = X |0\rangle = |1\rangle$$

While:

$$U_P |0\rangle = I |0\rangle = |0\rangle$$

Calculating the fidelity:

$$F = \left|\langle 0| X^\dagger I |0\rangle\right|^2 = |\langle 0| XI |0\rangle|^2 = |\langle 0| X |0\rangle|^2 = |\langle 0|1\rangle|^2 = 0$$

**Terrible fidelity!** The same gate appears perfect and terrible depending on which input state we choose. So asking "*what is the fidelity of this gate?*" is incomplete, instead we should ask "*for which input state?*".

**The Solution: Average Fidelity**

Instead of testing a single input state, we conceptually test **all possible input states**. Remember:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

Can be any point on the Bloch sphere. So rather than computing the fidelity for a single state (Equation 91):

$$F(|\psi\rangle) = \left|\langle\psi|\, U_P^\dagger U_I\, |\psi\rangle\right|^2$$

For one specific $|\psi\rangle$, we **average over all possible input states**:

$$F_{\text{avg}} = \int \left|\langle\psi|\, U_P^\dagger U_I\, |\psi\rangle\right|^2 \mathrm{d}\psi \tag{92}$$

This is the **Average Fidelity** of the quantum gate. The notation $\mathrm{d}\psi$ indicates a **uniform average over all possible quantum states**. For a single qubit, every pure state can be represented as a point on the Bloch sphere. Therefore, the average fidelity measures the performance of the gate over the entire Bloch sphere rather than for a specific input state.

In other words, we conceptually:

1. Choose a quantum state $|\psi\rangle$;
2. Apply both the ideal gate $U_I$ and the physical gate $U_P$;
3. Compute the fidelity;
4. Repeat the process for all possible states on the Bloch sphere;
5. Average the results.

The resulting value provides a single number that characterizes the overall quality of the gate, independently of the particular input state chosen.

**Can we really test all possible states?** In practice, this is **impossible because the Bloch sphere contains infinitely many states**. Fortunately, it can be shown (proof outside the scope of this course) that the average fidelity can be computed directly from the matrices describing the ideal and physical gates:

$$F_{\text{avg}} = \frac{\left|\mathrm{Tr}\left(U_P^\dagger U_I\right)\right|^2 + d}{d(d+1)} \tag{93}$$

Where Tr is the **trace of a matrix**, a sum of the diagonal entries:

$$\mathrm{Tr}(A) = \sum_i A_{ii}$$

And:

$$d = 2^n$$

Is the **dimension of the Hilbert space** and $n$ is the **number of qubits** involved in the gate.

**Meaning of $d$.** For a single-qubit gate:

$$d = 2^1 = 2$$

Because a qubit lives in a two-dimensional Hilbert space. For a two-qubit gate:

$$d = 2^2 = 4$$

Because the computational basis contains four states:

$$|00\rangle\,,\ |01\rangle\,,\ |10\rangle\,,\ |11\rangle$$

More generally, an $n$-qubit system has

$$2^n$$

Basis states, hence the dimension of its Hilbert space is

$$d = 2^n$$

**Interpretation.** The average fidelity replaces an impossible average over infinitely many quantum states with a simple calculation involving only the matrices of the ideal and physical gates. This allows hardware vendors and researchers to assign a single fidelity value to a gate and compare different quantum processors.

### 6.1.5 Fidelity and Decoherence

Imagine we start with a gate that takes time $t_{\text{gate}}$ to execute. During this time, decoherence attacks our qubit. Therefore, the longer the gate lasts, the more time decoherence has to damage the qubit's state. More decoherence means more errors, which means lower fidelity.

Precisely, the fidelity measures how accurately a physical gate reproduces the ideal gate (page 210). One important source of infidelity is decoherence (page 200). Since a gate requires a **finite execution time** $t_{\text{gate}}$, the qubit is exposed to environmental noise while the gate is running. The longer the gate duration compared to the **decoherence time** $t_{\text{dec}}$, the higher the probability that decoherence introduces errors. Consequently, **longer gates tend to have lower fidelity**. A **simple approximation** is:

$$p_{\text{gate}} \approx \frac{t_{\text{gate}}}{t_{\text{dec}}} \tag{94}$$

Where:

- $t_{\text{dec}}$ is the **Decoherence Time**. It measures how long a qubit can preserve its quantum state before the environment destroys it. For superconducting qubits, usually the decoherence time is on the order of microseconds ($\mu s$):

$$t_{\text{dec}} \approx 200\mu s$$

- $t_{\text{gate}}$ is the **Gate Time**. It measures how long a quantum gate takes to execute. Generally, $X$ gates take around 20 ns, $H$ gates around 30 ns, and CNOT gates around 300 ns. The important assumption is $t_{\text{gate}} \ll t_{\text{dec}}$, because otherwise the qubit would lose coherence while the gate is still running, leading to very low fidelity.

- $p_{\text{gate}}$ is the **Gate Error Probability**. It estimates the likelihood that a gate operation fails due to decoherence. The longer the gate duration relative to the decoherence time, the higher the error probability, which reduces the overall fidelity of the quantum operation.

For example, suppose $t_{\text{dec}} = 100\mu s$ and $t_{\text{gate}} = 1\mu s$. Then:

$$p_{\text{gate}} \approx \frac{1\mu s}{100\mu s} = 0.01$$

This means that the gate has a 1% chance of introducing an error due to decoherence, which corresponds to a fidelity of approximately 99%. If we can reduce the gate time to 0.5 $\mu s$, then:

$$p_{\text{gate}} \approx \frac{0.5\mu s}{100\mu s} = 0.005$$

This means the gate error probability drops to 0.5%, improving the fidelity to approximately 99.5%. Therefore, optimizing gate times is crucial for enhancing the fidelity of quantum operations.

☆ **Quality Ratio.** The **Quality Ratio** is a metric that answers the question: ***how many gate operations can we perform before decoherence becomes significant?*** It is defined as:

$$\text{Quality Ratio} = \frac{t_{\text{dec}}}{t_{\text{gate}}} \tag{95}$$

This is simply the inverse of the gate error probability:

$$\text{Quality Ratio} = \frac{1}{p_{\text{gate}}}$$

A higher quality ratio indicates that we can perform more gate operations before decoherence significantly degrades the qubit's state. For instance, if $t_{\text{dec}} = 100\mu s$ and $t_{\text{gate}} = 1\mu s$, then the quality ratio is:

$$\text{Quality Ratio} = \frac{100\mu s}{1\mu s} = 100$$

This means we can perform approximately 100 gate operations before decoherence significantly impacts the qubit's state. If we reduce the gate time to 0.5 $\mu s$, the quality ratio increases to:

$$\text{Quality Ratio} = \frac{100\mu s}{0.5\mu s} = 200$$

This means we can perform approximately 200 gate operations before decoherence becomes significant, effectively doubling the number of operations we can execute with high fidelity. Therefore, optimizing gate times not only reduces error probabilities but also increases the quality ratio, allowing for more complex quantum computations before decoherence limits performance. In general, a quality ratio of $10^3 - 10^4$ is the target for fault-tolerant quantum computing, as it allows for sufficient error correction to mitigate the effects of decoherence.

✔ **Updated Fidelity Formula.** The **Fidelity** of a quantum gate can be **approximated by considering the gate error probability due to decoherence**. If we assume that the only source of infidelity is decoherence, then the fidelity can be expressed as:

$$F_{\text{gate}} \approx 1 - p_{\text{gate}} = 1 - \frac{t_{\text{gate}}}{t_{\text{dec}}} \tag{96}$$

This formula indicates that the fidelity decreases linearly with the ratio of gate time to decoherence time. For example, if $t_{\text{dec}} = 100\mu s$ and $t_{\text{gate}} = 1\mu s$, then the fidelity is:

$$F_{\text{gate}} \approx 1 - \frac{1\mu s}{100\mu s} = 0.99$$

This means the gate has a fidelity of approximately 99%. If we reduce the gate time to 0.5 $\mu s$, the fidelity improves to:

$$F_{\text{gate}} \approx 1 - \frac{0.5\mu s}{100\mu s} = 0.995$$

This means the gate has a fidelity of approximately 99.5%. Therefore, optimizing gate times is crucial for enhancing the fidelity of quantum operations, as it directly reduces the error probability associated with decoherence.

**Threshold Theorem.** Every physical quantum gate introduces a small probability of error. Without error correction, these errors accumulate and eventually destroy the computation.

The **Threshold Theorem** states that if the physical **error rate** of the quantum hardware is **below a certain threshold value** (equivalently, if the gate fidelity is sufficiently high), then **quantum error correction can remove errors faster than they are generated**.

As a consequence, arbitrarily long quantum computations become possible using logical qubits and fault-tolerant protocols, even though the underlying physical qubits are noisy and have finite decoherence times.

For example, if every second 10 errors appear, and we can correct 20 errors per second, then the system stays healthy. But if every second 10 errors appear, and we can only correct 5 errors per second, then the system becomes unhealthy and eventually fails. Therefore, maintaining an error rate below the threshold is crucial for the viability of quantum computing.

## 6.2 Quantum Error Correction

Before discussing quantum compilation and transpiling, we must understand a fundamental limitation of current quantum hardware. Unlike classical computers, quantum computers are extremely sensitive to noise. Errors continuously affect qubits during computation, making it impossible to execute long quantum algorithms reliably without some form of error correction.

### 6.2.1 Quantum Computation Errors

Real quantum computers are physical devices and therefore interact with their surrounding environment. Even when carefully isolated, external disturbances can affect the quantum state of a qubit, as discussed in previous sections. Examples of these disturbances include electromagnetic interference, thermal fluctuations, and microscopic vibrations of surrounding particles.

**Example 1: Brownian motion**

The **Brownian Motion** is the **random motion of particles caused by thermal energy**.

In 1827, the Scottish botanist Robert Brown observed tiny pollen grains suspended in water. Instead of remaining still, they jittered randomly. At the time nobody understood why, but later it was discovered that water molecules were constantly hitting the pollen grain from random directions. The pollen grain was so small that these microscopic collisions became visible.

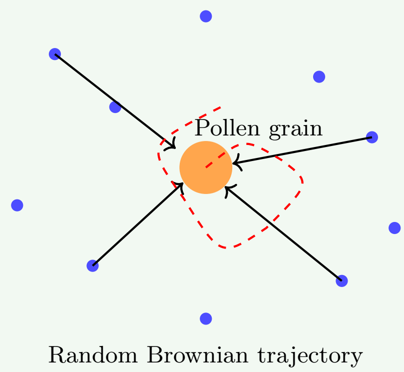



Illustration of Brownian motion. A microscopic pollen grain suspended in a fluid is continuously struck by surrounding molecules. Because the collisions occur randomly and from different directions, the grain follows an irregular trajectory (red dashed line). Brownian motion provides direct evidence that matter at finite temperature is constantly moving at the microscopic scale.

**How is linked to thermal energy?** Brownian motion is a direct consequence of the thermal energy of the surrounding environment.

At finite temperature, particles in a fluid are in constant motion due to their thermal energy. This motion leads to random collisions with suspended particles, causing the observed Brownian motion. The intensity of Brownian motion increases with temperature, as higher thermal energy results in more vigorous particle movement and more frequent collisions. Indeed, if we boil the water, the Brownian motion becomes more pronounced as the water molecules move faster and collide more frequently with the pollen grain.

**Amazing, but how is it relevant for quantum computing?** The actual **enemy is not Brownian motion** itself, but the fact that the **thermal environment constantly interacting with the qubit**. Brownian motion is simply an intuitive example showing that matter at finite temperature is never perfectly still, but rather in constant motion. This means that **qubits are continuously affected by their environment**, leading to errors in quantum computations. Even if we try to isolate the qubits as much as possible, they will still be subject to random disturbances from the surrounding environment, making it impossible to achieve perfect error-free quantum computation.

### Why Error Correction is necessary

Quantum algorithms often require millions of gate operations. Without a protection mechanism, continuous interaction with the environment would cause decoherence (page 200) and accumulate gate errors (page 199) during computation, making it **highly unlikely that the correct result would be obtained**.

However, there is hope. The **Threshold Theorem** (page 216) discussed previously provides the curcial insight that reliable **quantum computation is still possible** if errors are corrected continuously and the physical error rate remains below a certain threshold.

For this reason, modern quantum computers cannot rely solely on physical qubits (page 166). Instead, they require **Quantum Error Correction Codes** (QECCs), which encode quantum information redudantly across many physical qubits in order to detect and correct errors before they destroy the computation.

- Noise → Errors → Decoherence → Computation Failure
- Noise → Quantum Error Correction → Reliable Computation

### 6.2.2 Quantum Error Correction Codes (QECC)

After understanding that physical qubits suffer from decoherence and gate errors, we need a **mechanism to preserve quantum information during a computation**. Quantum Error Correction Codes (QECCs) achieve this by introducing **redundancy**: instead of storing information in a single physical qubit, the information is distributed across many qubits.

We previously discussed this concept on page 166, where we compared logical and physical qubits. Here, however, we present the concept with more detail because we now have a better understanding of possible errors and their impact on quantum computations.

- A **Physical Qubit** is a **real qubit implemented by hardware**, such as superconducting qubits, trapped-ion qubits or photonic qubits. Unfortunately, physical qubits are directly affected by decoherence (page 200), thermal noise (page 217), and gate imperfections (infidelity, page 199). Therefore, a **physical qubit is unreliable if used alone** for quantum computation, as it is prone to errors that can lead to incorrect results.
- To obtain reliable computation, quantum information is not stored in a single physical qubit. Instead, one creates a **Logical Qubit** $|q_L\rangle$ which is **encoded into many physical qubits**. For instance, a logical qubit can be encoded into 5, 7, or even hundreds of physical qubits, depending on the specific error correction code used.

  The user of the quantum computer thinks in terms of logical qubits, while the hardware manipulates the underlying physical qubits.

#### ⚠ How can errors be detected without collapsing the state?

Suppose our logical qubit is encoded into some physical qubits:

$$q_0,\, q_1,\, q_2,\, q_3,\, q_4$$

An error occurs on one of the physical qubits, say $q_2$. However, since the logical qubit is encoded across all five physical qubits, enough information remains in the qubits to recover the original state. The question is, ***how do we know that an error occurred and which qubit was affected?***

✖ The first naïve solution could be to measure all the physical qubits to check for errors. However, this approach is flawed because **measuring a qubit collapses its quantum state**, which would destroy the quantum information we are trying to protect.

At this point, we can **exploit the redundancy introduced by the logical encoding**. Consider a simple example where:

$$|0_L\rangle = |000\rangle \qquad |1_L\rangle = |111\rangle$$

Where $L$ stands for “logical”. In this encoding, the logical qubit $|0_L\rangle$ is represented by three physical qubits all in the state $|0\rangle$, and the logical qubit $|1_L\rangle$ is represented by three physical qubits all in the state $|1\rangle$. A generic logical qubit is therefore:

$$|\psi_L\rangle = \alpha\,|000\rangle + \beta\,|111\rangle$$

Notice that in a **valid encoded state all physical qubits agree with each other**: they are either all 0 or all 1.

Now, suppose that a **bit-flip error** affects the second physical qubit. The error acts on the entire quantum state:

$$\alpha\,|000\rangle + \beta\,|111\rangle \quad \longrightarrow \quad \alpha\,|010\rangle + \beta\,|101\rangle$$

In this case, the second qubit has flipped from 0 to 1 in the first term and from 1 to 0 in the second term. The resulting state is **no longer a valid encoded state** because the physical qubits do not all agree with each other.

**Solution: Syndrome.** Rather than asking "*what is the quantum state?*" which would destroy the computation, we only ask: "*are neighbouring qubits equal or different?*". For instance, we can check if $q_0$ and $q_1$ are the same, and if $q_1$ and $q_2$ are the same. This way, we can detect that $q_2$ is different from the others without measuring the actual state of any qubit. In other words, the **pattern of answers identifies the location of the error without revealing the coefficients** $\alpha$ and $\beta$ of the logical qubit. The collection of these answers is called the **syndrome**.

**Definition 2: Syndrome**

A **Syndrome** is a classical piece of information that indicates the **presence and type of error without revealing the logical quantum state**. A *syndrome* is obtained by performing a **Syndrome Measurement**, which simply collects information about the error, a sort of "*error diagnosis*". However, it is a **quantum measurement** and therefore **causes a collapse**. But it **collapses only the error-related degrees of freedom** (the syndrome subspace), not the logical quantum information encoded in the qubit. Therefore the logical state survives while the error information becomes known.

For instance, if we have the logical state:

$$\alpha\,|0\rangle + \beta\,|1\rangle$$

The syndrome does **not** tell us $\alpha$ or $\beta$, the syndrome only tells us things like "*error detected: yes*; *location: qubit 5*; *type: bit-flip*".

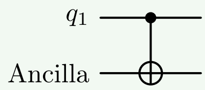
### How can we perform syndrome measurements without collapsing the logical state?

The key insight is that we can perform **indirect measurements** by coupling the data qubits to an additional qubit, called an **Ancilla Qubit**, and then measuring the ancilla. This way, we can extract information about the error without directly measuring the data qubits, thus preserving the logical quantum state.

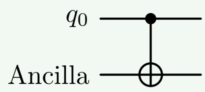
Usually the ancilla qubit starts in a known state, such as $|0\rangle$. We then apply a series of quantum gates that entangle the ancilla with the data qubits in such a way that the state of the ancilla after the interaction contains information about the syndrome.

**Example 2: Syndrome Measurement with an Ancilla Qubit**

We want to know if $q_0$ and $q_1$ are the same or different. We initialize the ancilla qubit in the state $|0\rangle$ and apply a CNOT gate with $q_1$ as the control and the ancilla as the target:




If $q_1$ is in the state $|0\rangle$, the ancilla remains in $|0\rangle$. If $q_1$ is in the state $|1\rangle$, the ancilla flips to $|1\rangle$. Next, we apply another CNOT gate with $q_0$ as the control and the ancilla as the target:




After these two gates, the state of the ancilla will be $|0\rangle$ if $q_0$ and $q_1$ are the same (both $|0\rangle$ or both $|1\rangle$) and $|1\rangle$ if they are different. The final value becomes Ancilla $= q_0 \oplus q_1$, which is the syndrome for this particular check.

Finally, we **measure the ancilla qubit**. The measurement outcome gives the syndrome, which can be used to determine whether an error occurred and, in many cases, which qubit was affected.

An important observation is that the syndrome does not exist beforehand. The ancilla qubit is the physical system that temporarily stores the syndrome information so that it can be measured without directly measuring the data qubits.

**Definition 3: Ancilla Qubit**

An **Ancilla Qubit** is an **auxiliary qubit** introduced to assist a quantum computation. In QECC, ancilla qubits are used to interact with data qubits, extract information about errors without collapsing the data qubits' states, and help correct errors. They **do not store the logical quantum information** being protected.

In summary, QECC makes reliable quantum computation possible, but at a **significant cost**. A logical qubit is not implemented by a single physical qubit. Instead, it requires many physical qubits, including both data qubits and ancilla qubits used for syndrome extraction and error correction.

As a consequence, the number of physical qubits required by a quantum computer grows dramatically. In practice, a **single logical qubit may require tens or even hundreds of physical qubits**. This phenomenon is known as the **Blow-Up Problem**.

**Definition 4: Quantum Error Correction Code (QECC)**

A **Quantum Error Correction Code (QECC)** is a **technique used to protect quantum information** from noise and decoherence. A logical qubit is encoded into multiple physical qubits, introducing redundancy that allows errors to be detected and corrected.

Error detection is performed through **syndrome measurements**, typically using **ancilla qubits**, so that information about errors can be extracted *without* directly measuring the logical quantum state.

By continuously detecting and correcting errors, QECC enables reliable quantum computations despite the presence of imperfect hardware.

### 6.2.3 Logical Gates

Originally, before Quantum Error Correction Codes (QECCs), life was easy: physical qubit, apply gate (e.g., Hadamard), get result. After QECC, things are different. A logical qubit is encoded into many physical qubits, and now the question is: "*how do we apply a gate to the logical qubit if the logical qubit is spread across many physical qubits?*".

#### Logical Gates

To apply a gate to a logical qubit, we need to define the gate operation on the logical qubit level. So, each fundamental gate needs to be defined on logical qubits. But the way the logical operation is performed depends on the logical qubit implementation. There is not one universal logical H gate. The implementation depends on the code used to encode the logical qubit.

**Definition 5: Logical Gate**

A **Logical Gate** is a quantum **operation acting on logical qubits**. Since a logical qubit is encoded into multiple physical qubits, a logical gate is implemented through a coordinated sequence of operations on the underlying physical qubits.

#### Physical Implementation of Logical Gates

In superconducting quantum computers, logical gates are ultimately implemented as **microwave pulses** sent through a cryogenic chip. The physical hardware consists of superconducting circuits, resonators, control lines, and Josephson junctions operating at temperatures close to absolute zero.

**Superconducting qubits** are the most common type of qubit used in current quantum computers (e.g., IBM, Google, Rigetti). The superconducting qubits are typically implemented as cryogenic qubits, which operate at extremely low temperatures $10-20$ millikelvins (0.01°C above absolute zero, $\approx -273.15$°C). At these temperatures, the superconducting materials exhibit almost zero thermal noise, allowing for coherent quantum operations (page 217). The gates are implemented via microwave pulses that manipulate the energy levels of the superconducting qubits.

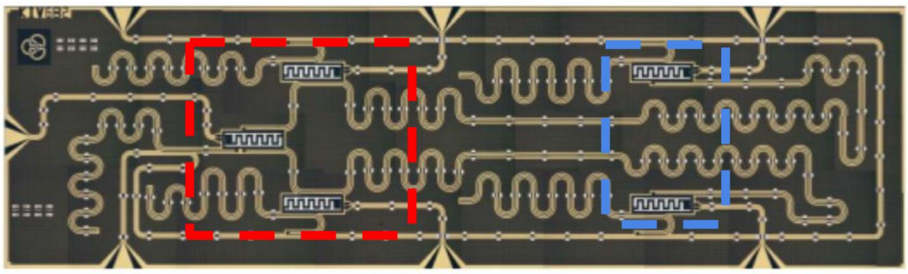


Figure 17: Photograph of a **superconducting quantum chip** used in quantum computing. [3] The squiggly lines are the microwave transmission lines/resonators that control the qubits. The highlighted regions contain the superconducting qubit circuitry and the microwave resonators/control structures used to implement quantum gates through microwave pulses.

## 6.3 Quantum Transpiling

### 6.3.1 The Quantum Computing Stack

Before studying placement, scheduling, routing, and optimization, we need to understand **where transpiling sits inside the quantum-computing workflow**. We can summarize the whole process with five layers: (1) the quantum processor, (2) the quantum programming language, (3) the quantum algorithm, (4) the quantum compilation (transpiling), and (5) the quantum execution. The role of the *transpiler* is to act as the **bridge between the abstract quantum algorithm and the physical hardware**. We will see that the **transpiler** is responsible for **taking a high-level quantum algorithm**, expressed in a quantum programming language, and **transforming it into a hardware-compatible circuit** that satisfies the constraints of a specific quantum processor.

In general, a quantum computer is not just a chip. Many software and hardware layers cooperate to transform a high-level algorithm into physical operations executed on real qubits. Let's briefly describe the main layers of the quantum computing stack:

1. **Quantum Processor**. The lowest layer is the **quantum processor**. A quantum processor is the **physical device** that **contains the qubits used for computation**. The processor is the actual hardware where quantum states evolve according to the laws of quantum mechanics.

2. **Quantum Programming Language**. Writing microwave pulses directly would be impossible for programmers. Instead, we use a **quantum programming language** (similar to classical programming languages). A quantum programming language allows us to describe a circuit using quantum instructions such as $H$, $CNOT$, and $X$. The language hides the physical details of the hardware and provides an abstract description of the computation.

3. **Quantum Algorithm**. A quantum algorithm is the **circuit that we want to execute**. At this stage we only think about the mathematical algorithm, without worrying about the hardware connectivity (i.e., which qubits can interact with which), gate durations (i.e., decoherence times), physical errors (i.e., noise), and qubit locations (i.e., where the qubits are physically located on the chip). The algorithm represents the **ideal computation**.

4. **Quantum Compilation** (Transpiling). This is the central topic of the chapter. A **quantum algorithm cannot usually be executed directly on hardware**. The circuit must first be adapted to the specific quantum processor. This adaptation process is called: **quantum compilation** or **Transpiling**. We will see four main tasks of the transpiler: (1) **placement**, (2) **scheduling**, (3) **routing**, and (4) **optimization** (use heuristics to merge gates or rearrange operations). In general, the transpiler takes an ideal circuit and trasnforms it into another circuit that performs the same computation, respects hardware constraints, minimizes errors, and minimizes execution time.

5. **Quantum Execution**. After transpilation, the circuit is finally converted into physical control signals. It is the translation of gate-level instructions into signals sent to the processor. For superconducting qubits this means: generating microwave pulses, sending them to specific qubits, controlling interactions between qubits, performing measurements. This is the **stage where the abstract circuit becomes a real physical experiment**.

In general, the **transpiler is the middle layer that makes the whole system work**.

### 6.3.2 Placement

In an ideal quantum circuit we freely write operations between any pair of qubits, like CNOT$(q_1, q_8)$. However, a real quantum processor is not fully connected. Physical qubits occupy fixed positions on a chip and a two-qubit gate can generally be executed only between adjacent qubits. Therefore, before execution, the transpiler must decide: ***which logical qubit should be assigned to which physical qubit?*** This task is called **placement**.

**Definition 6: Placement**

**Placement** is the process of **mapping the qubits of a quantum circuit onto the physical qubits of the processor**. In other words, it is the process of deciding which logical qubit should be assigned to which physical qubit. The result is the initial configuration used by the hardware-aware transpiler to perform the subsequent steps of routing and scheduling.

**The Goal of Placement.** A good placement tries to **position interacting qubits close together**. The reason of this goal is simple: if two qubits are far apart, they must later be moved using SWAP operations. SWAP operations are costly because they require multiple gates, larger circuit depth, more noise and lower fidelity. Therefore, placement tries to **reduce future routing costs** before routing even begins.

#### How do Transpilers understand how far apart qubits are?

To quantify how far two qubits are, the transpiler uses the **Manhattan distance**.

The **Manhattan distance**[10] is simply the **distance we travel if we can only move along the grid lines of the chip** (up/down and left/right). Suppose we have two qubits at positions $A = (x_1, y_1)$ and $B = (x_2, y_2)$. The Manhattan distance between them is given by:

$$d_M(A, B) = |x_1 - x_2| + |y_1 - y_2| \tag{97}$$

For example, if $A = (1, 2)$ and $B = (4, 6)$, then the Manhattan distance is:

$$d_M(A, B) = |1 - 4| + |2 - 6| = 3 + 4 = 7$$

This means that to move from qubit $A$ to qubit $B$, we would need to perform 7 operations if they are not directly connected.

**Why Manhattan distance?** Suppose a qubit must move four positions to reach another qubit. The qubit must perform approximately four hops. Since routing (page 237) is implemented through SWAP gates, the number of **hops**

[10] The Manhattan distance takes its name from the grid-like structure of the streets in Manhattan, where we can only move along the streets (up/down and left/right) to get from one point to another.

**is directly related to the number of additional operations** required. For this reason, minimizing Manhattan distance approximately minimizes routing cost.

In summary, the goal of the placement can be summarized as follows: find the placement that minimizes the sum of Manhattan distances over all qubit pairs involved in multi-qubit gates. Conceptually:

$$\text{Cost} = \sum_{\text{all interacting pairs}} d_M$$

A placement with a smaller total cost will usually require fewer SWAP operations later.

### 6.3.3 Scheduling

After placement (page 227), the transpiler knows **where each qubit is located** on the processor. However, another question remains: "***when should each gate be executed?***". At first sight, one might think that all gates could be executed simultaneously, but this is not the case. In fact, there are scheduling issues primarily caused by **data dependencies**.

#### Data Dependencies

A gate can only be executed if all the qubits it requires already contain the correct input state. For example:

$$|0\rangle \;-\!\!\boxed{H}\!\!-\!\!\boxed{CNOT}\!\!-$$

The CNOT cannot be executed before the Hadamard gate. Because the CNOT uses the state produced by the Hadamard gate. The second operation therefore **depends** on the first one. This relationship is called a **Data Dependency**. Therefore, the two gates must be executed sequentially, and the scheduler must respect this ordering.

Fortunately, not all gates depend on each other. For example:

$$|0\rangle \;-\!\!\boxed{H}\!\!-$$

$$|1\rangle \;-\!\!\boxed{X}\!\!-$$

The two gates act on different qubits. Neither gate needs the output of the other. Therefore they can be **executed simultaneously**. This is a source of parallelism in quantum circuits.

In general, **Out-of-Order execution is possible** as long as the computation remains correct. This means the scheduler can reorder operations only if the final quantum state remains unchanged. The scheduler cannot move gates around the circuit arbitrarily; it must always respect the dependencies between operations.

#### Scheduling Objectives

The scheduler tries to achieve two goals:

- **Exploit as much parallelism as possible** to minimize the circuit depth.
- **Reduce the effects of decoherence** (page 200) by executing the circuit as quickly as possible.

These goals sometimes conflict with each other, and for this reason different scheduling policies exist:

- **As Soon As Possible (ASAP) Scheduling**. Execute an operation immediately when all its required input qubits are available. In other words, when the gate becomes executable, the scheduler executes it right away.

  ✅ The main **advantages** of ASAP scheduling are: maximizes parallelism, reduces total circuit execution time, and keeps hardware busy

- **As Late As Possible (ALAP) Scheduling**. Instead of executing operations immediately, the scheduler delays them as much as possible while still preserving correctness. Conceptually, when the gate needs to be executed, the scheduler checks if it can be delayed without violating any dependencies. If it can be delayed, the scheduler postpones its execution until the last possible moment.

  ✅ The main **advantages** of ALAP scheduling are: reduces idle time of qubits, minimizes the lifetime of quantum states, and can help to **mitigate decoherence effects**.

### ❓ Which scheduling policy is better?

The choice depends by the final goal of the scheduling.

- **Total execution time of the circuit**. For this quantity, ASAP is usually better because it tries to execute everything immediately and therefore often minimizes the overall circuit depth. In simple terms, **ASAP** scheduling tries to **minimize total circuit duration**.

- **Lifetime of a specific quantum state**. Instead, for this quantity, ALAP is usually better because it tries to delay operations as much as possible, which can help to reduce the time that a particular quantum state needs to be maintained, thus reducing the effects of decoherence on that state. In simple terms, **ALAP** scheduling tries to **minimize idle lifetime of quantum states**.

  Let's suppose we have the following circuit:

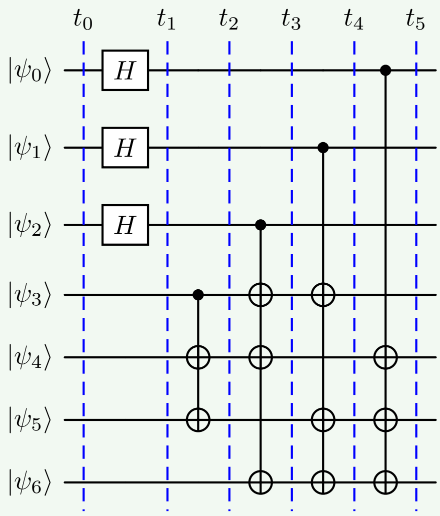
  $$|0\rangle \text{ — } \boxed{H} \text{ — } \boxed{CNOT} \text{ — }$$

  ❌ With **ASAP scheduling**, the Hadamard gate $H$ would be executed immediately at time 0, and the CNOT gate would be executed later at time 100 due to other operations that need to be executed in between. Then the superposition created by $H$ must survive for 100 time units, but this **increases the risk of decoherence**.

  ✅ Instead, **ALAP scheduling** would try to delay the execution of the Hadamard gate $H$ as much as possible, while still ensuring that the CNOT gate can be executed at time 100. This means that $H$ would be executed at time 90, just before the CNOT gate, which reduces the time that the superposition needs to be maintained and therefore **reduces the risk of decoherence**.

**Example 3: ASAP & ALAP Scheduling**

Consider the following quantum circuit:



Now, let's analyze how the scheduling would work under both policies. The operations must include the initialization of the qubits, the application of the Hadamard gates, and the execution of the CNOT gates.

### ASAP Scheduling

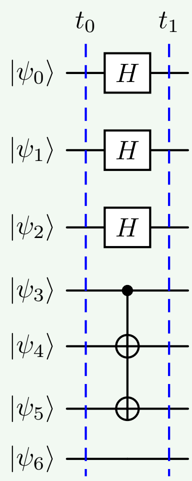
With **ASAP**, all qubits are initialized immediately at time 0 because there are no dependencies:

$$\text{INIT} \quad |\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle, |\psi_5\rangle, |\psi_6\rangle \quad \text{at time } 0$$

Once the qubits are initialized, every gate can be executed as soon as its dependencies are satisfied. The main idea is: "*if a gate can run now, run it now*".

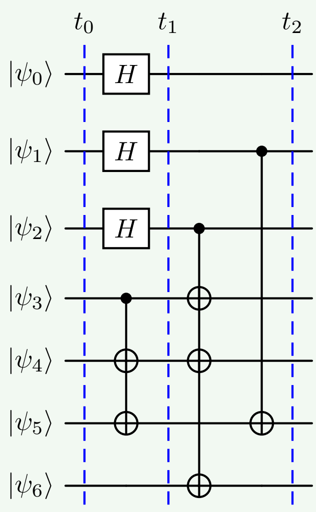
1. At time 1, the Hadamard gates on $|\psi_0\rangle$, $|\psi_1\rangle$, and $|\psi_2\rangle$ can be executed immediately because they only depend on the initialization of the qubits. So we execute them at time 1. Regarding the qubits $|\psi_3\rangle$, $|\psi_4\rangle$ and $|\psi_5\rangle$, we can execute the CNOT gates that depend on them as soon as the Hadamard gates on $|\psi_0\rangle$, $|\psi_1\rangle$ and $|\psi_2\rangle$ are executed.

$$\text{INIT} \quad |\psi_0\rangle, |\psi_1\rangle, |\psi_2\rangle, |\psi_3\rangle, |\psi_4\rangle, |\psi_5\rangle, |\psi_6\rangle$$
$$H(|\psi_0\rangle),\ H(|\psi_1\rangle),\ H(|\psi_2\rangle),\ \text{CNOT}(|\psi_3\rangle, |\psi_4\rangle),\ \text{CNOT}(|\psi_3\rangle, |\psi_5\rangle)$$



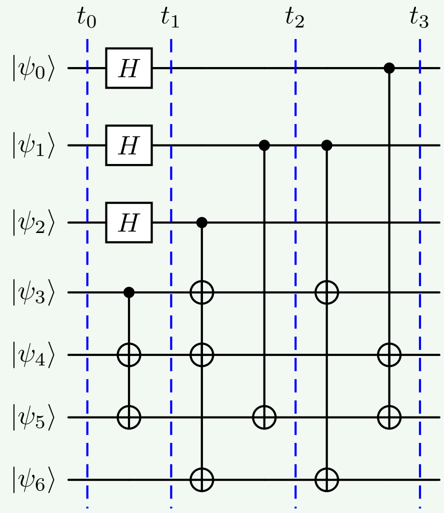

2. At time 2, we can execute the CNOT gates that depend on $|\psi_2\rangle$ and $|\psi_1\rangle$ as soon as the Hadamard gates on $|\psi_0\rangle$, $|\psi_1\rangle$ and $|\psi_2\rangle$ are executed. So we execute them at time 2. Regarding the qubits $|\psi_3\rangle$, $|\psi_4\rangle$ and $|\psi_5\rangle$, we can execute the CNOT gates that depend on them as soon as the CNOT gates that depend on them are executed. However, we cannot execute the CNOT gate that depends on $|\psi_6\rangle$ because it depends on the CNOT gate that depends on $|\psi_1\rangle$. And also, the CNOT gate that depends on $|\psi_1\rangle$ cannot be executed completely at time 2 because it depends on the CNOT gate that depends on $|\psi_2\rangle$. So we execute it partially at time 2, and we will execute the remaining part at time 3.

$$\text{CNOT}(|\psi_2\rangle, |\psi_3\rangle),\ \text{CNOT}(|\psi_2\rangle, |\psi_4\rangle),$$
$$\text{CNOT}(|\psi_2\rangle, |\psi_5\rangle),\ \text{CNOT}(|\psi_1\rangle, |\psi_6\rangle)$$



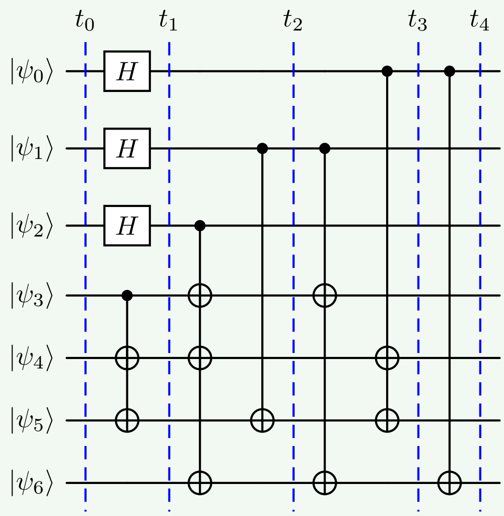
3. At time 3, we can execute the CNOT gates that depend on $|\psi_1\rangle$ and $|\psi_0\rangle$ as soon as the CNOT gates that depend on $|\psi_2\rangle$ and $|\psi_1\rangle$ are executed. Regarding the qubits $|\psi_3\rangle$, $|\psi_4\rangle$ and $|\psi_5\rangle$, we can execute the CNOT gates that depend on them as soon as the CNOT gates that depend on them are executed. However, we cannot execute the CNOT gate that depends on $|\psi_6\rangle$ because it depends on the CNOT gate that depends on $|\psi_1\rangle$. So we execute it partially at time 3, and we will execute the remaining part at time 4.

$$\text{CNOT}(|\psi_1\rangle, |\psi_3\rangle),\ \text{CNOT}(|\psi_1\rangle, |\psi_6\rangle),$$
$$\text{CNOT}(|\psi_0\rangle, |\psi_4\rangle),\ \text{CNOT}(|\psi_0\rangle, |\psi_5\rangle)$$




4. Finally, at time 4, we can execute the CNOT gate that depends on $|\psi_6\rangle$ as soon as the CNOT gate that depends on $|\psi_1\rangle$ is executed.

$$\text{CNOT}(|\psi_0\rangle, |\psi_6\rangle)$$

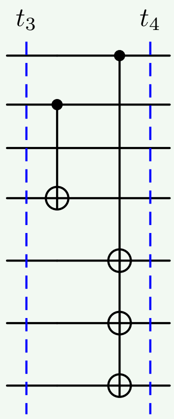


### ALAP Scheduling

With **ALAP**, the scheduler works with the opposite philosophy: "*if a gate can be dalyed safely, delay it*". Differently from ASAP scheduling, we simply create a schedule starting from the end of the circuit and moving backwards.

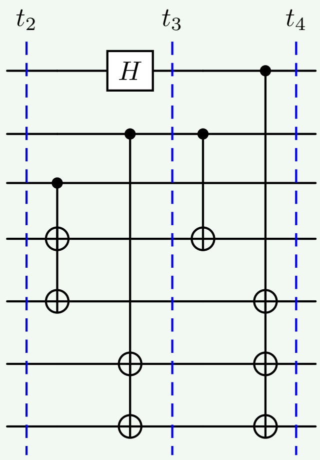
4. At time 4, we start with the last CNOT gate. The remaining free time slots are filled with the CNOT gates that depend on $|\psi_1\rangle$ and $|\psi_3\rangle$.

$$\text{CNOT}(|\psi_0\rangle, |\psi_4\rangle), \text{CNOT}(|\psi_0\rangle, |\psi_4\rangle), \text{CNOT}(|\psi_0\rangle, |\psi_4\rangle)$$
$$\text{CNOT}(|\psi_1\rangle, |\psi_3\rangle)$$



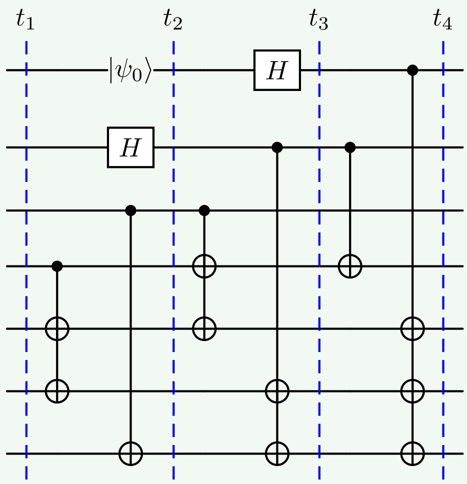

3. At time 3, we continue with the same logic, taking the left piece of the circuit and filling the remaining free time slots.

$$H(|\psi_0\rangle), \text{CNOT}(|\psi_1\rangle, |\psi_5\rangle), \text{CNOT}(|\psi_1\rangle, |\psi_6\rangle),$$
$$\text{CNOT}(|\psi_2\rangle, |\psi_3\rangle), \text{CNOT}(|\psi_2\rangle, |\psi_4\rangle)$$



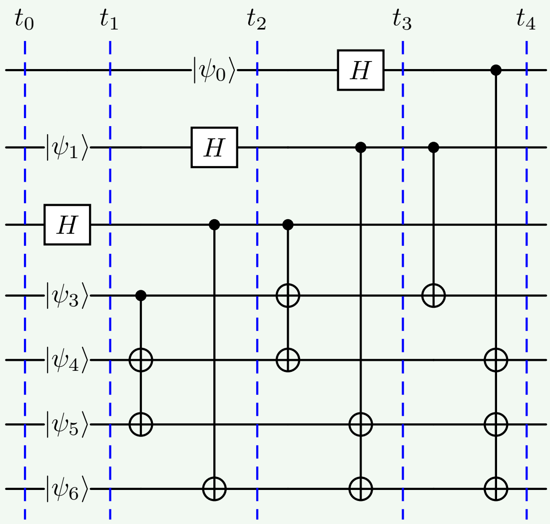
2. At time 2:

$$\text{INIT} \quad |\psi_0\rangle\,, H(|\psi_1\rangle), \text{CNOT}(|\psi_2\rangle\,, |\psi_6\rangle), \text{CNOT}(|\psi_3\rangle\,, |\psi_4\rangle), \text{CNOT}(|\psi_3\rangle\,, |\psi_5\rangle)$$



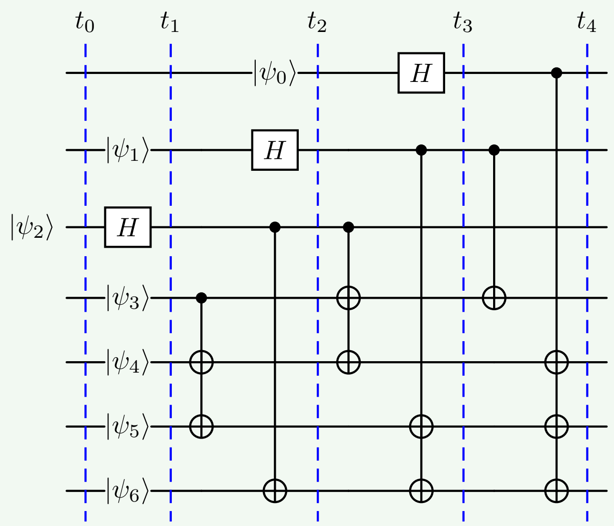

1. At time 1:

$$\text{INIT} \quad |\psi_1\rangle\,, |\psi_3\rangle\,, |\psi_4\rangle\,, |\psi_5\rangle\,, |\psi_6\rangle\,, H(|\psi_2\rangle)$$




0. At time 0:

$$\text{INIT} \quad |\psi_2\rangle$$




In general, the ASAP version is more aggressive because does everything as early as possible. Instead, the ALAP version is more conservative because delays operations until they are needed. Both schedules preserve the same logical computation. The difference is not **what** gates are executed, but **when** they are executed.

In summary, ASAP is useful to start operations early and exploit parallelism, while ALAP is useful to reduce idle time between the moment a qubit is prepared and the moment is used. That is why scheduling is not only a performance problem, but also a **fidelity problem** because it can affect the overall quality of the computation by influencing how long qubits need to maintain their quantum states, which in turn affects the likelihood of errors due to decoherence.

### 6.3.4 Routing

To understand routing, we need to recall the fundamental hardware constraint: a multi-qubit gate can only be executed between adjacent qubits (i.e., qubits that are directly connected in the hardware architecture). For example, suppose the algorithm requires a $\text{CNOT}(|\psi_3\rangle, |\psi_5\rangle)$ gate, but the placement step maps the qubits onto the processor such that $|\psi_3\rangle$ and $|\psi_5\rangle$ are not adjacent:

$$\begin{bmatrix} v_0 & v_1 & v_2 & v_3 \\ v_4 & v_5 & v_6 & v_7 \\ v_8 & v_9 & & \end{bmatrix}$$

Here $v_3$ is in the first row and $v_5$ is in the second row, so they are **not adjacent**. Therefore the CNOT cannot be executed directly.

**Solution: Routing.** If qubits are not adjacent, how do we move them? Using SWAP gates. A **SWAP gate exchanges the quantum states of two adjacent qubits** (page 107):

$$\text{SWAP}(|\psi_i\rangle, |\psi_j\rangle) = |\psi_j\rangle \otimes |\psi_i\rangle$$

Thus, the qubits themselves do not physically move, but their quantum states are exchanged. The **Routing** is the **process of inserting SWAP gates into the circuit** to ensure that **all multi-qubit gates operate on adjacent qubits**. The goal of routing is to find an efficient sequence of SWAP gates that minimizes the overhead in terms of additional gates and circuit depth.

**Why Routing is Expensive?** Routing solves the connectivity problem, but it introduces additional gates. For instance, to execute $\text{CNOT}(|\psi_3\rangle, |\psi_5\rangle)$, we might need to perform a sequence of SWAP gates to bring $|\psi_3\rangle$ and $|\psi_5\rangle$ adjacent to each other.

- Each SWAP gate increases the circuit depth (longer execution time);
- Every additional gate introduces also error probability. This means that the **fidelity** (page 215):

  $$F_{\text{gate}} \approx 1 - p_{\text{gate}}$$

  Decreases with the number of gates, so more gates means lower fidelity.
- A **deeper circuit takes longer to execute**, which means that qubits remain active for a larger fraction on their decoherence time, increasing the likelihood of errors due to decoherence.

**Connection with Placement.** Now we can finally understand why placement was important. Placement tries to minimize the Manhattan distance (page 227) between qubits that interact. *Why?* Because **smaller distances imply fewer SWAP gates**, which means **less routing overhead**, **higher fidelity**, and **shorter execution time**. Therefore, placement and routing are tightly connected: a good placement reduces the amount of routing required later.

In summary, routing is the process of inserting SWAP operations to bring non-adjacent qubits together so that multi-qubit gates can be executed on real hardware.

# 7 QUBO Formulation

## 7.1 QUBO Problems

In the previous sections of the course, we showed how quantum computers manipulate information using qubits and quantum gates. But a fundamental question remains: ***which problems are we trying to solve with these quantum computers?***

One important class of problems is **optimization problems**, where the goal is to find the best solution among many possible alternatives.

**Many optimization problems** arising in science, engineering, logistics, finance, and machine learning can be **reformulated into** a common **mathematical framework** called **Quadratic Unconstrained Binary Optimization (QUBO)**. The importance of QUBO is that a wide variety of seemingly different optimization tasks can be expressed using the same mathematical structure.

This common formulation is particularly useful in quantum computing because it can be directly connected to physical systems, Hamiltonians, and quantum annealing algorithms. For this reason, QUBO serves as a bridge between classical optimization problems and quantum optimization methods.

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### 7.1.1 Introduction to QUBO

Many real-world problems can be formulated as optimization problems. Given a set of possible solutions, the **goal is to find the one that minimizes** (or **maximizes**) a certain **objective function**.

Examples include: scheduling tasks, routing vehicles[11], portfolio optimization (finance), resource allocation (logistics), and machine learning optimization (training models). A **challenge** is that these **problems often appear very different from one another**. Instead of developing a completely different solution method for every problem, it is useful to express them using a common mathematical formulation.

One of the most important formulations is the **Quadratic Unconstrained Binary Optimization (QUBO) model**. In a QUBO problem, the solution is represented by a collection of binary variables:

$$x_i \in \{0, 1\}$$

And the objective is to find the assignment of these variables that minimizes a cost (or energy) function. So QUBO is a **mathematical framework** for

[11]The Vehicle Routing Problem (VRP) is an optimization problem that asks: "*how should a fleet of vehicles serve a set of customers at the lowest cost while satisfying operational constraints?*". It is one of the most important problems in logistics, transportation, and supply chain management. We are given a depot (warehouse, distribution center, etc.), a set of customers that need service, and a fleet of vehicles. We need to determine: (1) which customers each vehicle should visit; (2) the order in which each vehicle should visit them; and (3) when the visits should occur (in some variants).

expressing optimization problems in terms of binary variables and a quadratic cost function.

### Mathematical Formulation

The general QUBO formulation is[12]:

$$y = \sum_i a_i x_i + \sum_{i>j} q_{ij} x_i x_j \tag{98}$$

Or, more compactly, in matrix form:

$$\arg\min_x y = \mathbf{x} Q \mathbf{x}^T \tag{99}$$

Where:

- $\mathbf{x}$ is a vector of **binary variables**. The **variables** of a QUBO problem can **only assume two values**:

  $$x_i \in \{0, 1\}$$

  Each variable therefore behaves similarly to a boolean variable: 0 represents **false** and 1 represents **true**. The number of variables is denoted by $n$, which represents the size of the optimization problem.

  #### Useful Property of Binary Variables

  Since the variables are binary, $x_i \in \{0, 1\}$, we always have $x_i^2 = x_i$. Indeed $0^2 = 0$ and $1^2 = 1$. This seemingly simple property is extremely important because it allows many optimization expressions to be simplified and rewritten in QUBO form. We will see examples of this in the next sections.

- $Q \in \mathbb{R}^{n \times n}$ is a matrix containing the coefficients of the problem. It contains all **coefficients defining the optimization problem**. The matrix may be:
  - **Symmetric**, meaning $q_{ij} = q_{ji}$ for all $i$ and $j$ (i.e., the values above and below the diagonal are the same).
  - **Upper triangular**, meaning $q_{ij} = 0$ for all $i > j$ (i.e., only the values on and above the diagonal are non-zero).

  Also:

  - The diagonal elements $q_{ii} = a_i$ represent the **contribution of individual variables**;
  - While the off-diagonal elements $q_{ij}$ represent the **interaction between pairs of variables**.

- $y$ is the **objective function to be minimized**.

[12]The coefficients $a_i$ and $q_{ij}$ are sometimes called **biases** and **couplers**, respectively. The origin of this terminology will become clear when introducing the Ising model.

### Understanding the QUBO Acronym

The name **QUBO** stands for **Quadratic Unconstrained Binary Optimization**. Each word describes an important characteristic of the formulation. Understanding the meaning of the acronym helps clarify both the capabilities and limitations of QUBO models:

- **Quadratic**. A QUBO objective function can contain:
  - Linear terms $a_i x_i$, which represent the contribution of individual variables.
  - Quadratic terms $q_{ij} x_i x_j$, which represent the interaction between pairs of variables.

  However, it cannot contain higher-order terms such as cubic $x_i x_j x_k$ or quartic $x_i x_j x_k x_l$, or any polynomial term of degree greater than two. For this reason, QUBO problems are said to be **quadratic**. The restriction to quadratic interactions is particularly important because it allows the problem to be mapped naturally onto physical systems whose energy depends on pairwise interactions (e.g., Ising models in physics, we will see examples of this in the next sections).

- **Unconstrained**. In its pure form, a **QUBO problem does not include explicit constraints**. For example, expressions such as $x_1 + x_2 \leq 1$ or $x_1 = x_2$ cannot be imposed directly. Instead, constraints must be incorporated indirectly by modifying the objective function and introducing suitable penalty terms. This topic will be discussed in more detail in the next sections, page **??**.

  **Why can't QUBO include explicit constraints?** In a normal optimization problem, suppose we want to minimize $\min f(x)$ subject to $x_1 + x_2 = 1$. This is a **constrained optimization problem** because the solver knows objective function $f(x)$ and the constraints $x_1 + x_2 = 1$ and can use both to find the optimal solution.

  In contrast, the **QUBO solver** receives only $\min \mathbf{x} Q \mathbf{x}^T$ and has no information about any constraints. Therefore, the solver cannot enforce the constraint $x_1 + x_2 = 1$ directly. Instead, we need to modify the objective function to include a **penalty term that discourages solutions that violate the constraint**. For example, we could add a term like $P(x_1 + x_2 - 1)^2$ to the objective function, where $P$ is a large positive constant. This way, any solution that does not satisfy $x_1 + x_2 = 1$ will incur a large penalty, effectively enforcing the constraint indirectly.

- **Binary**. The **decision variables are binary** $x_i \in \{0, 1\}$. Each variable can therefore assume only two possible values: 0 or 1. This property makes QUBO particularly suitable for representing logical decisions such as: yes / no; true / false; selected / not selected; active / inactive. The binary nature of the variables will later allow a direct connection with qubits and spin systems.

- **Optimization**. The goal of a QUBO problem is to find the **variable assignment that minimizes the objective function**:

$$\arg\min_x y = \arg\min_x \mathbf{x} Q \mathbf{x}^T$$

Among all possible configurations of the binary variables, we search for the **one associated with the lowest cost** (or energy) value.

This interpretation is particularly important because later in the chapter the objective function will be viewed as an **energy landscape**, and solving the optimization problem will become equivalent to finding the **ground state** of a quantum system.

**Why is QUBO Important?** One of the **main advantages** of the QUBO formulation is that many **difficult optimization problems** can be **rewritten using this common framework**. Once a problem has been converted into QUBO form, the same optimization machinery can be applied regardless of the original application domain. In other words, the true strength of QUBO is the ability to serve as a universal language for optimization problems.

For this reason, technologies capable of solving QUBO problems efficiently may have a significant impact across many fields of science and engineering.

### 7.1.2 QUBO as an Energy Minimization Problem

A useful way to think about a QUBO problem is not in terms of optimization, but in terms of **energy**. Instead of saying "*find the variable assignment that minimizes the objective function*", we can equivalently say "***find the configuration of variables with the lowest energy***". In fact, the objective function:

$$y = \mathbf{x}Q\mathbf{x}^T$$

Can be interpreted as an **energy function** associated with a particular configuration of the binary variables. Every possible assignment of the variables corresponds to a different energy value.

#### Configurations and Energies

Consider a simple problem with two **binary variables**:

$$x_1, x_2 \in \{0, 1\}$$

There are **four possible configurations** of these variables:

$$(0, 0)\,,\ (0, 1)\,,\ (1, 0)\,,\ (1, 1)$$

For each configuration, the **QUBO function produces** a numerical value:

$$y(x_1, x_2)$$

This value can be interpreted as the **energy associated with that configuration**.

| Configuration | Energy |
|---|---|
| $(0, 0)$ | $y(0, 0) = E_1$ |
| $(0, 1)$ | $y(0, 1) = E_2$ |
| $(1, 0)$ | $y(1, 0) = E_3$ |
| $(1, 1)$ | $y(1, 1) = E_4$ |

The optimization problem then becomes the **search for the configuration with the smallest energy**. In physics, the state with the lowest possible energy is called the **Ground State**. Thus, solving a QUBO problem is equivalent to finding:

$$\mathbf{x}_{\text{opt}} = \arg\min_{\mathbf{x}} \mathbf{x}Q\mathbf{x}^T$$

Where $\mathbf{x}_{\text{opt}}$ is the optimal solution, i.e., the configuration corresponding to the ground state of the energy function. This terminology is extremely important because later we will map the QUBO objective function to a quantum Hamiltonian. At that point, the optimization problem will become simply the problem of finding the ground state of a quantum system. In other words, the **mathematical problem remains the same**; only the **interpretation changes**.

### Energy Landscape (useful for visualization)

It is often useful to visualize a QUBO problem as an **energy landscape**. Each possible **configuration of the variables** corresponds to a **point in this landscape**, while the **value** of the **objective function determines its height**:

- ✗ High energy: **undesirable** solution.
- Low energy: **desirable** solution.
- ✓ Minimum energy: **optimal** solution (ground state).

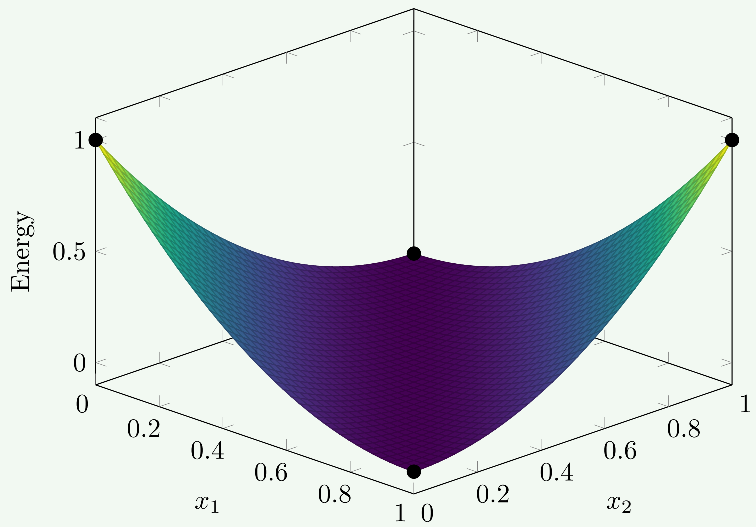
Conceptually, solving the optimization problem is like searching for the deepest valley in the landscape. The global minimum corresponds to the solution of the QUBO problem.

**Example 1: QUBO Energy Landscape**

For example, consider the following QUBO function:

$$E(x_1, x_2) = 1 - x_1 - x_2 + 2x_1x_2$$

This function assigns an energy value to each of the four configurations of the binary variables $x_1$ and $x_2$. We can visualize this energy landscape in a 3D plot, where the $x$ and $y$ axes represent the **binary variables**, and the $z$ axis represents the **energy value**. The points corresponding to the four configurations will have different heights based on their energy values, and the optimal solutions will be the points with the lowest height (energy).




In summary, the QUBO formulation can be viewed as an energy minimization problem, where the objective function represents an energy landscape. The goal is to find the configuration of binary variables that corresponds to the lowest energy, which is equivalent to solving the original optimization problem. This perspective is particularly useful when we later map QUBO problems to quantum Hamiltonians, as it allows us to leverage concepts from quantum mechanics to find optimal solutions.

### 7.1.3 Boolean Logic in QUBO

One of the **strengths** of the QUBO formulation is that logical conditions can be translated into energy functions. The **general strategy** is remarkably simple:

1. **Identify** the configurations that satisfy the desired condition.
2. **Assign** them the lowest possible energy.
3. **Assign** higher energy to all other configurations.
4. **Construct** a quadratic function that reproduces those energy values.

The **minimum of the resulting** energy function will then correspond exactly to the **solutions we want**.

#### Example: XOR

Let's see how this works in practice with a simple example: the **logical XOR** of two binary variables $x_1$ and $x_2$. The XOR condition is satisfied only when exactly one of $x_1$ and $x_2$ is 1: $a \neq b$. The truth table for XOR is:

| $x_1$ | $x_2$ | XOR$(x_1, x_2)$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

The goal is to find the assignments that make the boolean function true.

**From Logic to Energy.** Instead of directly solving the logical expression, we assign energies to each configuration. A natural choice is:

- **Energy 0** for the **desired** solutions (those that satisfy the XOR condition): $(0, 1)$ and $(1, 0)$.
- **Energy 1** for the **undesired** solutions (those that do not satisfy the XOR condition): $(0, 0)$ and $(1, 1)$.

This gives us the following energy landscape:

| $x_1$ | $x_2$ | XOR | Energy |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |

Now the optimization problem is equivalent to finding the configurations with minimum energy.

**Constructing the Energy Function.** We now need to find a quadratic function $E(a,b)$ that reproduces the energy values of the table. For this simple case, we can build it incrementally:

1. The configuration $(a,b) = (0,0)$ must have energy 1. So a constant temr $E = 1$ is a good starting point. However, all four configurations would then have energy 1, so additional terms are required.

2. Next, notice that the configurations $(0,1)$ and $(1,0)$ should have energy 0. Subtracting $a$ and $b$ lowers the energy whenever one of the variables becomes 1:
$$E(a,b) = 1 - a - b$$
But now the configuration $(1,1)$ has energy $-1$, which is too low.

3. To fix last configuration, we add a quadratic term $2ab$ that increases the energy when both variables are 1:
$$E(a,b) = 1 - a - b + 2ab$$
Now the energy function correctly assigns energy 0 to the desired configurations and energy 1 to the undesired ones:

| $x_1$ | $x_2$ | $E(x_1, x_2)$ |
|---|---|---|
| 0 | 0 | 1 - 0 - 0 + 2(0)(0) = 1 |
| 0 | 1 | 1 - 0 - 1 + 2(0)(1) = 0 |
| 1 | 0 | 1 - 1 - 0 + 2(1)(0) = 0 |
| 1 | 1 | 1 - 1 - 1 + 2(1)(1) = 1 |

Therefore, the logical problem of finding the XOR of two binary variables has been transformed into an optimization problem of minimizing the energy function:
$$E(a,b) = 1 - a - b + 2ab$$
The minimum energy configurations correspond exactly to the solutions of the original logical problem.

**Equivalence of Formulations.** In QUBO, the absolute value of the energy is usually irrelevant. What matters is where the minima are located. In other words, once we have a function that correctly identifies the desired configurations as minima, ***is the energy function unique?*** The answer is **no**. Different energy functions may represent the same optimization problem, provided that they preserve the same ground states.

In this example, the formulation could be modified by adding or subtracting a constant without changing the location of the minima. For instance, we could add a constant term $-1$ to the energy function:

$$E'(a,b) = E(a,b) - 1 = (1 - a - b + 2ab) - 1 = -a - b + 2ab$$

Evaluating the new function:

| $a$ | $b$ | $E'(a,b)$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | -1 |
| 1 | 0 | -1 |
| 1 | 1 | 0 |

These two formulations are equivalent in terms of optimization, since they have the same minima. The choice between them is a matter of convenience and does not affect the solution of the problem. More generally, **two QUBO formulations** are considered **equivalent** if they **preserve the same ground states**, even when their energy values differ.

This flexibility is extremely useful when constructing QUBO models. A modeler is free to choose the energy function that is easiest to derive, combine with constraints, or implement, as long as the desired solutions remain the ground states.

### Visualizing the Energy Landscape

The **energy values associated with the four possible assignments can be visualized as a discrete energy landscape**. Each point corresponds to one configuration of the binary variables, while the vertical axis represents the energy assigned by the QUBO function.

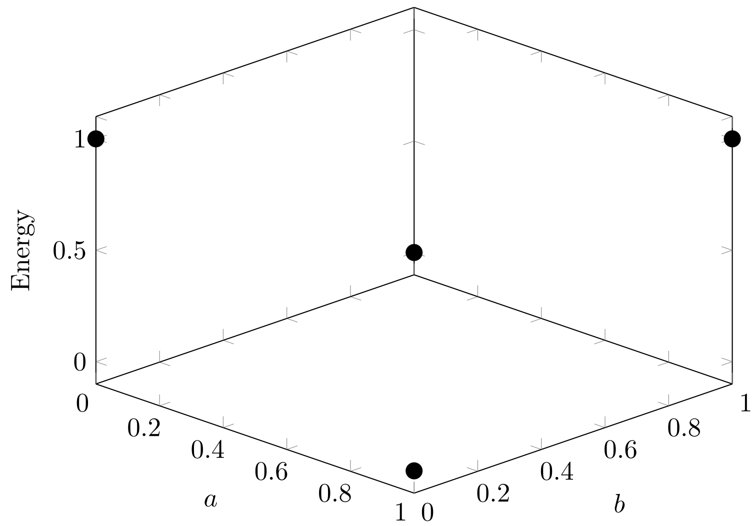



Figure 18: Discrete energy landscape of the QUBO formulation for the boolean function $a \neq b$. The ground states correspond to the minimum-energy configurations $(0,1)$ and $(1,0)$.

## 7.2 Continuous Variables and Constraints

### 7.2.1 Continuous Variables through Binary Expansion

This section addresses the first practical limitation of QUBO: **variables must be binary**. Up to now this was not a problem because we were working with Boolean variables $a, b \in \{0, 1\}$. However, most real-world optimization problems involve quantities such as temperatures, distances, costs, production levels, or resource allocations, which are usually **represented by continuous or integer-valued variables**. The natural question is therefore, "*how can a QUBO formulation represent variables that are not binary?*". The answer is **binary expansion**.

#### Binary Expansion

Recall how ordinary binary numbers are represented. For example, the number 13 can be represented in binary as:

$$13 = 1 \cdot 2^3 + 1 \cdot 2^2 + 0 \cdot 2^1 + 1 \cdot 2^0 = (1101)_2$$

The same principle can be used inside a QUBO formulation. Instead of representing a variable $x$ directly, we write:

$$x = \sum_k 2^k x_k \qquad x_k \in \{0, 1\} \tag{100}$$

Each binary variable $x_k$ determines whether the corresponding power of two contributes to the value of $x$.

Suppose we want to represent a real variable using binary variables. One possible encoding is:

$$x = \sum_{k=-2}^{2} 2^k x_k$$

Where $k$ ranges from $-2$ to 2. Expanding the powers of two, we get:

$$x = \frac{1}{4}x_{-2} + \frac{1}{2}x_{-1} + x_0 + 2x_1 + 4x_2$$

Each coefficient corresponds to a different binary weight.

- ***What values can be represented?*** Since each variable is binary $x_i \in \{0, 1\}$, the smallest representable value is obtained when all variables are zerp, $x_{\min} = 0$. The largest representable value is obtained when all variables are one, $x_k = 1$ for all $k$, which gives:

  $$x_{\max} = \frac{1}{4} + \frac{1}{2} + 1 + 2 + 4 = \frac{31}{4} = 7.75$$

  Therefore the representation can descibe values between 0 and 7.75.

- ***Precision***. The smallest coefficient is $\frac{1}{4}$, as a consequence, the representable values differ by multiplies of 0, 25. This quantity is called the **resolution** (or *precision*) of the representation. Adding more negative powers of two would increase the precision, while adding more positive powers of two would increase the maximum representable value.

**Who decides the range and precision of the representation?** The **modeler decides both the expansion and the precision**. For instance, suppose our original optimization problem contains a variable $x \in [0, 5]$. To convert the problem into QUBO, we must decide how to encode $x$ using binary variables. If we choose the same expansion as before, we can represent values up to 7.75, which is sufficient to cover the range of $x$. In general, **higher precision requires more binary variables**, which **increases the size of the QUBO problem** and can make it **more difficult to solve**. Therefore, there is a trade-off between precision and computational complexity that the modeler must consider when choosing the binary expansion.

In general, **Binary Expansion** is simply the idea of **representing a number as a sum of powers of two**. It is the binary equivalent of decimal notation. In the context of QUBO, it allows us to **replace one continuous** (or integer) **variable with several binary variables**.

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